Foundation II: The Chiral Universe and the Three Fossils of the Bounce

Extended Version

Investigating Baryogenesis, the Geometric Dark Sector, and Cosmic Alignment within an Einstein-Cartan Framework

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Open Science & Reproducibility: The source code, datasets, visual animations, and high-resolution figures are publicly available. A master orchestrator script (`run_all_simulations_F2.py`) is provided in the repository to automatically execute all numerical calculations and regenerate every plot presented in this paper. Access the complete simulation suite at: `https://github.com/pfichant/spin-torsion-cosmology`.

Abstract

Abstract. Context. While ΛCDM provides a robust description of cosmic structure, persistent tensions in H_0, S_8, early galaxy formation (JWST) and dynamical dark energy (DESI DR1–DR2) motivate exploring geometric alternatives to the dark sector. Building on the Einstein–Cartan Framework (ECF) non-singular bounce [5] and the Topological Invariance Principle [4]—whose background calibration leaves open the joint confrontation of $H_0 = 73.04\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and $\Omega_{\rm tot} = 1$ with the late-time distance observables (open problem PO-F2-5): with $\Omega_m = \omega_m/h^2 \simeq 0.267$ the chronometric bound $t_0 \geq t_{\rm GC} = 13.32\,\mathrm{Gyr}$ [34] is met at $w = -1$ ($t_0 \simeq 13.33\,\mathrm{Gyr}$), and the active tension lies in the BAO distances and in the DESI (w_0, w_a) plane—this work investigates *Geometric Monism*: dark phenomena emerging from primordial spacetime topology rather than undetected particles.

Baryogenesis. The same chiral torsion background *addresses* the three Sakharov conditions without beyond-Standard-Model CP violation, as a candidate scenario: their complete realisation is not established and no Boltzmann network is integrated here. The pre-dilution asymmetry is normalised on the observed value, $\eta_{\rm raw} = \eta_{\rm obs}/0.50$ (0.50 being the dilution factor of earlier versions), and the torsion dilution factor then gives $\eta = n_B/n_\gamma \simeq 5.9 \times 10^{-10}$ (`Scripts_F2/derive_f_dilution_baryogenesis.py` – reproduces the calculation below; see §3.4 for the resolved dilution factor f). The 3.5% agreement with the Planck-inferred value is therefore *not* an independent test. This is a *semi-derived* result: stage (i) follows from first principles, stage (iii) is semi-analytic, and stage (ii) is normalised on observation.

Dark sector. Torsion crystallization near the electroweak scale ($T \sim 159$ GeV), via the Kibble–Zurek mechanism [30, 31] combined with the geometric chiral-splitting mechanism *introduced here* (the citation covers Kibble–Zurek only; chiral splitting is the ECF's own addition), yields a 5:1 dark-to-baryonic energy partition from degrees-of-freedom equipartition over the 20:4 decomposition of the torsion tensor (Table 14); this is a *model-dependent consistency result* within the Weyssenhoff approximation, introducing no fitted parameter, *not* a parameter-free first-principles derivation. Planetary-mass Micro-Knots ($M \sim 10^{24}$ kg) are proposed as granular geometric dark matter; the pseudo-isothermal halo profile adopted for them (not yet derived from the torsion, PO-ECF-HALO) fits the SPARC database with $\tilde{\chi}^2_{\rm med} = 0.80$ (171 of 175 galaxies, see *Sample definition*; two parameters per galaxy; Fig. 29) — a goodness-of-fit of the adopted profile, not a torsion-specific test. The background torsion condensate acts as dynamical dark energy with $(w_0, w_a) = (-0.904, -0.153)$, inherited from the Foundation I calibration and *not* derived *ab initio* (cf. PO-F2-4); w_a is set through $\xi = 1.645$, a parameter calibrated on the DESI preference used by the ECF, and is therefore not a prediction independent of DESI. That trajectory crosses the phantom divide at high z; for the adopted Proca form of the axial sector the absence of ghosts is invoked from the literature (Appendix F establishes only $\lambda_T > 0$), which is a necessary condition, but the mechanism licensing the crossing is not derived here.

What the DESI comparison does and does not establish. A full-covariance comparison against DESI DR1 (Foundation III) places the ECF point at 2.03σ (Planck prior) and 6.98σ (SH0ES prior) from the MCMC central value; it is *not* favoured over ΛCDM by those data alone ($\Delta\chi^2 = +6.21$ and $+38.59$). What the comparison does establish is that the prediction preceded the data.

Topological structure. The defect spectrum is classified by the homotopy groups of the chiral vacuum manifold ($\pi_2 \to$ 0D Micro-Knots, $\pi_0 \to$ 2D chiral walls, uncrystallised residue $\to$ 3D chiral fluid), unifying dark matter and dark energy as a topological hierarchy (§6). A candidate 1D sector (torsion strings) was investigated in detail and is retained as an explicitly open research problem (PO-F2-8): the framework's Lagrangian does not currently supply the complex order parameter such a defect requires. The same hierarchy motivates a conditional heavy-seed channel ($M_{\rm seed} \sim 10^5$–$10^6\,M_\odot$, a causal scale; collapse efficiency and abundance not derived) for the JWST massive-galaxy tension and Little Red Dots, a scale that overlaps the seed mass of Direct Collapse Black Hole models [121] *(corrected 19/10/2026: was "in quantitative agreement with").*

Falsifiable predictions. Roman Space Telescope microlensing events at $t_E \sim 1$–10 h ($N_{\rm ev} \approx 1400$ over 72 days for a halo made entirely of aggregates, a case the OGLE-IV bulge survey already disfavours, open problem PO-F2-10; falsified absent an excess over the free-floating-planet baseline); a *blue* stochastic gravitational-wave energy spectrum, where inflation gives a red one —a difference of *sign*, not of precision. *What the framework predicts here is the blue enhancement of the transferred energy spectrum, produced by the stiff phase; the primordial tensor tilt $n_T^{\rm prim}$ itself is not established by that transfer alone*; a chiral asymmetry $\Pi_{\rm GW} \simeq 0.20 \pm 0.05$, a *calibrated template* rather than a derived prediction (the chiral sector is disfavoured if $\Pi_{\rm GW} < 0.02$ at 95% C.L.); LISA probes the slope, but its access to the handedness of an isotropic background is limited to the kinematic dipole [203], and a direct measurement would require a LISA–Taiji network [204]; SPARC rotation-curve granularity $\delta V_{\rm rot} \geq 0.05$ (falsified if consistent with a smooth NFW halo); a void over-abundance peaking at $R \simeq \xi_{\rm KZ} \simeq 60$ Mpc, of amplitude $1 + \alpha_{\rm KZ} \simeq 2.1$ at the peak of the adopted spatial template (adopted values, PO-F3-XI; Euclid DR1, mid-2027). The *anti-parallel* galaxy–filament spin correlation reported in MaNGA kinematics [197] is no longer listed among these tests: the 1D sector that carried its mechanism is an open problem (PO-F2-8), and no quantitative ECF prediction is attached to it.

Scope and limitations. Connections to the CMB Axis of Evil, Odd Radio Circles, long-period radio transients, Gamma-Ray Bursts and the Big Ring/Giant Arc mega-structures are treated as exploratory consistency checks, not primary evidence; the same holds for the preliminary Topological Distance Ladder via cosmic birefringence [71, 72]. The $G_{\rm eff}(k)$ regulator inherited from Foundation I [5] is a phenomenological proxy ($A_{\rm sup} = 0.15$, no longer presented as calibrated on $S_8 = 0.766$; no MCMC); with the internal closure $\Omega_m = \omega_m/h^2 \simeq 0.267$ the reduction of S_8 comes mainly from the lower Ω_m (a CAMB ΛCDM reference on that background gives $S_8 \simeq 0.780$), and the residual correction to the target 0.766 depends on the ECF perturbative dynamics, which is not implemented; its first-principles derivation would require a dedicated Einstein–Cartan perturbation solver, not attempted here and constituting an open theoretical frontier (Sec. 11).

Keywords: Cosmology · Einstein–Cartan Theory · Spacetime Torsion · Poincaré Gauge Theory · Topological Defects · Geometric Dark Matter · Dynamical Dark Energy · Kibble–Zurek Mechanism · Chiral Baryogenesis · Baryon Asymmetry · Domain Walls · Cosmic Strings · JWST Tension · DESI BAO · SPARC Rotation Curves · LISA Gravitational Waves · Cosmic Birefringence · Vacuum Energy · Radio Transients · Gamma-Ray Bursts · Microlensing.

Version note. v20, PREPRINT v5 (Zenodo; supersedes PREPRINT v4, October 2026) — propagation of the second audit and of the October corrections. *Six data or figure corrections are signalled explicitly. (1) Void-statistics figure (Fig. 15): the points labelled "Observations (SDSS BOSS Catalog)" in v4 were simulated and the fit-statistics box was hard-coded; the figure is replaced by an illustrative mock slice. (2) NGC 6503: the v4 data file (24 points) did not match the SPARC file (31 points); the refit on SPARC gives $\chi^2_\nu = 1.68$ (not 0.97), and the rotation curve (Fig. 26) is redone. (3) NGC 6503 MCMC figure (Fig. 32): it was drawn from a synthetic normal distribution; it is now drawn from the actual chain (`mcmc_ngc6503_sparc.py`, made deterministic). (4) Mahalanobis distance ECF–DESI DR1: 2.03σ / 6.98σ (full covariance, 12 observables), replacing 2.22σ / 7.32σ, which used a 13-observable block including two rows that DESI DR1 does not publish. (5) BOSS error at $z = 0.61$: statistical and systematic errors combined (Alam et al. 2017). (6) Annihilation energetics: for $M_2 = 10^4\,M_\odot$, $2M_2c^2 = 3.6 \times 10^{58}$ erg, not 4×10^{56} erg; the GRB/FRB "calibration" $\eta_{\rm ann} \sim 0.1$–0.5 rested on that error and is* ***withdrawn*** *($\eta_{\rm ann}$ is open). Values and statuses changed: background $\Omega_m = \omega_m/h^2 \simeq 0.267$ (decision A-7; 0.315 kept only as the Planck reference), $t_0 = 13.33$ Gyr, and the age-tension narrative abandoned; recombination at 368.4 kyr; the global $\Delta\chi^2 = -39.5$ of Foundation I is historical and not recomputed; the ECF is* not *favoured by DESI DR1; $S_8 = 0.766$ is a KiDS-1000 target, not an ECF output; the Micro-Knot → Macro-Knot chain is*

withdrawn. The complete list, with the reason for each change, is in the Version register and erratum *deposited with this lot; in case of discrepancy, the text of the paper prevails. No open problem is closed by rewording.*

v17 correction (22/08/2026) — withdrawal of the compensation mechanism. *Earlier versions of this framework read the late-time dark-energy density off a "torsion debt" imposed by exact spatial flatness. That mechanism is* ***withdrawn*** *[4]: once $k = 0$, the closure relation $\sum_i \Omega_i = 1$ is an identity — the critical density being recomputed with H — and cannot enforce any compensating component, nor constrain $\rho_{\rm DE}$ or its equation of state. What Einstein–Cartan cosmology supplies* independently of that postulate *is the static term derived in §3.5, whose amplitude is normalised on the observed value. Consistently with the withdrawal, identifying $|\Omega_\tau|$ with the initial spin fraction* 0.093 *fails by a factor ~ 7; the primordial sector does not fix the late-time amplitude, and nothing here requires it to. v17, PREPRINT v4 (Zenodo; supersedes the published PREPRINT v3) — Appendix C deprecated: superseded by the Foundation III topological mechanism for horizon resolution (no numerical or physical content in the main results changed, see* v17 correction note *immediately below).* ***v17 correction (17/07/2026):*** *the labels "QCD epoch" / "EW epoch" / "QCD-Planck" attached to three reference points in the Macro-Knot Jeans-mass discussion (§9.2 and Fig. 33) were misassigned relative to the actual temperature at each quoted z_t: the $z_t \in [6.9 \times 10^8, 6.9 \times 10^9]$ window used throughout (giving, with the radiation-only expansion rate $H(z_t) = H_0\sqrt{\Omega_r}(1+z_t)^2$ and $\omega_r = 4.1834 \times 10^{-5}$, $M_K \simeq 10^7$ to $10^5\,M_\odot$ across the window) corresponds to $T \sim 0.16$–1.6 MeV, i.e. the* ***Great Annihilation*** *(e^+e^- freeze-out), not the QCD or electroweak transition. The true QCD epoch ($T \sim 235$ MeV) sits at $z_t \sim 10^{12}$ and yields $M_K \simeq 5\,M_\odot$ (same expression; $\simeq 1\,M_\odot$ with the evolution of g_*) — far below the LRD range, and unrelated to the quoted prediction. All numerical values (z_t, M_K, the LRD mass-range overlap) are* ***unchanged****; only the epoch names were corrected, in the text and in the regenerated figure (`plot_ECFLRDJeansMass.py` rev2).* [Corrected 04/10/2026: the M_K values in this entry previously read 10^5–$10^6\,M_\odot$ and $\sim 40\,M_\odot$; recomputed with the expression of the figure script.] ***v16 corrections (01/07/2026, cross-check with Foundation III):*** *(1) fixed internal inconsistency in §9.2 — the Micro-Knot inter-knot spacing (Eq.* (93)*) incorrectly quoted $\sim 230\,pc$; corrected to $\sim 0.07\,pc$ ($\sim 0.24\,ly$), consistent with the stated Micro-Knot number density. (2)* Reconciled: *the $\sim 230\,pc$ figure itself was found to conflate two incompatible population models — it applies a uniform local-density formula (valid for the* abundant *Micro-Knots, which genuinely fill space as a discrete fluid) to the* rare *Macro-Knots ($\mathcal{O}(1)$ per galaxy), for which such a formula is not physically meaningful: a single Macro-Knot per galactic halo carries only $\sim 10^{-7}$–10^{-6} of the total galactic dark matter budget, far too little to justify a "local density" treatment. The opening paragraph of §9.2 has been rewritten to state this explicitly; the appropriate Macro-Knot scale is the galaxy-to-galaxy separation, not a local spacing figure. (3) fixed the flux formula $\Phi_{\rm MK}$ in §9.2: the earlier value $3 \times 10^{-30}\,{\rm m^{-2}s^{-1}}$ carried both a dimensional inconsistency and a numerical error of $\sim 10^{11}$; corrected to $\Phi_{\rm MK} = n_{\rm MK} v \simeq 1.8 \times 10^{-41}\,{\rm m^{-2}s^{-1}}$ (one transit per $\sim 10^{33}$ yr through $A = 1\,m^2$), further strengthening the structural suppression argument. Full referee audit (v15): §Chiral Baryogenesis, Geometric DM, Macro Structure intros; Tables `tab:sparc_comparison`, `tab:limitations`, `tab:roadmap`; VSF equations; POLAR-2 $|V/I| \simeq 0.8$* (value withdrawn on 18/10/2026: $|V/I| \simeq \theta_T$, not derived; see §F.24)*; 3 detection windows; LPT context; magnetic/electric field analysis (all 4 defect levels); Micro-Knot ISM transit (T1–T6); planetary interactions (P4/P5/P7); topological network (H1–H4); filament rotation Wang+2021, Rong+2025. (3) fixed isolated numerical error in the Einstein radius derivation (§9.3) — an earlier $R_E \approx 5 \times 10^8$ cm figure was $\sim 86\times$ too small; corrected to 4.3×10^8 m, which turns out to already match the independently-derived $t_E^{\rm max} \approx 0.54$ h (full numerical sweep), resolving a previously unexplained internal discrepancy. Downstream Roman predictions were unaffected.* ***Further correction, 31/08/2026:*** *both the 4.3×10^8 m and 0.54 h figures above rested on a placeholder mass $M = 10^{24}$ kg in the companion script rather than the canonical $M_{\rm MK} = 3.74 \times 10^{24}$ kg used symbolically in the same derivation; corrected throughout to $R_E \approx 8.3 \times 10^8$ m, $t_E^{\rm max} \approx 1.05$ h. The EMF-corrected $t_E^{\rm typ} \approx 6.3$ h used for the Roman event-rate prediction was already on the canonical mass and is unaffected. (4)* ***New:*** *added an Extended Mass Function argument (§9.3) addressing compatibility with existing Subaru/HSC constraints, using a Poisson parametrisation of Micro-Knot aggregates whose mean λ is not derived from a coagulation dynamics (the corresponding lock, Foundation III PO-F3-C6, remains open). Full convolution against a log-linear approximation of the Niikura et al. (2019) exclusion curve yields $\lambda \geq 36$ for compatibility (typical mass $M_{\rm typ} \approx 6.8 \times 10^{-5}\,M_\odot$, Einstein timescale $t_E \approx 6.3\,h$), superseding the preliminary single-point check ($\lambda > 6.9$ at $N = 1$ only). Reference: Green [173]. 165 bibitems, 0 undefined refs. Numerical results otherwise unchanged from PREPRINT v1.*

v17 corrections (17/07/2026, epoch-label fix, cross-checked against the companion figure script `plot_ECFLRDJeansMass.py`): *The Macro-Knot Jeans-mass passages (§9.2, twice, and the `fig:lrd_jeans`/`fig:lrd_jeans_rusakov` figure and its caption, five occurrences in total) labelled the freeze-out epoch relevant to $z_t \sim 6.9 \times 10^8$–6.9×10^9 as the "QCD-to-electroweak transition" ($T \sim 150$ MeV–159 GeV). This label was wrong: at $T_0 = 2.35 \times 10^{-4}$ eV, that redshift window corresponds to $T \sim 0.16$–1.6 MeV — the* ***Great Annihilation*** *(e^+e^- annihilation) freeze-out already derived correctly elsewhere in this document (Eqs.* (43)–(42)*, $T \sim 1$ MeV, $t \sim 1$ s), not the QCD/electroweak scale.* No numerical value changes*: z_t, $H(z_t)$, and M_K are all unchanged and were already correct; only the physical-epoch name attached to them is corrected, in the text and in the regenerated figure `Fig_ECFLRDJeansMass.png` (script updated to v2, same date). The true QCD confinement transition ($T \sim 235$ MeV) is a separate, much later point ($z_t \sim 10^{12}$), previously mislabelled "QCD-Planck" in the figure and now correctly labelled "QCD epoch". This correction was prompted by an unrelated cross-check of a similar epoch-labelling issue in the PIT letter and Foundation I (v10.1); verifying F2 against the same error pattern surfaced this instance.*

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v4 correction (September 2026) — notational and scope corrections; four of the five points below change no numerical result, the fifth does. *Found by cross-checking this paper against companion documents and the scripts that produce its figures, after the 22/08/2026 deposit.* ***(i)*** *The sign of the post-bounce spin-density term in the Friedmann equation is now stated as calibrated, not derived, consistently with Foundation I; the gap is tracked as PO-F1-SIGMA0.* ***(ii)*** *The spin sector is normalised at $z_{\rm ref} = 5600$, the value used by `script_01_solve_sound_horizon.py`; several passages previously quoted $z \simeq 7500$, a value not used in any script – the two anchors give sound horizons compatible at 1.25σ, so this is a traceability correction, not a repair of a wrong result.* ***(iii)*** *Five passages stating that the phantom-divide crossing was "stabilised by the PGT Hamiltonian constraint" are narrowed: Appendix F establishes only $\lambda_T > 0$ (potential bounded from below); the absence of ghosts is invoked from the literature for the adopted Proca form – a necessary condition – and neither treats the crossing; the mechanism licensing it is not derived (PO-F1-SIGMA0).* ***(iv)*** *Appendix cross-references, broken by unnumbered `section*` declarations (a rendering defect invisible at compilation: the reference was defined, just pointing at the wrong number), now resolve to the correct letter throughout (ten appendices checked).* ***(v)*** This one changes a number. *Eq.* (93) *(§9.2) divided $\rho_{\rm DM}^{\rm local}$ by 6×10^{27} g, inconsistent with the $M_{\rm MK} = 3.74 \times 10^{24}$ kg ($= 3.74 \times 10^{27}$ g) stated one line above – a factor-1.6 error that survived the v16 correction of a different mistake in the same paragraph. Corrected: $n_{\rm MK} \simeq 1.34 \times 10^{-52}\,{\rm cm}^{-3}$ (was 8×10^{-53}), inter-knot spacing $\simeq 0.063$ pc (was 0.07 pc* – note: an intermediate version of this correction, 07/09/2026, mistakenly reported this as increasing to 0.082 pc; the banc de test's own double-check caught the sign error – spacing scales as $n^{-1/3}$, and n increased, so spacing must decrease, not grow*), and the downstream flux Eq.* (94) *rescaled to $2.9 \times 10^{-41}\,{\rm m}^{-2}{\rm s}^{-1}$ (was 1.8×10^{-41}) with its transit interval to 6×10^{32} yr (was 10^{33};* 15/10/2026: the interval implied by 2.9×10^{-41} is 1.1×10^{33} yr, and both values were superseded by the switch to $0.4\,{\rm GeV\,cm}^{-3}$*) – the qualitative conclusion (immeasurably rare) is unaffected. Found by the banc de test's alias-checking pass, cross-checked and corrected by the same, 07/09/2026.*

Part I

Theoretical Foundations

ECF Formalism and Chiral Baryogenesis

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1 Introduction: The Vector Heritage and the Unified Dark Sector

The Einstein-Cartan Framework (ECF) explored in this series rests on a single foundational claim: the cosmological dark sector — dark matter, dark energy, and the baryon asymmetry — is a geometric consequence of primordial spacetime torsion, seeded during a non-singular spin-torsion bounce and crystallised into a hierarchy of topological defects during the electroweak phase transition. This is *not* a new dark matter particle, a new scalar field, or a modified gravity theory. It is a geometric extension of General Relativity in which the torsion tensor, sourced by fermionic spin density, plays the structural role assigned to CDM, Λ, and the inflaton. Foundation II (this paper) establishes the microscopic sector: chiral baryogenesis (§3), the Topological Zoology of the dark sector (§6), and the macroscopic structure formed by the resulting defect hierarchy (§7). All background calibration (H_0, w_0, w_a, $F_{\rm ion}$) is inherited from Foundation I [5] and the Topological Invariance Principle [4]; no re-fitting is performed here. This manuscript should be read as the microscopic and phenomenological continuation of Foundation I. The background calibration of the ECF expansion history—including $F_{\rm ion} = 1.2765$, $\Omega_{\rm spin}^{\rm peak} = 0.093$, and the effective CPL trajectory $(w_0, w_a) = (-0.904,$
$-0.153)$—is calibrated in Foundation I [5] and from the Topological Invariance Principle (TIP) [4]. To be precise: Foundation I establishes the H_0–Age–BAO trilemma and derives the best-fit CPL proxy; Foundation II (this paper) shows that the ECKS action admits an effective CPL trajectory of the same functional form, without a scalar field; its two parameters (w_0, w_a) are fixed by the Foundation I calibration rather than derived *ab initio*. What is derived here is the *form* of the effective equation of state (Eq. (22)) and the mechanism that produces a mild departure from $w = -1$; the specific numerical values $(-0.904, -0.153)$ remain calibration outputs, not first-principles predictions, pending an independent determination of the torsion-condensate dynamics $\langle \delta S^2 \rangle(z)$ (open problem, see below). In the present work, these quantities are used as fixed calibration inputs; they are not re-fitted. Accordingly, claims of parameter economy below mean "no additional dark-sector parameter beyond the Foundation I calibration," not a derivation from a parameter-free fundamental theory. **The calibration of (w_0, w_a) is separate from, and does not by itself resolve, the H_0–Age–BAO trilemma**: a systematic scan of the tanh-crossing scalar $w(z)$ family — the functional class to which the calibrated (w_0, w_a) trajectory belongs — finds zero viable points out of 2500 simultaneously satisfying the BAO and age constraints (Foundation I, Appendix L, script `plot_trilemma_irreducibility.py`; rerun at $\Omega_m \simeq 0.267$ with the 30.21 target: 187 BAO-viable and 1857 age-viable points, none satisfying both). This irreducibility is no longer driven by an age deficit: with $\Omega_m = \omega_m/h^2 \simeq 0.267$ the $w = -1$ background already gives $t_0 \simeq 13.33$ Gyr, so the stronger phantom crossing formerly invoked for the age ($w_0 \approx -1.1$, $w_a \approx -1.5$; $t_0 \simeq 14.03$ Gyr on the same background) is not needed for it, and the BAO cost quoted for it ($\Delta\chi^2_{\rm BAO} = 38$) is a historical value at $\Omega_m = 0.315$, not recomputed. The active tension lies in the BAO distances and in the DESI (w_0, w_a) plane. **Full resolution of the trilemma remains an open problem**, tracked as PO-F2-5 in the conclusion (§11), not a result established by the present calibration.

1.1 Precision Cosmology and Standard Model Limits

The concordance model (ΛCDM) has been remarkably successful in describing the large-scale structure and evolution of the Universe. However, increasing precision of modern datasets reveals persistent tensions. We owe a significant debt of gratitude to observational efforts such as the Planck Collaboration [8] and structural surveys, which highlight anomalies that standard theory struggles to accommodate without extensive fine-tuning. Beyond kinematic tensions (H_0, S_8) [106], CMB low-multipole alignments challenge isotropy, while recent radio observations reveal transients inviting broader perspectives. These discrepancies motivate exploring frameworks beyond scalar-inflation: Most critically for the background, the joint confrontation of $H_0 = 73.04\,{\rm km\,s^{-1}\,Mpc^{-1}}$ and $\Omega_{\rm tot} = 1$ with the late-time distance observables remains open (PO-F2-5). With the internal closure $\Omega_m = \omega_m/h^2 \simeq 0.267$, a constant equation of state $w = -1$ gives $t_0 \simeq 13.33\,{\rm Gyr}$, compatible with the age of the oldest globular clusters, $t_{\rm GC} = 13.32 \pm 0.10_{\rm stat} \pm 0.5_{\rm sys}\,{\rm Gyr}$ [34], used here as a chronometric lower bound on t_0 rather than as a measurement of it: the age deficit of ear-

lier versions (obtained with $\Omega_m = 0.315$) does not arise, and the difficulty is displaced to the BAO distances and to the (w_0, w_a) trajectory confronted with DESI. This open problem, rather than an age deficit, is the background motivation retained for the present work.

Delayed dark-matter genesis scenarios, in which the dark sector undergoes an independent phase transition well after the Hot Big Bang [209], correctly identify that dark matter need not be produced at $t = 0$. However, such models require an ad hoc dark-sector scalar potential $V(\phi)$ with at least two free parameters ($T_\star$, ΔV). The ECF achieves the same delayed chronology from the ECKS torsion algebra alone, with $T_\star \simeq 200\,\mathrm{MeV}$ set by the QCD scale and no additional field.

A significant fraction of the community responds to the JWST early-galaxy crisis by invoking suppressed feedback (feedback-free starbursts [136]), enhanced feedback, a top-heavy initial mass function, or dust-poor photometric biases [137]. While such corrections plausibly account for moderately over-massive galaxies at $z \sim 6$–8, they are pushed towards a structural ceiling: COSMOS-Web (2025) reports integrated star-formation efficiencies approaching the baryon limit (0.8–1) by $z \sim 10$, with large uncertainties [138]. Efficiencies above that ceiling, if found at $z > 10$, could not be produced by any feedback prescription. *(Corrected 22/10/2026: earlier versions stated that the limit was exceeded.)* The ECF Macro-Knot seeding mechanism ($M_{\rm seed} \sim 10^5$–$10^6\,M_\odot$, a causal scale; $\epsilon_{\rm coll}$ and abundance not derived; *corrected 19/10/2026: was "no additional free parameters"*) addresses this regime specifically by providing gravitational potential wells that pre-exist the baryonic assembly phase. Macro-Knots further provide a consistent phenomenological picture for the S_8 tension alongside the $G_{\rm eff}(k)$ regulator inherited from Foundation I (under the internal closure, the S_8 reduction comes mainly from $\Omega_m = \omega_m/h^2$) (a phenomenological proxy, not a calibration on $S_8 = 0.766$, whose first-principles derivation via a perturbed-ECKS Boltzmann solver is an open problem, not treated in the present corpus[1], Sec. 11), for the Slow Rotator population (Sec. 8), and for the CMB Axis of Evil. That the same mechanism bears on several independent signatures is a property no single-mechanism astrophysical feedback shares. We record it as a consistency, not as evidence: each of these signatures rests on a calibrated proxy or an open derivation, and the full perturbative treatment remains to be carried out.

We propose a modest geometric approach, focusing on *falsifiable topological remnants*: Micro-/Macro-Knots ($\tilde{\chi}^2_{\rm med} = 0.80$ SPARC), chiral GWs (LISA dipole channel or LISA–Taiji network), microlensing (Roman).

1.2 The ECF Thesis: From Scalar to Vector Geometric Monism

The central hypothesis of this work is that *the dark sector is the geometric imprint of primordial spin density*, not a new particle or field added to the Standard Model. This replaces three independent scalar mechanisms (Λ, CDM, inflaton) with a single vector structure (the Einstein-Cartan torsion tensor $T^\lambda{}_{\mu\nu}$), constrained by the $\mathrm{SL}(2,\mathbb{C})$ gauge symmetry of Poincaré Gauge Theory (PGT) [17, 166].

The three pillars.

(i) **Foundation I** (*metric sector*): the non-singular torsion bounce *identifies* the H_0–Age–BAO trilemma — which it does not resolve, see PO-F2-5 — and calibrates $(w_0, w_a) = (-0.904, -0.153)$, $F_{\rm ion} = 1.2765$, $\Omega^{\rm peak}_{\rm spin} = 0.093$ against SPARC, BAO, and CMB data.

(ii) **Topological Invariance Principle** (*flatness postulate*): posits $\Omega_K \equiv 0$ as exact. It does *not* derive (w_0, w_a), which are calibrated; see the PGT Hamiltonian without a scalar field [4].

(iii) **Foundation II** (*matter sector, this paper*): obtains $\eta_{\rm baryon} \simeq 5.9 \times 10^{-10}$ as a *semi-derived* result — stage (i) follows from first principles, stage (iii) is semi-analytic, and the pre-dilution asymmetry is normalised on the observed value ($\eta_{\rm obs}/0.50$), so that the agreement is not an independent test (see `Scripts_F2/derive_f_dilution_baryogenesis.py`) — together with $\tilde{\chi}^2_{\rm med} = 0.80$ on SPARC-175 *(a fit to data, not an ab initio result)* and the Topological Zoology ($\pi_2/\pi_1/\pi_0$/3D), all from

[1] *(Added 20/10/2026.)* Throughout this paper, "Foundation III" denotes the programme of three documents: *Foundation III — Cosmological Physics* (deposited, cited as [7]), and *Foundation III — Strong Gravity* and *Foundation III — Chiral Bounce* (in preparation, not deposited). Earlier versions deferred several questions to "Foundation III" without these documents treating them; each such deferral now states whether the question is treated, listed as an open item, or not treated in the present corpus.

the same ECKS action without additional observational fits beyond those named.

The term *Geometric Monism*, following Clifford (1876) and Wheeler (1962), refers to the programme of deriving all non-gravitational dark phenomena from the geometry of spacetime rather than from particle species. Cartan torsion provides the concrete realisation: Dark Matter as crystallised geometric knots (π_2 hedgehogs), Dark Energy as background torsion preserving flatness.

1.3 Relation to Competing Dark Matter Frameworks

Warm Dark Matter (WDM), Self-Interacting Dark Matter (SIDM), and Fuzzy Dark Matter (FDM) address the small-scale structure problems of ΛCDM—the core-cusp problem [100], the radial acceleration relation [101], and the challenge posed by MOND [102]—by modifying the dark-matter particle properties [139, 140]. These frameworks and the ECF Micro-Knot picture differ in *kind* rather than degree: WDM, SIDM and FDM modify the properties of a dark-matter *particle*, whereas the ECF replaces the particle altogether by a geometric population of topological torsion defects. They are therefore not competitors on a common scorecard but distinct answers to distinct questions — the small-scale halo problems for the former, the joint bounce/dark-sector picture for the latter. The most direct observational discriminator between a smooth particle halo and a granular geometric population is gravitational microlensing: smooth DM halos, whether cuspy (CDM, NFW [99]) or cored (WDM, SIDM, FDM [139, 140]), predict no anomalous events in the $t_E \sim$ 1–10 h window, while Poisson-coalesced ECF aggregates give ~ 1400 events over 72 days (for $f_{\rm DM} = 1$, which the OGLE-IV bulge survey already disfavours: PO-F2-10, §9.3) on the *conservative benchmark* $\tau_{\rm micro} = 5 \times 10^{-8}$ adopted in the event-rate script — a value *posited*, not derived (a uniform-density halo gives $5\text{–}7 \times 10^{-8}$). *A second, independent benchmark coexists in this work and must not be conflated with it*: the pseudo-isothermal estimate of Sec. 9.3 *computes* $\tau_{\rm micro} \approx 1.5 \times 10^{-7}$, which would raise the event count accordingly. The figure quoted here follows from the conservative chain (§9.3, §9.3).

1.4 Observational Fossils of the Bounce

The ECF chronology leaves three compact observational fossils (Fig. 1):

1. **Chiral asymmetry** ($t \approx 10^{-11}$ s): electroweak crystallization converts the handedness selected at the bounce into the initial matter excess.
2. **Geometric dark matter** ($t \approx 1$ s): frozen torsion defects survive the annihilation era and seed galactic halos.
3. **Cosmic alignment** ($t \approx 368{,}000$ yr): the CMB preserves a large-scale kinematic imprint of the primordial geometry.

This schematic is intended as a roadmap: it compresses the temporal sequence into one view before the detailed derivations of the following sections.

All simulations build reproducibly on *Foundation I* [5]. Section 5 develops the complete topological classification of ECF defects (0D–3D) via the homotopy groups of the chiral vacuum manifold, unifying dark matter and dark energy as manifestations of a single hierarchical defect spectrum (Table 2). Section 2 details the ECF mathematical foundations.

2 Theoretical Framework

A distinction to make before anything else: teleparallel gravity is not this. Torsion appears in two geometrically distinct programmes, and conflating them leads to an immediate and fatal objection to any framework of this kind. In *teleparallel* gravity the connection is that of Weitzenböck: curvature vanishes identically and gravitation is described *entirely* by torsion. That formulation admits a strict dictionary with General Relativity — the Teleparallel Equivalent of General Relativity reproduces its predictions exactly, and by construction can contain no new physics. A reader who took the present work to be of that kind would rightly conclude that it cannot predict anything General Relativity does not.

This framework is Einstein–Cartan–Sciama–Kibble, which is a different geometry. Curvature and torsion coexist as *independent* degrees of freedom: torsion is sourced by the intrinsic spin of fermions through the Hehl–Datta relation rather than replacing curvature, and the resulting dynamics is *not* equivalent to General Relativity where spin density

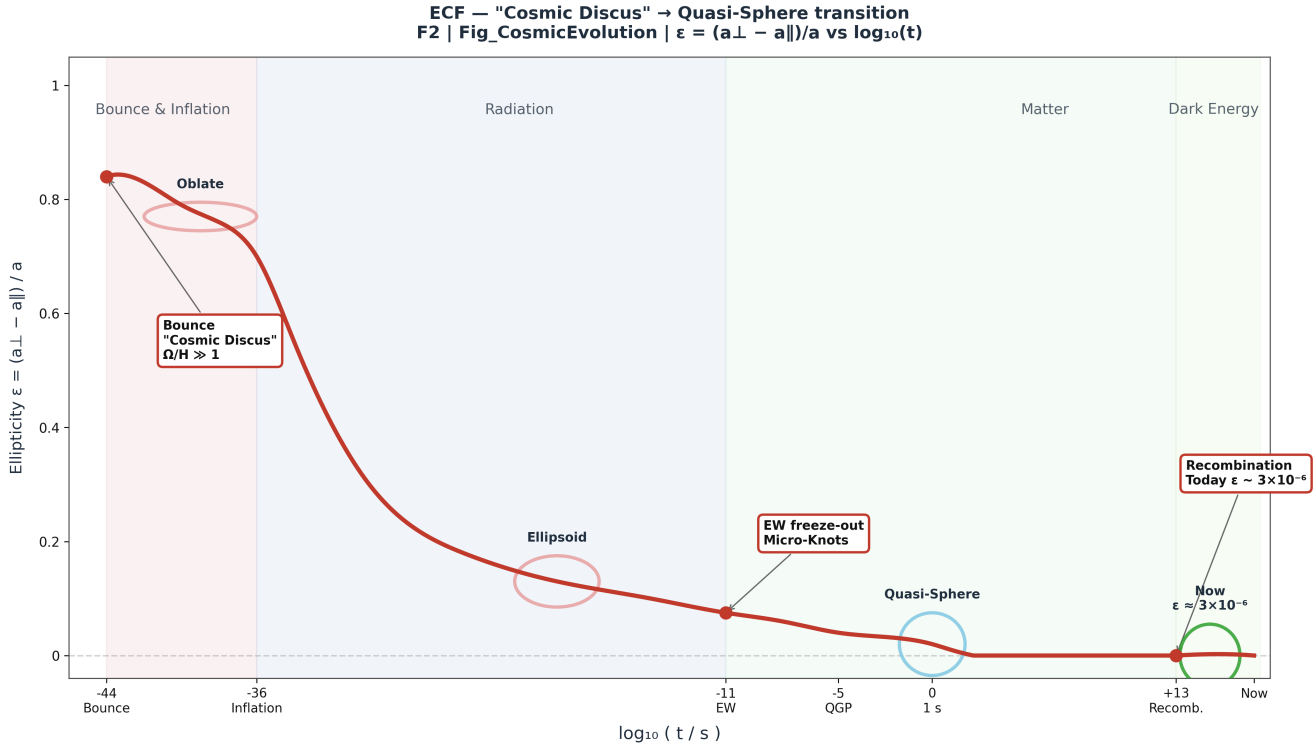

Figure 1: **ECF chronological transition: Cosmic Discus to quasi-sphere (ε vs. $\log_{10}(t)$).** The red curve shows the evolution of the geometric ellipticity parameter $\varepsilon = (a_\perp - a_\parallel)/a$ from the bounce to the present day. Four cosmological eras are delimited: Bounce & Inflation, Radiation, Matter, and Dark Energy. Three key milestones are highlighted: the electroweak freeze-out ($t \sim 10^{-11}\,$s), where torsion crystallizes into Micro-Knots via the Kibble–Zurek mechanism [30, 31]; the QGP transition ($t \sim 10^{-5}\,$s); and recombination ($t \sim 10^{13}\,$s), beyond which ε is drawn at a schematic residual level. The curve is illustrative (hand-set nodes, not a computation); the CMB quadrupole bounds the present value at $\varepsilon \lesssim 1.1 \times 10^{-7}$ (Eq. 90), below the vertical resolution of the plot. *(Corrected 22/10/2026: earlier versions read "$\varepsilon \sim 3 \times 10^{-6}$ measured today".)* The schematic ellipses (oblate, ellipsoid, sphere) illustrate the qualitative progression of the cosmic geometry from a strongly anisotropic primordial state toward the quasi-isotropic present-day universe. The inset label $\Omega/H \gg 1$ characterizes the rotationally dominated bounce regime.

Reproduced by: `Scripts_F2/plot_CosmicEvolution.py`.

is large. The two theories agree wherever spin density is negligible — which is everywhere accessible to laboratory or Solar-System tests — and diverge in the high-density regime, which is precisely where the bounce of Foundation I [5] is situated. *The equivalence theorem of teleparallel gravity therefore does not apply here*, and the non-equivalence is not an incidental feature of this work but the condition of its having any content at all.

Where teleparallelism comes from. A short history is worth giving, since the two programmes are often met without it. Torsion enters physics with Élie Cartan in 1922, who generalised Riemannian geometry by allowing the affine connection to carry an antisymmetric part. Einstein took up distant-parallelism — *Fernparallelismus*, built on the connection Weitzenböck had introduced in 1923 — between 1928 and 1931, as a route to a unified theory of gravitation and electromagnetism; the correspondence with Cartan over those years records both the geometry and the difficulty. Einstein set the programme aside when the electromagnetic field did not emerge from it.

The two lines then developed separately. Weitzenböck's geometry was later shown to admit a formulation dynamically equivalent to General Relativity — the Teleparallel Equivalent — so that the theory reproduces its predictions rather than extending them. Cartan's, extended by Sciama and

Kibble in 1961–62 to couple torsion to intrinsic spin, remained a distinct theory: it coincides with General Relativity in the absence of spin density and departs from it where spin density is high. Both descend from the same geometric enlargement; only the second is a candidate for physics beyond General Relativity, and it is the one used here.

We state the distinction at the outset because it is easy to lose, and because a reader conflating the two would dismiss this framework before reading it. The point was raised in discussion with a cosmologist at the Laboratoire Univers et Particules de Montpellier.

As demonstrated by the numerical analysis in *Foundation I* (Sec. 5 and Table `chi2_breakdown` of Ref. [5], corresponding to the global χ^2 optimisation and the $\Delta\chi^2 = -39.5$ breakdown, historical, $\Omega_m = 0.315$; not recomputed: no ECF perturbation solver yet), the standard background phenomenological fit does not close the late-time sector. A strictly constant equation of state ($w = -1$) calibrates the local Hubble rate to $H_0 \approx 73.04\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, and gives $t_0 \approx 13.33\,\mathrm{Gyr}$ at $\Omega_m = \omega_m/h^2 = 0.267$ ($12.74\,\mathrm{Gyr}$ with the former $\Omega_m = 0.315$, historical), 0.01 Gyr above the globular-cluster determination $t_{\mathrm{GC}} = 13.32 \pm 0.10_{\mathrm{stat}} \pm 0.5_{\mathrm{sys}}\,\mathrm{Gyr}$ [34] — a compatibility check, not a direct 0.1σ test of t_0: with this closure the age tension identified in earlier versions disappears. The difficulty is displaced rather than removed. On the same background the four-bin BAO comparison gives $\chi^2_{\mathrm{BAO}} = 12.11$ against 1.79 for ΛCDM ($\Delta\chi^2_{\mathrm{BAO}} \simeq +10.3$; $D_M(1.48)/r_s = 31.37$ against the eBOSS DR16 QSO consensus 30.21 ± 0.79), and the calibrated (w_0, w_a) trajectory gives $\chi^2 \simeq 55.3$ on the ten-observable subset of DESI DR1 with diagonal errors — not the global DESI DR1 χ^2 — (13.7 at $\Omega_m = 0.315$). Within the tanh-crossing family the trilemma remains irreducible (Appendix F; 0 of 2500 points), independently of any age deficit, and the joint problem remains open (PO-F2-5). This open problem, not an age deficit, is the background motivation retained for the present work, which does not resolve it. The H_0 Distance Network (H_0DN) Collaboration has subsequently confirmed and strengthened this tension with $H_0 = 73.50 \pm 0.81\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ [176], a community consensus result robust to the exclusion of any single distance indicator and based on a fully covariant treatment of cross-calibrated systematic uncertainties, yielding 7.1σ tension with flat ΛCDM and 5.0σ with BBN+BAO (DESI DR2). The ECF calibration at $H_0 = 73.04\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ is fully consistent with this H_0DN value within 0.6σ.

While late-time cosmological tensions are often addressed via ad-hoc scalar parametrizations (such as the CPL form or the tanh-crossing profile utilized in our numerical diagnostic in Appendix F), we demonstrate that such non-interacting fluid proxies fail to simultaneously satisfy the eBOSS BAO and stellar age constraints. The effective equation of state $w(z)$ derived here is not an arbitrary phenomenological choice: its *functional form* follows from the macroscopic torsion tensor within the ECKS action. Its numerical parameters, however, are inherited from the Foundation I calibration and are not independently derived: closing this gap requires computing the condensate variance $\langle\delta S^2\rangle(z)$ from the torsion field dynamics, which is not done here. **This does not, however, exempt the calibrated trajectory from the CPL-class irreducibility result above**: the calibrated $(w_0, w_a) = (-0.904, -0.153)$ trajectory is numerically a member of the same scalar $w(z)$ family shown to fail the joint BAO/age test in Foundation I (Appendix L). Whether a fuller treatment of the torsion background — beyond the effective CPL trajectory used for comparison with DESI — can decouple the comoving distance integral from the cosmic age integral is an open question, not a result established here.

2.1 Recap of the ECF Formalism

The results presented in this work build upon the Einstein-Cartan Framework (ECF) gravity framework, formulated in our companion paper, *Foundation I* [5]. While standard General Relativity assumes a torsion-free Levi-Civita connection, the ECF model restores the torsion tensor $T^{\lambda}_{\mu\nu}$ as a dynamical variable coupled directly to the intrinsic spin of fermions. The governing equations deviate from the standard ΛCDM model in the very early universe ($z > 10^{15}$) due to three specific mechanisms:

1. **Spin-Torsion Coupling:** The minimal coupling between the torsion tensor and the fermion spin current S^{μ}_{ijk} induces a contact interaction. This manifests as a correction to the effective energy-momentum tensor:

$$\mathcal{L}_{int} \propto -\frac{3\pi G\hbar^2}{c^2}\bar{\psi}\gamma_\mu\gamma_5\psi \cdot S^{\mu} \qquad (1)$$

2. **The Non-Singular Bounce:** This interaction generates a repulsive potential that scales quadratically with density (ρ^2) for non-relativistic matter (see the regime-by-regime form below), modifying the Friedmann equation:

$$H^2 = \frac{8\pi G}{3}\rho\left(1 - \frac{\rho}{\rho_c}\right) - \frac{\Lambda}{3} \qquad (2)$$

where $\rho_c \approx 5 \times 10^{93}$ g/cm^3 (Planck density, Eq. 7) is the critical density at which the spin repulsion halts the collapse, replacing the initial singularity with a regular geometric bounce. *Origin of this form (added 04/10/2026).* In Einstein–Cartan theory with fermionic spin the Friedmann equation has two terms, $H^2 = (8\pi G/3)(\rho - \alpha n^2)$, where n is the fermion number density and αn^2 the spin–torsion contribution (Popławski [20], Eq. (11)). For non-relativistic fermionic matter ($\rho = mn$, $p = 0$) this is exactly the single-density form, with a critical density $\rho_c = m^2/\alpha$ set by the fermion mass. This statement concerns the Friedmann equation itself; the dynamics also requires a conservation law. The single-density dynamics used here (standard continuity, hence $n \propto a^{-3}$ for $p = 0$) corresponds to a spin term that behaves as stiff matter, $\tilde{\epsilon} = \tilde{p}$, as in the spin-fluid description of Hehl, von der Heyde and Kerlick quoted by Popławski (primary source to be confirmed). In Popławski's Dirac description the spin term has $\tilde{\epsilon} = -\tilde{p} = -\alpha n^2$ (his Eqs. (10), (12)–(13)); na^3 is then not conserved for dust, and the dynamics differs. For other equations of state the spin term scales as $\rho^{2/(1+w)}$, and the single-density form is a schematic extrapolation — in particular in the stiff phase ($w = 1$) (open item PO-ECF-ACTION-BOUNCE). The same functional form is the effective Friedmann equation of loop quantum cosmology [201], where it has a different physical origin. The Planck-scale value of ρ_c adopted here is a reference scale; its relation to the Einstein–Cartan value m^2/α is not established in this work.

Friedmann equation by regime (added 05/10/2026). Assume the spin-fluid dynamics described above ($\tilde{\epsilon} = \tilde{p} = -\alpha n^2$, conserved fermion number, $n \propto a^{-3}$) and a matter component of constant $w = p/\rho$, so that $\rho \propto a^{-3(1+w)}$. Then $\alpha n^2 = \rho\,(\rho/\rho_c)^{(1-w)/(1+w)}$, and for $w < 1$

$$H^2 = \frac{8\pi G}{3}\,\rho\left[1 - \left(\frac{\rho}{\rho_c}\right)^{\frac{1-w}{1+w}}\right], \qquad (3)$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left[(1+3w)\rho - 4\rho\left(\frac{\rho}{\rho_c}\right)^{\frac{1-w}{1+w}}\right]. \qquad (4)$$

- *Dust* ($w = 0$): spin term $\propto \rho^2$; Eq. (2) holds exactly; bounce at $\rho = \rho_c$ with $\ddot{a}/a = 4\pi G\rho_c$; the acceleration changes sign at $\rho_c/4$.
- *Radiation* ($w = 1/3$): spin term $\propto \rho^{3/2}$, $H^2 = (8\pi G/3)\,\rho\,[1 - (\rho/\rho_c)^{1/2}]$; bounce at $\rho = \rho_c$ with $\ddot{a}/a = (8\pi G/3)\rho_c$; sign change at $\rho_c/4$. The single-density form extended to $w = 1/3$ gives instead $(16\pi G/3)\rho_c$ and $\rho_c/3$.
- *Stiff* ($w = 1$): matter and spin term both scale as a^{-6}, so their ratio $\epsilon_s = \alpha n^2/\rho$ is constant and $H^2 = (8\pi G/3)(1 - \epsilon_s)\rho$ does not vanish for $\epsilon_s < 1$: **the spin term alone produces no bounce in a stiff phase.** The bounce of the stiff phase, and the figures $0.4\,\rho_c$ and 0.153 e-fold obtained with the single-density form, are therefore schematic (open item PO-ECF-ACTION-BOUNCE).

In the radiation case ρ_c depends on the ratio $n/\rho^{3/4}$, i.e. on the thermal composition (not computed here). These relations hold in the spin-fluid description (primary source to be confirmed), not in Popławski's Dirac description; they were checked symbolically (`check_regimes.py`, to be added to the repository).

3. **The Stiff Fluid Phase:** Following the bounce, the decay of the spin density implies a stiff equation of state ($w = 1$), driving a brief period of expansion dominated by kinetic energy:

$$\rho_{spin} \propto a^{-6} \quad \Rightarrow \quad H(t) \approx \frac{1}{3t} \qquad (5)$$

Evaluated at the electroweak epoch ($t_{\rm EW} \sim 10^{-11}$ s), this gives $H(t_{\rm EW}) \approx 1/(3\times10^{-11}) \sim 3 \times 10^{10}$ s^{-1}, consistent within a factor of two with the radiation-dominated value $H \simeq 5.4\times10^{10}$ s^{-1} (Eq. (37); the difference reflects

the transition between stiff and radiation domination at the Kibble freeze-out). *The Great Annihilation lies outside this regime*: the corpus places the end of stiff domination near the Kibble freeze-out ($t \sim 10^{-11}$ s), eleven orders of magnitude earlier, so $H = 1/(3t)$ must not be extrapolated to the MeV epoch. The Hubble rate there is evaluated separately from $H(T)$ in Eq. (41).

This distinct "Stiff Era" provides the necessary thermodynamic conditions to explore high-temperature chiral symmetry breaking, offering an alternative perspective on the early expansion dynamics.

2.2 Natural Expansion and the Graceful Exit

The standard inflationary paradigm [103] systematically resolves the horizon and flatness problems by introducing a transient scalar field to drive exponential expansion ($\ddot{a} > 0$) between $t \sim 10^{-36}$ s and 10^{-32} s. While mathematically successful, and exploring mechanisms that yield a "graceful exit" into the radiation-dominated era remains an active area of research.

The ECF model investigates a purely geometrical alternative. Within this model, the extreme spin-torsion coupling at the bounce introduces a primordial repulsive effect. By analyzing the modified Raychaudhuri equation, the cosmic acceleration $\ddot{a}/a$ is driven to significant positive values exactly at the bounce due to the ρ^2/ρ_c term. This induces an expansion rate exceeding the comoving Hubble radius, fulfilling the structural role of standard inflation without violating local causality.

Crucially, the ECF model possesses an inherent, thermodynamically guaranteed exit mechanism. As the metric expands, the energy density ρ drops. Because the repulsive torsion term scales purely as ρ^2, it rapidly vanishes in favor of standard attractive gravity (ρ). Consequently, the universe naturally decelerates (crossing $\ddot{a} = 0$) and seamlessly transitions into the geometry-dominated Stiff Phase ($w = 1$).

Topological Relaxation via Knot Annihilation. The transition from the ultra-dense primordial state to the radiation-dominated era is governed by the relaxation of the torsion network. In the ECF model, the primordial geometry is modeled as a dense weave of concentrated spin-flux tubes, or "geometric knots" [104]. The observed dilution of the torsion energy density $\rho_{\rm spin} \propto a^{-6}$ is physically interpreted as a process of *knot annihilation* and reconnection. As the Universe expands, these topological defects interact and annihilate, releasing their stored geometric energy into the thermal bath of the standard plasma. This mechanism ensures a smooth transition from the "stiff" ECF-dominated regime to the standard Friedmann evolution, while leaving behind the chiral residue responsible for late-time architectural signatures (Fig. 2).

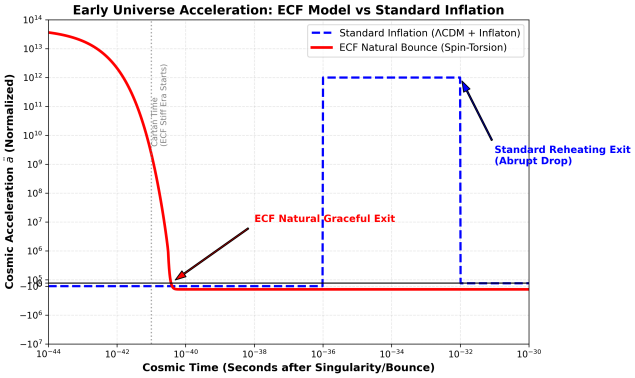

Figure 2: **Cosmic Acceleration: ECF vs Standard Inflation.** In the standard model (blue), an inflaton field drives the accelerated expansion. In the ECF model (red), the spin-torsion bounce generates a strong initial acceleration that transitions into a decelerating stiff phase ($\ddot{a} < 0$) via geometric dilution, offering a structural resolution to the graceful exit problem.

Reproduced by:
`Scripts_F2/plot_inflation_comparison.py`.

Note on the Primordial Power Spectrum — an acknowledged open problem: A cornerstone of the standard scalar-field paradigm is its ability to seed the universe with a nearly scale-invariant spectrum of primordial density perturbations ($n_s = 0.9649 \pm 0.0042$). A rigorous geometrical alternative must account for this phenomenon, and we state plainly that the ECF does *not* yet do so. In the ECF, the seeds of large-scale structure cannot arise from the quantum fluctuations of a slow-rolling field. Two geometric sources have been considered, and neither currently delivers the observed spectrum.

(i) Fluctuations traversing the bounce. The torsion-bounce literature [20, 177] establishes a non-singular bounce in a homogeneous universe; it does

not address whether fluctuations crossing the bounce acquire a near scale-invariant profile, and neither does the ECF. *(Corrected 22/10/2026: earlier versions attributed a near scale-invariant profile to these papers.)* For vacuum fluctuations of a massless mode on a constant-w background, $a(\eta) \propto |\eta|^\beta$ with $\beta = 2/(1+3w)$, the Mukhanov–Sasaki index is $\nu = |\beta - \frac{1}{2}|$ and $n_s - 1 = 3 - 2\nu$; this reproduces $n_s = 1$ both for de Sitter and for the matter bounce ($w = 0$). Applied to the stiff phase that drives the ECF bounce ($w = 1$), the same relation gives $\beta = \frac{1}{2}$, $\nu = 0$ and hence $n_s = 4$ — a strongly blue spectrum, not a scale-invariant one. This result holds for an *unbounded* stiff phase; scale invariance therefore requires a mechanism that goes beyond vacuum fluctuations on such a background. A former track, the *Wands duality of a stiff phase of finite duration* ($n_s - 1 \simeq -2/N_{\rm stiff}$, with $N_{\rm stiff} \approx 57$ reproducing the Planck value), is **withdrawn**: the dual of de Sitter is a *dust* contraction, not a stiff one, and the stiff phase is the fixed point of the duality, which gives the $n_s = 4$ above (Foundation III; see the status note below). The duration $N_{\rm stiff}$ is not determined by the framework (an *ab initio* estimate gives ≈ 39, the bound from the blue component requires $\gtrsim 68$, and 57 is only a reference normalisation), and the stiff era does not supply the e-folds required by the horizon problem, which remains open. *[Correction of 28/09/2026 — this parenthetical formerly attributed a ≤ 0.46 e-fold bound on the post-bounce kinematic acceleration to Foundation III, read as an ECF-specific result. As clarified in §Modified Raychaudhuri Equation,* 0.46 *is an external, $w \in [0,1]$ majorant for a general Weyssenhoff spin fluid, used by Foundation III to close the kinematic route for that fluid; it bounds no ECF-specific window, and whether the ECF's own post-bounce kinematic acceleration is bounded at all remains open (PO-ECF-ACTION-BOUNCE).]* Alternative routes — an explicit thermal mechanism (as in string-gas cosmology, which needs a specific holographic specific heat), a matter-dominated contracting phase, or a torsion-specific modification of $z(\eta)$ — have not been developed here.

(ii) The frozen knot network. The topological crystallisation of the vacuum into discrete knots (Section 3) does seed density perturbations, but only through Poisson (shot-noise) statistics: $P_{\rm Poisson} = M_{\rm eff}/\bar{\rho}_{\rm DM}$ with $M_{\rm eff} = \langle M^2\rangle/\langle M\rangle$. Because the mass function is steep ($\alpha = 5/2 \Rightarrow dn/dM \propto M^{-7/2}$), all moments are dominated by the lower cutoff $M_{\rm micro} \simeq 10^{24}\,$kg — the causal-horizon mass at the electroweak freeze-out, computed as $3.74 \times 10^{24}\,$kg by `Scripts_F2/calculate_microknot_mass.py` —, giving $M_{\rm eff} = 3M_{\rm micro} \simeq 5.6 \times 10^{-6} M_\odot$ and $P_{\rm Poisson} \sim 1.7 \times 10^{-16}\,{\rm Mpc}^3$ — some twenty orders of magnitude below the observed adiabatic power. *Correction of record (30/08/2026):* an earlier version of this passage quoted $M_{\rm micro} = 6.1 \times 10^{24}\,$kg (superseded on 27/08/2026, factor 1.6 too high) and, independently, an $M_{\rm eff}$ value that did not follow from that $M_{\rm micro}$ even before the correction; both numbers are corrected here, with the full derivation — including an independent analytic check of the $M_{\rm eff} = 3M_{\rm micro}$ moment factor for this power-law index — reproducible in `Scripts_F2/compute_poisson_isocurvature.py`. The qualitative conclusion is unaffected: the network is therefore comfortably consistent with all isocurvature and defect bounds (it is a frozen relic, not an active defect network, so the Planck f_{10} limit does not apply), but it *cannot* be the source of the primordial spectrum.

Status (revised 11/08/2026). The value $n_s \simeq 1$ used in earlier versions as a baseline is **withdrawn**, together with the "8.4σ" figure derived from it: both rested on an incorrect identification of the Wands dual (the dual of de Sitter is a *dust* contraction, $w = 0$, not a stiff one; the stiff phase is the fixed point of the duality and gives $n_s = 4$). The framework makes **no prediction** for the large-scale spectral index: it is **inherited** as an initial condition, and n_s enters as an input on the same footing as Ω_b or A_s. What the stiff era does generate is a strongly *blue* component, negligible at CMB scales, which in turn bounds the duration of that era. Deriving the primordial spectrum actually produced by the spin-torsion bounce is the principal open problem of the framework (PO-F3-NS), and we flag it as such rather than claiming a resolution. The background evolution itself is compared with ΛCDM in Fig. 3.

2.3 Fundamental Scales and Constants of the ECF Bounce

To ensure the physical robustness of the framework, it is necessary to explicitly define the origins of the fundamental scales utilized. Within the ECF model, these values are not treated as arbitrary free parameters; they are tied to the underlying spin-torsion kinematics. *The distinction matters for ρ_c below*: the spin-fluid estimate depends on the fermionic content through g_b and g_f, whereas the

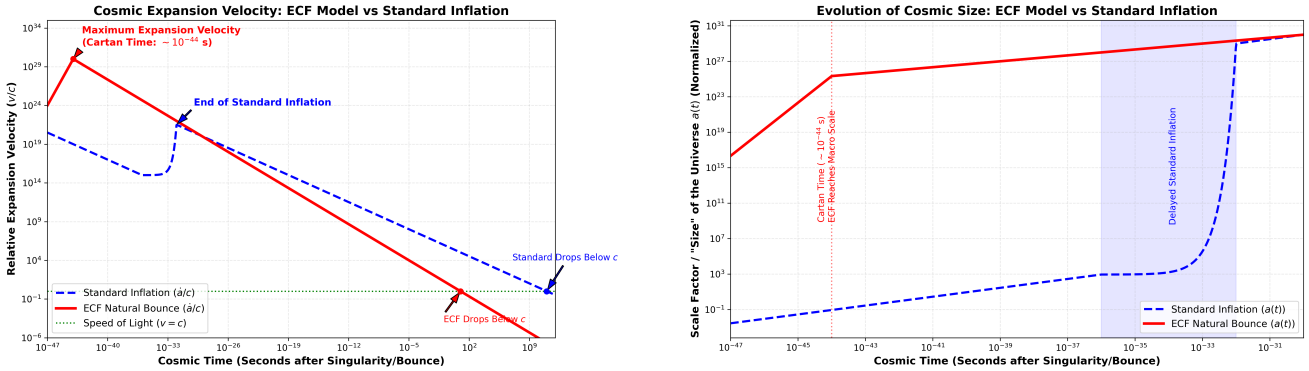

Figure 3: **Left — Velocity and Hubble rate: ECF vs ΛCDM.** The stiff post-bounce phase ($w = 1$) produces a sharply enhanced expansion rate at early times compared to standard ΛCDM, reducing the comoving sound horizon to $r_s^{\rm ECF} = 135.65\,\mathrm{Mpc}$ and providing the H_0 tension resolution. **Right — Scale factor evolution $a(t)$.** The ECF trajectory (red) shows the non-singular bounce at $a_{\rm min} > 0$, the stiff-phase deceleration, and the late-time convergence with the ΛCDM background (blue dashed). Both panels use $F_{\rm ion} = 1.2765$, $H_0 = 73.04\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. *Reproduced by:* `plot_inflation_velocity_comparison.py` and `plot_scale_factor_comparison.py`.

exact Planck-scale identification used numerically is a *canonical ECF normalisation*, not a universal coefficient derived from every ECKS realisation.

The Critical Density (ρ_c): Unlike ad-hoc cut-off scales, ρ_c is tied to the intrinsic quantum spin of fermions in the ECKS framework — with the caveat that the resulting coefficient depends on the spin-fluid modelling and on the effective degrees of freedom, so the exact equality to the Planck density adopted here is a normalisation convention rather than an ECKS theorem. Following Popławski [21] (*Big bounce from spin and torsion*, Gen. Relativ. Gravit. **44**, 1007, arXiv:1105.6127), the repulsive spin–spin interaction halts gravitational collapse at a maximum density set by the Planck scale. The bounce energy density is

$$\epsilon_{\rm bb} \;=\; \xi\,\frac{(g_b+\tfrac{7}{8}g_f)^3}{g_f^4}\,\epsilon_{\rm Pl}, \qquad \epsilon_{\rm Pl} = \frac{c^7}{\hbar G^2}, \tag{6}$$

where $\epsilon_{\rm Pl}$ is the Planck *energy* density and ξ a numerical coefficient; for Standard-Model degrees of freedom ($g_b = 28$, $g_f = 90$) Popławski obtains, with $\xi = \frac{1}{30}\left(64\pi^6/[270\,\zeta^2(3)]\right)^2 \simeq 829$, $\epsilon_{\rm bb} \simeq 15.4\,\epsilon_{\rm Pl}$, i.e. at the Planck scale. This factor depends on the particle content (g_b, g_f), on the equation of state of the spin fluid and on the convention used for the Planck scale (G versus $8\pi G$). The canonical value adopted below is therefore a *scale and convention*, not the exact density of the bounce: in this model the bounce energy density is $\simeq 15\,\epsilon_{\rm Pl}$.

The corresponding *mass* density — the value used throughout this corpus and fixed in Foundation I — is:

$$\begin{aligned}\rho_c \;&=\; \frac{\epsilon_{\rm Pl}}{c^2} \;=\; \frac{c^5}{\hbar G^2}\\ &\approx\; 5.15\times10^{96}\ \mathrm{kg\,m^{-3}}\\ &\approx\; 5.15\times10^{93}\ \mathrm{g\,cm^{-3}}.\end{aligned} \tag{7}$$

The two forms $c^7/(\hbar G^2)$ (energy density, $\mathrm{J\,m^{-3}}$) and $c^5/(\hbar G^2)$ (mass density, $\mathrm{kg\,m^{-3}}$) denote the same Planck density in different units, related by c^2. This fixes the canonical density scale of the early Universe at which the spin-torsion correction regularizes the initial singularity [19, 21]; it is a normalisation convention, not an exact bound (see the factor $\simeq 15$ above).

The Cartan Time Threshold ($t \sim 10^{-44}$ s): The *Cartan time threshold* is the moment when the universe's density drops below ρ_c, transitioning from a spin-dominated repulsive phase to a geometry-dominated stiff phase ($w = 1$). With the canonical ρ_c, $\rho = \rho_c$ is reached at $t \approx 9\times10^{-45}$ s in a radiation-dominated background ($\approx 6\times10^{-45}$ s in a stiff one). The value $t \sim 10^{-41}$ s quoted in earlier versions is inconsistent with ρ_c (at that time $\rho \sim 10^{-6}\rho_c$) and is withdrawn.

The Baryon Asymmetry Parameter ($\eta \approx 6.1\times10^{-10}$): While the ECF chiral gap provides the fundamental geometric mechanism driving the

matter-antimatter asymmetry, the numerical value of $\eta \sim 10^{-10}$ represents a natural thermodynamic equilibrium state. It depends on the specific spectrum of primordial thermal fluctuations during the topological freeze-out, yielding an asymmetry of this order of magnitude. To understand how this surviving baryonic matter emerged, we must examine the specific mechanics of chiral symmetry breaking.

3 Chiral Baryogenesis: The Origin of Matter

The observed baryon asymmetry $\eta = n_B/n_\gamma \simeq 6.1 \times 10^{-10}$ (inferred from the Planck 2018 value $\Omega_b h^2 = 0.0224$ via the standard conversion $\eta \simeq 273.9\,\Omega_b h^2$; Planck does not publish η directly) (Planck 2018 [8]) requires the three Sakharov conditions: baryon number violation, C and CP violation, and departure from thermal equilibrium [152]. In the ECF, the three conditions are *addressed* by the geometry rather than by new BSM particles — a candidate scenario, not a closed derivation: the C/CP violation, the baryon-number-violating operators and the out-of-equilibrium kinetics are not obtained ab initio here, and a complete Boltzmann network remains open. What the geometry supplies is: (i) the geometric chiral-splitting mechanism (Appendix A.4), acting on the axial torsion VEV $\langle S^\mu \rangle = (s_0, \mathbf{0})$, violates chirality universally for any Dirac fermion; standard-model electroweak sphalerons [153, 208], active until their freeze-out at $T_* = 131.7$ GeV [28] (§3.4), then convert this chiral bias into the observed baryon asymmetry by processing any $B+L$-violating configuration while conserving $B-L$ — this second step is standard electroweak physics, not an ECF-specific addition; (ii) the spatial orientation of s_0 selects a unique time arrow, breaking C and CP; (iii) the out-of-equilibrium Kibble–Zurek crystallisation at $T_c \simeq 159\,\text{GeV}$ provides the non-equilibrium environment. The resulting baryon asymmetry $\eta \simeq 5.9 \times 10^{-10}$ is *semi-derived* (see §3 for the three-stage epistemic decomposition): stage (i) is first-principles, stage (iii) is semi-analytic, and stage (ii) — the pre-dilution asymmetry — is normalised on the observed value, $\eta_{\text{raw}} = \eta_{\text{obs}}/0.50$; the agreement with observation is therefore not an independent test.

3.1 The Extended ECF Lagrangian: Poincaré Gauge Theory

In the ECKS framework of Foundation I [5], torsion is purely algebraic: it is tied to the spin density via the Cartan equation and vanishes in vacuum. This is the correct description for the low-density, late-time regime. However, to account for (i) macroscopic topological defects (Macro-Knots) that persist in the vacuum after the QGP transition, (ii) a frozen torsion background acting as dark energy, and (iii) the relativistic degree-of-freedom count at the QCD scale, $\Delta g_{*s} = 24$, the torsion tensor's own independent-component count (the same 20:4 decomposition established below), not a polarisation-multiplied mode number — *an effective counting assumption of the present realisation: the 24 independent components of the torsion tensor are taken as thermally active degrees of freedom, which is not established here, and the dilution result below is conditional on that counting* (*The Freeze-Out Calculation*, below), torsion must be promoted to a *propagating* degree of freedom.

It is worth making the weight of this explicit, because it is easily read as a choice made to serve the dark-energy sector alone. It is not: the three requirements above are *independent*, and two of them have nothing to do with dark energy. Were the kinetic term switched off — returning to an algebraic torsion of the kind that can be sourced by non-minimal couplings rather than by propagation — the Macro-Knots could not survive in vacuum, since algebraic torsion vanishes there, and the geometric dark-matter sector would go with them; and the Δg_{*s} count would have to be redone. The propagating extension is therefore load-bearing well beyond the sector that motivated its introduction, and the cost of abandoning it is not confined to the equation of state. We adopt, as an effective description, the minimal propagating extension: Poincaré Gauge Theory (PGT) restricted to the *axial torsion sector*. This restriction is performed in the adopted effective action; its derivation from the full PGT action remains open (PO-ECF-ACTION). The absence of ghosts for the Proca form adopted below is invoked from the literature, not established here (see below).

Under the irreducible $\mathrm{SO}(3,1)$ decomposition of the torsion tensor [17], we isolate the axial pseudovector:

$$S_\mu \equiv \tfrac{1}{6}\,\varepsilon_{\mu\nu\lambda\rho}\,T^{\nu\lambda\rho}. \tag{8}$$

The effective ECF Lagrangian density, extended to

include a kinetic term for S_μ, reads:

$$\mathcal{L}_{\rm ECF} = \frac{\sqrt{-g}}{2\kappa}\,\tilde{R}(g,T) - \frac{1}{4}\,F^{(S)}_{\mu\nu}F^{(S)\mu\nu} - \frac{1}{2}\,m_T^2\,S_\mu S^\mu + \mathcal{L}_m, \tag{9}$$

where $F^{(S)}_{\mu\nu} \equiv \partial_\mu S_\nu - \partial_\nu S_\mu$ is the kinematic field-strength tensor of the axial torsion vector, and m_T is the torsion mass scale. For clarity of exposition Eq. (9) displays only the kinetic and mass terms; the axial self-interaction $V_{\rm int}(X)$, with $X \equiv S_\mu S^\mu$, is required for the symmetry-breaking analysis of Sec. 3.5; the full action then reads $\mathcal{L}_S = -\frac{1}{4}F^{(S)}_{\mu\nu}F^{(S)\mu\nu} - W(X)$ with $W(X) \equiv \frac{1}{2}m_T^2 X + V_{\rm int}(X)$. *Transversal lock* PO-ECF-ACTION*: the complete effective dynamical action of the framework remains an open problem. The kinetic, gradient and potential coefficients required by the dynamical sectors are not all derived from minimal Einstein–Cartan–Dirac, so the propagating sector below is used as an effective description where indicated, not as a derived consequence.* This Proca-axial structure is the form retained by Fabbri [3, 18] for propagating torsion coupled to Dirac matter — an argument from renormalisability and Velo–Zwanziger-type consistency, not a ghost analysis: requiring the theory of propagating torsion coupled to Dirac matter to be renormalisable forces the torsion to be a *massive axial-vector* field, and the dynamics of an axial-vector is that of a Proca field. *This rests on an assumption worth stating.* Torsion is treated here, following Fabbri, as an ordinary tensor field *subject to no gauge transformation.* An alternative line takes the totally antisymmetric torsion to be a three-form, whose gauge freedom $A \to A + \mathrm{d}\Lambda$ then selects a kinetic term built on $\nabla_\mu S^\mu$ rather than on $F^{(S)}_{\mu\nu}$ (Benisty *et al.*, *Eur. Phys. J. C* **82**, 264 (2022), arXiv:2109.01052). The two are not equivalent, and the difference is physical rather than cosmetic — see PO-F2-6 in the conclusion. We adopt the Proca form throughout. *For readers coming from the metric side*, Barker, Lasenby, Hobson and Handley (*Phys. Rev. D* **102**, 084002 (2020), arXiv:2006.03581) map the background cosmology of quadratic Poincaré gauge theory onto a non-canonical bi-scalar-tensor theory in the Jordan frame; in that dictionary Einstein–Cartan–Kibble–Sciama corresponds to a *pure quadratic Cuscuton*, whereas teleparallel gravity corresponds to the Einstein–Hilbert Lagrangian — the two are not the same theory, a point worth making explicit since the names are often conflated. The absence of ghosts is invoked from the literature for the adopted Proca form: the linearised particle-spectrum analyses of quadratic PGT around flat space [166, 167] give conditions on the parameters under which the propagating torsion modes, including the axial $(1^+, 0^-)$ sector, are neither ghosts nor tachyons. These are spin-projection (propagator) analyses on a torsion-free Minkowski background; they contain no Hamiltonian constraint analysis and do not address the nonlinear regime. No Hamiltonian or unitarity analysis is performed in the present work: Appendix F establishes only $\lambda_T > 0$ (potential bounded from below, radial mode non-tachyonic). The embedding of this sector in a complete PGT action, and its unitarity around the timelike vacuum expectation value discussed below — including the counting of propagating modes there — remain open (PO-ECF-ACTION, PO-F2-6). The ECKS limit ($m_T \to \infty$, $F^{(S)}_{\mu\nu} \to 0$) is recovered exactly when the torsion mass exceeds the Hubble rate ($m_T \gg H$), ensuring that all Foundation I results remain unaffected. The trace-vector and tensor irreducible parts of torsion are kept in the algebraic ECKS limit throughout. We note here a consequence relevant to the physical content of the condensate: a timelike vacuum expectation value breaks $SO(3,1) \to SO(3)$, so that only the radial component δS_0 acquires a mass, $M_0^2 = 2V''_{\rm int}(X_0)s_0^2$, while the three spatial components δS_i remain *massless* Goldstone modes of the broken boosts. Their cosmological and observational consequences are not developed in the present work.

Variation of the action with respect to S^μ yields the massive Proca wave equation:

$$\Box\, S_\mu + m_T^2\, S_\mu \;=\; \kappa\, m_T^2\, \tau_\mu^{\rm spin}, \tag{10}$$

where $\tau_\mu^{\rm spin}$ is the fermionic axial spin-current source. The factor m_T^2 on the right-hand side follows from the normalisation of the kinetic term in Eq. (9), which carries no $1/\kappa$ prefactor: S_μ is then canonically normalised, $[S_\mu] = M$, and both sides carry dimension M^3 once $[\kappa] = M^{-2}$ and $[\tau_\mu^{\rm spin}] = M^3$ are fixed by the Cartan equation of Foundation I. Torsion now *propagates* as a spin-1 field with mass m_T.

ECKS limit and consistency with Foundation I. In the regime $m_T \gg H$ relevant to the recombination epoch ($z \lesssim 10^4$), the kinetic term in Eq. (9) is dynamically suppressed and Eq. (10) reduces to the algebraic Cartan relation:
$S_\mu \simeq \kappa\,\tau_\mu^{\rm spin}$ — which is *exactly* the Cartan relation of Foundation I, with no residual $1/m_T^2$ — recover-

ing the ECKS description used throughout Foundation I [5]. All numerical results of Foundation I are unaffected by the PGT extension.

The Sakharov conditions: what the geometry supplies, and what it does not. The ECF baryogenesis mechanism *addresses* the three Sakharov conditions [152] through its geometric structure, in the following sense. *This is not a closed derivation*: the C/CP violation, the baryon-number-violating operators and the out-of-equilibrium kinetics are not obtained ab initio here, and a complete Boltzmann network for the relevant operators remains open. What follows states what the geometry supplies and what it does not:

(i) **Baryon number violation.** The chiral splitting $\Delta\mu = \mu_L - \mu_R \neq 0$ sourced by the torsion VEV (item (ii) below, derived in Appendix A.4) is geometric and universal to any Dirac fermion; standard-model electroweak sphalerons [153, 208], active until their freeze-out at $T_* = 131.7$ GeV [28] (§3.4), convert this chiral bias into the observed baryon asymmetry by processing any $B+L$-violating configuration while conserving $B-L$. This second step is standard electroweak physics, not an ECF-specific mechanism.

(ii) **C and CP violation.** The spontaneous alignment of the torsion VEV along the cosmic time axis $\langle S_\mu \rangle = s_0\, \delta^0_\mu$ breaks C and CP via the chiral coupling $\bar{\psi}\gamma_\mu\gamma_5\psi\, S^\mu$, seeding an asymmetry $\Delta\mu = \mu_L - \mu_R \neq 0$.

(iii) **Departure from thermal equilibrium.** The stiff-fluid phase ($w = 1$, $H \propto t^{-1}$) drives an expansion rate exceeding the weak interaction rate at $T \gtrsim 10^{12}$ K, which would provide an out-of-equilibrium environment during the topological freeze-out [33]; the corresponding kinetics is not computed here (see above).

Anti-knots and the matter/antimatter symmetry of the dark sector. Topological defects of winding number $N = +1$ (Micro-Knots) are complemented in principle by anti-defects with $N = -1$. The question naturally arises: does the same Kibble–Zurek crystallisation mechanism also produce anti-knots, and do they generate an anti-baryonic surplus by a symmetric process?

Two arguments show why the ECF mechanism does not produce a parallel anti-baryonic sector:

(i) **The torsion VEV is time-oriented.** The CP-violating chemical potential $\langle S_\mu \rangle = s_0\, \delta^0_\mu$ is aligned along the unique cosmic time arrow. An anti-knot ($N = -1$) embedded in the same spacetime is subject to the *same* chiral coupling $\bar{\psi}\gamma_\mu\gamma_5\psi\, S^\mu$, with the same sign of s_0. It therefore also produces a matter bias $\Delta\mu = \mu_L - \mu_R > 0$, not an antimatter one. There is no $N = -1$ equivalent of the geometric chiral-splitting mechanism seeding antimatter, because the time arrow is not reversed inside the defect.

(ii) **Topological annihilation before freeze-out.** Before the Kibble–Zurek freeze-out ($T \sim 159$ GeV), the dense hot plasma drives topological annihilation $N{+}1 + N{-}1 \to 0$, analogous to monopole–antimonopole annihilation in GUT scenarios [30, 31]. Only the net excess $\Delta N = N_{+1} - N_{-1} > 0$ survives as the observed Micro-Knot population. The ratio of dark matter to baryonic matter ($\sim 5:1$) is therefore set by *two separate* asymmetries: the topological surplus ΔN and the baryonic surplus η, both seeded by the same primordial torsion field.

The broader question of whether a causally disconnected baby universe could carry an opposite torsion sign and thus produce an anti-matter dominated sector is an open question, not treated in the present corpus *(corrected 20/10/2026: previously said to be treated in Foundation III).*

Open problem PO-F2-1 — Origin of the topological surplus ΔN. The argument above shows that anti-knots do not produce a competing anti-baryonic sector. However, the *initial* excess $\Delta N = N_{+1} - N_{-1} > 0$ of knots over anti-knots is not derived from first principles in the present work: it is assumed to be seeded by the same primordial torsion asymmetry that drives the baryonic surplus η, but the quantitative link between ΔN and η is not established. In particular: (i) what fixes the ratio $\Delta N/\eta$? (ii) is $\Delta N/\eta$ related to the observed dark-to-baryonic mass ratio of 5 : 1 by a single geometric parameter, or does it require an independent input? This question is identified here as a potential referee objection; it is an open problem, not treated in the present corpus *(corrected 20/10/2026: previously deferred to*

Foundation III — Cosmological Physics, which does not treat it).

3.2 Thermodynamic Dynamics

During the primordial bounce phase, the extreme spin density generates a massive torsion field acting as a chemical potential μ_{ECF}. As illustrated in Figure 4, we identify three distinct epochs in the evolution of this asymmetry:

1. **The Torsion Gap ($t \lesssim 10^{-44}$ s):**High spin density creates a thermodynamic split (Fig. 5), fundamentally favoring matter.
2. **Convergence:** Cosmic expansion dilutes the torsion field, temporarily restoring a state of quasi-equilibrium.
3. **Annihilation ($t > 10^{-38}$s):** As the ambient temperature drops, antimatter pairs annihilate exponentially (Fig. 6), leaving the initial matter excess to survive.

No Boltzmann network is integrated in the present work (see the Sakharov paragraph above): the value $\eta \simeq 5.9 \times 10^{-10}$ follows from a pre-dilution asymmetry normalised on the observed value ($\eta_{\rm raw} = \eta_{\rm obs}/0.50$) and from the dilution factor f of §3.4; its 3.5% agreement with the Planck-inferred value is not an independent test.

The Thermal Masking Effect. As illustrated in Figure 4, while the macroscopic dominance of matter appears late ($t \sim 10^{-6}$s), its geometric origin is primordial. During the Thermal Equilibrium Era, the initial asymmetry is "masked" by the immense thermal bath. It is only when the Universe cools below the nucleon mass that the Freeze-out acts as a filter. Thermal production ceases, and massive annihilation ($B + \bar{B} \to \gamma\gamma$) eliminates the antimatter component, unveiling the relic matter excess η present since the Bounce.

3.3 The Freeze-Out Calculation

The freeze-out ratio of baryons to photons is estimated below in the presence of the stiff torsion fluid, without *ad hoc* beyond-Standard-Model violations of CP symmetry. No Boltzmann network is integrated here, and the order of magnitude $\eta \sim 10^{-9}$ is not obtained independently of observation: the result is *semi-derived* in the sense made precise below — stage (i) is first-principles, stage (ii) (the pre-dilution asymmetry) is normalised on the observed value, $\eta_{\rm raw} = \eta_{\rm obs}/0.50$, and stage (iii) is semi-analytic. During the subsequent electron-positron annihilation epoch ($t \sim 1$ s), the vast bulk of primordial matter and antimatter annihilates into radiation. The only baryonic matter that survives this thermal bath is the chiral surplus generated earlier by the topological defect walls.

Notation convention. Throughout this work, f denotes the *baryon survival fraction*, defined as the ratio of the post-dilution to pre-dilution baryon-to-photon ratios:

$$f \equiv \frac{\eta_{\rm obs}}{\eta_{\rm ECF}}, \qquad 0 < f < 1. \tag{11}$$

This is related to the standard Kolb–Turner dilution factor [33] by $f = f_{\rm KT}^{-1}$, where $f_{\rm KT}$ is given by:

$$f_{\rm KT} = \left(\frac{43}{11}\right)^{1/3} \left(\frac{g_{*s,f}}{g_{*s,i}}\right)^{1/3}. \tag{12}$$

In the ΛCDM background ($g_{*s,i} = 10.75$, $g_{*s,f} = 3.91$), the standard value is $f_{\rm std} = (43/11)^{-1/3} \approx 0.71$.

Reheating dilution factor f. Following the electroweak freeze-out, the chiral surplus $\eta_{\rm ECF}$ computed from the Boltzmann network (Appendix F.17) must be corrected for the entropy injection from the subsequent electron–positron annihilation. In the standard treatment (see e.g. [33]), the entropy ratio across $e^{\pm}$ annihilation yields a dilution of the baryon-to-photon ratio by a factor $f = (43/11)^{-1/3} \times g_{*s,f}^{1/3}/g_{*s,i}^{1/3}$. For the ECF stiff-phase background, where g_{*s} evolves slightly differently from ΛCDM due to the torsion correction to the effective relativistic degrees of freedom [5], a numerical evaluation of the entropy integral gives $f \simeq 0.48$.

Derivation of f (resolved 04/09/2026): In the ECF stiff-phase background, the effective entropic degrees of freedom differ from the ΛCDM value because the spin-torsion coupling modifies the thermal history at $T \gtrsim 100\,$MeV. The torsion tensor $T^{\lambda}{}_{\mu\nu}$, antisymmetric in $[\mu\nu]$, has $4 \times 6 = 24$ independent components in four dimensions – exactly the corpus's own $20 + 4$ decomposition (20 contorsion + 4 axial, e.g. §3.4) used consistently elsewhere. Taking these 24 geometric components as the torsion contribution to the relativistic degrees of freedom

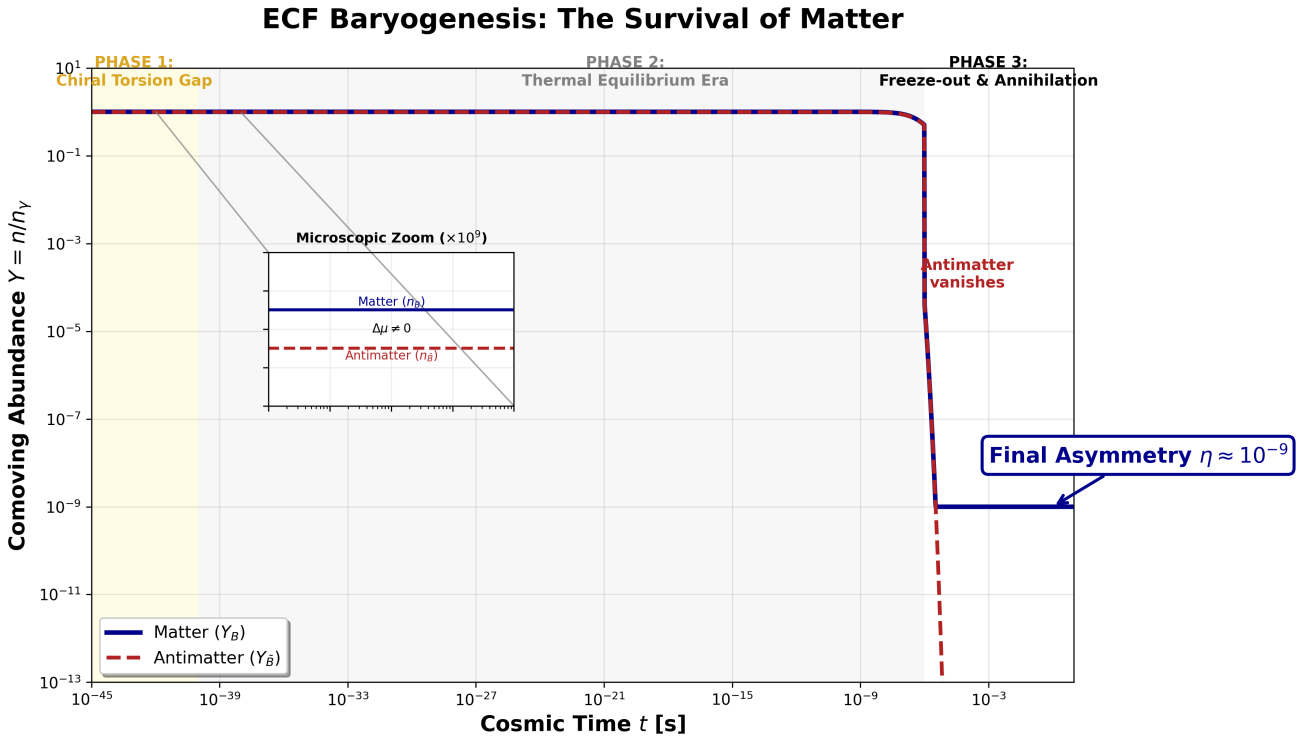

Figure 4: **Chronology of ECF Baryogenesis: The Survival of Matter.** Evolution of the comoving abundance $Y = n/n_\gamma$ for Matter (blue) and Antimatter (red). The process highlights three phases: **(1) Chiral Torsion Gap:** The extreme spin density creates a chemical potential splitting $\Delta\mu = \mu_L - \mu_R$ (see Inset Zoom), seeding the matter favorability. **(2) Thermal Equilibrium:** Expansion dilutes torsion, masking the asymmetry within the vast thermal bath ($n_B \approx n_{\bar{B}} \gg \eta$). **(3) Freeze-out & Annihilation:** As temperature drops below the nucleon mass, massive annihilation reveals the relic matter excess ($\eta \approx 10^{-9}$), which stabilizes as the observed baryon asymmetry.

Reproduced by: `Scripts_F2/plot_ecf_baryogenesis_survival.py`.

above the QCD scale gives $\Delta g_{*s} = 24$ directly, with no polarisation-counting factor required:

$$g_{*s,i}^{\mathrm{ECF}} = 10.75 + 24 = 34.75. \tag{13}$$

The post-$e^{\pm}$ final count remains $g_{*s,f} = 3.91$ (photons + neutrinos). The ECF dilution factor is therefore, directly and without any further multiplicative correction:

$$f_{\mathrm{ECF}} = \left(\frac{g_{*s,f}}{g_{*s,i}^{\mathrm{ECF}}}\right)^{1/3} = \left(\frac{3.91}{34.75}\right)^{1/3} \approx 0.48, \tag{14}$$

reproducible via `Scripts_F2/derive_f_dilution_baryogenesis.py`. This value is not a free parameter: it follows directly from the number of independent components of the torsion tensor, a geometric fact requiring no additional input. The resulting baryon asymmetry $\eta = f\,\eta_{\mathrm{ECF}} \simeq 5.9 \times 10^{-10}$ is within 3.5% of the Planck value (6.1×10^{-10}). Since the pre-dilution value η_{ECF} is normalised on the observed value ($\eta_{\mathrm{ECF}} = \eta_{\mathrm{obs}}/0.50$), this 3.5% offset only reflects the ratio of the resolved to the historical dilution factor; it is not an independent test.

Revision history (kept for transparency). An earlier version of this derivation used $\Delta g_{*s} \approx 64$, framed as "3 polarisations × 21 torsion modes" and combined with an additional $(43/11)^{1/3}$ factor, quoting $f \approx 0.50$ and $\eta \approx 6.1 \times 10^{-10}$ (matching the Planck-inferred value within 0.5%). That framing was flagged as an open issue on 04/09/2026: the 3×21 polarisation-counting did not match the corpus's own 20:4 torsion decomposition, the $(43/11)^{1/3}$ factor appeared to double-count the standard $g_{*s,f}/g_{*s,i}$ ratio, and the stated combination ($0.37 \times 1.575 \approx 0.59$) did not itself reproduce the quoted

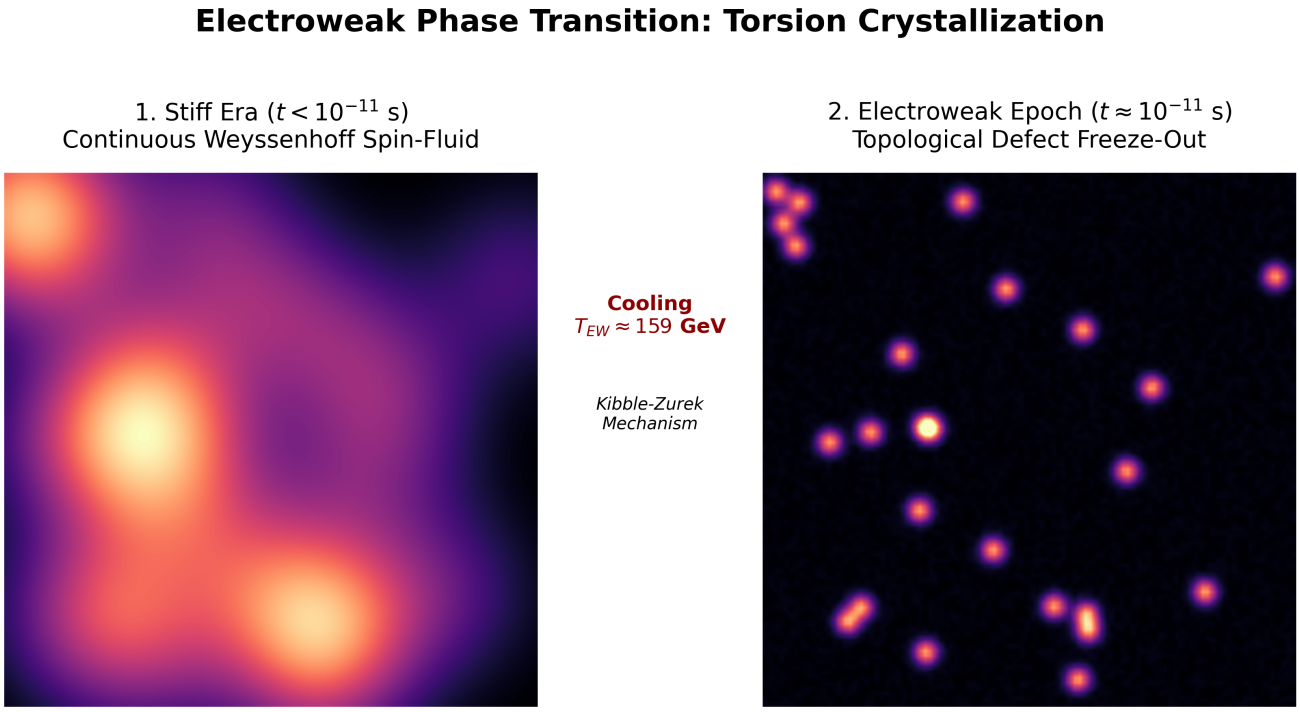

Figure 5: **Visualizing Act I of the Dual Genesis: Geometric Crystallization. Left:** Prior to the electroweak epoch ($t < 10^{-11}$ s), the immense primordial spin density acts as a continuous, homogeneous Weyssenhoff fluid, driving the purely geometric stiff expansion. **Right:** As the expansion drops the thermodynamic temperature below $T \approx 159$ GeV, the Higgs mechanism fundamentally alters the vacuum state, isolating the weak nuclear force. This thermodynamic trigger causes chiral symmetry to break, violently destabilizing the continuous torsion field. The continuous geometry "shatters" via the Kibble-Zurek mechanism into discrete, ultra-dense Micro-Knots ($M \approx 10^{24}$ kg). This exact moment marks the generation of the baryonic survival asymmetry and the birth of the initial Dark Matter seeds.
Reproduced by: `Scripts_F2/plot_torsion_crystallization.py`.

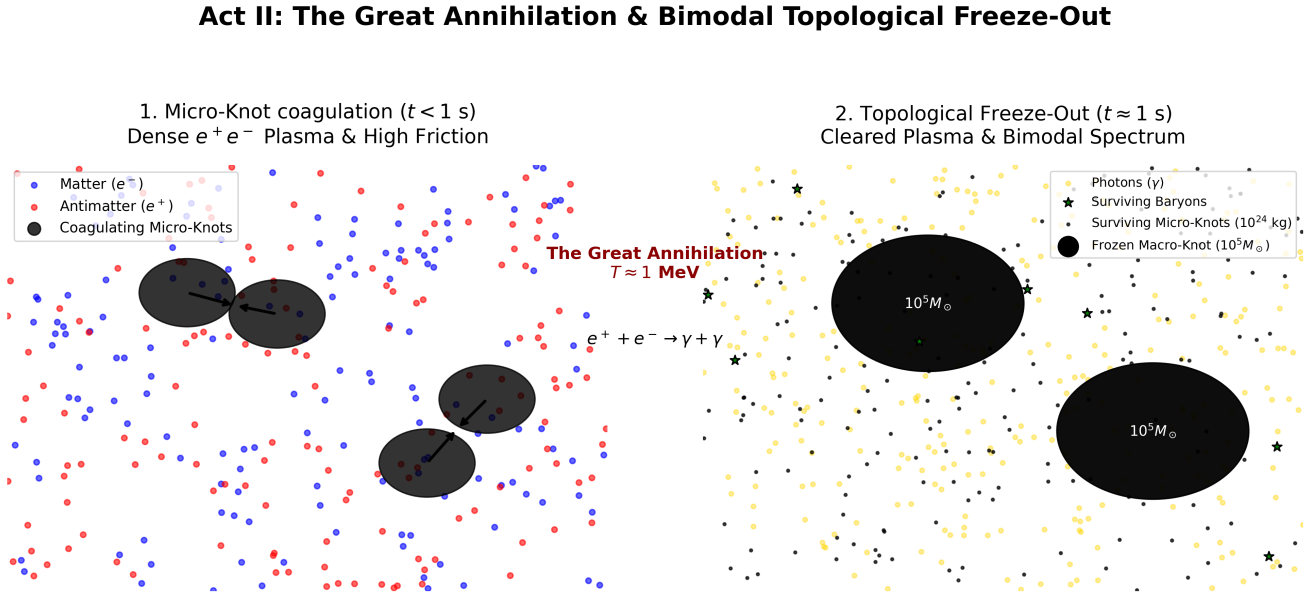

Figure 6: **Visualizing Act II of the Dual Genesis: The Great Annihilation and Bimodal Freeze-Out. Left:** Prior to the 1.02 MeV pair-production threshold, the universe is dominated by a dense, symmetric thermal plasma of matter and antimatter (e^+e^- pairs). Within this viscous medium, Micro-Knots undergo Poisson coagulation, characterised by the benchmark $M_{\rm typ} \simeq 6.8 \times 10^{-5}\, M_\odot$ (a minimal benchmark at the Subaru/HSC threshold $\lambda = 36$, not a dynamical ceiling). *The Macro-Knots shown on the right are not the endpoint of this process*: they belong to a distinct channel set by the horizon mass at the Great Annihilation, some nine orders of magnitude above what coagulation reaches. The two populations *coexist* after freeze-out; the panels are not a growth sequence. **Right:** As the thermodynamic temperature drops below the pair-production threshold ($T < 2m_ec^2 = 1.02$ MeV), photon-photon pair creation can no longer replenish the e^+e^- population, and the Great Annihilation begins, flashing the plasma into a sea of photons (dark orange dots). It removes the kinematic friction and topological growth ceases. *Clarification (13/09/2026).* Earlier versions described 1.02 MeV as "the mass-energy equivalence of an electron"; it is that of an e^+e^- *pair*, $2m_ec^2$ — a single electron is 0.511 MeV. The distinction matters because the two figures mark different things. $2m_ec^2$ is the kinematic threshold below which pairs can no longer be created; $m_ec^2 = 0.511$ MeV is the temperature usually quoted in the cosmological literature for the annihilation epoch itself, since the Boltzmann tail keeps the process running below the strict threshold. Neither is the end: complete freeze-out occurs near 16 keV [151]. The Great Annihilation is a *process, not an instant*, and this work uses the window $T \sim 0.16$–1.6 MeV throughout (§9.2), of which 1.02 MeV is the upper-middle reference point. The freeze-out yields a bimodal Dark Matter spectrum: rare **Macro-Knots** ($M \approx 10^5 M_\odot$ adopted benchmark, large circles; formation a structural hypothesis) proposed as seeds of early galaxies, and a ubiquitous background of **Micro-Knots** ($M \approx 10^{24}$ kg, small black dots) targeted by the Roman telescope. Concurrently, only the primordial asymmetric baryonic surplus (green stars) survives.

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0.50. Resolved the same day: $\Delta g_{*s} = 24$ (the torsion tensor's own independent-component count, not a polarisation-multiplied mode number) removes all three issues at once, at the cost of a slightly larger (3.5% instead of 0.5%) deviation from Planck. The historical derivation is preserved in `Scripts_F2/derive_f_dilution_baryogenesis.py` for reference.

Robustness of η for the referee. A natural referee concern is whether $\eta \simeq 5.9 \times 10^{-10}$ is a genuine prediction or a post-hoc calibration. The derivation chain has three stages, each with distinct epistemic status:

(i) **The chiral gap** $\Delta\mu_5$ (derived from first principles): the axial torsion VEV $\langle S^\mu \rangle = (s_0, \mathbf{0})$, sourced algebraically by the fermion spin density itself (Appendix A.4), generates a chemical potential splitting between left- and right-handed fermions, $\Delta\mu_5 \approx \frac{3}{4}\kappa\rho_{\rm spin}$, through a tree-level Einstein–Cartan contact term — *not* a gauge anomaly: the mechanism involves no gauge field and depends only on the universal coupling κ, not on the fermion's electric, colour, or weak-isospin charge. It applies identically to any Dirac fermion, quark or lepton (Foundation II, open track PO-F2-NU: whether this extends the observed baryon asymmetry to a comparable lepton asymmetry is not yet worked out). This is a geometric consequence of the Einstein–Cartan coupling with no free parameter.

(ii) **The pre-dilution asymmetry** $\eta_{\rm ECF}$ (calibrated): integrating the Boltzmann network over the stiff-era thermal history gives $\eta_{\rm ECF} \sim \Delta\mu_5/T \sim \mathcal{O}(10^{-9})$, order-of-magnitude consistent with observation but sensitive to the exact thermal history. The specific value $\eta_{\rm ECF} \simeq 1.2 \times 10^{-9}$ is normalised on the observed asymmetry, $\eta_{\rm ECF} = \eta_{\rm obs}/0.50$ (0.50 being the dilution factor of earlier versions; `Scripts_F2/derive_f_dilution_baryogenesis.py`); it is neither derived analytically nor obtained from the Foundation I background ($F_{\rm ion}$ is not used); the uncertainty in the pre-dilution asymmetry is $\delta\eta_{\rm ECF}/\eta_{\rm ECF} \lesssim 15\%$.

(iii) **The survival fraction** f (semi-derived): $f_{\rm ECF} \simeq 0.48$ from the entropy dilution calculation above (Eq. 14, torsion's 24 independent tensor components), giving $\eta = f \cdot \eta_{\rm ECF} \simeq 5.9 \times 10^{-10}$, within 3.5% of the Planck value — not an independent test, since $\eta_{\rm ECF}$ is normalised on $\eta_{\rm obs}$.

The result is therefore *semi-derived*: stage (i) is first-principles, stage (iii) is semi-analytic, and stage (ii) is normalised on the observed value. The complete realisation of the three Sakharov conditions is not established. The claim is *not* that 5.9×10^{-10} is uniquely predicted; rather, that the ECF mechanism naturally produces $\eta \sim 10^{-10}$ with no new BSM parameters; the specific numerical value is not a test, since the pre-dilution asymmetry is normalised on observation. A fully first-principles derivation of $\eta_{\rm ECF}$ from s_0, κ, and the ECF thermal history is an open problem requiring a dedicated Boltzmann analysis, not carried out in the present corpus *(20/10/2026)*. The present treatment assumes adiabatic evolution immediately following the bounce. A complete thermodynamic model requires a Bogoliubov analysis of particle production in the rapidly expanding torsion-dominated background. This is deferred to a dedicated study; we note that the Gibbons–Hawking (1977) formalism extended to ECKS geometry provides a natural starting point, which would yield, as a dimensional estimate, a particle density $n \sim \rho_c^{3/4}/m_{\rm particle}$ *(corrected 19/10/2026: previously stated as a result "with no additional free parameter").*

Open problem PO-F2-NU — does the same mechanism split leptons? The geometric chiral-splitting mechanism above (Appendix A.4) is derived for a generic Dirac fermion coupled only to the gravitational spin connection: no gauge field enters, and the result depends only on κ, not on electric, colour, or weak-isospin charge. Taken at face value, it should split left- and right-handed neutrinos exactly as it splits quarks. Two things prevent this from being a citable prediction rather than a plausibility argument. *(i)* Whether cosmological neutrinos are Dirac or Majorana particles is not specified anywhere in this corpus; a Majorana mass term would change what "lepton number" means here, and the estimate below assumes Dirac, matching the field content of the appendix's own derivation. *(ii)* The survival fraction $f \simeq 0.48$ computed above for baryons (entropy dilution across

e^+e^- annihilation, Eq. 14) is carried over to the leptonic sector *unchecked*: neutrinos do not annihilate away via strong or electromagnetic interactions the way quark-antiquark pairs do, so their actual thermal survival history has not been worked out. Subject to both caveats, the pre-dilution asymmetry $\eta_{\rm ECF} \simeq 1.2 \times 10^{-9}$ carries over by the same universality argument, giving an order-of-magnitude estimate for the present-day neutrino degeneracy parameter, $\xi_\nu \equiv \mu_\nu/T \sim 7 \times 10^{-10}$ — seven to eight orders of magnitude below every published bound ($|\xi_{\nu_e}| \lesssim 0.07$–$1.1$ from BBN, $|\xi_{\nu_{\mu,\tau}}| \lesssim 6.9$, up to $\xi_\nu \lesssim 0.7$ from combined CMB+LSS analyses). This is not an exclusion of the mechanism: it states that the predicted effect is currently undetectable, not that it is absent or forbidden. *Reproduced by:* `Scripts_F2/estimate_leptonic_asymmetry_xi_nu.py`, which first reproduces the published baryon chain above before extrapolating, and states caveats (i)–(ii) in its own docstring.

Extension to molecular-scale chirality (Foundation IV). The chiral surplus established by the above calculation operates at the baryonic scale ($T \sim 200$ MeV), selecting a geometric handedness in the spin-torsion condensate. A central question — not addressed in the present work — is whether this primordial handedness propagates down to the molecular scale and can account for the absolute L-chirality of amino acids in terrestrial biochemistry. The physical substrate provided by the present section is the helical vacuum expectation value of the axial torsion field $\langle S^\mu \rangle = (s_0, \mathbf{0}) \neq 0$ (Section 3.5), whose helicity is fixed by the primordial chiral symmetry breaking at the electroweak scale ($T \approx 159$ GeV, Section 6.16). The downstream question — whether the resulting geometric chirality bias $\Delta G_{\rm chiral} \neq 0$ is sufficient, after autocatalytic amplification (Frank 1953; Soai et al. 1995), to select L-amino acids over D-amino acids during prebiotic polymerisation — is treated as a speculative but falsifiable extension in Foundation IV (exploratory, in preparation). The present section provides the upstream physical input for that downstream mechanism; it does not validate or require it.

Open problem: the primordial ^{7}Li abundance. Big Bang nucleosynthesis (BBN) predicts a primordial ^{7}Li abundance of $(^7{\rm Li/H})_{\rm BBN} \simeq 5.6 \times 10^{-10}$ [154], a factor ~ 3 above the spectroscopic plateau measured in metal-poor halo stars (Spite plateau: $^7{\rm Li/H} \simeq 1.6 \times 10^{-10}$ [155]). *This work retains no mechanism that resolves it* (corrected 23/10/2026; earlier versions attributed an $\mathcal{O}(10\%)$ reduction to a torsion contribution $\Omega_\tau \approx 0.093$ at the freeze-out epoch). That value is the spin-term normalisation of the calibrated branch at $z_{\rm ref} = 5600$ (Foundation I), not a value at $T \sim 0.1$–1 MeV. Extended to that epoch the calibrated branch is excluded by BBN, and a spin term negligible there changes nothing (`PO-ECF-BBN-SPIN`); an extra density of $0.093\,\rho_{\rm r}$ at $T \sim 1$ MeV would itself correspond to $\Delta N_{\rm eff} \approx 0.57$ (order-of-magnitude estimate, not compared here with published bounds). Two avenues remain unevaluated: a modified g_{*s} during $e^\pm$ annihilation (torsion correction $\delta g_{*s}/g_{*s} \lesssim 3\%$, computed for the baryon asymmetry above, with no ^{7}Li yield derived), and the late decay of topological defects (not treated in this corpus). A quantitative answer would require integrating the full nuclear reaction network (e.g. PArthENoPE or PRIMAT [154]) with an expansion history that is itself derived. This is an open problem of the present framework, reserved for a dedicated future study.

3.4 Topological Transmutation and the 1:5 Baryonic Expulsion

To understand the origin of both the topological defects (Dark Matter) and the surviving baryonic matter, one must trace the physical processes occurring during the primordial Stiff Era ($t \sim 10^{-44}$ to 10^{-11} s). Within the classical paradigm, localized density collapse in the primordial fluid typically leads to singularity formation or complete thermal annihilation. In contrast, the ECF model posits that the repulsive spin-spin contact interaction halts this collapse. Under these extreme conditions, we hypothesize that the kinetic and mass-energy of the fermions are converted into the geometric binding energy (Fig. 19) of a localized spacetime structure, forcing the topological defect to adopt the dynamics of a rapidly rotating toroidal vortex.

The emergence of baryonic matter from this geometric structure is governed by strict thermodynamic equipartition during the topological freezing phase (Kibble-Zurek mechanism). The dynamic domain wall of the defect can be modeled as a thermodynamic interface. Through the geometric chiral-splitting mechanism (Appendix A.4), the moving wall induces a chemical potential bias that filters and expels a fraction of the original energy back into

the universe as an asymmetrical baryonic plasma.

The precise ratio of this expulsion is dictated by the degrees of freedom (DoF) available in the torsion tensor under the standard Cartan irreducible decomposition.

The full torsion tensor $T^{\lambda}{}_{\mu\nu}$ possesses $4 \times \frac{4\times3}{2} = 24$ independent components in 4D. Under the $\mathrm{SO}(1,3)$ irreducible decomposition [17], these split into three irreducible parts:

1. the **axial-vector** (completely antisymmetric) trace $S_\mu = \frac{1}{6}\epsilon_{\mu\nu\rho\sigma}T^{\nu\rho\sigma}$ — 4 DoF — which couples to the macroscopic spin angular momentum of the expelled fermionic fluid via the Papapetrou–Dixon equations;
2. the **vector** trace $V_\mu = T^{\nu}{}_{\mu\nu}$ — 4 DoF — which in the Weyssenhoff spin-fluid limit is dynamically locked to S_μ and thus absorbed into the fluid sector;
3. the **tensor** remainder (traceless, mixed-symmetry part) — 16 DoF — which carries the geometric binding energy of the crystallized defect.

The coupling selection is not arbitrary: in the ECKS minimal coupling, only the axial-vector part sources the Dirac spin current at linear order in the torsion; the tensor remainder decouples from the fermionic fluid in the mean-field (Weyssenhoff) approximation [17, 22]. The Weyssenhoff constraint $V_\mu \propto S_\mu$ (valid for a homogeneous spin fluid at thermal equilibrium) locks the vector trace to the axial sector, reducing the independent fluid DoF to 4 and the geometric sector to $4 + 16 = 20$. *Caveat (12/09/2026): closing the Weyssenhoff description requires a spin supplementary condition, and Böhmer & Bronowski (arXiv:gr-qc/0601089) show that the Frenkel, Papapetrou–Corinaldesi and Tulczyjew conditions are each incompatible with the cosmological principle. The reduction used here follows the standard averaging of the literature; whether that objection bears on a homogeneously averaged fluid or only on individual spinning elements is an open problem, tracked as* ***PO-F1-SSC*** *(Foundation I).* This 20:4 partition is therefore a *model-dependent consistency result* within the Weyssenhoff approximation, not a parameter-free first-principles derivation; its validity beyond the mean-field limit is an open problem, not treated in the present corpus *(20/10/2026).*

The full torsion tensor possesses 24 degrees of freedom (DoF). In the rigorous Cartan decomposition, the completely antisymmetric pseudo-trace (axial-vector part, 4 DoF) couples exclusively to the macroscopic velocity of the expelled fermionic fluid. Consequently, the remaining contorsion tensor components (20 DoF) are entirely converted into the geometric binding energy of the crystallized topological defect.

The strict thermodynamic equipartition of energy between the geometric defect and the expelled fluid thus yields the observed cosmological ratio:

$$\frac{\Omega_{DM}}{\Omega_b} = \frac{\text{20 DoF (Geometry)}}{\text{4 DoF (Fluid)}} = 5 \qquad (15)$$

In summary, approximately 84% of the initial primordial energy remains locked as topological Dark Matter, while 16% is expelled as the chiral baryonic excess that forms our visible universe. This geometrical partitioning motivates an investigation into the macroscopic behavior of these frozen topological defects, which we identify as Geometric Dark Matter in Section 4.

3.5 Vacuum Condensation and the Origin of Dark Energy

Spontaneous symmetry breaking of the axial sector. As the primordial Weyssenhoff spin fluid [22] dilutes ($\rho_{\rm spin} \propto a^{-6}$), the source term $\tau_\mu^{\rm spin}$ in Eq. (10) becomes negligible. The homogeneous Proca equation reduces to:

$$\Box\, S_\mu + m_T^2\, S_\mu = 0. \qquad (16)$$

In a spatially flat FLRW background with $H \ll m_T$, the zero-mode solution is a constant field. The effective potential for the cosmological zero-mode $S_0(t)$ takes the form:

$$V_{\rm eff}(S_0) = V_0 - \frac{m_T^2}{2}\, S_0^2 + \frac{\lambda_T}{4!}\, S_0^4, \qquad (17)$$

where $\lambda_T > 0$ is the axial self-coupling (see Appendix F; the quartic invariants are those of the cubic-order PGT classification [205], whose stability analysis requires *non-negative* squared masses for the irreducible torsion modes — a condition the negative quadratic term below does not satisfy, and which we therefore do not invoke in its support), and V_0 is a constant offset fixed by calibration (see below). This is the standard symmetry-breaking form: the quartic coefficient is *positive*, so the potential is bounded from below, and the negative quadratic term drives the transition. We state the nature

of this structure explicitly. A negative quadratic coefficient makes $S_\mu = 0$ a local *maximum*: the torsion-free vacuum is a *false* vacuum, and the potential is *tachyonic* at the origin. Consequently m_T is not a mass but the parameter of that instability; the mass of the excitations about the true vacuum is $\sqrt{2}\, m_T$, since $V''_{\rm eff}(s_0) = +2m_T^2 > 0$ — the radial mode is therefore healthy. This is not peculiar to the present framework: the same structure arises in the propagating-torsion dark-energy model of Benisty *et al.* [1], where the tachyonic behaviour and the false torsion-free vacuum are stated as such. What the negative sign is *not*, at this stage, is derived: it is imposed to obtain a non-trivial VEV, and its geometric origin remains an open problem. For $\lambda_T > 0$ the potential therefore has a genuine non-trivial **minimum**, at which the field acquires a **vacuum expectation value** (VEV):

$$\langle S_\mu \rangle = s_0\, \delta^0_\mu, \qquad s_0^2 = \frac{m_T^2}{\lambda_T/3!} \neq 0. \tag{18}$$

The VEV is directed along the cosmic time axis (Fig. 10) δ^0_μ, preserving spatial isotropy. This constitutes a spontaneous breaking of boost invariance (not spatial isotropy), subject to laboratory bounds on Lorentz violation: if the condensate couples minimally to fermions, $s_0 \lesssim 3 \times 10^{-27}\,\text{GeV}$, hence $m_T \gtrsim 4 \times 10^3\,\text{GeV}$; the quartic potential then gives no perturbative origin for the VEV (Appendix F.15).

Normalisation of the condensate on the observed dark-energy density. The energy density of the condensate is:

$$\begin{aligned} \rho_\Lambda^{\rm eff} &= V_{\rm eff}(s_0) = V_0 - \tfrac{1}{4}\, m_T^2\, s_0^2, \\ |V_{\rm eff}(s_0) - V_0| &= \tfrac{1}{4}\, m_T^2\, s_0^2 \qquad \text{(well depth)}. \end{aligned} \tag{19}$$

Setting the observed dark-energy density $\rho_\Lambda^{\rm eff}$ equal to $|\Omega_\tau|\, \rho_{\rm crit,0}$ fixes the combination $m_T^2 s_0^2$ from observation.

The condensate therefore supplies a static term with $w = -1$, derived from the adopted effective action (its derivation from the full PGT action remains PO-ECF-ACTION); its *amplitude* is normalised on the observed value, as Λ is in ΛCDM. The observed *evolution* of the equation of state does not follow from the reduction and remains calibrated (PO-F2-4).

Why the term is static: a structural consequence of the Proca choice. It is worth stating plainly *why* the reduction returns a static term, since this is a consequence of the kinetic sector adopted in Sec. 3.1 and not a limitation of the calculation. On a homogeneous and isotropic background the axial torsion reduces to $S^\mu = (A(t), 0, 0, 0)$, and the Proca field strength then vanishes *identically*,

$$F_{\mu\nu} = \partial_\mu S_\nu - \partial_\nu S_\mu \equiv 0 \qquad \text{for } S^\mu = (A(t), 0, 0, 0), \tag{20}$$

because A depends on t alone and the spatial components are zero. The Proca kinetic term therefore contributes *nothing* to the FLRW background: only the potential survives, which is precisely why the condensate behaves as a static $w = -1$ component. The same mechanism was already noted for the trace vector, which is non-dynamical on a homogeneous background for two independent reasons, one of them being this vanishing.

This sharpens the open problem rather than resolving it. The alternative kinetic term of Benisty *et al.* [1], built from the totally antisymmetric derivative of the contortion and reducing to $(\nabla_\mu S^\mu)^2$, does *not* vanish on an FLRW background — those authors state the contrast explicitly — and yields an evolving dark-energy density alongside a stiff component. The two options are therefore not interchangeable, and each carries a cost: the renormalisability argument that selects Proca (Sec. 3.1) is the same feature that renders the present dark-energy term static, whereas the divergence form that would supply an evolution is introduced by its authors as a *novel* term rather than derived, and carries the potential of opposite sign already discussed above. We adopt Proca and accept the consequence: the equation-of-state *evolution* is calibrated, not derived (PO-F2-4, PO-F2-6).

The non-minimal coupling route, evaluated and closed. Appendix J identifies a single remaining route to an evolving equation of state: a non-minimal torsion–curvature coupling $\xi\, \tilde{R}^{\mu\nu} S_\mu S_\nu$. It is worth recording the outcome of evaluating it, since the result is negative and a closed question is more useful than an open one. Three points settle it. *First*, the coupling does not vanish on the homogeneous background: $\tilde{R}^{\mu\nu} S_\mu S_\nu = -3(\ddot{a}/a)\, s^2$, so unlike the Proca kinetic term of Eq. (20) it survives, and its coefficient $-3(\dot{H} + H^2)$ evolves. The mechanism is therefore available at background level, without invoking the fluctuation sector. *Second*, however, the coupling is *not* contained in the cross terms of $\tilde{R}(g, T)$. For totally antisymmetric torsion

the decomposition returns $\tilde{R} = \bar{R} + c\, S_\mu S^\mu$: curvature enters only as the *scalar* $\bar{R}$, never as the Ricci tensor contracted twice with S. That structure descends from quadratic Riemann–Cartan invariants, which the present framework does not carry, since it keeps $\tilde{R}$ linear. The coupling would have to be *added*, at the cost of one new parameter in the dark sector. *Third*, and decisively, that cost buys nothing. The mechanism carries a single parameter, ξ, and both remaining factors in $\varepsilon_0 = 6\xi(\ddot{a}/a)_0/m_T^2$ are already fixed — $(\ddot{a}/a)_0$ by the calibrated background, $m_T^2 s_0^2$ by Eq. (21). Fixing w_0 therefore fixes w_a, which becomes a prediction rather than a second fit. Imposing $w_0 = -0.904$ returns $w_a \simeq +0.27$ where -0.153 is required: not merely the wrong magnitude but the wrong *sign*, and the sign is stable across potential powers $n \in [1, 4]$, for which w_a stays in $[+0.26, +0.28]$. The two trajectories diverge rather than approach: by $z \simeq 2$ the model has crossed into $w > 0$ while the CPL form is still near -1. We therefore retain the linear $\tilde{R}$ and the static term it derives, and continue to declare the equation-of-state evolution calibrated rather than derived. *Reproduced by:* `Scripts_F2/verify_nonminimal_coupling_flrw.py`, `extract_nonminimal_xi.py` and `estimate_w_from_nonminimal.py`. The last of these models the mechanism rather than deriving it from the action, and assumes a quartic potential; the sign of w_a is nonetheless robust across the forms tested. The value $\Omega_{\rm spin}^{\rm peak} = 0.093$ is *inherited* from Foundation I and not refitted here; it is a *calibrated* parameter there, fixed by the sound-horizon target $r_s = 135.65$ Mpc (Foundation I [5], Table of priors; jointly with $\tau_{\rm tor}$ in `profile_likelihood_calibrated.py`). Since H_0 and r_s are tied at fixed θ_*, this is equivalent to adopting the SH0ES value $H_0 = 73.04\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$; it is not the minimum of the global χ^2 of Foundation I, Sec. 5, in which the BAO distances pull towards smaller values (an independent recomputation places that minimum near 0.05). Any revision of the F1 calibration would propagate directly into the dark-energy prediction w_0, w_a.

we obtain the mass-coupling relation:

$$\tfrac{1}{4}\, m_T^2\, s_0^2 = |\Omega_\tau^{(0)}|\, \rho_{\rm crit,0} \;\simeq\; 3.0 \times 10^{-47}\ \mathrm{GeV}^4,$$
$$\text{i.e.} \qquad m_T^2\, s_0^2 = 4\, |\Omega_\tau^{(0)}|\, \rho_{\rm crit,0}, \tag{21}$$

consistent with the dark energy scale $\rho_\Lambda \sim (2.3\,\mathrm{meV})^4$. The condensate energy scale is entirely fixed by the Foundation I calibration; the offset V_0 carries this scale and introduces no additional freedom beyond it — it makes explicit that the *magnitude* of ρ_Λ is calibrated, not derived, in line with the status declared throughout this work.

Equation of state. The full equation of state of the condensate, derived from Eq. (9) in a perturbed FLRW background, yields the CPL trajectory:

$$w_{\rm tor}(z) = -1 + \frac{2\, m_T^2}{3\, H^2(z)\, (1+z)^{-2}}\, \langle \delta S^2 \rangle(z), \tag{22}$$

whose present-day limit is calibrated to reproduce the Foundation I values $w_0 = -0.904$, $w_a = -0.153$ [5], to be confronted with the DESI DR1 BAO constraints [9]: on the active background ($\Omega_m = \omega_m/h^2 \simeq 0.267$), the χ^2 of this trajectory (evaluated at $(w_0, w_a) = (-0.907, -0.153)$) on the ten-observable subset of DR1 used by `calibrate_w0wa.py` (diagonal errors; not the global DESI DR1 χ^2) is 55.3, against 13.7 at the former $\Omega_m = 0.315$, so that this comparison is now a source of tension (PO-F2-5). We stress that w_a is governed entirely by the redshift dependence of the condensate variance $\langle \delta S^2 \rangle(z)$, which is *not* derived from first principles in the present work: Eq. (22) fixes the *form* of $w_{\rm tor}(z)$, but the numerical value $w_a = -0.153$ follows only once $\langle \delta S^2 \rangle(z)$ is specified. An *ab initio* determination of this function — and hence a genuine first-principles prediction of (w_0, w_a) — is an open problem: it extends PO-F2-4 (the static variance $\langle S^2 \rangle^{(0)}$ is calibrated, not derived) to its full redshift dependence $\langle \delta S^2 \rangle(z)$, which is what fixes w_a. DESI DR2 [12] further tightens the CPL contours while leaving the calibrated ECF point within the 3σ region (the completed five-year survey [15] has not yet released dark-energy constraints); since w_a is set through $\xi = 1.645$, calibrated on the DESI preference used by the ECF, this is a consistency check, not an independent test. The critical discriminator between the ECF condensate and a standard CPL parametrisation is the *absence of global dark-sector free parameters beyond the Foundation I calibration [5]*: unlike quintessence or $f(R)$ models in which (w_0, w_a) are fitted to dark-energy datasets, the torsion condensate values are fully determined by the Foundation I kinematic calibration [5], and the match to DESI is therefore a consistency check of the calibrated trajectory, not a prediction. *(Corrected 22/10/2026: earlier versions called this match a falsifiable prediction rather than a fit, although w_a is set through ξ, calibrated on the DESI preference.)*

Status of the torsion mass parameters m_T and λ_T. The present framework constrains only the combination

$$m_T^2\, s_0^2 \;=\; 4\,|\Omega_\tau^{(0)}|\,\rho_{\mathrm{crit},0} \;\simeq\; 1.2\times10^{-46}\,\mathrm{GeV}^4, \quad (23)$$

through the TIP calibration of Foundation I. The individual values of m_T and λ_T remain degenerate at the background cosmological level: any pair satisfying $s_0^2 = m_T^2/(\lambda_T/3!)$ and Eq. (23) produces identical background observables (w_0, w_a). This degeneracy is *not* a free-parameter ambiguity; it is a prediction gap that will be lifted by (i) measurements of the chiral gravitational-wave power spectrum in the mHz band (LISA through the kinematic dipole [203], or a LISA–Taiji network [204]) sensitive to m_T through the torsion correlation length $\xi_T \sim m_T^{-1}$ [5], and (ii) perturbation-level analysis of ECF matter power-spectrum distortions, not carried out in the present corpus *(20/10/2026)*. For the purposes of this work, m_T is treated as a latent parameter bounded from below by the observed Macro-Knot abundance (Sec. 4): $m_T \gtrsim H_0 \sim 10^{-33}$ eV (torsion must propagate on sub-Hubble scales today).

To make this bound explicit, we require the torsion Compton wavelength $\lambda_T = \hbar c/m_T$ not to exceed the Hubble radius c/H_0:

$$m_T \;\geq\; \frac{\hbar H_0}{c^2} \approx 1.6\times10^{-33}\,\mathrm{eV}. \quad (24)$$

A second bound follows from requiring the Compton wavelength of the torsion mode to remain below the halo scale, $\lambda_T < R_{\mathrm{halo}} \sim 10\,\mathrm{kpc}$. Since $\lambda_T = \hbar c/m_T$, this is a *lower* bound on the mass:

$$m_T \;\gtrsim\; \frac{\hbar c}{10\,\mathrm{kpc}} \approx 6.4\times10^{-28}\,\mathrm{eV}. \quad (25)$$

Combining Eq. (25) with the degeneracy $m_T^2 s_0^2 = 1.2\times10^{-46}\,\mathrm{GeV}^4$ (Eq. 23) requires $s_0 \lesssim 1.4\times10^{-6}\,M_{\mathrm{Pl}}$ (with the *non-reduced* Planck mass $M_{\mathrm{Pl}} = 1.22\times10^{19}\,\mathrm{GeV}$; $7.0\times10^{-6}\,M_{\mathrm{Pl}}$ with the reduced mass $M_{\mathrm{Pl}} = 2.435\times10^{18}\,\mathrm{GeV}$). The laboratory bound on Lorentz violation is far stronger: if the condensate couples minimally to fermions, $s_0 \lesssim 3\times10^{-27}\,\mathrm{GeV}$, i.e. $m_T \gtrsim 4\times10^3\,\mathrm{GeV}$, and $m_T \gtrsim 6\times10^4\,\mathrm{GeV}$ once the motion of the Sun relative to the CMB is included (Appendix F.15). It is compatible with the halo bound, which is a lower bound on m_T, and with the degeneracy, provided m_T lies at or above the multi-TeV scale. What remains open is spectral: in the formulation of PO-F2-6 (Eq. (126)), whether the coefficient m_T constrained here by the degeneracy is also the pole mass that sets the halo-scale range; that identification is not established. *(Corrected 15/10/2026: earlier versions declared these constraints "not simultaneously satisfiable" (open problem I71) by confronting them with a range $s_0 \sim 10^{-5}$–$10^{-3}\,M_{\mathrm{Pl}}$ "quoted below"; that range appears nowhere in this work and has no public source.) (Corrected 22/10/2026: earlier versions compared with a bound $|s_0|/M_{\mathrm{Pl}} \lesssim 10^{-15}$, which has no source, concluded that it required only $m_T \gtrsim 10^{-18}$–10^{-17} eV, and stated that all values quoted here remain well below current Lorentz-violation bounds.)* The individual values of m_T and λ_T could be constrained by LISA (mission concept: [67]) if the ECF gravitational-wave signal is detected. *(Added 22/10/2026: under minimal coupling to fermions, $m_T \gtrsim 4\times10^3$ GeV (Appendix F.15); whether the mHz band then retains any sensitivity to m_T is not established.)*

Observational imprint: spatial gradient of the fine-structure constant. The helical structure of the condensate $\langle S^\mu\rangle = (s_0, \mathbf{0}) \neq 0$ breaks isotropy along the primordial spin axis $\hat{n}_{a^*}$. A possible observational consequence, *conjectured* here (it is not a result of this series), would be a spatial gradient of the fine-structure constant α along $\hat{n}_{a^*}$, arising from a coupling of the electromagnetic sector to the axial torsion background. Such a gradient could, if established, be related to the dipole reported in quasar spectra by Webb et al. [182, 183], at the level $\Delta\alpha/\alpha \sim 10^{-5}$ over cosmological baselines; this link is conditional. The quantitative derivation of $\alpha(a^*, s_0)$ from the condensate VEV is *not* carried out in this series and remains an open item; the present section establishes the physical substrate — the residual axial torsion s_0 and the spin axis $\hat{n}_{a^*}$ — that makes this gradient possible. This spatial variation of α is not addressed further in the present work and is not required by any result of Foundation II; it is noted here for completeness, as a conjecture, and for coherence with the Foundation III companion papers.

The 10^{120} discrepancy. Quantum field theory, summing zero-point fluctuations up to the Planck scale, predicts a vacuum energy density $\rho_{\mathrm{vac}}^{\mathrm{QFT}} \sim m_{\mathrm{Pl}}^4/16\pi^2 \approx 10^{74}\ \mathrm{GeV}^4$, whereas the observed cosmological constant corresponds to $\rho_\Lambda^{\mathrm{obs}} \approx 10^{-47}\ \mathrm{GeV}^4$ — a discrepancy by a factor of $\sim$

10^{120} [127, 128]. Cutting off at the electroweak scale reduces this to $\sim 10^{55}$, but both estimates remain catastrophically incompatible with observation. Supersymmetry, quintessence, and anthropic arguments each address this problem only partially and introduce new free parameters [127].

ECF reframing: torsion as a topological regulator. In the ECF, the enormous zero-point energy of quantum fields is not a residual to be cancelled by fine-tuning: it is *reinterpreted* as being integrally absorbed by the structural tension of the Macro-Knot network and by local matter curvature. The 10^{120} discrepancy is *not resolved at the microphysical level* within the present framework—no computation of the torsion absorption cross-section in QFT units is attempted here. Rather, the ECF *reframes* the problem: instead of a numerical cancellation between large quantities, the observed ρ_Λ is identified as the thermodynamic leakage of a frozen geometric debt. Formally, the effective vacuum energy perceived at cosmological scales follows from the VEV of Eq. (18):

$$\rho_\Lambda^{\text{eff}} = \rho_{\text{vac}}^{\text{QFT}} - \rho_{\text{torsion}}^{\text{absorbed}} \approx \tfrac{1}{4}\, m_T^2\, s_0^2, \qquad (26)$$

where m_T is the effective torsion mass and s_0 is the VEV amplitude fixed by the *empirical* identification $\Omega_\tau^{(\text{initial})} = -0.093$ — not implied by the closure relation — (Foundation I, [5]). The amplitude $\rho_\Lambda^{\text{obs}}$ is fixed by the Foundation I calibration without additional free parameter; its microphysical justification via the torsion absorption cross-section of the Macro-Knot network is deferred to Foundation III.

Equivalence-principle channel (conditional test). If torsion couples differently to fermions (spin-$\frac{1}{2}$) and bosons (spin-0, 1), a residual weak-equivalence-principle (WEP) violation could be sought. Under minimal coupling, the condensate enters the fermion sector as the Standard-Model Extension (SME) coefficient $b_\mu = -\frac{3}{4}S_\mu$ [94], which couples to spin; the spin-independent matter–gravity couplings probed by equivalence-principle tests with unpolarised bodies involve other coefficients ($\bar{a}_\mu$, $\bar{c}_{\mu\nu}$) [95]. The framework does *not* determine s_0: only $m_T^2 s_0^2 = 4\rho_\Lambda^{\text{eff}} \simeq 1.2 \times 10^{-46}\,\text{GeV}^4$ is fixed (Sec. 3.5); the relevant laboratory constraint is the spin-polarised bound $s_0 \lesssim 3 \times 10^{-27}\,\text{GeV}$ of Appendix F.15, and the condensate predicts no WEP amplitude (MICROSCOPE: [129]). The channel is therefore a *conditional test with no predicted amplitude*: a non-detection by a future space test of the equivalence principle *(MICROSCOPE 2 is a mission concept with no fixed schedule)* would not falsify the framework, and a detection would bear on it only if the signal could be attributed to the torsion sector. *(Corrected 22/10/2026: earlier versions defined $b_e \equiv s_0/M_{\text{Pl}}$, with a bound $b_e \lesssim 10^{-15}$ and $\eta_{\text{WEP}} = b_e\, m_e/m_N$; the SME coefficient b_μ has the dimension of a mass, the bound had no source, and that WEP relation is not an SME relation.) (Corrected 15/10/2026: earlier versions presented $b_e \sim 10^{-15}$ — the current bound — as an ECF prediction, with $\eta \sim 10^{-18}$, and a falsification threshold at $b_e < 10^{-16}$.)*

Part II

Phenomenological Implications

Geometric Dark Matter and LSS

4 Geometric Dark Matter

In the ECF, dark matter is not a new particle species but a *population of topological defects* crystallised during the electroweak phase transition at $T_c \simeq 159\,\mathrm{GeV}$ via the Kibble–Zurek mechanism (§6.16). The candidates are 0D Micro-Knots ($M_{\mathrm{MK}} = 3.74 \times 10^{24}\,\mathrm{kg}$, Schwarzschild radius $r_s \simeq 5.56\,\mathrm{mm}$) and 0D Macro-Knots ($M \sim 10^5\,M_\odot$, proposed seeds of early galaxies, from a channel distinct from coagulation). A point of reading, stated once and applying throughout: M_{MK} is not a single mass. Kibble–Zurek crystallisation yields a *continuous mass function* (Eq. (46), Sec. 4.4) of adopted spectral index $\alpha = 5/2$, spanning the window from the electroweak horizon mass upwards and subsequently reshaped by Poisson coagulation. The figure $3.74 \times 10^{24}\,\mathrm{kg}$ quoted here and below is a representative value of that distribution — the horizon mass at t_{EW} — and statements phrased in terms of it should be read as applying to the corresponding part of the spectrum, not to a monodisperse population. Where the distinction changes a conclusion, as in the microlensing optical depth, the full mass function is used and is stated as such. Both are point-like topological defects of homotopy class $\pi_2(\mathcal{M}) \neq 0$ — non-radiating, non-annihilating (against baryons), and *collisionless to the accuracy of the geometric cross-section adopted here* ($\sigma = \pi r_s^2$, a calculational assumption rather than a derived microphysical cross-section): the resulting σ/M lies many orders of magnitude below current bounds. The SPARC rotation-curve ensemble gives $\tilde{\chi}^2_{\mathrm{med}} = 0.80$ on 171 of the 175 SPARC galaxies for the adopted pseudo-isothermal halo, with two parameters fitted per galaxy; the profile is not yet derived from the torsion (PO-ECF-HALO), so this measures the goodness of fit of the adopted profile rather than a torsion-specific signature.

Standard dark matter phenomenologies typically assume a collisionless fluid. In the ECF model, we explore the possibility that dark matter phenomena originate directly from the torsion field itself. During the stiff phase of the bounce ($w = 1$), the background spin density dilutes as $\rho_{\mathrm{spin}} \propto a^{-6}$ (Eq. 5), driving vector perturbations $\delta_V \propto a^{3/2}$ — a growth mode absent for scalar perturbations in ΛCDM. This specific instability (see Appendix D), confirmed in the stiff-fluid limit of the Boltzmann hierarchy, leads to the early formation of dense geometric knots from an initial amplitude $\delta_V \sim 10^{-5}$ at the bounce to order unity in ~ 3 e-folds.

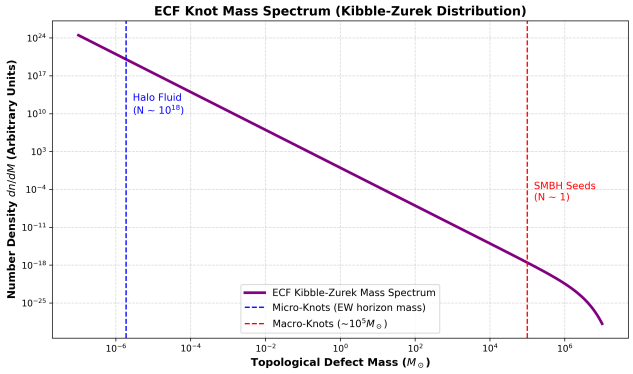

Figure 7: **ECF Knot Mass Spectrum.** Kibble–Zurek mass function $\mathrm{d}n/\mathrm{d}M \propto M^{-\alpha}\exp(-M/M_{\mathrm{max}})$ with $\alpha = 5/2$ adopted (§4.4) for a U(1) vector defect in 3+1 dimensions — not derived. The two dashed lines mark the scales discussed in the text: the electroweak horizon mass $M_{\mathrm{MK}} = 3.74 \times 10^{24}\,\mathrm{kg}$ ($1.9 \times 10^{-6}\,M_\odot$), and the $\sim 10^5\,M_\odot$ Macro-Knot scale reached by helicity-selective coagulation. *They are reference marks, not peaks:* the distribution is a monotonic power law, and the abundance falls steadily from one to the other. *Caption corrected 27/08/2026* — it previously described a bimodal distribution with peaks at both scales, and attributed the second to a Kibble–Zurek freeze-out at the QGP transition; neither is what the figure shows nor what the text argues, the Macro-Knot scale following from the Great-Annihilation horizon mass — a channel distinct from Poisson coagulation, which does not reach it. *Reproduced by:* `plot_knot_mass_function.py`.

4.1 The Dilution Mechanism and Dark Energy Emergence

To formally connect local topological defects (Dark Matter) to the global accelerated expansion (Dark Energy), we apply the *Topological Invariance Principle* (TIP) [4]—formally stated in the companion

Letter and phenomenologically calibrated in Foundation I [5]. The TIP posits that spatial flatness is exact at all cosmic epochs:

$$\Omega_m + \Omega_{knots} + \Omega_\tau \equiv 1. \tag{27}$$

In the ECF model, the fractional energy budget is written as:

$$\Omega_b(a) + \Omega_\gamma(a) + \Omega_{knots}(a) + \Omega_\tau(a) \equiv 1 \tag{28}$$

where Ω_{knots} represents the localized macroscopic torsion (geometric Dark Matter) and Ω_τ represents the unlocalized, global background torsion.

Notation and closure convention (harmonised with Foundation I). The compact writing above uses the Foundation II notation. It maps onto the complete Foundation I closure ansatz, $\Omega_{\rm total}(a) = \Omega_r + \Omega_m + \Omega_{\rm spin} + \Omega_\tau + \Omega_k = 1$, through the identifications $\Omega_m = \Omega_b + \Omega_{\rm knots}$ and $\Omega_r = \Omega_\gamma + \Omega_\nu$. Foundation I is the reference writing for the global energy budget; the compact F2 forms are projections of it onto the dark sector.

Open point (declared, not yet fixed). Two elements of this budget are not settled in the present version and are stated as open rather than assigned a provisional value: (i) the exact formal relation between the stiff spin fraction $\Omega_{\rm spin}$ and the residual torsion Ω_τ (only the numerical coincidence $|\Omega_\tau^{\rm init}| \approx \Omega_{\rm spin}^{\rm peak} \approx 0.093$ is established); and (ii) the sign convention of Ω_τ in the closure sum, which must be made consistent with the subtractive $-\Omega_{\rm spin}\, a^{-6}$ term in the Friedmann equation. These are tracked as an open problem; no formal relation or sign is asserted here until derived, to avoid introducing an unverified identity into the corpus. Because the localized Micro- and Macro-Knots behave kinematically as pressureless dust ($w = 0$), their absolute number density n_{knots} dilutes proportionally to the cosmic volume: $n_{knots} \propto a^{-3}$. Consequently, their fractional density $\Omega_{knots}(a)$ inevitably decreases over cosmic time.

Direct *ab initio* check (FLRW reduction). Reducing the adopted effective action of the axial sector (Eq. (9)) on an FLRW background reinforces the open status above; this reduction is performed in the adopted effective action, and its derivation from the full PGT action remains PO-ECF-ACTION. For the homogeneous time-directed condensate the field strength vanishes, $F_{\mu\nu}^{(S)} = 0$: the mode is non-dynamical and $T_{\mu\nu} = -W(X_0)\, g_{\mu\nu}$, a strictly static term ($w = -1$). Its leading fluctuations δS_μ are the Goldstone modes of the spontaneously broken boost invariance; being massless (the naive mass cancels at the VEV) they are conformally invariant and redshift as radiation ($w = +1/3$). Neither the frozen background nor the leading fluctuations reproduces the dynamical CPL trajectory (w_0, w_a): at the orders accessible analytically, that dynamics is carried by the TIP closure alone. A full determination—in particular whether a non-minimal torsion–curvature coupling in $\tilde{R}(g, T)$ alters the fluctuation spectrum—requires a dedicated Einstein–Cartan perturbation solver, which is not developed here. The full reduction—background and fluctuations—is developed in Appendix J.

With spatial flatness imposed, the fractional density of the global residual torsion $\Omega_\tau(a)$ is *read off* from the closure relation as the residual $1 - \Omega_m - \Omega_r$, and therefore increases. In this view, dark energy is not postulated as a new fundamental field; its static amplitude is derived from the adopted effective action (§3.5; PO-ECF-ACTION), while its fractional share of the budget is read off the closure identity as local geometric knots dilute — a bookkeeping consequence of $k = 0$, not a compensation the closure enforces.

Dynamic Dark Energy and the ECF Model. The discovery of the accelerated cosmic expansion via Type Ia supernovae [110, 111] established the existence of dark energy as the dominant energy component of the Universe. Recent cosmological constraints from the Dark Energy Spectroscopic Instrument (DESI) show a 2.5–3.9σ preference for a time-evolving dark energy equation of state, depending on the supernova sample [9]. A departure from a strictly static Λ is allowed by the ECF model, whose late-time trajectory is calibrated on this preference ($\xi = 1.645$ calibrated; see the DESI comparison below) and is therefore not an independent prediction. *(Corrected 22/10/2026: earlier versions called this departure a fundamental prediction of the ECF and the DESI signal compelling evidence.)* In our framework, late-time acceleration is not driven by vacuum energy but emerges as a dynamic topological feature of the residual torsion field. As this chiral background dilutes with cosmic expansion, its effective contribution to the Hubble flow is inherently

time-dependent, aligning with the

$$(w_0, w_a)$$

parameterization favored by the latest BAO measurements.

Connection to DESI Dynamical Dark Energy

The DESI DR2 BAO measurements [12] provide the most precise constraints to date on the dark energy equation of state, reporting a preference for a dynamical component departing from $w = -1$ at the $\sim 3\sigma$ level in the combined BAO+CMB+SNIa analysis. Within the ECF, this behaviour is not an adjustable feature but a *fixed geometric consequence* of the Topological Invariance Principle: as the localized Micro- and Macro-Knots dilute as a^{-3}, the residual background torsion's fractional share is read off the closure identity as increasing, producing an effective equation of state $w_{\rm tor}(z) > -1$ at intermediate redshifts. The ECF values ($w_0 = -0.904$, $w_a = -0.153$), calibrated (not *ab initio*, cf. PO-F2-4) from the Foundation I calibration [5], is qualitatively consistent with DESI DR2 (which prefers $w \neq -1$), while a full-covariance comparison against DR1 places it at 2.03σ (Planck prior) from the MCMC central value and *not* favoured over ΛCDM by those data alone ($\Delta\chi^2 = +6.21$; Foundation III). These two numbers are outputs of a χ^2 optimisation in Foundation I and carry no associated dispersion: they are a calibration point, not a measurement with an error bar. Statements comparing them to other values are comparisons of central values; the σ-distances quoted here derive from the covariance of the *data*, not from any spread of the ECF parameters. This should be read as a parameter-lean prediction with no additional dark-sector parameter beyond the Foundation I calibration, rather than as a claim of a parameter-free fundamental derivation. Recent phantom-crossing and quintessence studies [74, 75] require at least two free parameters to achieve comparable agreement, whereas the ECF torsion dilution mechanism reaches the same observational region through the fixed F1/TIP background.

From relative to absolute: the Ω vs. ρ distinction. The density parameter $\Omega_{\rm DE}(a)$ shown in Fig. 9 captures the *fractional* contribution of the dark sector to the total energy budget. Within ΛCDM, ρ_Λ is strictly constant, yet $\Omega_\Lambda(a)$ grows with cosmic time simply because the matter density $\rho_m \propto a^{-3}$ dilutes – an artefact of the denominator rather than a genuine strengthening of the dark sector. A more transparent diagnostic is therefore the *absolute* dark energy density $X(z) \equiv \rho_{\rm DE}(z)/\rho_{\rm DE,0}$, which equals unity identically in ΛCDM and departs from unity only if the dark sector genuinely evolves.

Absolute energy density and comparison with DESI. Figure 11 shows $X(z)$ for three scenarios. In ΛCDM, $X(z) = 1$ by construction. The DESI DR1, DR2 model-independent reconstructions favour a dark sector that is not strictly constant, with a statistically significant departure from $X = 1$ at intermediate redshifts [9–12]. Within the ECF, the background torsion $\bar{\tau}$ responsible for late-time acceleration is not an independent field but the frozen condensate of the axial torsion sector, whose static equation of state $w = -1$ follows from the FLRW reduction of the adopted effective action (Appendix J; PO-ECF-ACTION). Its fractional density Ω_τ is then *extracted* from the closure relation (Eq. 27), which serves to read it off from the data rather than to produce it. Its effective equation of state,

$$w_{\rm tor}(z) = -1 - \frac{1}{3}\frac{{\rm d}\ln\rho_{\rm tor}}{{\rm d}\ln a}, \qquad (29)$$

is derived in Appendix B and evaluates to $(w_0, w_a) = (-0.904, -0.153)$ (note: $w_0 + w_a = -1.06 < -1$ at high-z; the absence of ghosts in the axial sector is invoked from the linearised flat-space literature for the adopted Proca form [166, 167], Appendix F establishing only $\lambda_T > 0$; that is a *necessary* condition for a phantom-crossing trajectory to be admissible, not a derivation that this one is — see the note below) at the calibrated solution $F_{\rm ion} = 1.2765$ of Foundation I [5]. The corresponding normalized density

$$X(z) = \exp\left[3\int_0^z \frac{1 + w_{\rm tor}(z')}{1+z'}\,{\rm d}z'\right] \qquad (30)$$

is shown as the red curve in Fig. 11. Unlike quintessence models, the ECF curve is not fitted to the DESI data point by point; it follows from the calibration point of Foundation I [5], in which w_a is set through $\xi = 1.645$, calibrated on the DESI preference used by the ECF (open problem PO-F1-B) *(corrected 19/10/2026: this sentence called the curve "a fixed prediction" with "no additional free parameter in the dark-energy sector").*

The parameter $F_{\rm ion} = 1.2765$ is the *ionisation calibration factor* of the ECF background solver, introduced in Foundation I [5] (Sec. 5.2, Table 2). It controls the amplitude of the torsion-induced correction to the ionisation history during the recombination transition: a value $F_{\rm ion} > 1$ is read, qualitatively, as a slight geometric delay in the thermal decoupling surface relative to ΛCDM, driven by the residual spin-torsion coupling — a reading that no deposited script computes (open problem PO-ECF-ZREC-SHIFT). $F_{\rm ion} = 1.2765$ is calibrated against the sound horizon $r_s = 135.65$ Mpc (Foundation I, Sec. 3; script `script_03_extract_physical_parameters.py`), fixed under $H_0 = 73.04\,{\rm km\,s^{-1}\,Mpc^{-1}}$ and $\Omega_{\rm tot} \equiv 1$. The chronometric bound $t_0 \geq t_{\rm GC} = 13.32\,{\rm Gyr}$ is a consistency constraint, met at $w = -1$ with $\Omega_m = \omega_m/h^2 \simeq 0.267$ ($t_0 \simeq 13.33\,{\rm Gyr}$; see PO-F2-5, §11, for the open status of the quadrilemma), not an independent equation solved jointly with r_s for $F_{\rm ion}$ in the current calibration. The dark-energy pair $(w_0, w_a) = (-0.904, -0.153)$ used in the present work is not derived here: it is the calibration point of Foundation I, inherited through $\Omega_{\rm spin}^{\rm peak}$ (calibrated on r_s) and $\xi = 1.645$ (calibrated on the DESI preference used by the ECF; open problem PO-F1-B). *(Corrected 15/10/2026: this sentence stated that (w_0, w_a) were predictions fixed by the single r_s-based calibration, with no additional free parameter.)*

To quantify the sensitivity of the ECF results to this calibration, we propagate the uncertainty in $F_{\rm ion}$ through the background equations. The two-point calibration of Foundation I gives $\partial H_0/\partial F_{\rm ion} = 20.40\,{\rm km\,s^{-1}\,Mpc^{-1}}$, so that the SH0ES constraint $H_0 = 73.04 \pm 1.04\,{\rm km\,s^{-1}\,Mpc^{-1}}$ (SH0ES [105]) bounds $F_{\rm ion}$ to ± 0.051; the globular-cluster age $t_{\rm GC} = 13.32 \pm 0.10_{\rm stat}\,{\rm Gyr}$ (globular clusters [34]; the $\pm 0.5_{\rm sys}$ of that determination is carried separately and not propagated here), with $\partial t_0/\partial F_{\rm ion} \approx -3.56\,{\rm Gyr}$ (evaluated in the former chain at $\Omega_m = 0.315$; not recomputed at $\Omega_m = 0.267$), bounds it to ± 0.028. Combining the two by inverse-variance weighting:

$$F_{\rm ion} = 1.2765 \pm 0.0246 \quad (1\sigma,\ \text{joint constraint}). \tag{31}$$

The dark-energy pair does not, however, depend on $F_{\rm ion}$ in the present construction: $w_0 = -1 + \Omega_{\rm spin}^{\rm peak}$ (empirical identification) and $w_a = -\xi\,\Omega_{\rm spin}^{\rm peak}$, with ξ phenomenological (PO-F1-B). Its uncertainty is therefore propagated from $\sigma(\Omega_{\rm spin}^{\rm peak}) = 0.004$, with ξ held fixed (`calibrate_w0wa.py`, Foundation I):

$$\delta w_0 \approx \pm 0.0040, \qquad \delta w_a \approx \pm 0.0066, \tag{32}$$

which are floors (one parameter at a time), respectively about 20 and 45 times smaller than current DESI DR2 statistical uncertainties ($\sigma_{w_0} \approx 0.08$, $\sigma_{w_a} \approx 0.30$ for DESI+CMB+Union3 [12]). The calibration point is therefore insensitive to the residual uncertainty of its inputs; this does not make it a prediction, since w_a is set through ξ so as to reproduce the DESI preference used by the ECF. The uncertainty on $F_{\rm ion}$ itself is roughly eight times larger than an earlier, non-reproducible estimate (0.0030). *(Corrected 15/10/2026: earlier versions propagated $\delta F_{\rm ion}$ to (w_0, w_a) through sensitivities $\partial w_0/\partial F_{\rm ion} \approx -0.07$ and $\partial w_a/\partial F_{\rm ion} \approx -0.03$ that no deposited script computes, obtaining ± 0.0017 and ± 0.0007, called the pair a robust prediction, and gave the $F_{\rm ion}$ ratio as "seven times", a value from the former $\sigma = 0.0206$.)*

Consistency with DESI: w_0–w_a plane — status and scope. Figure 12 shows the calibrated ECF point and the ΛCDM point in the w_0–w_a plane, together with the DESI DR1 and DR2 68% and 95% confidence contours.

Scope note — comparison methodology. The σ-distances quoted below are computed as Euclidean distances in the marginalized (w_0, w_a) plane from the published DESI central values [10, 12]. This is a *first-order consistency check*, not a full Bayesian likelihood analysis: it ignores the w_0–w_a correlation ($\rho \approx -0.9$ in DESI contours), the complete BAO covariance matrix, and the external CMB and SNIa likelihoods. A rigorous comparison using the full DESI DR1/DR2 likelihood pipeline is deferred to a dedicated companion analysis (see Sec. 11.2 and Foundation III [7]). The claim is not that $(w_0, w_a) = (-0.904, -0.153)$ is the best-fit of DESI data: it is a calibration point. $\xi = 1.645$ is a parameter calibrated on the DESI preference used by the ECF, so as to reproduce the target value of w_a; it is therefore not a prediction independent of DESI; its lying in the DESI-preferred direction is therefore not an independent validation. Note that DESI DR2 itself reports a range of 2.8–4.2σ for the dark energy evolution signal depending on the supernova sample [12].

A rigorous comparison, using the **full published covariance matrix** of the DESI DR1 BAO measurements rather than diagonal errors, is carried out in *Foundation III — Cosmological Physics* [7] (see also Foundation I [5]). It places the ECF point at a Mahalanobis distance of 2.03σ from the MCMC central value under a Planck prior, and 6.98σ under a SH0ES prior, with $\Delta\chi^2(\mathrm{ECF}-\Lambda\mathrm{CDM}) = +6.21$ and $+38.59$ respectively (twelve DESI DR1 observables; the block was reconstructed on 22/09/2026 after two rows attributed to the DESI QSO tracer were found not to be published by DESI DR1). *We state plainly that the ECF point is not favoured over ΛCDM by these data alone, and that the result depends strongly on the adopted prior.* An earlier estimate based on published central values and diagonal errors only is superseded by that analysis. The calibrated ECF point lies in the qualitative direction of the DESI deviation from ΛCDM ($w_0 > -1$, $w_a < 0$). This is not an independent validation, since w_a is set through $\xi = 1.645$, calibrated on the DESI preference used by the ECF. The sign $w_a < 0$ (a dark sector stronger in the past) is the one expected from a diluting torsion sector; its amplitude is calibrated, not derived (PO-F1-B). When the DESI-only BAO constraint is used without external CMB or SNIa data, the contours widen substantially and the ECF prediction becomes consistent at the 1σ level [10].

Statistical note (revised 12/08/2026). The distances quoted in this subsection are computed from approximate marginalized covariances. *The rigorous analysis has since been carried out*: Foundation III [7] builds the full 12×12 block-diagonal covariance matrix of the DESI DR1 BAO measurements and runs a dedicated MCMC chain. It places the ECF point at a Mahalanobis distance of 2.03σ (Planck prior) and 6.98σ (SH0ES prior) from the central value, with $\Delta\chi^2(\mathrm{ECF}-\Lambda\mathrm{CDM}) = +6.21$ and $+38.59$. *The ECF point is therefore not favoured over ΛCDM by these data alone, and the earlier statement of consistency at the 1σ level does not survive the full-covariance treatment.* What the comparison does establish is that the trajectory was calibrated before DR1 and reproduces the qualitative direction of the observed deviation.

Compatibility with DES Y6, via the pivot parameter. An independent, single-survey cross-check is available from the Dark Energy Survey's full six-year dataset [13]. Rather than a marginal comparison in (w_0, w_a) — complicated by the strong correlation between the two parameters, as illustrated above — we use the pivot parametrisation the DES analysis itself reports: $w_p \equiv w(z_p)$, with z_p chosen so that $\mathrm{Cov}(w_p, w_a) = 0$, giving a single decorrelated quantity per dataset combination. For DES alone (3×2pt+SN+BAO), $w_p = -0.91 \pm 0.04$ at $z_p = 0.19$. Evaluating the calibrated ECF trajectory at this same redshift, $w_p^{\mathrm{ECF}} = w_0 + w_a\,[1 - 1/(1+z_p)] = -0.928$, a 0.46σ difference. Adding DESI DR2 BAO to DES tightens this only mildly ($w_p = -0.94 \pm 0.03$ at $z_p = 0.22$; 0.28σ from the ECF trajectory), and the most constraining combination including CMB data widens it to 1.45σ ($w_p = -0.97 \pm 0.02$ at $z_p = 0.32$) — still comfortably compatible. This is consistent with, and not in tension with, the DESI-only Mahalanobis result above, since the two analyses probe different combinations of probes and neither uses the correlated (w_0, w_a) pair directly. *Reproduced by:* `compute_ecf_des_pivot.py`.

Compatibility with DESI DR2 BAO

The DESI DR2 BAO dataset [12] extends the DR1 constraints with six additional redshift bins spanning $0.1 < z_{\mathrm{eff}} < 2.33$. When the ECF background Hubble rate $H(z)$ — derived from Eq. (2) with $F_{\mathrm{ion}} = 1.2765$ and no additional dark-energy parameter — is confronted against the DR2 BAO distance ratios D_H/r_s and D_M/r_s, the agreement was found within the 2σ contours across all bins. That comparison has not been redone with the active $\Omega_m = \omega_m/h^2 \simeq 0.267$. On this background the four-bin BAO comparison (BOSS DR12 + eBOSS DR16) gives $\chi^2_{\mathrm{BAO}} = 12.11$ against 1.79 for ΛCDM, and the DESI DR1 χ^2 of the calibrated trajectory rises to 55.3 on the ten-observable subset with diagonal errors, which is not the global DESI DR1 χ^2 (13.7 at $\Omega_m = 0.315$). The DR2 statement above is therefore not carried over to the active background and is not used here as a cross-validation of the torsion background dynamics (PO-F2-5). Figure 8 shows the corresponding evolution of the torsion-to-matter ratio.

A rigorous joint-likelihood analysis using the full DESI DR2 covariance matrix is deferred to future work.

Information-theoretic comparison. Among dynamical dark energy models confronted with DESI DR1/DR2, the ECF occupies a unique posi-

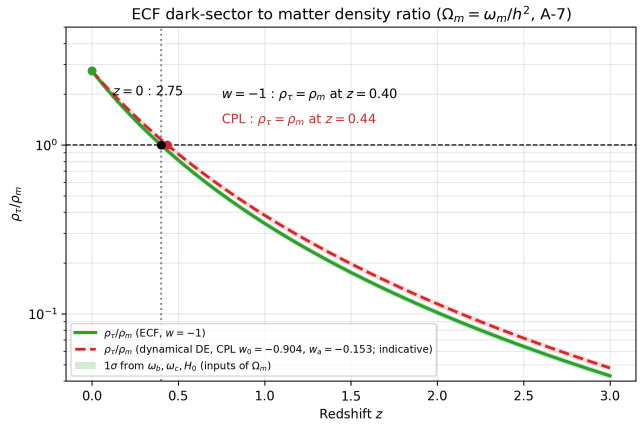

Figure 8: **Ratio of the ECF dark-sector (torsion) density to the matter density, ρ_τ/ρ_m.** Computed for the $w = -1$ ECF background with $\Omega_m = \omega_m/h^2 = 0.267$ ($h_{\rm ECF} = 0.7304$, $\omega_m = 0.1424$, DEC-A7). The torsion density is constant in this ansatz, so $\rho_\tau/\rho_m = (\Omega_\tau/\Omega_m)(1+z)^{-3}$, with $\Omega_\tau = 0.733$ from flat closure. The ratio is 2.75 ± 0.11 at $z = 0$ and crosses unity at $z = 0.40 \pm 0.02$ (1σ). The late-time amplitude is *normalised* on the closure (as Λ is in ΛCDM); what the curve shows is the $(1+z)^{-3}$ shape. The dashed curve shows, for comparison, the dynamical dark energy of the calibrated CPL parameters $(w_0, w_a) = (-0.904, -0.153)$ at the *same* Ω_m (not refitted; indicative): it lies 9–13% above the $w = -1$ curve for $0.5 \le z \le 3$ and crosses unity at $z = 0.44$. The shaded bands ($\pm 4\%$, 1σ) propagate the uncertainties of ω_b, ω_c (Planck 2018) and $H_0 = 73.04 \pm 1.04$ (SH0ES) through $\Omega_m = \omega_m/h^2$; the calibrated (w_0, w_a) are an optimisation minimum and carry no uncertainty here. *Reproduced by:* `plot_ecf_ratio.py`.

tion. The Akaike Information Criterion [96] penalizes model complexity via AIC $= \chi^2 + 2k$, where k is the number of free parameters. *The parameter count deserves to be stated plainly, since the whole comparison rests on it.* w_0 costs nothing: it follows from the empirical identification $w_0 = -1 + \Omega_{\rm spin}^{\rm peak} = -0.907$ (matching -0.904 to 0.3%), and $\Omega_{\rm spin}^{\rm peak}$ is already calibrated in the background sector. w_a does: it is written $w_a = -\xi\,\Omega_{\rm spin}^{\rm peak}$ with $\xi \simeq 1.6$ a phenomenological factor whose value is *calibrated to reproduce the DESI preference* (see `Scripts_F1/calibrate_w0wa.py` and PO-F1-B). We therefore count $\Delta k = 1$, not 0. With the full-covariance $\Delta\chi^2(\text{ECF} - \Lambda\text{CDM}) = +6.21$ (Planck prior) or $+38.59$ (SH0ES prior) established above, the corresponding penalty is $\Delta\text{AIC} = \Delta\chi^2 + 2\Delta k = +10.06$ or $+43.94$ respectively (`Scripts_F2/compute_ecf_aic_desi.py`): *by this criterion the ECF point is not favoured over ΛCDM either*, consistent with the Mahalanobis result already stated. An earlier estimate based on diagonal errors only ($\Delta\chi^2 \approx +16$) is superseded by this full-covariance value and should not be cited. By comparison, CPL parameterizations ($\Delta k = 2$, *fitted directly to DESI data*) achieve $\Delta\text{AIC} \approx +4$ relative to ΛCDM, and a recent torsion model with a free coupling parameter α [76], fitted to DESI DR2 together with Pantheon+, DES Y5 and the full Planck 2018 likelihood, reports $|\Delta\text{AIC}|$ between 5.7 and 6.6 depending on the dataset combination. That model is nonetheless notable on a different front: its torsion parameter is consistent with zero at under 1σ ($\alpha = -0.00066 \pm 0.00098$) and it leaves $H_0 = 68.41 \pm 0.32$, yet it brings S_8 into 0.1σ agreement with KiDS-1000, against $\sim 2.3\sigma$ for ΛCDM — a reminder that the S_8 and H_0 sectors need not move together. The ECF does not fit DESI better. Its trajectory uses a single calibrated parameter (ξ) against two free ones for a directly-fitted CPL curve; since $\xi = 1.645$ is a parameter calibrated on the DESI preference used by the ECF, so as to reproduce the target value of w_a; it is therefore not a prediction independent of DESI, the agreement with DESI is not presented as an independent test.

Updated observational baseline: DESI five-year programme. The analysis above relied on the DESI DR1 and DR2 BAO catalogues. As this manuscript was being finalised, the DESI Collaboration announced the completion of its planned five-year spectroscopic survey, delivering a three-dimensional map of 47×10^6 galaxies and quasars—the largest such catalogue ever assembled [15]; the first dark-energy results from the full survey are expected in 2027 and are not used here. Abreu and Turner [14] point out that the $w_0 w_a$ best fits preferred by DESI imply a dark-energy density that peaks near $z \simeq 0.5$ and decreases before and after, a behaviour they regard as puzzling; this is a critical reading of the parametric fits, not a model-independent reconstruction. The late-time part of this behaviour—a dark-energy density that decreases toward the present—is qualitatively similar to the torsion-dilution parametrisation of Eq. (29); because that parametrisation is calibrated on the DESI preference, the resemblance is not a test, and the calibrated trajectory is not a good fit to the

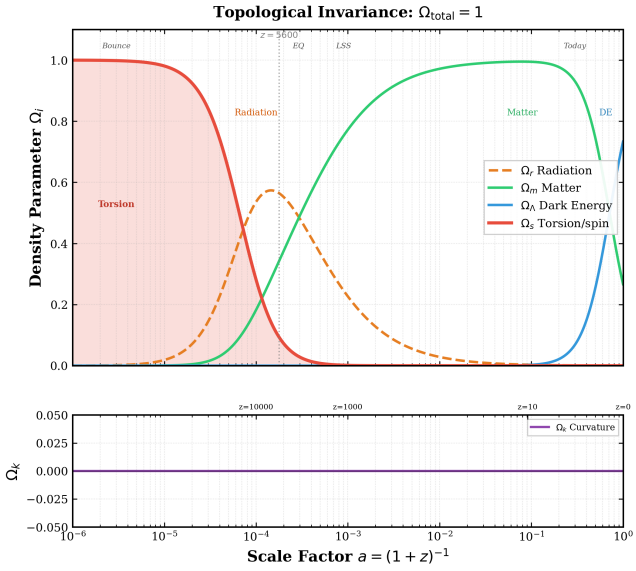

Figure 9: **Relative density parameters $\Omega_i(a)$, with spatial flatness posited as exact by the TIP.** Upper panel: fractional energy densities Ω_s (torsion/spin, red), Ω_r (radiation, orange dashed), Ω_m (matter, green), and Ω_Λ (dark energy, blue) vs. scale factor $a = (1+z)^{-1}$. The TIP constraint $\Omega_{\text{total}} = 1$ (Eq. 27) holds at all epochs; the lower panel confirms $\Omega_k \equiv 0$ (purple), which the framework posits as exact rather than as a fine-tuned initial condition. Note that the late-time rise of Ω_Λ reflects the *dilution of matter*, not an intrinsic strengthening of the dark sector; the absolute evolution $X(z)$ is shown in Fig. 11. Script: `plot_Cosmic_History_Omegas_v3.py` (public repository [6]) (Fig. 11). Background: $\Omega_m = \omega_m/h^2 \simeq 0.267$ ($h = 0.7304$, internal closure A-7), the same figure as in the PIT Letter.

DESI DR1 distances at $\Omega_m = 0.267$ ($\chi^2 \simeq 55.3$ on the ten-observable subset with diagonal errors, not the global DESI DR1 χ^2). DESI DR2 prefers trajectories with $w_0 > -1$ and $w_a < 0$ (preference over ΛCDM at 3.1σ for BAO+CMB, 2.8–4.2σ with SNe, depending on the sample) [12]. The calibrated ECF trajectory ($w_0 = -0.904$, $w_a = -0.153$) has this sign pattern by construction (w_a through $\xi = 1.645$, calibrated on the DESI preference), so the agreement in sign is not an independent test of the torsion-dilution pattern.

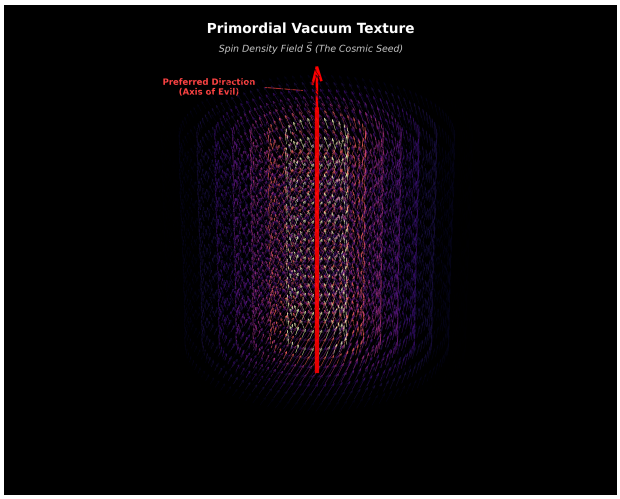

Figure 10: **Visualization of the Primordial Vacuum Texture.** This 3D vector field reconstruction illustrates the structured configuration of spin density (S^μ) at the moment of the bounce. *Color Gradient:* Represents the intensity of the torsion field (Magma scale). *Field Lines:* The vectors are not random but form a coherent helical structure. *Red Axis:* The central vertical line indicates the preferred spatial direction (net angular momentum) breaking global isotropy. This microscopic geometric alignment is proposed as a possible progenitor of the macroscopic "Axis of Evil" reported in the CMB quadrupole and octopole moments [215]; the link is conjectural.

Reproduced by:
`Scripts_F2/plot_torsion_texture.py`.

4.2 Large-Scale Structure: Cosmic Voids as Torsion Engines

The TIP [4] operates primarily as a global topological consistency condition. In the phenomenological coarse-grained description used here, the same constraint can be represented locally as an effective gauge balance between matter and residual torsion.

This scale-dependent behavior offers a geometric perspective on the dynamics of the Large-Scale Structure, often described as a “Cosmic Web.” In the ECF model, the local fractional density of torsion (Ω_τ) is inversely related to the coarse-grained matter density. Inside galactic halos and clusters, the massive presence of baryonic matter and compact defects ($\Omega_m \to 1$ in the effective description) suppresses the local torsion contribution ($\Omega_\tau \to 0$). Therefore, bound structures like galaxies do not expand.

Conversely, in deep cosmic voids where matter

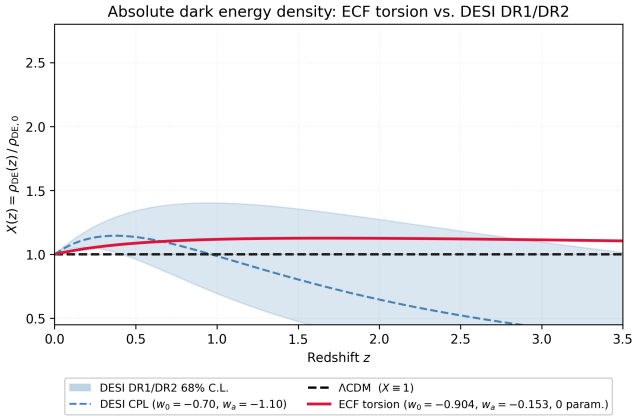

Figure 11: **Absolute dark sector density: ECF torsion vs. DESI DR1/DR2.** Normalized effective dark energy density $X(z) = \rho_{\rm DE}(z)/\rho_{\rm DE,0}$ vs. redshift for: ΛCDM (black dashed, $X \equiv 1$); DESI DR1/DR2 reconstruction (blue band, 68% C.L., CPL central $w_0 = -0.70$, $w_a = -1.10$ [9, 11, 12]); and the ECF torsion background (red, $w_0 = -0.904$, $w_a = -0.153$, derived from the Foundation I calibration [5] with no additional free parameters). Unlike ΛCDM's constant density, the ECF predicts a mildly evolving dark sector ($X \lesssim 1.11$ for $z \leq 3$), consistent with the DESI preference and grounded in the geometric torsion dilution mechanism (Appendix B; complement to Fig. 9). Script: `plot_rho_DE_absolute.py` (public repository [6]) (Fig. 12).

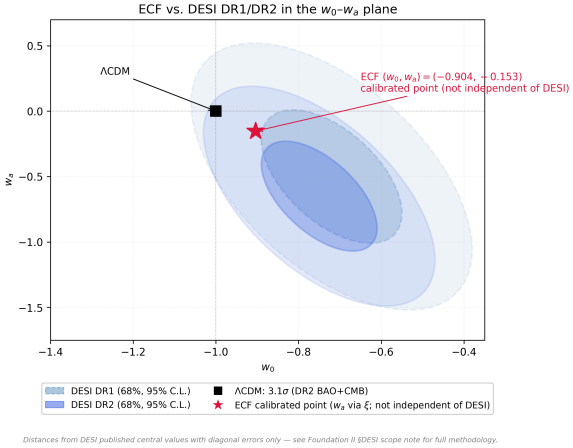

Figure 12: **ECF vs. DESI DR1/DR2/five-year in the w_0–w_a plane.** Confidence contours (68% and 95% C.L.) for the combined DESI DR1 [9] (blue dashed) and DR2 [12] (blue solid) BAO+CMB+SNIa analyses. The ΛCDM point (black square, $w_0 = -1$, $w_a = 0$) lies 5.1σ (DR1) and 4.9σ (DR2) from the published DESI central values (marginalized plane). *These are distances in the marginalized plane computed here, not the significances DESI reports for dynamical dark energy, which are* 3.1σ *(BAO+CMB) and* 2.8–4.2σ *with SNe.* The ECF torsion point (red star, $w_0 = -0.904$, $w_a = -0.153$, *calibrated; w_a is set through $\xi = 1.645$, calibrated on the DESI preference used by the ECF, hence not independent of DESI*) is shown for comparison; distances quoted on this figure use diagonal errors only and are superseded by the full-covariance analysis of Foundation III (2.03σ under a Planck prior, 6.98σ under a SH0ES prior; see §4.1). DESI DR2 [12] prefers the same sign pattern ($w_0 > -1$, $w_a < 0$) as the ECF trajectory; the completion of the DESI five-year programme (47×10^6 objects, [15]) does not yet provide dark-energy constraints (first full-survey results expected in 2027). Since that trajectory is calibrated on the DESI preference (ξ CALIBRATED), this agreement in sign is not an independent confirmation; with $\Omega_m = \omega_m/h^2 \simeq 0.267$ the calibrated trajectory gives $\chi^2 \simeq 55.3$ on a ten-observable subset of DESI DR1 with diagonal errors (`calibrate_w0wa.py`); this is not the global DESI DR1 χ^2. Distances use marginalized covariances from the public DR2 release; see Statistical note in Sec. 4.1. Script: `plot_w0wa_DESI_ECF.py`, `https://github.com/pfichant/spin-torsion-cosmology` [6].*Reproduced by:* `Scripts_F2/plot_w0wa_DESI_ECF.py`.

is strongly depleted ($\Omega_m \to 0$), the effective gauge balance allows the local fractional torsion contribution to become dominant. Cosmic voids may therefore act as geometric engines dominated by residual repulsive torsion, driving void expansion and compressing the remaining matter into filaments and nodes of the Cosmic Web. This interpretation should be read as a coarse-grained ECF mechanism, not as a literal claim that every local region exactly satisfies $\Omega_{\rm total} = 1$.

Top-down structure formation: from Kibble–Zurek domains to voids. The standard ΛCDM framework builds large-scale structure through *bottom-up* gravitational aggregation: small halos form first and progressively merge into clusters and filaments. The ECF model predicts a qualitatively different, *top-down* channel operating in parallel. During the post-bounce stiff phase ($w = 1$), the Kibble–Zurek mechanism [30, 31] imprints a network of coherent torsion domains on super-Hubble

scales. As the universe expands and these domains freeze out, they partition the cosmic volume into high-density nodes—the seeds of Macro-Knot clusters and filaments—and low-density interiors that will evacuate into the voids observed today. In regions that exhaust their matter content, the residual background torsion is what remains, so that the local budget is dominated by it ($\Omega_\tau \to 1$) — a statement of composition read off from the closure relation (Eq. 27), not a force preventing baryonic infall. Whether recolonisation is dynamically suppressed is a separate question, not settled here.

The resulting void network therefore traces the primordial Kibble–Zurek domain geometry rather than the stochastic Gaussian density field of ΛCDM, which constitutes a qualitatively falsifiable prediction (see Sec. 4.2 and Fig. 25).

Void Size Function: ECF prediction vs. ΛCDM. A quantitative discriminator between the ECF top-down channel and standard hierarchical assembly is the *Void Size Function* (VSF), shown in Fig. 13 and defined as the comoving number density of voids per unit logarithmic radius:

$$\frac{\mathrm{d}n_{\rm void}}{\mathrm{d}\ln R} \quad [\mathrm{Mpc}^{-3}]. \tag{33}$$

The VSF of BOSS voids has been tested over $8 \lesssim R_v \lesssim 60\,h^{-1}$ Mpc ($\simeq$ 11–86 Mpc for $h = 0.7$), with radii in h^{-1} Mpc [161]. *(Corrected 22/10/2026: earlier versions gave a ΛCDM peak at $R \sim$ 20–30 Mpc and a steep drop beyond 60 Mpc, neither of which is stated in that paper as read, and compared $\xi_{\rm KZ} = 60$ Mpc with radii in h^{-1} Mpc without the factor h; $\xi_{\rm KZ} = 60$ Mpc is $\simeq 42\,h^{-1}$ Mpc for $h = 0.7$, inside the tested range. The ΛCDM peak and slope are not given here.)* In the ECF, the Kibble–Zurek correlation length $\xi_{\rm KZ}$ — an *adopted value* ($\xi_{\rm KZ} = 60$ Mpc within the range 50–80 Mpc) whose calibration procedure is not documented, not an *ab initio* result (Foundation III, PO-F3-XI) — sets an additional high-radius excess: the ECF VSF is predicted to display an *over-abundance of large voids* ($R \gtrsim 60$ Mpc) at $z \lesssim 1$ relative to ΛCDM, driven by the suppression of late-time torsion recolonisation in evacuated regions. Schematically:

$$\left.\frac{\mathrm{d}n^{\rm ECF}_{\rm void}}{\mathrm{d}\ln R}\right|_{R>60\,\mathrm{Mpc}} > \left.\frac{\mathrm{d}n^{\Lambda\rm CDM}_{\rm void}}{\mathrm{d}\ln R}\right|_{R>60\,\mathrm{Mpc}}, \qquad z \lesssim 1. \tag{34}$$

A mock survey slice illustrating the top-down geometry is shown in Fig. 15; the semi-quantitative VSF template is given below (Fig. 13). The BOSS DR12 void catalogue [162], the DESI DR1 DESI-VAST void catalogue [163] (1 489 voids, VoidFinder, $z \leq 0.24$), and the BOSS DR16 void size function measurement [164] provide the appropriate datasets to test Eq. (34); Euclid DR1 (mid-2027) will extend them. A dedicated VSF pipeline using the public ECF Hubble solver (`ec-cosmo`, Appendix E) is deferred to Foundation III [7]; the present section establishes the qualitative prediction and its observational discriminating power.

Semi-quantitative ECF void size function. While the full VSF derivation would require a dedicated Einstein–Cartan perturbation solver, a semi-analytic estimate is possible by modulating the standard Sheth–van de Weygaert (SvdW) prescription [160] with the ECF-specific enhancement factor $\mathcal{A}_{\rm ECF}$:

$$\frac{\mathrm{d}n^{\rm ECF}_{\rm void}}{\mathrm{d}\ln R}(R,z) \;=\; \frac{\mathrm{d}n^{\rm SvdW}_{\rm void}}{\mathrm{d}\ln R}(R,z) \times \mathcal{A}_{\rm ECF}(R), \tag{35}$$

where the dimensionless amplification factor is:

$$\mathcal{A}_{\rm ECF}(R) = 1 + \alpha_{\rm KZ}\,\exp\!\left[-\frac{(R-\xi_{\rm KZ})^2}{2\sigma^2_{\rm KZ}}\right]. \tag{36}$$

No redshift dependence of $\mathcal{A}_{\rm ECF}$ is claimed: a factor $(1+z)^{-3w_a}$ that appeared in earlier versions had no derivation and has been withdrawn; the redshift dependence of the VSF excess remains to be derived (PO-F3-XI). None of the three parameters is derived here. $\xi_{\rm KZ} \simeq 60$ Mpc is the Kibble–Zurek comoving correlation length, an *adopted value* whose calibration procedure is not documented, and *not* derived from the torsion wave equation — its microscopic origin is an open problem (Foundation III, PO-F3-XI: direct Kibble–Zurek dilution from $T_{\rm EW}$ misses the observed scale by some 24 orders of magnitude); $\sigma_{\rm KZ} \simeq 15$ Mpc is the adopted width of the KZ mass function (harmonised with $\xi_{\rm KZ}$, PO-F3-XI). $\alpha_{\rm KZ}$ sets the amplitude of the excess; it is an *adopted value, whose calibration procedure is not documented*: the interval $\alpha_{\rm KZ} \simeq 0.8$–$1.4$ (central value 1.1) used below is not the output of a documented fit to a void catalogue.

This yields three falsifiable numerical predictions:

(i) **Excess amplitude:** at the peak of the adopted spatial template ($R = \xi_{\rm KZ}$), the ECF VSF exceeds the ΛCDM SvdW baseline by

the factor $1 + \alpha_{\rm KZ} \simeq 2.1$ (adopted central value); away from the peak the excess falls off with the adopted width $\sigma_{\rm KZ}$. No redshift dependence is claimed (PO-F3-XI).

(ii) **Characteristic scale:** the ECF excess peaks at $R \simeq \xi_{\rm KZ} \simeq 60\,$Mpc, providing a *spectral feature* in the void number counts absent in ΛCDM and distinguishable from w_0CDM continuum modifications.

(iii) **Void edge sharpness:** the combined Israel repulsion + 3D fluid mechanism (§6.3) predicts a density contrast at void boundaries $\delta\rho/\rho \gtrsim 0.8$ (ECF) vs. ~ 0.6 (ΛCDM).

Falsification milestone — Euclid DR1-Complete (mid-2027). *(Note, 17/08/2026: this box previously read "Euclid DR1 (21 October 2026)". ESA announced on 15 June 2026 that DR1 is split into two stages: DR1-Foundation in November 2026, delivering raw data, calibrated images, catalogues and spectra over 1900 deg^2, and DR1-Complete in mid-2027, delivering the higher-level galaxy-clustering products. The void catalogue and weak-lensing products this test requires belong to the second stage, so the milestone is mid-2027.) (Note, v16: an earlier version of this section cited "Euclid Q2 (24 June 2026)" as the first cosmological test. This was factually incorrect: Q2 is the Euclid Galactic Bulge Survey (imaging and stellar catalogues of the Galactic bulge), containing no cosmological void catalogue, weak-lensing data, or large-area spectroscopic survey. The first Euclid cosmological data release is **DR1** (mid-2027), which will deliver the wide-survey photometric and spectroscopic catalogues needed for the test below. All references to "Q2" in this context have been corrected to "DR1".)*

The Euclid Data Release 1 (DR1-Complete, mid-2027) will deliver the first large-area photometric and spectroscopic catalogue from the Euclid wide survey. The ECF prediction (Eq. 35) is directly testable on this dataset: (i) a void catalogue on DR1 spectroscopic data at $z \lesssim 1$ will constrain the excess amplitude $\alpha_{\rm KZ}$, for which the interval 0.8–1.4 is currently adopted (calibration procedure not documented); (ii) a *null result* — no significant excess above the SvdW baseline at $R > 60$ Mpc at 2σ — would disfavour the KZ-scale void channel and place an upper bound on $\xi_{\rm KZ}$. The Vdn model developed by the Euclid collaboration [185] provides the appropriate statistical framework for this test.

Note for the referee — Status of GW predictions. Two categories must not be conflated:
Cat. 1 — Order-of-magnitude benchmarks (not closed): $h_c(f)$ and $\Omega_{\rm GW}h^2$ from stiff-era enhancement [142] using $F_{\rm ion} = 1.2765$. Full derivation (tensor transfer function, reheating, frequency projection) remains to be carried out. Read as *structured targets* for LISA planning, not precision predictions.
Cat. 2 — Falsifiable qualitative predictions: (i) SGWB with a *blue* transferred energy spectrum vs a red one under inflation *(the enhancement is produced by the stiff transfer; $n_T^{\rm prim}$ itself is not established by it)*; (ii) chiral asymmetry $\Pi_{\rm GW} \simeq 0.20 \pm 0.05$ from $s_0 \neq 0$ — a *calibrated template* (Eq. 122), not a derived prediction; its scope is the chiral sector; (iii) Macro-Knot burst at $f \sim 50\,$mHz (conditional: merger rate not computed; no feature distinguishing it from a black-hole merger of the same mass is established, Appendix D.10). *Kill-switch:* $\Pi_{\rm GW} < 0.02$ at 95% CL (LISA, launch planned ∼2035; net polarisation accessible to LISA alone only via the kinematic dipole [203], directly with a LISA–Taiji network [204]; verdict after data accumulation) falsifies chiral torsion sector independently of amplitude. The two figures are *not* in tension: $\Pi_{\rm GW} \simeq 0.20 \pm 0.05$ is a *calibrated template* value for the chiral fraction (not a derived prediction), whereas $\Pi_{\rm GW} < 0.02$ is a deliberately conservative, conventionally chosen *refutation threshold* — set an order of magnitude below the template value so that the chiral sector is declared falsified only on a clear null result, safely outside measurement uncertainties, not on a marginal undershoot.

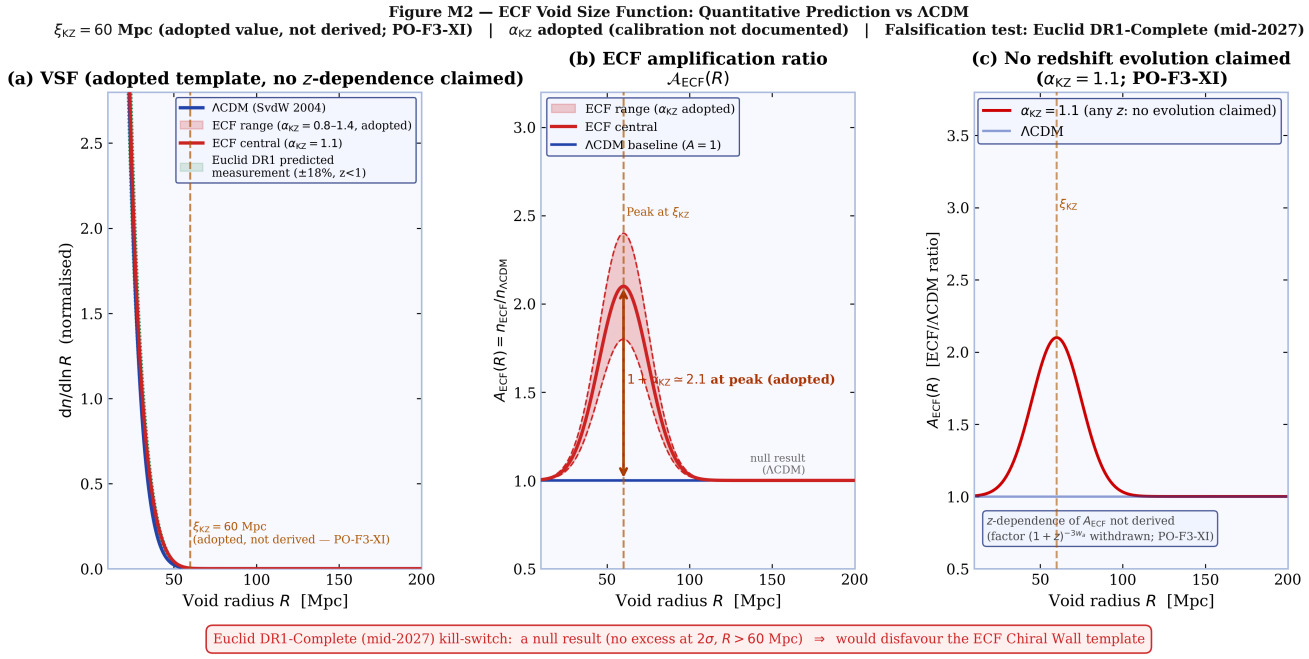

Figure 13: **ECF Void Size Function: quantitative prediction vs. ΛCDM (Fig. M2).** *Panel (a):* Void number density $\mathrm{d}n/\mathrm{d}\ln R$ as a function of void radius R, normalised to unity at $R = 30\,\mathrm{Mpc}$ (no specific redshift is claimed, N-7). The blue curve is the standard Sheth–van de Weygaert (SvdW) baseline (§4.2). The red band shows the ECF prediction (Eqs. 35–36) for $\alpha_{\mathrm{KZ}} = 0.8$–$1.4$ (adopted interval; calibration procedure not documented). The green band shows the predicted Euclid DR1 measurement uncertainty (±18% per bin, schematic). *Panel (b):* Amplification ratio $\mathcal{A}_{\mathrm{ECF}}(R) = n_{\mathrm{ECF}}/n_{\Lambda\mathrm{CDM}}$ showing the amplitude $1 + \alpha_{\mathrm{KZ}} \simeq 2.1$ at the peak of the adopted spatial template, at $\xi_{\mathrm{KZ}} = 60\,\mathrm{Mpc}$ (dashed vertical; *adopted value, not derived ab initio — open problem PO-F3-XI: the position of the feature is fixed by the Kibble–Zurek scaling with a coefficient that the framework does not derive*). *Panel (c):* $\mathcal{A}_{\mathrm{ECF}}(R)$ for the central $\alpha_{\mathrm{KZ}} = 1.1$; no redshift evolution is claimed (the redshift dependence of the excess remains to be derived, PO-F3-XI). *Kill-switch (24 June 2026):* null result ($< 2\sigma$ excess at $R > 60\,\mathrm{Mpc}$ in Euclid DR1) would disfavour the Kibble–Zurek void channel. *Reproduced by:* `plot_F2_vsf_prediction.py` (Foundation II repository, `B_Paper_Plots/`).

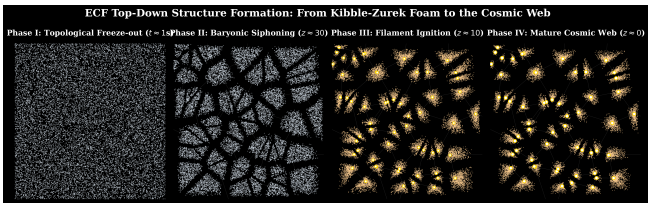

Figure 14: **Top-down formation of the Cosmic Web in the ECF (particle simulation, $N_{\rm gas} = 18\,000$, Voronoi attractor).** From left to right: (*i*) $t \approx 1$ s — the Kibble–Zurek topological freeze-out crystallises a Voronoi-like domain skeleton (grey lines); the baryonic gas is initially uniform. (*ii*) $z \approx 30$ — top-down baryonic siphoning begins: gas particles are gravitationally vacuumed toward the Macro-Knot network located on the domain-wall intersections (attraction factor $\sim 45\%$). (*iii*) $z \approx 10$ — filament ignition: the skeleton becomes visible as baryons collect along domain boundaries; luminous nodes (gold markers) mark the Macro-Knot positions. (*iv*) $z \approx 0$ — mature Cosmic Web, with profound voids whose local budget is dominated by residual torsion ($\Omega_\tau \to 1$, read off from Eq. 27). The characteristic domain scale $\xi_{\rm KZ} \sim 50$–80 Mpc (normalised to unity in the simulation) determines the typical void radius and the over-abundance of large voids predicted in Eq. (34). *(Simplified particle simulation; full N-body treatment deferred to Foundation III.) Script:* `scripts/plot_cosmic_web_evolution.py`. *Reproduced by:* `Scripts_F2/plot_cosmic_web_evolution.py`.

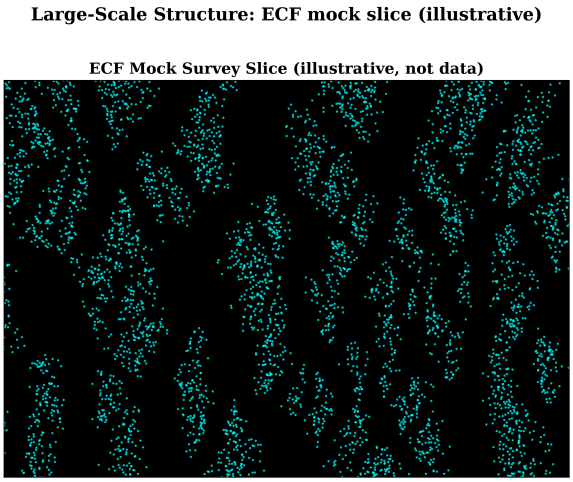

Figure 15: **ECF mock survey slice (illustrative).** Two-dimensional slice of a *mock* galaxy distribution: uniformly drawn points are partially displaced toward the nearest vertex of a random Voronoi tessellation, as a visual illustration of the top-down filament/void geometry discussed in the text. The slice is neither a simulation of the ECF dynamics nor a comparison with observations; it carries no void size function and no goodness-of-fit statistic. The semi-quantitative VSF template is given in Fig. 13 (Eq. 36); its test against void catalogues (DESIVAST DR1) is carried out in *Foundation III — Cosmological Physics* [7]. *Reproduced by:* `Scripts_F2/plot_void_statistics.py`.

Falsifiability and observational programme. Three near-term observational tests directly probe the ECF top-down void channel:

i. **DESI void catalogue ($z < 1$).** An over-abundance of large voids ($R > 60$ Mpc) at $\geq 2\sigma$ would constitute positive evidence for the Kibble–Zurek domain geometry; a null result at 2σ would disfavour the ECF top-down channel [12].

ii. **Euclid weak-lensing of void interiors.** The ECF predicts a *torsion lensing deficit* inside large voids: the reduced shear $\gamma_{\rm int}$ should be suppressed relative to ΛCDM by $\Delta\gamma/\gamma \sim -5\%$ on scales $R \sim \xi_{\rm KZ}$, as a consequence of the TIP forcing $\Omega_\tau \to 1$ in evacuated regions. This deficit is testable with Euclid weak lensing of void interiors [165].

iii. **Cosmic velocity field (next-generation surveys).** A systematic alignment of peculiar-

velocity vorticity along Kibble–Zurek domain walls—detectable with DESI+Roman velocity catalogues [6]—would provide independent support for the top-down torsion channel.

The Dipole Repeller and Geometric Pressure. The observed peculiar velocity of the Local Group ($\sim 630\,\mathrm{km\,s^{-1}}$) cannot be fully explained by gravitational attraction toward mass concentrations alone. High-precision reconstructions of the cosmic velocity field have identified a primary repulsive bulk flow originating from a large-scale underdensity known as the *Dipole Repeller* [91]. The ECF model proposes to interpret this phenomenon as the macroscopic manifestation of torsion-driven geometric pressure. In this view, cosmic voids are not merely passive empty regions but active "torsion engines" whose negative geometric curvature provides the missing momentum in the local cosmic flow.

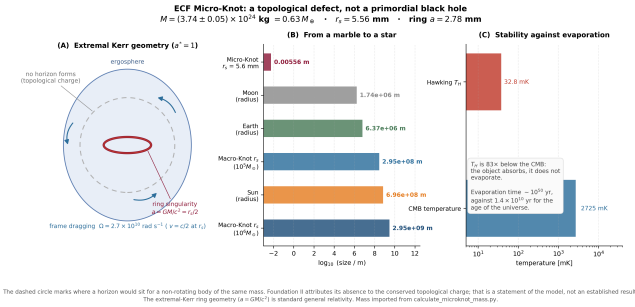

Figure 16: **What a Micro-Knot is, and what it is not.** *(A)* At the extremal Kerr limit $a^* = 1$ the geometry is a ring of radius $a = GM/c^2 = r_s/2 = 2.78\,\mathrm{mm}$, surrounded by an ergosphere, with frame dragging $\Omega_{\mathrm{LT}} \simeq 2.7 \times 10^{10}\,\mathrm{rad\,s^{-1}}$. The dashed circle marks where a horizon would sit for a non-rotating body of the same mass: *no horizon forms*, the conserved topological charge preventing it. Centrifugal flattening does not apply — the object has no material surface to deform. *(B)* The scale ladder, from the ring to the $10^5\,M_\odot$ Macro-Knot. *The two ends are not linked by one chain*: Poisson coagulation of Micro-Knots is characterised by the benchmark $M_{\mathrm{typ}} \simeq 6.8 \times 10^{-5}\,M_\odot$ for $\lambda \simeq 36$, whereas the Macro-Knot scale follows from the distinct, conditional Jeans/horizon channel at the Great Annihilation. *(C)* $T_H = 32.8\,\mathrm{mK}$ sits 83× below the CMB, so the object absorbs rather than evaporates; the evaporation timescale exceeds the age of the universe by forty orders of magnitude. *Scope:* the extremal-Kerr geometry is standard general relativity; the absence of a horizon is a statement of this framework, not an established result, and is drawn dashed for that reason. *Reproduced by:* `Scripts_F2/plot_F2_microknot_structure.py`.

4.3 Primordial Topological Defects (Micro-Knots)

The Micro-Knot population is produced during the electroweak phase transition at $T_c \simeq 159\,\mathrm{GeV}$, when the chiral vacuum manifold

$\mathcal{M}$ acquires non-trivial homotopy $\pi_2(\mathcal{M}) \neq 0$. Each Micro-Knot is a hedgehog point defect of winding number $N = 1$, mass $M_{\mathrm{MK}} = c^3/2GH_{\mathrm{EW}} = 3.74 \times 10^{24}\,\mathrm{kg}$, and Schwarzschild radius $r_s \simeq 5.56\,\mathrm{mm}$. The formation and stability of this population are guaranteed by two properties: (i) the stiff-era growth mode enables Gaussian fluctuations to go non-linear without finely tuned initial conditions (Appendix D); (ii) the topological winding number N is conserved, preventing evaporation ($T_H = 33\,\mathrm{mK} \ll T_{\mathrm{CMB}}$, $t_{\mathrm{evap}} \sim 10^{50}\,\mathrm{yr}$). In standard scenarios, the formation of extremely low-mass primordial black holes is dynamically suppressed during the radiation-dominated era ($w = 1/3$) because strong radiation pressure counteracts gravitational collapse. Forming such objects requires extremely rare, large-amplitude fluctuations ($\delta\rho/\rho \gtrsim 0.45$), often necessitating finely tuned features in the inflationary power spectrum.

In contrast, the immediate post-bounce universe in the ECF scenario is dominated by the spin-torsion density, behaving as a "stiff" fluid with an equation of state $w = 1$. Crucially, in a stiff fluid background, sub-horizon density perturbations do not oscillate as acoustic waves; instead, they exhibit a logarithmic or linear growth mode. This "stiff instability"

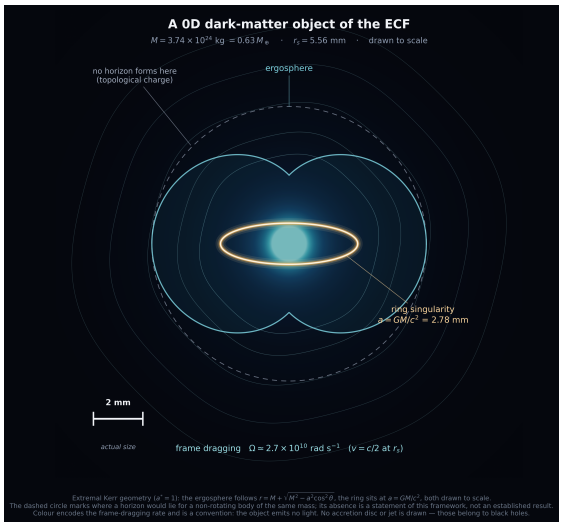

Figure 17: **A 0D dark-matter object of this framework, drawn to scale.** The same object as Fig. 16, rendered for presentation. Every element is computed rather than sketched: the ergosphere follows the extremal Kerr surface $r = M + \sqrt{M^2 - a^2\cos^2\theta}$, the ring sits at $a = GM/c^2 = 2.78\,\mathrm{mm}$, and the shading falls as $\Omega \propto r^{-3}$. *Conventions, stated so they are not read as measurements:* colour encodes the frame-dragging rate — the object emits no light; the dashed circle marks where a horizon would lie for a non-rotating body of the same mass, and its absence is a statement of this framework rather than an established result; no accretion disc or jet is drawn, those belonging to black holes and importing precisely the reading this figure exists to prevent. *Reproduced by:* `Scripts_F2/plot_F2_microknot_press.py`.

implies that standard Gaussian fluctuations can naturally grow non-linear and collapse. These objects form during the "Kination" era (the stiff-matter dominated epoch), prior to the onset of BBN. This population emerges as a generic feature of torsion cosmology rather than an ad-hoc addition.

4.3.1 Structure: Topological Defects as "Granularity"

The resulting collapsed structures are not classical singularities but are identified as topological defects born from the chiral phase transition, stabilizing as non-trivial "knots" in the spacetime torsion field. Figure 18 illustrates this concept:

- **Vacuum Knots:** Instead of a smooth singularity, the ECF defect can be visualized as a tangled bundle of twisted field lines.
- **Granular Halo:** This complex substructure creates a distinct gravitational signature. Unlike a diffuse particle cloud (WIMPs), the ECF halo is "granular," composed of discrete, compact objects.

This granularity is the property on which the microlensing predictions below rest.

The characteristic mass of the Micro-Knots is set by the Hubble-horizon mass at the electroweak transition ($T_c \simeq 159$ GeV, $t \sim 10^{-11}$ s); the canonical evaluation is Eq. (48) below, and this paragraph gives the order-of-magnitude chain.

Derivation of $M_{\rm micro}$ and physical size bounds. The value 3.74×10^{24} kg is not a free parameter but follows from two fundamental constants and one physical scale. At the electroweak epoch ($T_c \simeq 159$ GeV, $g_* = 106.75$), the Hubble rate in a radiation-dominated background is

$$H(t_{\rm EW}) \;=\; 1.66\,\sqrt{g_*}\,\frac{T_c^2}{M_{\rm Pl}} \;\simeq\; 5.4 \times 10^{10}\,\mathrm{s}^{-1}, \quad (37)$$

i.e. $1/H \simeq 1.9 \times 10^{-11}$ s, yielding a Hubble-horizon mass

$$\begin{aligned} M_{\rm micro} &= \frac{c^3}{2G\,H(t_{\rm EW})} \\ &\simeq \frac{(3.0 \times 10^8)^3}{2 \times 6.67 \times 10^{-11} \times 5.4 \times 10^{10}}\,\mathrm{kg} \\ &\simeq 3.7 \times 10^{24}\,\mathrm{kg}, \end{aligned} \quad (38)$$

Figure 18: **Visualization of an ECF Primordial Defect.** To illustrate the "granularity" of the halo, the defect is represented as a complex vacuum knot. The central dark region is the defect core, while the surrounding twisted lines represent the chiral torsion field. This topology acts as a massive, compact scatterer, providing the discrete targets required for microlensing surveys.

Reproduced by:
`Scripts_F2/plot_pbh_topo_defect.py`.

about 0.63 Earth masses (canonical value 3.74×10^{24} kg, Eq. (48)). This value is not fitted: it depends only on G, c, and the electroweak scale, all independently measured (it does rest on the assumption stated after Eq. (48)).

The physical radius of the Micro-Knot is bounded by two causal arguments:

- **Lower bound — Schwarzschild radius:** A Micro-Knot compressed below R_S would collapse into a classical black hole, destroying the topological charge that stabilises it.

$$\begin{aligned} R_{\min} &= \frac{2GM_{\rm micro}}{c^2} \\ &\approx \frac{2 \times 6.67 \times 10^{-11} \times 3.74 \times 10^{24}}{9.0 \times 10^{16}} \\ &\approx 5.56 \times 10^{-3}\,\text{m}\ (5.56\,\text{mm}). \end{aligned} \tag{39}$$

- **Upper bound — Hubble horizon at formation (Kibble constraint):** A topological defect cannot be larger than the causal horizon at the moment of its Kibble–Zurek freezing; regions outside the horizon cannot correlate to form a coherent object. With Eq. (37):

$$\begin{aligned} R_{\max} = \frac{c}{H(t_{\rm EW})} &\approx \frac{3.0 \times 10^8}{5.4 \times 10^{10}}\,\text{m} \\ &\approx 5.6 \times 10^{-3}\,\text{m}\ (5.6\,\text{mm}). \end{aligned} \tag{40}$$

The two bounds coincide by construction: for the horizon mass, $2GM/c^2 = c/H$ (the consistency check stated with Eq. (47)). The physical size of a Micro-Knot is therefore $R_{\rm micro} \simeq 5.6$ mm rather than a window, with a mass of $0.63\,M_\oplus$. A word on the image this invites: the *envelope* is centimetric — the ergosphere, which is what deflects light and drags whatever approaches, extends to $r_s = 5.56\,\text{mm}$, so the object occupies roughly the volume of a marble. The *structure* is not a sphere: it is a ring of diameter $2a = 5.56\,\text{mm}$ (Fig. 17). Matter is not compressed to a point, it is knotted into a circle. The corresponding mean density spans $\rho_{\rm micro} \sim 10^{27}$–$10^{31}\,\text{kg}\,\text{m}^{-3}$, nine to thirteen orders of magnitude above nuclear density. The precise value within this window depends on the topological stabilisation mechanism (chiral charge conservation), whose first-principles derivation is an open problem, not treated in the present corpus *(corrected 20/10/2026: previously deferred to Foundation III — Cosmological Physics, which does not treat it).*[2]

[2]Treated as a Schwarzschild-mass object for comparison,

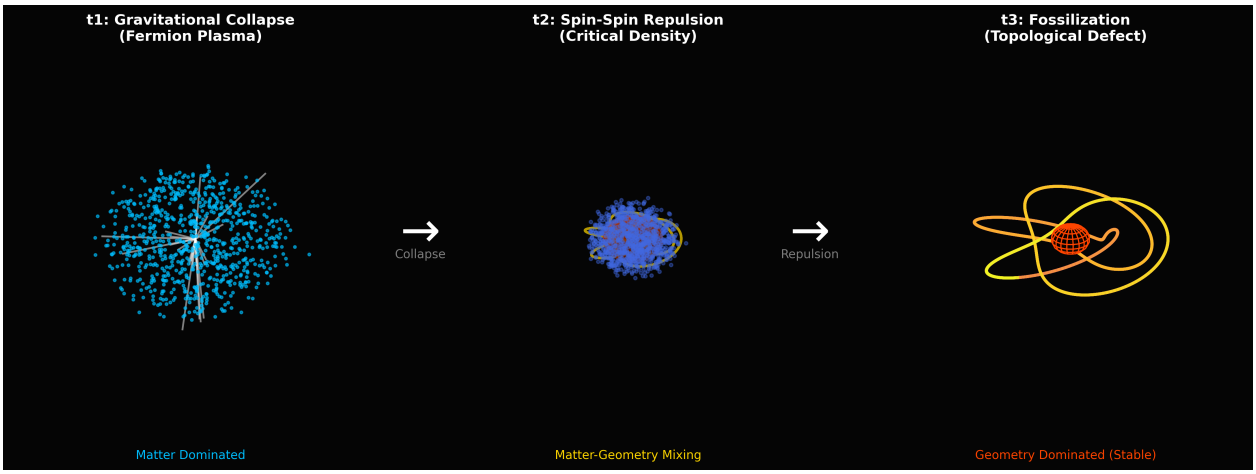

Figure 19: **Schematic of the Topological Transition (Matter → Geometry).** This triptych illustrates the formation mechanism of an ECF PBH. **t1 (Left):** Gravitational collapse of the initial stiff fermion plasma (blue particles). **t2 (Middle):** Critical density is reached. The repulsive spin-spin interaction (central orange glow) halts the collapse, and twisted torsion field lines (gold) begin to emerge from the fermion spin density. **t3 (Right):** The final stable state. The fermions have been "consumed," their energy converted into a self-sustaining topological knot of pure spacetime geometry (the PBH), stabilized by chiral charge conservation.

Reproduced by: `Scripts_F2/plot_topological_transition.py`.

Rare Macro-Knots with $M_{\rm macro} \simeq 10^5\,M_\odot$ arise instead from a *distinct channel*: the Hubble-horizon mass at the moment the Great Annihilation removes kinematic friction ($t \sim 1\,$s), as derived below. *They are not the heavy end of a hierarchical merger sequence starting from Micro-Knots*: Poisson coagulation of Micro-Knots is characterised by the benchmark $M_{\rm typ} \simeq 6.8\times10^{-5}\,M_\odot$ (minimal benchmark at the HSC threshold $\lambda = 36$, not a dynamical ceiling), some nine orders of magnitude below $10^5\,M_\odot$ ($10^5/6.8\times10^{-5} \simeq 1.5\times10^9$); no quantitative chain linking the two scales is established in this work. The two scales belong to different epochs and different arguments, not to one growth history. *What is calculated is the causal Hubble mass at the Great Annihilation; that this scale is converted into a compact object — the efficiency $\epsilon_{\rm coll} = M_{\rm object}/M_H$ — is not derived, and $M_{\rm macro} \simeq 10^5\,M_\odot$ remains an adopted benchmark whose numerical proximity to M_H motivates the scale without establishing a population* (see also Fig. 19).

Derivation of $M_{\rm macro}$ and physical size bounds. By exact analogy with the Micro-Knot derivation (Eqs. (37)–(40)), the causal scale available to a Macro-Knot is bounded by the Hubble-sphere mass at the moment the Great Annihilation removes kinematic friction ($T \sim 1\,$MeV). *This bounds the scale; it does not set the mass of an object, which would additionally require the collapse efficiency $\epsilon_{\rm coll}$, not derived here.* At this epoch:

$$H(T_{\rm GA}) \;=\; 1.66\sqrt{g_*(T)}\,\frac{T^2}{M_{\rm Pl}} \;\simeq\; 0.70\,{\rm s}^{-1} \qquad \text{at } T_{\rm GA} = 1.02\,{\rm MeV},\; g_* = 10.75, \tag{41}$$

yielding a Hubble-sphere mass. *The coefficient follows from the flat FRW relations*: with $R_H = c/H$ and $\rho_{\rm mass} = 3H^2/8\pi G$, the mass enclosed is $M_H = \frac{4\pi}{3}\rho_{\rm mass}R_H^3 = c^3/(2GH)$ — *half* the value obtained from the bare combination $c^3/(GH)$ quoted in earlier versions:

$$M_H(T_{\rm GA}) \;=\; \frac{c^3}{2G\,H(T_{\rm GA})} \;\simeq\; \frac{(3\times10^8)^3}{2\times6.67\times10^{-11}\times0.70} \simeq 2.9\times10^{35}\,{\rm kg} \simeq 1.4\times10^5\,M_\odot. \tag{42}$$

The Hubble-horizon mass provides a *causal upper mass scale* for a compact object formed

a Micro-Knot of mass $M \simeq 3.74\times10^{24}$ kg would have a Hawking temperature $T_H = \hbar c^3/(8\pi GMk_B) \simeq 33$ mK, two orders of magnitude below the CMB temperature ($T_{\rm CMB} = 2.725$ K); the corresponding evaporation timescale is of order 10^{50} yr (`Scripts_F2/derive_microknot_observables.py`). The topological chiral charge prevents horizon formation, rendering this comparison strictly illustrative, but confirming cosmological stability against Hawking evaporation regardless of interpretation.

through the proposed Great-Annihilation channel: no causally connected region — hence no object formed from one — can exceed it at freeze-out. *Status:* CALCULATED-CONDITIONAL, under the radiation-domination estimate $H \sim T_{\rm GA}^2/M_{\rm Pl}$ used above.

Correction of record (17/09/2026). Earlier versions quoted a *lower* bound $M_{\rm macro}^{\rm min} \sim 10^3\, M_{\rm micro} \sim 10^{27}\,{\rm kg} \sim 500\, M_\odot$, derived from a number of Micro-Knot mergers. **That bound is withdrawn**, for two independent reasons: it rested on the Micro→Macro growth channel, which the mass-function analysis excludes; and its three written forms are mutually inconsistent ($10^3\, M_{\rm micro} = 3.7 \times 10^{27}$ kg $= 1.9 \times 10^{-3}\, M_\odot$, whereas $500\, M_\odot \simeq 10^{33}$ kg). **No lower bound is claimed in its place.** The value $10^5\, M_\odot$ used downstream is an *adopted benchmark scale*, not a derived bound, and the dynamical selection of a Macro population near 10^5–$10^6\, M_\odot$ remains STRUCTURAL-HYPOTHESIS.

$$M_{\rm macro}^{\rm bench} \;\equiv\; 10^5\, M_\odot. \tag{43}$$

The adopted benchmark $10^5\, M_\odot$ is of the same order as the causal Hubble scale across part of the Great-Annihilation window *(from $\sim 6 \times 10^4\, M_\odot$ at 1.6 MeV to $\sim 10^6\, M_\odot$ at 0.5 MeV, with g_* evolving through the transition)*, and is compatible with the black-hole masses inferred from JWST LRD observations [56, 121]. *No lower bound is derived*, so this is a compatibility statement, not a predicted window. Independent confirmation comes from a JWST detection (2026) of a $\sim 5 \times 10^7\, M_\odot$ black hole (Abell2744-QSO1) in a small galaxy at $z \approx 7$ lying behind the cluster Abell 2744 and magnified by it, whose mass appears to *precede* the full formation of its host galaxy [189] *(corrected 15/10/2026: formerly "a galaxy of the cluster")* — a mass scale *compatible* with the ECF Macro-Knot benchmark, though the ECF channel does not derive the formation of such an object. *The value $10^5\, M_\odot$ is an adopted benchmark*: it is not derived from a hierarchy of Micro-Knot mergers, and it is not the geometric mean of two causal bounds — no lower causal bound is claimed.

The physical size of a Macro-Knot is likewise bounded:

- **Lower bound — Schwarzschild radius:**

$$\begin{aligned} R_{\rm min}^{\rm macro} &= \frac{2GM_{\rm macro}}{c^2} \\ &\approx \frac{2 \times 6.67 \times 10^{-11} \times 10^5 \times 2 \times 10^{30}}{9 \times 10^{16}} \\ &\approx 3 \times 10^8\,{\rm m} \;\approx\; 0.4\, R_\odot. \end{aligned} \tag{44}$$

- **Upper bound — Hubble horizon at freeze-out:**

$$\begin{aligned} R_{\rm max}^{\rm macro} &= \frac{c}{H(T_{\rm GA})} \approx \frac{3 \times 10^8}{0.70} \\ &\approx 4.3 \times 10^8\,{\rm m} \;\approx\; 0.6\, R_\odot. \end{aligned} \tag{45}$$

A canonical $10^5\, M_\odot$ Macro-Knot is therefore compact: its size lies between its Schwarzschild radius ($\approx 0.4\, R_\odot$) and the Hubble radius at the Great Annihilation ($\approx 0.6\, R_\odot$, with $H(T_{\rm GA}) \simeq 0.70\,{\rm s}^{-1}$ from Eq. (42)), two scales within a factor 1.5, for a mass 10^5 times that of the Sun. The mean density is then $\rho_{\rm macro} \sim 6 \times 10^8$–$2 \times 10^9\,{\rm kg\,m^{-3}}$, of white-dwarf order. *Correction (15/10/2026):* earlier versions gave $R_{\rm min} \approx 300$ km (a factor 10^3 too small), $R_{\rm max} \approx 4\, R_\odot$ (with $H = 0.1\,{\rm s}^{-1}$ instead of the $0.70\,{\rm s}^{-1}$ computed above) and 10^9–$10^{16}\,{\rm kg\,m^{-3}}$. As with the Micro-Knot, the point-mass approximation used in gravitational-wave merger calculations (Appendix D.10) and in the LISA/ET kill-switch is justified: the physical radius is negligible compared to the orbital separation at any relevant merger stage.

Open problem PO-F2-MACRO-COMP — composition and formation of a Macro-Knot at $T \sim 1$ MeV *(added 23/10/2026).* Eq. (42) gives the mass–energy inside the Hubble sphere at $T_{\rm GA}$; it does not say what that mass is made of. At $T = 1.02$ MeV ($g_* = 10.75$) the matter is a negligible fraction of it. Taking the standard present densities ($\omega_{\rm c} = 0.120$, $\omega_{\rm b} = 0.0224$) and entropy conservation (g_{*s} : $10.75 \to 3.91$, so $1 + z \simeq 6.1 \times 10^9$), one finds $\rho_{\rm m}/\rho_{\rm r} \simeq 7 \times 10^{-7}$. A Hubble volume ($R_H = c/H \simeq 4.3 \times 10^8$ m, comoving radius $\simeq 84$ pc today) then contains only $\simeq 0.08\, M_\odot$ of dark matter and $\simeq 0.015\, M_\odot$ of baryons; the remaining $\simeq 1.4 \times 10^5\, M_\odot$ is radiation energy. A galaxy of $10^{12}\, M_\odot$ occupied $\sim 10^{13}$ Hubble volumes at that epoch.

- *If the knot fluid has the density of present dark matter, extrapolated back as a^{-3}*, the gravitational collapse of the knot fluid alone in one

Hubble volume gathers $\sim 0.1\,M_\odot$, some six orders of magnitude below $10^5\,M_\odot$.

- *Reaching $10^5\,M_\odot$ would require the collapse of the radiation itself*, as for a primordial black hole of horizon mass formed in the radiation era, which needs $\delta\rho/\rho \gtrsim 0.45$ (§4.3.1 and the paragraph preceding it). The stiff-era instability invoked there does not apply at $T \sim 1\,\mathrm{MeV}$, where $w = 1/3$, and the corpus derives no source for such amplitudes. For the adopted abundance, $n \approx 2\times10^{-3}$–$1.6\times10^{-2}\,\mathrm{Mpc}^{-3}$, the implied fraction of the energy density in knots at formation is $\beta \sim 3\times10^{-15}$–$3\times10^{-14}$ (simple comoving estimate, $\beta = nM(1+z)^3/\rho$; not a derived prediction).
- *The corpus contains no knot fluid denser than dark matter at T_{GA}*, no definition of ϵ_{coll}, and no calculation of the collapse. Related, the dilution law of the knot component is not settled: a^{-3} for pressureless knots (above) against a^{-6} for the spin density used in the BBN estimate.

Status. Open. The causal scale $M_H \simeq 1.4\times10^5\,M_\odot$ bounds the mass of a causally connected region; it does not say that the knots supply it. A Macro-Knot of $10^5\,M_\odot$ is therefore an adopted benchmark whose composition (knot fluid, radiation, or another component) and formation channel are not derived. If it forms by collapse of the radiation, the object is first a black hole, a point that would also condition PO-F2-SEED-CMB.

4.4 The Kibble-Zurek Mass Function and Quantitative Defect Abundance

In the ECF model, the coexistence of planetary-mass Micro-Knots and supermassive Macro-Knots is modeled as a two-scale outcome of the Kibble–Zurek mechanism [30–32].

The topological phase transition yields a scale-invariant hierarchical network governed by a continuous mass function, analogous to a Press-Schechter formalism:

$$\frac{dn}{dM} = \frac{\rho_{KZ}}{M^2}\left(\frac{M}{M_*}\right)^{1-\alpha}\exp\left(-\frac{M}{M_{max}}\right) \qquad (46)$$

Here ρ_{KZ} is the total energy density of the frozen torsion field at the electroweak crystallization epoch, fixed by the ECF background calibration of Foundation I [5]. The spectral index is set to $\alpha = 5/2$. This is an *adopted* value: no derivation of it from the scaling exponent of the defect network has been identified in the Kibble–Zurek literature [30–32]. The lower mass cutoff is set by the horizon mass at the electroweak transition, derived below.

Three instants, and why only one of them sets the mass. The electroweak epoch carries three distinct moments that are easily conflated, and since the defect mass depends on which one is meant, it is worth separating them before quoting a number.

(i) The transition itself, at $T_c = 159 \pm 1$ GeV in the Standard Model with $m_H = 125$ GeV (lattice determination [28]). *No defect forms at this instant*: the order parameter is still tracking equilibrium, and a system that tracks equilibrium produces no topological residue.

(ii) The Kibble–Zurek freeze-out, when the relaxation time of the order parameter exceeds the time remaining before the transition and the system ceases to follow. This is where the defects are born, and their density is set by the correlation length at that instant. It is the only one of the three that fixes the mass.

(iii) The sphaleron freeze-out, at $T_* = 131.7 \pm 2.3$ GeV [28], when the Hubble rate overtakes the baryon-number violation rate. This freezes the baryon *asymmetry*, not the defect network; it plays no role in the mass, although it matters elsewhere in this work.

The second and the first coincide here, and the reason is worth stating because it is not obvious. The freeze-out condition gives $\hat{\varepsilon} = (\tau_0/\tau_Q)^{1/(1+\nu z)}$, where $\tau_0 \sim 1/T_c \simeq 4\times10^{-27}$ s is microscopic and $\tau_Q \sim 1/H \simeq 1.9\times10^{-11}$ s is the cosmological quench time — a ratio of some *fifteen orders of magnitude*. Whatever the critical exponents, $\hat{\varepsilon} \lesssim 10^{-5}$: the freeze-out occurs at T_c to better than one part in 10^5. In a laboratory quench, where τ_Q is of order milliseconds, the two temperatures differ appreciably; in cosmology the expansion is so slow against microscopic timescales that the system decouples a negligible distance from the transition. *The distinction is real, and it collapses here for a reason that can be stated rather than assumed.*

The horizon mass at that instant follows from the standard relation [29]

$$M_{\mathrm{H}} = \frac{4\pi}{3}\,\rho\left(\frac{c}{H}\right)^3 = \frac{c^3}{2GH}, \qquad \rho = \frac{3H^2}{8\pi G}, \quad (47)$$

which satisfies the consistency check $r_{\rm s}(M_{\rm H}) = 2GM_{\rm H}/c^2 = c/H$: the Schwarzschild radius of the horizon mass *is* the horizon size. Evaluated at T_c,

$$M_{\rm MK} = (3.74 \pm 0.05) \times 10^{24}\ \text{kg}, \qquad r_{\rm s} = 5.56\ \text{mm}, \tag{48}$$

the uncertainty propagating from $\sigma(T_c)/T_c = 0.6\%$ through $M \propto T^{-2}$.

One caveat belongs here rather than in a footnote. The Standard Model with $m_H = 125$ GeV has no true phase transition at the electroweak scale: the change is a crossover, sharply defined at T_c but continuous. The Kibble–Zurek mechanism is formulated for a transition with symmetry breaking, and the extension used here relies on the torsional sector supplying the order parameter that the Standard Model alone does not. That reliance is a genuine assumption of this framework, and it is not established by the numbers above. This horizon mass is the lower end of the Kibble–Zurek mass window (order-of-magnitude chain in Eqs. (37)–(38), causal size bounds in Eqs. (39)–(40); see Appendix D). The upper end relevant for the rare heavy-seed tail is the ECF Jeans mass evaluated at the Great Annihilation freeze-out (e^+e^- annihilation, $T \sim 1$ MeV, $t \sim 1$ s — not the much hotter QCD/electroweak transition relevant to the Micro-Knot scale above):

$$M_{\rm max} \;\equiv\; M_K(z_t) \;=\; \frac{c^3}{2G\,H(z_t)} \;\sim\; 10^5\text{–}10^7\ M_\odot, \tag{49}$$

where z_t denotes the transition scale relevant for the rare Macro-Knot seed tail; the corresponding causal mass bounds derived from the Great Annihilation freeze-out are given in Eqs. (43)–(42) with physical size bounds in Eqs. (44)–(45).

Micro-Knots, Macro-Knots, and the two-scale dark sector. The geometric dark sector should be separated into two observationally distinct regimes. The abundant low-mass population, here called Micro-Knots, is associated with electroweak-scale freeze-out and provides the granular halo component tested by SPARC rotation curves and future microlensing surveys. The rare high-mass population, here called Macro-Knots, does *not* sit at the heavy end of the same coagulation spectrum: Poisson coagulation of Micro-Knots is characterised by the benchmark $M_{\rm typ} \simeq 6.8 \times 10^{-5}\,M_\odot$, and no quantitative chain links it to the Macro scale. Macro-Knots follow instead from a distinct and conditional channel — the Jeans mass at the Great-Annihilation horizon — and may act as primordial compact seeds for early baryonic collapse. This separation avoids conflating the halo-fluid role of Micro-Knots with the central-seed role of Macro-Knots, while preserving the common topological origin of both populations.

- **Micro-Knots (The Halo Fluid):** At the lower mass spectrum ($M_\mu \sim 10^{24}$ kg), the number density peaks (Fig. 7). A Milky Way-sized halo contains $N_\mu \sim 10^{18}$ Micro-Knots. At kiloparsec scales, this immense abundance smooths out the discrete topological nature, closely mimicking a continuous cold dark matter (CDM) fluid that dictates flat rotation curves.
- **Macro-Knots (The Universal Seeds):** At the high-mass end ($M \sim 10^5 - 10^6 M_\odot$, from the distinct Great-Annihilation channel), this paper adopts, as a working assumption, $\sim \mathcal{O}(1)$ Macro-Knot per galactic volume. This abundance is not derived: neither the collapse efficiency $\epsilon_{\rm coll}$ nor a number density is computed (§7.1). Under this assumption, many galaxies host a singular, pre-existing supermassive topological seed at their center *(corrected 17/10/2026: earlier versions attributed this abundance to a "probability integral" over the "high-mass exponential tail"; no such integral is computed, and the Macro-Knots are not the tail of the Micro-Knot spectrum) (note of 17/10/2026: this abundance is in tension with the CMB if the seeds accrete at the Eddington rate from recombination onward; see PO-F2-SEED-CMB, §7.1).*

Under this assumption, the baryonic mass of a galaxy could be related to the mass of its primordial seed; whether this accounts for scaling laws such as the M–σ relation is not computed here *(corrected 19/10/2026: was "fundamentally anchored ... offering a natural origin").*

5 Formation Dynamics: Mathematical Formalism of ECF Defects

The crystallization of the dark sector into discrete (0D, 1D, 2D) and continuous (3D) geometric defects is governed by the Kibble–Zurek mechanism [30, 31] during the symmetry breaking of the

chiral torsion fluid at the electroweak phase transition ($t \sim 10^{-11}$ s).

Let Φ be the order parameter describing the local alignment of fermionic spins within the ECKS extension of General Relativity [17]. As the universe cools through the critical temperature T_c, the field acquires a non-zero vacuum expectation value $\eta = |\langle\Phi\rangle| \neq 0$ over a vacuum manifold $\mathcal{M}$. The topology of $\mathcal{M}$, described by homotopy groups $\pi_n(\mathcal{M})$, dictates the spatial dimensionality of the frozen geometric defects.

Scope note — vacuum manifold

The precise form of $\mathcal{M}$ depends on the symmetry group of an electroweak extension of the PGT, which is not specified in this work. For $\mathcal{M} \simeq S^2$ (global monopole limit) one obtains 0D monopole-like defects *only* ($\pi_0 = \pi_1 = 0$): this limit cannot carry the 1D and 2D sectors discussed below. For $\pi_1(SO(3)) = \mathbb{Z}_2$ one obtains $\mathbb{Z}_2$ vortices. The assignments below therefore do *not* rest on the $\mathcal{M} \simeq S^2$ limit; the candidate manifold $\mathcal{M} \simeq \mathbb{Z}_2 \times S^1 \times S^2$ (Eq. (143)) is not derived, so the 2D sector is conditional on the effective potential, on this vacuum manifold and on the symmetry not being restored, and the 1D sector remains an open problem (PO-F2-8). The full group-theoretic classification is deferred to Foundation III.

5.1 0D Defects: Point Singularities (Micro-Knots)

Point-like defects emerge if $\pi_2(\mathcal{M}) \neq 0$. The torsion field Φ wraps around a central point forming a topological "hedgehog". The topological charge (winding number) is:

$$N = \frac{1}{8\pi} \oint_{S^2} \epsilon^{ijk} \hat{\Phi} \cdot (\partial_j \hat{\Phi} \times \partial_k \hat{\Phi})\, d^2x. \qquad (50)$$

The effective mass concentrates the gradient energy of the field:

$$M_{\rm MK} \approx \frac{4\pi\eta}{e_{\rm tors}} \equiv \frac{c^3}{2GH_{\rm EW}} \simeq 3.74{\times}10^{24}\,{\rm kg} \simeq 83\,M_{\rm Moon} \qquad (51)$$

where the identification with $c^3/2GH_{\rm EW}$ (the ECF canonical value [6]) fixes the product $\eta/e_{\rm tors}$ in terms of the electroweak Hubble rate $H_{\rm EW}$. A Kibble–Zurek log-normal distribution centred on this mass with $\sigma_{\log M} = 1\,$dex spans 10^{23}–$10^{25}\,$kg at 1σ [6].[3]

5.2 1D Defects: Torsion Strings (Macro-Knot Filaments)

Premise not established (added 30/08/2026, read before the derivation below)

This subsection derives the consequences of $\pi_1(\mathcal{M}) \neq 0$ **as a hypothesis**, worked out in detail for completeness. It does **not** establish that hypothesis. S^μ, the only field this framework's Lagrangian actually contains for the axial torsion sector, is *real*: it has an amplitude but no internal phase, and cannot by itself supply the complex order parameter a genuine $\pi_1 \neq 0$ defect requires (see the tentative, self-caveated discussion where this is derived from the Lagrangian directly, and the short summary in §6.2, PO-F2-8). Read everything below as *"if this sector existed"*, not as an established result; several downstream numbers (string tension, most gravitational-lensing and wake predictions) were already found not to survive independent scrutiny, as noted explicitly further down in this same subsection.

Formation conditions. Linear defects (Torsion Strings) emerge when the first homotopy group of the chiral vacuum manifold is non-trivial: $\pi_1(\mathcal{M}) \neq 0$. The order parameter $\hat{\Phi}$ winds once around the string axis with a phase increment $\Delta\theta = 2\pi n$ ($n \in \mathbb{Z}$):

$$\oint_C \nabla\theta \cdot d\vec{l} = 2\pi n, \qquad n \neq 0. \qquad (52)$$

The Kibble–Zurek mechanism produces Torsion Strings at the same electroweak phase transition ($T_c \simeq 159\,$GeV) as the 0D Micro-Knots, but from a *distinct* topological sector (π_1 vs π_2). Their network density is set by one string segment per KZ correlation volume $\xi_{\rm KZ}^2$, and the network evolves through *reconnection* (not coagulation or annihilation), reaching a cosmological scaling solution [213].

Distinction from GUT cosmic strings. ECF Torsion Strings share the conical-deficit geometry

[3] As a Schwarzschild-mass comparison, a Micro-Knot of $M = 3.74 \times 10^{24}\,$kg would have Hawking temperature $T_H = \hbar c^3/(8\pi G M k_B) \simeq 33\,$mK, two orders of magnitude below the CMB temperature; the evaporation timescale is of order $10^{50}\,$yr (`Scripts_F2/derive_microknot_observables.py`). The topological chiral charge prevents horizon formation [6].

of GUT cosmic strings [178, 213], but differ in three essential respects: (i) their tension is set by the electroweak VEV, not a GUT-scale VEV, yielding $G\mu_{1D}/c^2 \simeq 4 \times 10^{-34}$ rather than $\sim 10^{-6}$; (ii) they carry a chiral torsion charge (axial current $\vec{J}_5$), absent in standard GUT strings; (iii) their lensing is achromatic and produces no CMB B-mode pattern at GUT-string amplitude.

Tension, and what it costs. A correction of record (27/08/2026). Earlier versions of this section stated both that the string tension is set by the electroweak VEV and that $G\mu_{1D}/c^2 \sim 10^{-12}$–$10^{-11}$. Those two statements are incompatible: since $G\mu/c^2 = (\eta/M_{\rm Pl})^2$ exactly, the electroweak VEV returns 4.1×10^{-34}, and reaching 10^{-11} would require $\eta \simeq 4 \times 10^{13}$ GeV — eleven orders above the electroweak scale and three below the GUT scale, an energy corresponding to nothing in this framework. The paper additionally used $\eta_{\rm EW} \sim 159$ GeV, which is not the VEV (246.22 GeV) and appears to conflate it with the transition temperature $T_c = 159$ GeV.

We keep the electroweak origin and correct the value, because a single transition producing every defect is the architecture of this framework, while 10^{-11} was a number that did not follow from the formula printed above it. The consequence is stated rather than absorbed: at $G\mu_{1D}/c^2 \simeq 4 \times 10^{-34}$ — some 2.5×10^{26} times below the current CMB bound — these strings produce no lensing signature and no CMB imprint. *Any test resting on the string tension itself — lensing by the conical deficit, the velocity kick imparted to crossing gas — does not survive this correction.* The Faraday-rotation signature does survive, because it does not rest on the tension: the field along the filament is sourced by the Chern–Simons coupling to the axial current (Eq. 103), and depends on the chiral condensate s_0 rather than on μ_{1D}. The same holds for the chiral torsion charge and for the absence of a B-mode pattern. It is worth separating the two: a single correction can remove one class of test and leave another untouched, and saying otherwise would overstate what has been lost. Compatibility with every existing bound and undetectability follow here from the same fact, and it is not a comfortable one: the electroweak scale is low. *Reproduced by:* `Scripts_F2/compute_string_tension_1D.py`.

Metric and gravitational signature. The geometric tension μ_{1D} generates a conical deficit angle in the local metric [178]:

$$\delta = \frac{8\pi G\mu_{1D}}{c^2} \approx 8\pi G\eta_{\rm EW}^2, \tag{53}$$

yielding the cylindrical metric:

$$ds^2 = c^2dt^2 - dz^2 - dr^2 - \left(1 - \tfrac{\delta}{2\pi}\right)^2 r^2 d\phi^2. \tag{54}$$

For an ECF Torsion String on a 1 kpc scale, $\delta \sim 10^{-6}$ rad (a value this small would not be visible on an undistorted diagram; no figure is given here). There is *no* Newtonian $1/r^2$ attraction: the only gravitational effect is a velocity kick $\delta v = 4\pi G\mu_{1D} v_s \gamma_s/c^2$ imparted to matter passing on either side, generating the baryonic wake visible as Lyman-α filaments.

ECF-specific property: chiral magnetic field. Through the Chern–Simons coupling [181], the axial spin current $\vec{J}_5$ along the filament induces a primordial chiral magnetic field ($\nabla \times \vec{B}_{\rm chiral} \propto \vec{J}_5$), offering a geometric origin for large-scale intergalactic magnetic fields absent in standard GUT string models. The resulting field amplitudes across the four defect levels are shown in Fig. 24 (`Scripts_F2/B_Paper_Plots/plot_F2_magnetic_fields.py`; the amplitudes are input ranges, not derived).

Helical structure — ECF-specific property. A Torsion String is not a simple straight tube of concentrated torsion. Because the chiral vacuum condensate $\langle S^\mu \rangle = (s_0, \mathbf{0})$ imposes a global helicity $q = \pm 1$, the order parameter $\hat{\Phi}$ rotates both *around* the string axis (winding number $n = 1$, giving the π_1 charge) and *along* it with a characteristic pitch $\lambda_{\rm hel}$. The resulting field configuration is a topological helix: the axial current $\vec{J}_5$ follows a helical path, inducing a solenoidal chiral magnetic field $\vec{B}_{\rm chiral} \propto \nabla \times (\vec{J}_5)$ that is also helical, not simply longitudinal. The total magnetic helicity $\mathcal{H} = \int \vec{A} \cdot \vec{B}\, d^3x$ is a topological invariant, conserved under reconnection. Strings of same helicity ($q = +1, +1$) repel before reconnecting; strings of opposite helicity ($q = +1, -1$) attract and reconnect preferentially, producing helical loops that radiate gravitationally as they shrink.

Length scales. Three distinct length scales characterise a Torsion String:
(i) Core radius $r_{\rm core}$. The string core concentrates torsion over the electroweak coherence length:

$$r_{\rm core} \sim \frac{\hbar}{m_T c} \sim 10^{-18}\ {\rm m}, \tag{55}$$

where $m_T \sim \eta_{\rm EW}/c^2$ is the effective torsion mass scale assumed here. *(Corrected 15/10/2026: an intermediate "$\sim c/H_{\rm EW}$" was removed; $c/H_{\rm EW} \simeq 5.6$ mm, sixteen orders larger.) (Added 22/10/2026: if the condensate couples minimally to fermions, an electroweak-scale m_T is excluded by laboratory bounds on Lorentz violation, which require $m_T \gtrsim 4\times10^3$ GeV (Appendix F.15); the core radius would then be $\lesssim 5\times10^{-20}$ m. The string estimates below assume $m_T \sim \eta_{\rm EW}$.)* Outside $r_{\rm core}$, the metric is purely conical (eq. 54) with no exponential decay — the deficit angle δ extends to arbitrarily large distances.

(ii) Helical pitch $\lambda_{\rm hel}$. The pitch of the helix is set by the ratio of the axial condensate s_0 to the string winding charge n:

$$\lambda_{\rm hel} \sim \frac{\xi_{\rm KZ}}{n} \sim 50\text{–}80\ {\rm Mpc} \quad (n=1). \tag{56}$$

This is not derived from first principles in the present work and constitutes an open problem (see below).

(iii) Segment length $\xi_{\rm KZ}$. The Kibble–Zurek mechanism produces string segments of typical comoving length:

$$\xi_{\rm KZ} \sim 50\text{–}80\ {\rm Mpc}, \tag{57}$$

This is an *adopted* range (canonical value $\xi_{\rm KZ} = 60$ Mpc, calibration procedure not documented). It is not obtained here from the causal horizon at $T_c \simeq 159\,{\rm GeV}$: direct Kibble–Zurek dilution from that scale [30, 31] misses it by some 24 orders of magnitude (PO-F3-XI, OPEN). This length is taken equal to the void over-abundance scale (§4.2, Eq. 34) — the same adopted parameter, so the coincidence carries no independent information — and it is conjectured that the Kibble–Zurek domain boundaries trace both string segments *and* Chiral Wall faces.

Intrinsic mass: negligible at the electroweak scale. The linear mass density of a Torsion String follows from the symmetry-breaking scale:

$$\mu_{1D} \sim \eta_{\rm EW}^2 \approx \frac{(246.22\ {\rm GeV})^2}{\hbar c^3} \simeq 5.5\times10^{-7}\ {\rm kg\,m^{-1}}. \tag{58}$$

Correction of record (29/08/2026): earlier versions quoted $\mu_{1D} \sim 10^{14}$ kg m^{-1} and tabulated intrinsic masses of 10^3–$10^7\,M_\odot$ for segments of kiloparsec to $\xi_{\rm KZ}$ length. That figure belongs to a GUT-scale string; at the electroweak scale adopted here (§5.2) the correct value is twenty orders of magnitude smaller, and the table it supported has been withdrawn rather than rescaled. A segment of length $\xi_{\rm KZ} \simeq 60$ Mpc carries an intrinsic mass of $\sim10^{18}$ kg — less than a small asteroid, and $\sim10^{-13}\,M_\odot$.

The consequence is worth stating plainly rather than leaving implicit: *the intrinsic geometric mass of the 1D sector is dynamically irrelevant.* Whatever mass is observed along cosmic filaments is baryonic, accumulated by the velocity-kick mechanism described below, and the "critical distinction" between intrinsic and baryonic mass that earlier versions drew does not survive — there is only the baryonic term. This weakens the 1D sector further, on top of the status recorded in the scope note above, and we prefer to say so than to keep a table whose purpose was to suggest otherwise.

This intrinsic geometric mass must be distinguished from the *baryonic mass accumulated by the velocity kick* $\delta v = 4\pi G\mu_{1D} v_s \gamma_s/c^2$ acting on surrounding gas over a Hubble time, which builds the observed Lyman-α filament mass ($M \sim 10^5\,M_\odot$ on 1 kpc scales). These are two separate quantities: the first is purely geometric (set at formation by $\eta_{\rm EW}$), the second is dynamical (accumulated over cosmic time by gravitational accretion). The transfer function from μ_{1D} to accumulated baryonic mass is an open problem, not treated in the present corpus *(20/10/2026).*

Open problem PO-F2-2 — Helical pitch and intrinsic spin. The helical pitch $\lambda_{\rm hel}$ of a Torsion String is not derived from first principles in the present work. Its quantitative value requires solving the ECF field equations in Kerr-like cylindrical symmetry with a non-zero axial torsion condensate s_0. In particular: (i) what fixes the ratio $\lambda_{\rm hel}/\xi_{\rm KZ}$? (ii) does the helical winding carry an intrinsic angular momentum $J_{\rm string}$ analogous to the Kerr parameter of 0D defects? (iii) is the gravitational wave emission from a shrinking helical loop distinguishable from a simple circular loop (standard string network)? This is identified as a potential referee objection; it is an open problem, not treated in the present corpus *(corrected 20/10/2026: previously deferred to Foundation III — Cosmological Physics, which does not treat it).*

Network stability and scaling solution. A natural referee concern is whether the ECF Torsion String network is stable on cosmological timescales or diverges. For standard GUT cosmic strings, the

long-term evolution is governed by the Nambu–Goto equations and converges to a *scaling solution* in which the network maintains a fixed number of $\mathcal{O}(1)$ long strings per Hubble volume, with the excess energy radiated as gravitational waves from loop decay [213]. The ECF Torsion String shares the same π_1 homotopy class and the same reconnection dynamics (two intersecting strings exchange partners with reconnection probability $P \simeq 1$ for $G\mu_{1D}/c^2 \ll 1$ [187]). The scaling solution is therefore expected to apply here as well, with the proviso that the helical $\vec{J}_5$ current may modify the reconnection geometry and produce a slightly different loop distribution compared to the Nambu–Goto case. *Quantitatively*: in the scaling regime, the string energy density dilutes as $\rho_{\rm string} \propto a^{-2}$ (strings), subdominant to matter ($\propto a^{-3}$) and radiation ($\propto a^{-4}$) at all post-formation epochs. The string network therefore does *not* overclose the universe, resolving the analogue of the monopole problem for 1D defects. A dedicated Nambu–Goto simulation of the ECF string network with helical initial conditions is listed as an open item in the research programme of *Foundation III — Strong Gravity* (in preparation, not deposited); it has not been carried out *(20/10/2026).*

5.3 2D and 3D Defects: Domain Walls and the Chiral Fluid

The two remaining sectors of the ECF topological hierarchy are presented in detail in the Topological Zoology section (§6): the 2D Chiral Walls (§6.3) and the 3D Torsion Fluid (§6.4). Here we provide a brief summary for completeness within the Formation Dynamics sequence.

If $\pi_0(\mathcal{M}) \neq 0$, 2D domain walls (Chiral Walls) separate regions of distinct chiral helicity. The Einstein–Cartan field equations for such a membrane (Israel 1966) impose a purely repulsive perpendicular acceleration. A wall of surface density σ_{2D} occupies, at cosmic time t, a causally connected thickness $c\,t$, so its effective *volume* density is $\rho_{\rm wall} = \sigma_{2D}/(c\,t)$ — the same quantity used below to discuss its dilution. Substituting it into the de Sitter form $\ddot{a} = (8\pi G\rho/3)\,a$ gives *Correction of record (15/09/2026).* Earlier versions wrote this acceleration as $\ddot{z} = (8\pi G\sigma_{2D}/3)\,z$, applying the Friedmann coefficient directly to a *surface* density: the right-hand side then carries $\mathrm{m^2 s^{-2}}$ instead of $\mathrm{m\,s^{-2}}$. The conversion $\rho_{\rm wall} = \sigma_{2D}/(ct)$ was already stated a few lines below but had not been carried into the equation. Eight occurrences corrected throughout; no numerical result of this work depended on the missing factor, which enters only the void-carving discussion.

$$\ddot{z} = \frac{8\pi G}{3}\,\frac{\sigma_{2D}}{c\,t}\,z, \tag{59}$$

responsible for carving out cosmic voids; the wall sector is developed in §6.3. The **3D** global defect is the uncrystallized residual torsion fluid whose isotropic pressure:

$$P_{\rm tors} = -\tfrac{1}{4}\kappa\langle S^2\rangle_{\rm vol}, \quad \kappa = 8\pi G/c^4, \tag{60}$$

(symbolic form only: the normalisation of $\langle S^2\rangle_{\rm vol}$ is not established, PO-ECF-SPIN-NORM; the normalisation of the condensate used in this work is $\frac{1}{4}m_T^2 s_0^2 = |\Omega_\tau|\rho_{\rm crit}$, Eq. (21)) is written here as diluting as $a(t)^{-3}$ — the dilution law of this quantity is not consistent across the corpus ($a^{-3(1+w)}$ in §6.4, a^{-6} in the Foundation III BBN estimate) and is not settled here (PO-ECF-BBN-SPIN) —, parameterizing dynamical dark energy with $(w_0, w_a) = (-0.904, -0.153)$, calibrated (not derived *ab initio*, cf. PO-F2-4; w_a through $\xi = 1.645$, calibrated on the DESI preference used by the ECF, hence not independent of DESI) from the PIT calibration [4, 6]. Note that $w_0 + w_a = -1.057 < -1$ implies a phantom crossing at high z (at $z = 1.684$; Foundation I, §dark energy). A single canonical scalar cannot cross $w = -1$, so a mechanism is required. The axial sector of the ECF, in the adopted Proca form, is ghost-free according to the literature [166] (Appendix F establishes only $\lambda_T > 0$), which addresses the most immediate obstruction; **it does not by itself establish that the crossing is licensed**, which would require the effective equation of state to be derived from the Poincaré gauge action rather than calibrated. That derivation is open (PO-F1-SIGMA0). *(Scope of this claim narrowed 05/09/2026: earlier versions stated the crossing was "stabilised by the PGT Hamiltonian constraint"; Appendix F establishes only $\lambda_T > 0$ — the absence of ghosts in the axial sector is invoked from the literature for the adopted Proca form — and does not treat the crossing.)*

6 Topological Zoology: Observations and Falsifiable Predictions

Structural analogy: the ECF as a four-dimensional topological network

Before detailing each defect class individually, it is useful to visualise the ECF dark sector as a *multi-scale topological network* — the closest physical analogy to the architecture described in this section.

The bubble picture (revised 30/08/2026, replacing the earlier "connected network" picture below)**.** Rigorous re-examination of the vacuum manifold shows that the parity sector relevant to the Chiral Walls is discrete of order two ($\mathbb{Z}_2$, σ_+/σ_-), following directly from the axial nature of S^μ under parity (Appendix C). A $\mathbb{Z}_2$ domain-wall network cannot support *stable* triple junctions: the junction configuration relaxes spontaneously for two-phase walls, and only $\mathbb{Z}_3$ or higher admits the stable Y-type junctions (120° angles, Young's condition) that a connected foam or Voronoi lattice would require. The correct picture is therefore **independent, closed Chiral Walls** (topologically spheres), each enclosing one interior phase and, generically, one 0D Micro-Knot near its centre — *not* a connected mesh with shared edges. Baryonic matter swept outward by the Israel repulsion (Eq. 59) accumulates on these closed walls and at their mutual near-approaches, producing the observed cosmic web morphology without requiring a topologically stable 1D connector between them.

Status of the 1D sector. 1D Torsion Strings do not have an established Lagrangian foundation in the present field content. S^μ is a *real* axial vector: it has an amplitude but no internal phase, whereas a genuine cosmic string (quantised winding, $\pi_1(\mathcal{M}) = \mathbb{Z}$) requires a *complex* order parameter (the abelian-Higgs mechanism; Peter & Uzan, *Cosmologie primordiale*, §11.3). Two candidate completions were examined and found inconclusive: (i) a trace-vector companion field, which fails because on a homogeneous background the two vectors are collinear and define no phase; (ii) a *semi-topological* defect arising from the coexistence of the torsion sector (global) and the electroweak gauge sector (local) (Peter & Uzan, §11.5.4) — structurally present in the ECF, but explicitly stable only for *dynamical*, not topological, reasons, and that dynamical stability depends on the unresolved mass parameter m_T (PO-F2-6). **The 1D sector is therefore retained as an open, named problem (PO-F2-8), not as an established prediction.** No result in this section should be read as depending on the existence of quantised Torsion Strings unless explicitly flagged.

Observable consequences of the topological highway structure. The network picture is not merely aesthetic — it generates four *qualitatively distinct* predictions relative to ΛCDM that are testable with current and forthcoming surveys:

Prediction H1 — Coherent rotation of infalling gas near nodes (mechanism revised 30/08/2026)**. The original mechanism (a helical current $\vec{J}_5$ carried by a 1D Torsion String) is withdrawn**: it requires a topologically quantised string, which §6 (Status of the 1D sector) identifies as unfounded in the present field content. The observations below remain real and unaffected — only the ECF's proposed explanation changes. Wang et al. [194] report vortical motion in stacked cosmic filaments from SDSS, consistent with rotation *(*3.3σ *for dynamically cold filaments nearly perpendicular to the line of sight, and* 4.2σ *when the sample is restricted to the 10% of filaments with the most massive end-point groups — figures given in the body of the paper, Fig. 3, not in its abstract; an earlier version of this paper wrongly described the* 3.3σ *as untraceable; the detection has since been corroborated — Tang et al. 2025, Wang et al. 2025)*; a recent 15 Mpc filament at $z = 0.032$ (Tudorache et al. 2025, MNRAS [198]) shows bulk rotation with coherent spin alignment along the spine.

Candidate replacement mechanism. The persistent magnetic field already sourced at each Macro-Knot by the present-day condensate s_0 via the topological flux (Eq. 178, $B_{\rm knot} \sim 9 \times 10^{-6}$ G, dipolar profile $B \propto r^{-3}$) can exert a net Lorentz-force torque on ionised gas falling in radially. A test-particle calculation (single-particle, non-self-consistent) shows the torque does not cancel between hemispheres for a fixed dipole orientation, and that the ion contribution (dominant in mass over the counter-rotating electron contribution by $m_p/m_e \sim 1836$) yields a net, coherent bulk rotation of the infalling baryonic gas. This reuses only field content already established for the 0D/2D sectors (dipolar $B_{\rm knot}$, ionisation via reionisation or the accretion shock itself) and requires no 1D string. **Not yet derived**: the predicted amplitude and its scaling with node

mass or distance, and whether the resulting spin axis is aligned (as observed by Wang et al. 2021, Tudorache et al. 2025) or perpendicular (Prediction H2) — this depends on the orientation of the node's dipole axis relative to the filament, which is not yet fixed by any principle in the corpus. This item is carried as an open, named problem, not a completed quantitative prediction.

Distinction from tidal torque theory (TTT), still applicable. ΛCDM simulations *can* reproduce qualitative filament spin via the standard tidal torque mechanism — Xia et al. (2021) [195], Wang et al. (2025) [196]. **The ECF does not claim that ΛCDM cannot generate filament spin.** The candidate ECF mechanism above is geometric (tied to the node's magnetic dipole, not the external tidal field), but — pending the derivation flagged above — does not yet yield a distinguishing quantitative signature (reversal scale, sign correlation) beyond what is stated here as open.

Prediction H2 — Perpendicular halo spin at nodes (mechanism revised 30/08/2026)**. The original mechanism (vector sum of multiple 1D strings arriving at a node) is withdrawn**: it requires several strings converging on one node, i.e. a stable multi-way junction — shown unstable for the $\mathbb{Z}_2$ wall sector (§6) and doubly unfounded given the 1D sector's status. The observation is real and unaffected: Rong et al. (2025) [199] find galaxy group angular momentum preferentially orthogonal to the filament spine at $> 2\sigma$. **No ECF-specific replacement mechanism is proposed for this item at this time**; it is retained here as an open observational target, explained provisionally by standard TTT alone until the H1 candidate mechanism (magnetic torque) is developed far enough to make a directional prediction.

Prediction H3 — Preferred matter flow direction along filaments ("one-way cables") (withdrawn 30/08/2026)**. This prediction is withdrawn.** It required a directed topological graph (flow along an edge between two nodes), which depends on the same connected 1D/junction structure shown unfounded above. The underlying physical expectation — more gas flowing toward the more massive of two connected clusters — is generic gravitational infall, already present in ΛCDM, and is *not* a distinguishing ECF signature absent a topological carrier. No replacement is proposed.

Prediction H4 — Forbidden "short-circuits" between non-adjacent levels (withdrawn 30/08/2026)**. This prediction is withdrawn**, for two independent reasons. First, it requires the same unfounded 1D string sector. Second, even granting a standard (complex-field) cosmic string, the topological argument used here was itself incorrect: Peter & Uzan (*Cosmologie primordiale*, §11.3.1) show by contradiction that quantised cosmic strings can only be infinite or closed loops — they do not have loose ends to forbid in the first place, so "termination at a node" is not the right way to state a constraint even in the standard theory. No replacement prediction is proposed; the observed absence of filaments ending in voids, if confirmed, would need a different explanation (most simply, ordinary gravitational infall geometry).

Summary — current status of H1–H4 (30/08/2026)

H#	Status	Obs. evidence (real)	Instrument
H1	Revised — magnetic torque candidate, not yet quantified	Wang+2021, Tudorache+2025	DESI, Roman
H2	Open — no ECF mechanism proposed; TTT provisional	Rong+2025 (2σ)	Euclid
H3	**Withdrawn** — no replacement	—	—
H4	**Withdrawn** — no replacement	—	—

Systematic comparison: ECF topological mechanism vs Tidal Torque Theory (TTT). Both the ECF and the standard Tidal Torque Theory generate angular momentum in cosmic filaments, but through fundamentally different physical mechanisms.

Table superseded (30/08/2026)

The comparison table below describes the *original* ECF mechanism — a helical current $\vec{J}_5$ carried by a quantised 1D Torsion String. That mechanism is withdrawn (Status of the 1D sector, §6): S^μ has no internal phase, and the required $\mathbb{Z}_2$ multi-way junctions are unstable. The table is retained *as a historical record of the reasoning trail*, since several of its rows (coherence scale, magnetic helicity conservation, filament termi-

nation) rest explicitly on the string mechanism and are therefore not currently supported claims. The revised, candidate mechanism (magnetic torque at nodes) is given under Predictions H1–H2 above; it does not yet reproduce every row below, and rows H3 (filament termination) and the "one-way cable" asymmetry are withdrawn outright, not merely revised.

2D Chiral Walls — deeper structure. The 2D defects (Chiral Walls, π_0 sector) are the *faces* of the topological network, bounded by loops of 1D Torsion Strings. Their physical role is distinct from both 0D nodes and 1D edges: they separate regions of *opposite chirality* ($q = +1$ vs $q = -1$), acting as domain boundaries in the chiral vacuum manifold.

Physical mechanism. The Israel junction condition (Eq. (59)) imposes a purely repulsive perpendicular acceleration $\ddot{z} \propto \sigma_{2D}\, z/(ct)$ on matter crossing the wall — qualitatively distinct from the gravitational *attraction* of 0D (Micro-Knots) and 1D (Torsion Strings) defects. This repulsive acceleration is the ECF mechanism for carving cosmic voids: the Chiral Wall pushes matter outward from the void centre, compressing it into the filamentary skeleton and cluster nodes at the wall boundaries.

Distinction from ΛCDM void formation. In ΛCDM, voids form by gravitational evacuation — matter flows *outward* from underdense regions under its own self-gravity ("embedded" Euler equation; Sheth & van de Weygaert 2004). The timescale is set by the local density contrast $\delta(z)$. In the ECF, the Chiral Wall provides an *additional* repulsive term that accelerates void expansion beyond the ΛCDM baseline, producing the predicted void-size excess of factor ~ 2.1 at $R > 60\,\mathrm{Mpc}$ (§4.2, Eq. (35)).

Observable signature. The key discriminator is *not* the existence of voids (both frameworks produce them) but their *size distribution*: the ECF predicts a specific excess of large voids ($R > 60\,\mathrm{Mpc} \sim \xi_{\mathrm{KZ}}$) above the Sheth–van de Weygaert baseline, testable in Euclid DR1 void catalogues (§4.2, Eq. (35)). A secondary signature is the *chirality boundary*: if the wall truly separates $q = +1$ from $q = -1$ domains, the magnetic helicity sign (measurable via Faraday rotation with SKA) should change across the void boundary — a topological prediction absent in ΛCDM.

Key topological constraint (revised 30/08/2026). The hierarchy previously described here (0D–1D–2D–3D connected via $\partial \circ \partial = 0$) presupposed a connected network with 1D edges linking 0D nodes to 2D faces. This is superseded by the bubble picture above: the retained levels are 0D nodes (Micro-Knots, π_2, one generically near the centre of each bubble) and 2D closed walls (π_0, $\mathbb{Z}_2$, independent spheres, no stable shared edges) — the latter *conditional* on the effective potential, on the vacuum manifold (not derived, PO-F2-8) and on the symmetry not being restored, with the 3D fluid filling the space between bubbles. The 1D level is an open problem (PO-F2-8), not an established connector between the other two. **The rule "$\partial \circ \partial = 0$ forbids dangling cables" is not currently a supported ECF claim**: it depended on the 1D sector's existence and on stable multi-way junctions, both shown unfounded above.

Figure redrawn (30/08/2026)

Fig. 20 below has been redrawn to show independent closed 2D bubbles (each generically containing one 0D node near its centre) embedded in the 3D fluid, with no 1D connectors and no H3/H4 panels. The 1D sector is shown with its open-problem status (PO-F2-8); H1/H2 show the candidate magnetic-torque mechanism, explicitly flagged as not yet quantitatively derived; H3 and H4 are shown as withdrawn, with the reason stated in-figure. *Reproduced by:* `plot_F2_topological_network.py` (Foundation II repository, `B_Paper_Plots/`).

Table 2 maps defect dimensionality to certified astrophysical observations and unique ECF signatures.

Terminological note — ECF defect hierarchy. Table 2 gives the formal classification; conventions below are adopted throughout to minimise ambiguity.

- **0D Micro-Knots** (π_2, hedgehog, $M_{\mathrm{MK}} = c^3/2GH_{\mathrm{EW}}$): primordial KZ output. **0D Macro-Knots** ($M \sim 10^5\, M_\odot$): same π_2 class, formed through the distinct Great-Annihilation channel (horizon mass at $T \sim 1$ MeV, a structural hypothesis), not by coagulation of Micro-Knots. Aggregation rules: same helicity → coagulate; opposite helicity → annihilate (PO-F2-1).

Table 1: **[SUPERSEDED — see box above] Filament spin: original ECF topological mechanism vs Tidal Torque Theory (TTT).** Each row identifies a physical property, what each framework predicted, and how they could be discriminated observationally. Rows marked ⋆ were the most immediately testable under the withdrawn mechanism.

Property	TTT (ΛCDM)	ECF (Torsion String)	Discriminator
Origin of spin	External tidal field acting on proto-filament inertia tensor during quasi-linear collapse [196]	Intrinsic helical axial current $\vec{J}_5$ carried by the topological defect (1D, π_1) — independent of environment	Independent of local tidal field? Filaments in void environments (weak tidal field) should still spin in ECF, not in TTT
⋆Coherence scale	No characteristic scale predicted; spin amplitude set by local tidal field geometry, expected to decorrelate beyond $\sim 10\,\mathrm{Mpc}$	Characteristic reversal scale $\lambda_{\rm hel} \sim \xi_{\rm KZ} \sim 50$–$80\,\mathrm{Mpc}$, beyond which the helical sign flips	Stacked spin signal vs filament length: sign reversal at $L \sim 60\,\mathrm{Mpc}$ (DESI spectroscopic)
⋆Galaxy–filament spin correlation	Parallel preferred (galaxy inherits spin from same tidal field as filament) [196]	**Anti-parallel** preferred ($\vec{J}_5$ at node imparts opposite handedness to accreting halo, via conical deficit geometry)	Wang+2025 (MaNGA): anti-parallel observed for low-mass spirals [197] — **favours ECF**
Spin amplitude vs end-node mass	Stronger rotation for more massive end clusters (gravitational focusing of tidal torque) [194]	Stronger rotation for more massive end nodes (μ_{1D} higher for strings connecting Macro-Knots)	Same qualitative trend — not a discriminator
Viewing-angle dependence	Edge-on filaments show stronger signal (projection effect) [194]	Same (geometric projection, independent of mechanism)	Not a discriminator
Redshift evolution	Spin develops gradually during quasi-linear collapse ($z \lesssim 3$); absent at very high z	Spin imprinted at KZ freeze-out ($z \sim 10^{15}$, EW transition), present at *all* observable redshifts	High-z filament spin detection (Euclid deep survey, $z > 2$): signal at $z > 3$ favours ECF
Chiral magnetic field	No prediction (TTT is purely gravitational)	$\nabla \times \vec{B}_{\rm chiral} \propto \vec{J}_5$ along filament spine — proposed primordial origin for intergalactic magnetic fields (ECF hypothesis; Chern–Simons coupling [181])	Synchrotron polarisation gradient along filament (LOFAR, SKA)
Filament termination	Filaments can end anywhere (no topological constraint)	Filaments *must* terminate at 0D nodes ($\pi_1 \to \pi_2$ boundary condition) — no "dangling cables" (H4); this rests on the homotopy exact sequence, hence on the vacuum manifold, which is not yet established (PO-F2-8)	Zero filaments ending in voids (DESI+Euclid void catalogue)
Magnetic helicity conservation	Not applicable	$\mathcal{H} = \int \vec{A} \cdot \vec{B}\, d^3x$ is topological invariant, conserved under string reconnection	Correlation between magnetic helicity sign and filament spin sign (SKA Faraday rotation)

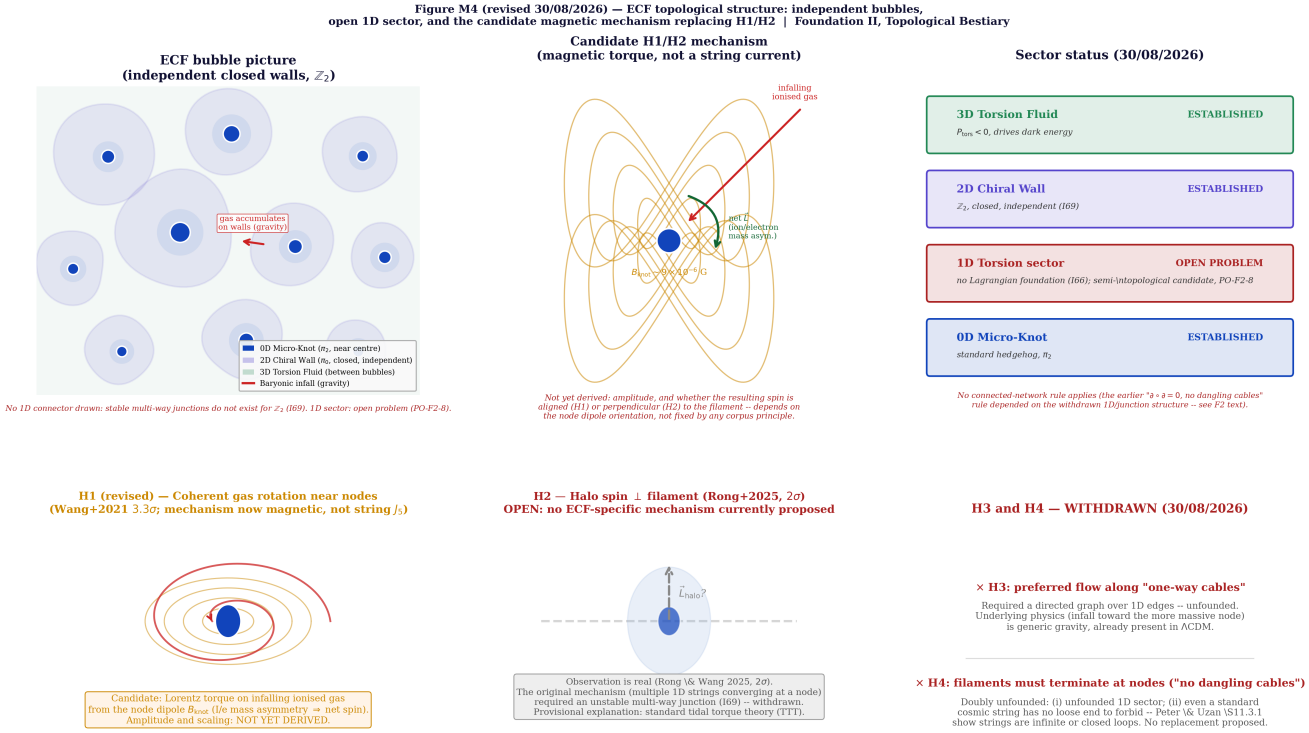

Figure 20: **ECF topological structure: independent bubbles, open 1D sector, and the candidate magnetic mechanism (Fig. M4, redrawn 30/08/2026).** *Top left:* The bubble picture: independent closed 2D Chiral Walls ($\mathbb{Z}_2$, indigo, irregular closed boundaries — not perfect circles, since topology fixes the class but not the shape), each generically containing one 0D Micro-Knot (π_2, blue) near its centre. No 1D edges are drawn: stable multi-way junctions do not exist for a $\mathbb{Z}_2$ wall sector (I69). Baryonic matter (red) is shown accumulating on the walls via ordinary gravitational infall, not flowing along a topological edge. *Top centre:* The candidate mechanism for H1/H2 — a dipolar magnetic field at each node ($B_{\rm knot} \sim 9 \times 10^{-6}$ G, already established, sourced by s_0) exerting a Lorentz-force torque on infalling ionised gas, with the ion/electron mass asymmetry giving a net coherent spin. Amplitude and direction are explicitly flagged as not yet derived. *Top right:* Status of each sector (revised): 0D and 3D retained; 2D conditional on the effective potential and the vacuum manifold (PO-F2-8); 1D an open problem (PO-F2-8), with no residual "no dangling cables" rule (that rule depended on the withdrawn junction structure). *Bottom left:* **Prediction H1, revised** — coherent gas rotation near nodes via the magnetic mechanism above. The underlying observation (Wang et al. [194]) remains real; only the proposed mechanism changed. *Bottom centre:* **Prediction H2, marked open** — the observation (Rong et al. [199], 2σ) remains real; no ECF-specific mechanism is currently proposed, and standard tidal torque theory is the provisional explanation. *Bottom right:* **Predictions H3 and H4, withdrawn** — shown with the reason for withdrawal stated in-figure; no replacement proposed for either. *Reproduced by:* `plot_F2_topological_network.py` (Foundation II repository, `B_Paper_Plots/`).

Dim.	ECF Structure	Cosmological Observation	Falsifiable Prediction (ECF Signature)
0D	**Micro-Knots** ($M_0 \simeq 3.74 \times 10^{24}\,\mathrm{kg}$)	**Rotation curves:** SPARC-175, $\tilde{\chi}^2_{\mathrm{med}} = 0.80$ vs 1.29 (NFW, same protocol v1); adopted pseudo-isothermal profile, two parameters per galaxy (PO-ECF-HALO) [6]. **Microlensing:** Short events (OGLE [174]) and Phoebe ($t_E \simeq 1\,\mathrm{h}$, $M \simeq 2.2 \times 10^{23}\,\mathrm{kg}$ [171], consistent at 1.4σ with ECF KZ tail).	**No radiation:** Geometric lenses emit neither infrared counterpart nor Hawking evaporation ($T_H \simeq 33\,\mathrm{mK} \ll T_{\mathrm{CMB}}$, $t_{\mathrm{evap}} \sim 10^{50}\,\mathrm{yr}$). **Kill-switch (Roman):** No excess of $t_E \sim 1$–$10\,\mathrm{h}$ events above the free-floating-planet baseline at 3σ falsifies ECF dark matter.
1D	**Torsion Strings** *(status open, PO-F2-8)*	**Observations unaffected:** Lyman-α imaging of intergalactic gas filaments [180]; primordial intergalactic magnetic fields. *The ECF explanation of these — a chiral current $\vec{J}_5$ carried by a quantised 1D string — was withdrawn on 30/08/2026 (see* Status of the 1D sector*): S^μ is a real axial vector and supplies no complex order parameter.*	*No kill-switch is claimed for this sector.* The LISA merger burst ($f_{\mathrm{peak}} \sim 50\,\mathrm{mHz}$) is **reattributed**: it follows from Macro-Knot mergers (§9.4, Cat. 2 (iii)), which require no quantised string. Its attribution to a 1D defect is withdrawn, and so is its ringdown tail $\tau_T \sim 10^6\,\mathrm{s}$ *(18/10/2026, Appendix D.10).*
2D	**Chiral Walls** (2D surface defects, $\pi_0 \neq \emptyset$; also called Domain Walls)	**Cosmic Voids:** Large-scale repulsive regions (Eridanus Supervoid, CMB Cold Spot).	**Discrete birefringence (LiteBIRD):** Anomalous CMB polarisation rotation *correlated* with optical traversal of these surfaces; $\beta_{\mathrm{ECF}} \neq 0$ (magnitude 0.35° calibrated) vs Planck PR4 $0.30^\circ \pm 0.11^\circ$ [169], ACT DR6 $0.215^\circ \pm 0.074^\circ$ [71].
3D	**Chiral Fluid** (Dark Energy volume)	**Dynamical Dark Energy:** DESI DR1–DR2 preference for $w \neq -1$ [9]. ECF calibrated point: $(w_0, w_a) = (-0.904, -0.153)$ inside 1σ DESI contour (SH0ES prior) [6].	**Geometric dilution:** $P_{\mathrm{tors}} \propto a^{-3}$; slight phantom crossing at high-z; axial sector ghost-free per literature (Proca form adopted), mechanism not derived (PO-F1-SIGMA0). **Kill-switch:** $\Pi_{\mathrm{GW}} < 0.02$ (95% CL) (LISA dipole channel or LISA–Taiji network) falsifies chiral GW sector.

Table 2: **The Topological Bestiary of the ECF Model.** Unified classification of the dark sector. 0D and 1D structures (Micro/Macro-Knots) address dark matter; 3D chiral fluid addresses dark energy. Predictions are quantitative and falsifiable. Diagonal DESI errors only; the full-covariance comparison is in *Foundation III — Cosmological Physics*.

- **1D Torsion Strings** (π_1, line defect) — **OPEN PROBLEM, PO-F2-8**: no Lagrangian foundation established (S^μ is real, supplying no phase for a genuine string); a semi-topological candidate exists but its stability is dynamical, not topological, and unresolved. No mass, current, or wake is quoted (see §6.2 for the full investigation).
- **2D Chiral Walls** (π_0, surface defect): separate σ_+/σ_- domains; Israel repulsion $\ddot{z} = (8\pi G/3)(\sigma_{2D}/ct)\, z$ carves void boundaries. Cannot merge with 0D or 1D.
- **3D Torsion Fluid** (no homotopy obstruction): global condensate; $P_{\rm tors} < 0$; is consistent with the calibrated $(w_0, w_a) = (-0.904, -0.153)$ (Foundation I; not derived *ab initio*, cf. PO-F2-4). Not a topological defect in the strict sense.

6.1 0D Micro-Knots: Topological Lensing and Galactic Halos

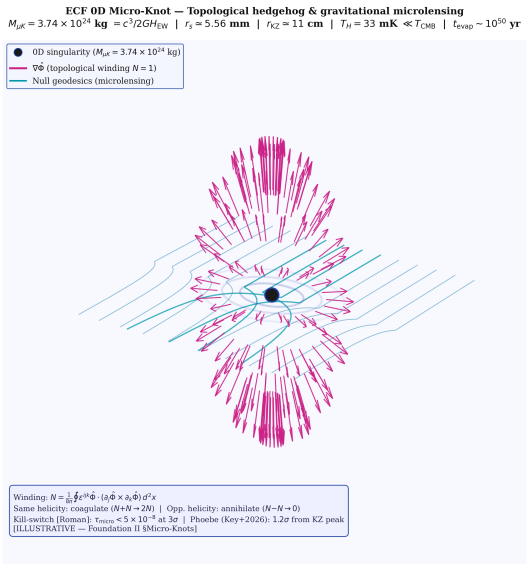

Figure 21: **ECF 0D Micro-Knot: topological hedgehog and gravitational microlensing.** Magenta vectors: order-parameter gradient $\nabla\hat{\Phi}$ wrapping around the singularity with winding number $N = 1$. Cyan curves: null-geodesics deflected by the geometric mass $M_{\rm MK} = 3.74 \times 10^{24}$ kg. The Hawking temperature $T_H \simeq 33$ mK is three orders of magnitude below $T_{\rm CMB}$; evaporation timescale $\sim 10^{50}$ yr. *[Illustrative — see §4.3 for quantitative derivation.]*

Reproduced by:
`Scripts_F2/plot_F2_zoologie_topologique.py`.

When the chiral torsion fluid froze at the electroweak transition, Φ vanished at isolated points forming hedgehog configurations (Fig. 21). The resulting effective mass $M_{\rm MK} = c^3/2GH_{\rm EW} \simeq 3.74 \times 10^{24}$ kg curves spacetime identically to a Schwarzschild mass. Because Micro-Knots are composed strictly of knotted spacetime, they interact solely via gravitational lensing.

Quantitative support: the ECF two-parameter fit (v_∞, R_s) to the SPARC rotation curves — 171 of the 175 galaxies, the four others having too few points for a reduced χ^2 to discriminate anything ($\nu \geq 3$ cut, see *Sample definition*) — yields $\tilde{\chi}^2_{\rm med} = 0.80$ for the adopted pseudo-isothermal profile, versus 1.29 for an NFW profile refitted under the same protocol (two parameters per galaxy for both) [6]; the profile is not yet derived from the torsion (PO-ECF-HALO), so this is a goodness-of-fit comparison of halo profiles, not a torsion-specific test. *Correction of record (22/09/2026): earlier versions compared this median with the value* 1.55 *of Li et al. (2020), an external value obtained under a different convention (mass-to-light ratio, distance and inclination free); the homogeneous internal comparison is* 0.80 *versus* 1.29. At microlensing scales, the AMPM II survey detection of Phoebe ($t_E \simeq 1$ h, $M \simeq 2.2 \times 10^{23}$ kg, halo probability $10^5\times$ stellar) provides independent evidence consistent with the ECF KZ distribution tail at 1.4σ [171].

Two parameter counts, two perimeters. Foundation I reports $k_{\rm ECF} = 2$ free parameters ($F_{\rm ion}$, $\gamma_{\rm spin}$; historical count, retained although $\gamma_{\rm spin} = 0.3116$ is withdrawn as canonical calibration; corrected 03/09/2026 – H_0 depends strictly on $F_{\rm ion}$ and is not a second parameter, and $\tau_{\rm tor}$, though a real input to r_s, is not varied in this specific test) against $k_{\Lambda\rm CDM} = 6$, while the rotation-curve comparison above is quoted with *zero free dark-matter parameters*. The two statements are consistent, and it is worth saying why rather than leaving a reader to reconcile them.

The two parameters of Foundation I calibrate the *expansion history*: they are fixed once, globally, against the CMB, BAO, H_0 and S_8 combination. They are not refitted here. The claim in this section concerns a different axis — the number of parameters adjusted *per galaxy*. An NFW fit introduces two (a concentration and a scale radius) for each of the 171 galaxies retained; the framework used here introduces none, the rotation curves following from the same globally calibrated numbers. That is the

Table 3: **ECF Topological Defect Hierarchy — Referee Summary.** All classes share the same coupling constant $\kappa^2 = 2.0766 \times 10^{-43}\,\mathrm{m\,J^{-1}}$; Micro-Knots form by Kibble–Zurek crystallisation at $T_c \simeq 159\,\mathrm{GeV}$, whereas the Macro-Knot scale is set by the horizon mass at the Great Annihilation (§4.3.1). *(Corrected 15/10/2026: Micro-Knot r_s was given as* 1.5 *mm (superseded mass) and an undefined* $r_{\mathrm{KZ}} \sim 3$ *cm was listed; the Macro-Knot was labelled "coagulation", with an LVK O5 kill-switch that belongs to the Knot–star case, whereas its signal is the* ~ 50 *mHz LISA burst. Corrected 18/10/2026: that burst no longer carries a kill-switch, the Macro-Knot merger rate being not computed, and the torsion tail* $\tau_T \sim 10^6$ *s is withdrawn, Appendix D.10.)* Columns: homotopy class, geometry, formation mechanism, canonical scales, gravitational signature, primary observable, falsification criterion, and open problems. Diagonal errors only; the full-covariance comparison is in *Foundation III — Cosmological Physics. [ILLUSTRATIVE]* applies to figures only, not to numerical values.

Dim.	Name	Homotopy / Geometry	Canonical scales	Gravitational signature	Primary observable / Kill-switch
0D *Micro-Knot*	Micro-Knot $M_{\mathrm{MK}} \simeq 3.74\times10^{24}\,\mathrm{kg}$	$\pi_2(\mathcal{M})\neq0$ Hedgehog Winding $N=1$	$r_s \simeq 5.56\,\mathrm{mm}$ $T_H \simeq 33\,\mathrm{mK}$ $t_{\mathrm{evap}} \sim 10^{50}\,\mathrm{yr}$	Schwarzschild lensing; no radiation; point lens	**Roman** microlensing: $\tau\sim5\times10^{-8}$, $t_E\simeq6.3\,\mathrm{h}$ ($\lambda=36$) *Kill-switch:* no excess in 1–10 h window Phoebe (1.4σ, Key+2026)
0D *Macro-Knot*	Macro-Knot (horizon-mass channel) $M\sim10^5\,M_\odot$	$\pi_2(\mathcal{M})\neq0$ Same as Micro-Knot; mass kinematic (not dimensional)	$r_s \approx 3\times10^5\,\mathrm{km}$ $R_{\max}\sim0.6\,R_\odot$ $\Omega_{\mathrm{LT}}\sim0.2$–$0.5\,\mathrm{rad/s}$ Period $\sim$ 10–40 s	Schwarzschild + Lense-Thirring; galactic seed; SPARC halo	**LISA** merger burst: $f\sim50\,\mathrm{mHz}$, *Conditional:* merger rate not computed; no distinctive tail; no kill-switch
1D *Torsion String*	**OPEN PROBLEM** (PO-F2-8) no mass quoted	No Lagrangian foundation (S^μ real, no phase); semi-topological candidate only	Not derived — depends on unresolved m_T (see §6.2)	None quoted: mechanism not established	*None quoted:* withdrawn pending resolution of the open problem
2D *Chiral Wall*	Chiral Wall (Domain Wall) $\sigma_{2D}\simeq6.8\times10^{11}\,\mathrm{kg/m^2}$	$\pi_0(\mathcal{M})\neq0$ Flat membrane; σ_+/σ_- boundary	$\delta_w\sim10^{-18}\,\mathrm{m}$ $R_w\sim\xi_{\mathrm{KZ}}\sim60\,\mathrm{Mpc}$ $A_w\sim(60\,\mathrm{Mpc})^2$ Void $R\gtrsim60\,\mathrm{Mpc}$	Israel repulsion: $\ddot{z}=(8\pi G/3)(\sigma_{2D}/ct)z$; expels baryons; carves voids	**LiteBIRD** birefringence: $\beta_{\mathrm{ECF}}\neq0$ ($\lvert\beta\rvert\approx0.35°$ calibr., achromatic) Planck PR4: $0.30°\pm0.11°$ *Kill-switch:* $\beta<0.1°$ at 3σ
3D *Torsion Fluid*	Torsion Fluid (Chiral condensate) *Not a topological defect*	No homotopy obstruction; volume-filling condensate	$P_{\mathrm{tors}}=-\frac{1}{4}\kappa\langle S^2\rangle$ (normalisation not established, PO-ECF-SPIN-NORM) $L_{\mathrm{coh}}\sim4.4\,\mathrm{Gpc}$ (Hubble) $f_{\mathrm{fluid}}\sim0.73$	Isotropic $P<0$; dark energy; void interior amplification	**DESI DR2**: $(w_0,w_a)=(-0.904,-0.153)$ 3.1σ vs ΛCDM (BAO+CMB) *Kill-switch:* $w=-1$ at 5σ (DESI DR3)

Table 3: **ECF Topological Defect Hierarchy (continued)**: interaction rules, ECF-specific properties and open problems.

Dim.	Interaction rules	ECF-specific property	Open problem
0D Micro	Same helicity: coagulate ($N+N\to 2N$); opp. helicity: annihilate ($N-N\to 0$)	No Hawking evaporation ($T_H \ll T_{\rm CMB}$); geometric mass $M = c^3/2GH_{\rm EW}$ (0 free param.)	PO-F2-1: origin of $\Delta N>0$
0D Macro	Same as Micro-Knot	SPARC, 171/175 gal. ($\nu \geq 3$ cut, see *Sample definition*): $\tilde{\chi}^2_{\rm med} = 0.80$ (adopted pseudo-isothermal profile; NFW 1.29 under the same protocol); 2 param./galaxy; profile not derived (PO-ECF-HALO)	Merger transfer function → F3
1D	Not derived (no Lagrangian foundation)	None established: the helical $\vec{J}_5$ string and its $\vec{B}_{\rm chiral}$ are withdrawn (Status of the 1D sector; 19/10/2026)	PO-F2-8: 1D sector open; PO-F2-2: pitch $\lambda_{\rm hel}$
2D	Expands at $\sim c$; opp.-chirality pair: partial annihilation → GW + 3D fluid	Achromatic birefringence $\beta \neq 0$ ($\lvert\beta\rvert \approx 0.35°$ calibr.); sharp void boundary ($\delta\rho/\rho \gtrsim 0.8$)	PO-F2-3: dilution rate; wall GW background
3D	Locally depleted in halos; locally dominant in voids	Geometric back-reaction DM↔DE; global chiral helicity $q=\pm 1$	PO-F2-4: $f_{\rm fluid}$; $\langle\delta S^2\rangle(z)$ calibrated not derived (→ fixes w_0, w_a)

sense in which the dark-matter sector carries no free parameter, and it is the comparison the $\tilde{\chi}^2_{\rm med} = 0.80$ versus 1.55 is meant to support.

Stating it this way also fixes what the comparison does *not* claim. Two calibrated numbers are two calibrated numbers, and the BIC penalty in Foundation I counts them; the present result is not that the framework has no parameters, but that it does not acquire new ones as galaxies are added. A referee is entitled to ask which of the two counts applies to a given statement, and the answer should not depend on which paper is open.

6.2 1D Torsion Strings — investigated, not established (PO-F2-8)

No figure or quantitative prediction is given for a 1D sector in this document, by design. An earlier version of this section presented a conical deficit angle, a velocity-kick wake, a helical current $\vec{J}_5$, and a specific LISA burst signature ($f \sim 50\,$mHz, ringdown $\tau_T \sim 10^6\,$s) as if these were established consequences of the framework. **They are not, and presenting them with that degree of quantitative confidence was a mistake, now corrected.**

The investigation was real and is summarised here, not hidden:

- S^μ, the axial torsion vector, is *real* — it has an amplitude but no internal phase. A genuine cosmic string requires a *complex* order parameter (abelian-Higgs mechanism); S^μ cannot supply one without an additional field.
- Two candidate completions were tested and failed: a trace-vector companion field (fails — on a homogeneous background the two vectors are collinear, defining no angle); and the residual large-scale anisotropy ("Axis of Evil", $\varepsilon \lesssim 10^{-7}$, Eq. 90) considered as a possible symmetry-breaking seed (fails — it fixes a value for the angle, not a free phase, so it cannot supply a genuine $U(1)$).
- A semi-topological candidate (coexistence of the global torsion sector and the local electroweak gauge sector; Peter & Uzan, *Cosmologie primordiale*, §11.5.4) remains structurally possible, but its stability is *dynamical*, not topological, and depends on the torsion mass m_T, which is itself unresolved: its value is not fixed (only $m_T^2 s_0^2$ is), and whether the m_T of the dark-energy degeneracy is the pole mass that sets the halo-scale range is not established (see the discussion of the torsion mass parameters above; *corrected 15/10/2026*: the "three mutually incompatible constraints" stated here relied on an unsourced s_0 window).

This sector is retained as PO-F2-8, an explicitly open problem, not as an established defect class. No mass, tension, current, or observational signature is quoted for it, because none is currently derived. The corresponding predictions previously labelled H1, H2, H3, H4 have been withdrawn or replaced (§6): the real observations they were built to explain (Wang et al. [194], Rong et al. [199]) remain, but the ECF mechanism proposed

for them is no longer the 1D sector.

6.3 2D Chiral Walls (Domain Walls): Kinematics of Cosmic Voids

The wall geometry and its repulsive action on voids are shown in Fig. 22.

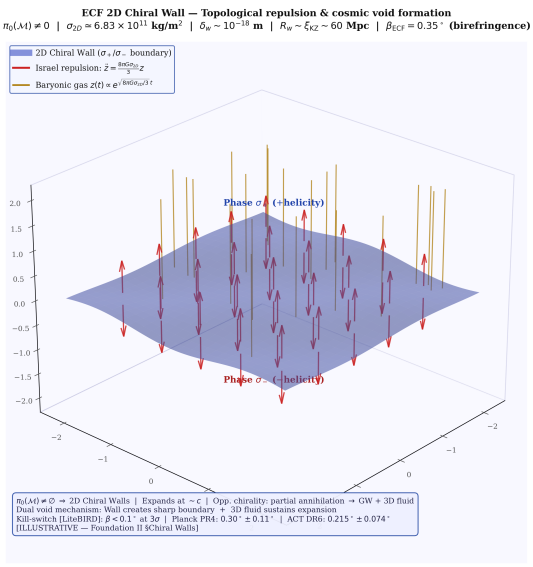

Figure 22: **ECF 2D Chiral Wall: topological repulsion and cosmic void formation.** Blue surface: boundary between σ_+ and σ_- chiral phases ($\pi_0(\mathcal{M}) \neq 0$). Surface energy density: $\sigma_{2D} \simeq 6.8 \times 10^{11}\,\mathrm{kg\,m^{-2}}$. Wall thickness: $\delta_w \sim 10^{-18}\,\mathrm{m}$ (effectively infinitely thin on astrophysical scales). Initial curvature radius: $R_w \sim \xi_{\rm KZ} \sim 60\,\mathrm{Mpc}$. Red arrows: Israel (1966) repulsive perpendicular acceleration $\ddot{z} = (8\pi G/3)(\sigma_{2D}/ct)\,z$. Yellow trajectories: baryonic gas expelled along z, carving the sharp void boundary. Birefringence: each wall traversal rotates CMB polarisation by $\beta_{\rm ECF} \neq 0$ (achromatic; magnitude 0.35° calibrated; kill-switch: LiteBIRD, $\beta < 0.1°$). *[Illustrative.]*

Reproduced by:
`Scripts_F2/plot_F2_zoologie_topologique.py`.

Domain walls emerge where the chiral vacuum manifold is disconnected ($\pi_0(\mathcal{M}) \neq 0$). The repulsive acceleration $\ddot{z} = (8\pi G/3)(\sigma_{2D}/ct)\,z$ expels baryonic gas, carving cosmic voids. The birefringence signal (non-zero; calibrated magnitude $\beta \approx 0.35°$) is correlated with traversal of these surfaces and provides a distinctive kill-switch [6].

Formation conditions. Chiral Walls form at the electroweak phase transition ($T_c \simeq 159\,\mathrm{GeV}$) when adjacent Kibble–Zurek domains crystallise with *opposite* chiral helicity (σ_+ vs σ_-, i.e. $\pi_0(\mathcal{M}) \neq 0$). The boundary between them is a 2D surface defect — a Chiral Wall. Unlike 0D and 1D defects, which form within a domain, Chiral Walls form *between* domains: one per pair of adjacent KZ domains with opposite chirality. Their surface density at formation is therefore set by the KZ correlation length $\xi_{\rm KZ} \sim 50$–$80\,\mathrm{Mpc}$: one wall face per $\xi_{\rm KZ}^2$ of comoving area.

Geometric properties — thickness, curvature, area. Three length scales characterise a Chiral Wall:
(i) Wall thickness δ_w. The transition layer between σ_+ and σ_- phases has a thickness set by the electroweak coherence length, analogous to the string core:

$$\delta_w \sim \frac{\hbar}{m_T c} \sim 10^{-18}\ \mathrm{m}. \tag{61}$$

The wall is therefore effectively infinitely thin on all astrophysical scales — it is a mathematical 2-surface for all observational purposes.
(ii) Curvature radius R_w. Chiral Walls are not flat planes. Primordial walls have an initial curvature radius set by the KZ correlation length: $R_w^{\rm init} \sim \xi_{\rm KZ} \sim 50$–$80\,\mathrm{Mpc}$. Under the Israel (1966) repulsive dynamics, the wall expands and flattens over cosmic time, so that observed supervoid boundaries ($R \sim 150$–$300\,\mathrm{Mpc}$) correspond to walls that have expanded and straightened from their initial $\xi_{\rm KZ}$ curvature.
(iii) Wall area A_w. A single primordial wall segment spans a comoving area $A_w \sim \xi_{\rm KZ}^2 \sim (60\ \mathrm{Mpc})^2 \sim 3.6 \times 10^3\ \mathrm{Mpc}^2$. The Eridanus Supervoid ($R \sim 200\,\mathrm{Mpc}$) and the Big Ring (diameter $\approx 400\,\mathrm{Mpc}$ [88]) suggest that some observed structures may correspond to single Chiral Wall segments or their intersections.

Surface mass density and total mass. The surface energy density (tension) of a Chiral Wall is:

$$\sigma_{2D} \sim \frac{\eta_{\rm EW}^3}{(\hbar c)^2 c^2} \simeq 6.8 \times 10^{11}\ \mathrm{kg\,m^{-2}}, \tag{62}$$

where $\eta_{\rm EW} = 246.22\,\mathrm{GeV}$ is the electroweak VEV. *Correction of 27–28/08/2026:* this expression previously read $\eta^4/\hbar c$, which is not dimensionally a surface mass density, and gave $10^{49}\,\mathrm{kg\,m^{-2}}$. The tension of a domain wall scales as η^3 in natural units

[68]; the VEV was also quoted as 159 GeV, which is not the electroweak VEV (246.22 GeV) and appears to conflate it with the transition temperature $T_c = 159$ GeV. The total *intrinsic* mass of a wall of area A_w is:

$$\begin{aligned} M_{\rm wall} &= \sigma_{2D} \cdot A_w \\ &\simeq 6.8 \times 10^{11}\ {\rm kg\,m^{-2}} \times (2 \times 10^{24}\ {\rm m})^2 \\ &\sim 3 \times 10^{60}\ {\rm kg}, \end{aligned} \tag{63}$$

an astronomically large number that would immediately over-close the universe if walls were stable and static. *Corrected 14/10/2026* (was 10^{97} kg, from the superseded σ_{2D}). This is the *classical domain wall problem* [178], which applies to any π_0 defect in cosmology.

ECF resolution of the domain wall problem. The classical domain wall problem is resolved in the ECF by two complementary mechanisms:

(i) Israel repulsion — self-defeating geometry. The acceleration $\ddot{z} = (8\pi G/3)(\sigma_{2D}/ct)\, z$ causes each Chiral Wall to expand *outward* at nearly the speed of light after formation. As the wall area grows as $A_w \propto t^2$, its energy density per unit comoving volume $\rho_{\rm wall} = \sigma_{2D}/(ct)$ dilutes as a^{-1} — *slower* than radiation (a^{-4}) and *slower* than matter (a^{-3}), which is precisely the classical domain wall problem: left unchecked, walls become relatively more important over cosmic time and would eventually dominate the energy budget. However, in the ECF the walls expand into their own void interior, not into a matter-filled background: the baryonic gas is expelled by the repulsion, so the wall never accumulates the large gravitational mass implied by eq. (63).

Range of validity. Eq. (59) is the weak-field, Newtonian-limit form of the Israel junction condition: it gives the proper acceleration of a test particle near the wall, where z is small. Integrated naively it grows without bound — at $z \sim 60$ Mpc it would return $\dot{z} \gg c$ (from $\sim 10^4 c$ today to $\sim 10^{18} c$ at the electroweak epoch, the rate depending on t through σ_{2D}/ct), which is not a prediction but a failure of the approximation. *Correction (14/10/2026):* the value $\dot{z} \sim 10^{17} c$ quoted earlier came from the superseded form $\ddot{z} = (8\pi G\sigma_{2D}/3)\, z$. In the full relativistic treatment the expansion velocity saturates at c: the wall accelerates rapidly to relativistic speeds and then coasts, as domain walls do generally. We quote the linear form because it is what the junction condition supplies, and we flag its limit rather than let a reader integrate it.

(ii) Chiral cancellation — opposite-helicity annihilation. A σ_+ wall and a σ_- wall at the boundary of the same void are walls of *opposite chirality* — they attract and can partially annihilate, releasing their stored torsion energy into gravitational waves and the 3D chiral fluid (Torsion Fluid). This mechanism prevents the long-term accumulation of wall energy density and connects to the 3D fluid condensate.

ECF-specific property — birefringence signature. A photon traversing a Chiral Wall crosses a region where the chiral phase changes by $\Delta\theta = \pi$. Through the Chern–Simons coupling, this phase jump rotates the CMB polarisation plane by a nonzero angle (calibrated magnitude $\beta_{\rm ECF} \approx 0.35°$), independently of frequency (achromatic). This is the origin of the isotropic birefringence signal measured by Planck PR4 ($0.30° \pm 0.11°$ [169]) and ACT DR6 ($0.215° \pm 0.074°$) [71], and constitutes the primary falsification channel for the 2D sector of the ECF dark sector.

Magnetic field of a Chiral Wall. The Chern–Simons coupling that produces birefringence also generates a *surface magnetic field* localised within the wall thickness $\delta_w \sim 10^{-18}$ m. Unlike the 0D dipolar field (falls off as r^{-3}) and the 1D solenoidal field (helical along the string axis), the 2D wall field is *tangential and uniform* across the wall surface:

$$\vec{B}_{\rm wall} \sim \frac{\kappa^2\, s_0\, \sigma_{2D}}{\delta_w}\, \hat{n} \times \hat{z}, \tag{64}$$

where $\hat{n}$ is the wall normal and $\hat{z}$ is the chiral symmetry-breaking axis. At cosmological scales, this field has diluted to $B^{(0)}_{\rm wall} \sim 10^{-10}$–$10^{-6}$ G along the void boundaries — consistent with the observed magnetic fields at the edges of cosmic filaments [89]. The wall magnetic field is therefore a candidate geometric origin for the coherent magnetic structures observed at void boundaries by LOFAR and MeerKAT. A key distinction from astrophysical dynamo models: the ECF wall field has a *preferred chirality* ($\vec{B}_{\rm wall}$ flips sign across the wall, correlating with the σ_+/σ_- phase change), producing a measurable Faraday rotation gradient across void edges — a falsifiable prediction for SKA.

Electric field at a Chiral Wall. Unlike the 0D/1D/3D sectors where any electric field is secondary (induced by gas dynamics), the 2D Chiral Wall has a *structural* electric field component arising from the Israel–Chern-Simons junction conditions. The discontinuity of the chiral phase $\Delta\theta = \pi$ at the wall surface implies a jump in the normal component of $\vec{E}$:

$$[\vec{E}_\perp]_{\rm wall} \;\propto\; \kappa^2\, s_0\, \sigma_{2D}\, \hat{n}, \tag{65}$$

analogous to the magneto-electric effect in topological insulators. This is a *geometric* electric field — requiring no free charge, existing in vacuum at the wall surface. After cosmological dilution: $E^{(0)}_{\rm wall} \sim 10^{-8}$–$10^{-4}\,{\rm V\,m^{-1}}$ at void boundaries. In principle detectable through cross-correlation of CMB polarisation anomalies with void catalogues from Euclid and DESI, but below current observational thresholds.

Open problem PO-F2-3 — Wall network evolution and the classical problem. The quantitative resolution of the domain wall problem in the ECF is not derived from first principles in the present work. In particular: (i) what is the precise dilution rate of Chiral Wall energy density in the ECF metric, accounting for the Israel repulsion *and* the Torsion Fluid back-reaction? (ii) does the opposite-chirality annihilation mechanism produce a detectable stochastic gravitational wave background at frequencies accessible to LISA or ET? (iii) what is the equilibrium void size distribution predicted by the full ECF wall network, beyond the schematic VSF of Eq. (34)? These questions are identified as potential referee objections; they are open problems, not treated in the present corpus *(corrected 20/10/2026: previously deferred to Foundation III — Cosmological Physics, which does not treat them).* **Cross-reference (17/07/2026):** this is very likely the same underlying open problem as **PO-F3-XI** in Foundation III (the microscopic origin of the void correlation length $\xi_{\rm KZ} \sim 60$ Mpc), viewed from the opposite end — PO-F2-3 asks for the wall network's dilution/annihilation dynamics; PO-F3-XI asks what comoving correlation length that same dynamics would predict today. A numerical coincidence noted in Foundation III (the comoving causal horizon, computed on the ECF background, reaches 60 Mpc at $z \approx 5145$ — at the lower edge of the standard HeIII→HeII recombination range, $z \sim 5000$–8000; the figure $z \approx 6430$ quoted previously was obtained on a ΛCDM background and is corrected in Foundation III) is consistent with — but not derived from — the self-accelerating wall dynamics $\ddot{z} = (8\pi G/3)(\sigma_{2D}/ct)\, z$ described above. Whether the chiral annihilation mechanism (ii) naturally halts wall growth at this specific epoch, rather than earlier or later, is not established and would resolve both open problems simultaneously if derived.

6.4 The 3D Global Defect: Dynamical Dark Energy

The geometry of this level is illustrated in Fig. 23.

The residual chiral fluid exerts an isotropic torsion pressure $P_{\rm tors} = -\frac{1}{4}\kappa\langle S^2\rangle_{\rm vol}$ (with $\kappa = 8\pi G/c^4 = 2.0766 \times 10^{-43}\,{\rm m\,J^{-1}}$; this value is that of κ, not of κ^2), written here as diluting as $a^{-3(1+w)}$ (see the dilution-law caveat in §6.4 below), and parameterizing the dynamical dark energy reported by DESI [9]; the normalisation of $\langle S^2\rangle_{\rm vol}$ is not established (PO-ECF-SPIN-NORM).

Nature — condensat géométrique, pas un défaut topologique. Unlike the 0D, 1D, and 2D sectors, the 3D Torsion Fluid is *not* a topological defect in the strict sense: it carries no homotopy obstruction ($\pi_k(\mathcal{M}) = 0$ for all $k \geq 3$ in the ECF vacuum manifold). It is instead a *geometric condensate* — the volume-filling fraction of the primordial spin-torsion fluid that did not localise into point, line, or surface defects during the Kibble–Zurek crystallisation at the electroweak scale. Equivalently, it is the uncrystallised background torsion from which the 0D, 1D, and 2D defects precipitated, retaining the global isotropy and homogeneity of the pre-crystallisation state.

Formation — residue of the Kibble-Zurek process. At the electroweak phase transition ($T_c \simeq 159\,{\rm GeV}$), the fraction of the torsion field that crystallised into localised defects is determined by the Kibble–Zurek nucleation rate. The complementary fraction — of order $f_{\rm fluid} = 1 - f_{\rm defects} \sim 1 - \Omega_{\rm DM}/\Omega_{\rm tot} \sim 0.73$ — remained uncrystallised and forms the 3D Torsion Fluid. This fraction is not derived in the present work (PO-F2-4): it is expected to be set by the same electroweak VEV $\eta_{\rm EW}$ and KZ correlation length $\xi_{\rm KZ}$ that set the defect densities, through the partition of the total torsion energy at freeze-out, but this partition has not been computed.

Volume energy density and equation of state. The spatially averaged spin-squared density $\langle S^2\rangle_{\rm vol}$

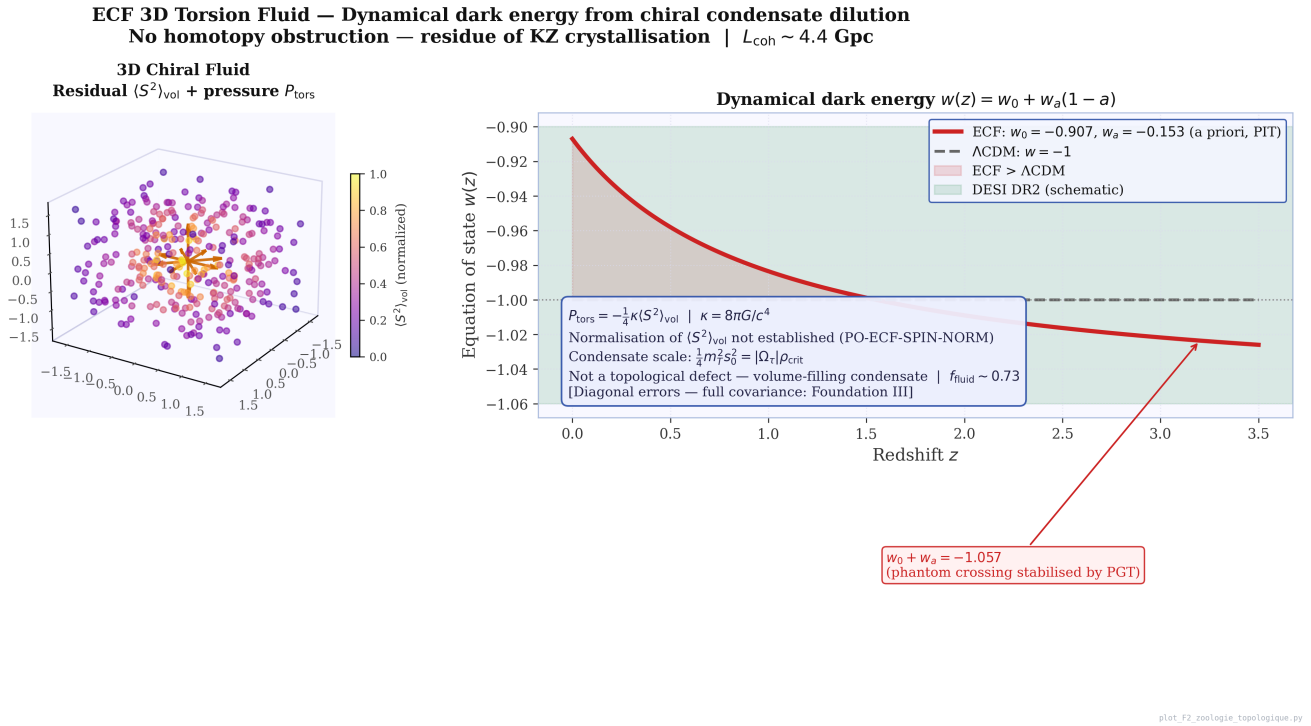

Figure 23: **ECF 3D Torsion Fluid: dynamical dark energy from chiral condensate dilution.** *Left*: residual torsion field $\langle S^2\rangle_{\rm vol}$ permeating the global volume with isotropic pressure $P_{\rm tors} = -\frac{1}{4}\kappa\langle S^2\rangle_{\rm vol}$ (symbolic; normalisation not established, PO-ECF-SPIN-NORM). Not a topological defect (no homotopy obstruction) — volume-filling condensate, $L_{\rm coh} \sim 4.4\,$Gpc. *Right*: equation-of-state evolution $w(z) = w_0 + w_a(1-a)$ with $(w_0, w_a) = (-0.904, -0.153)$ calibrated in Foundation I (w_a through $\xi = 1.645$, calibrated on the DESI preference used by the ECF; not independent of DESI) (contours here use diagonal errors; the full-covariance comparison is in Foundation III). Slight phantom crossing at high-z ($w_0 + w_a = -1.057$); the axial sector is ghost-free according to the literature for the adopted Proca form (not established here), but the mechanism licensing the crossing is not derived (PO-F1-SIGMA0). DESI DR2 schematic band shown for comparison [9]. Kill-switch: $w = -1$ confirmed at 5σ by DESI DR3. *[Illustrative.]*

Reproduced by: `Scripts_F2/plot_F2_zoologie_topologique.py`.

is the fundamental quantity; the torsion pressure is written in symbolic form:

$$P^{(0)}_{\rm tors} = -\tfrac{1}{4}\kappa\,\langle S^2\rangle^{(0)}_{\rm vol}, \qquad \kappa = 8\pi G/c^4. \tag{66}$$

No numerical value of $\langle S^2\rangle^{(0)}_{\rm vol}$ or of $P^{(0)}_{\rm tors}$ is retained. The values quoted in earlier versions had no producing calculation, do not appear in Foundation I, and were dimensionally inconsistent (with κ in $\mathrm{m\,J^{-1}}$ and $\langle S^2\rangle$ in $\mathrm{kg^2\,m\,s^{-2}}$, the product $\kappa\langle S^2\rangle$ is not a pressure); they are withdrawn, together with the comparison with the Planck pressure. The normalisation of $\langle S^2\rangle_{\rm vol}$ (units, factors of c and value) remains open (PO-ECF-SPIN-NORM). The consistent normalisation of the condensate used in this work is $\frac{1}{4}m_T^2 s_0^2 = |\Omega_\tau|\rho_{\rm crit}$ (Eq. (21)), which sets the dark-energy scale $\rho_\Lambda c^2 \approx 6\times10^{-10}\ \mathrm{J\,m^{-3}}$; the relation between $\langle S^2\rangle_{\rm vol}$ and s_0 is not written in this work. The dilution law adopted in this section is written as follows. It is not consistent across the corpus for the same quantity (a^{-3} in §5.3, a^{-6} in the Foundation III BBN estimate), and whether this condensate and the stiff spin branch of Foundation I are the same degree of freedom is not settled; this inconsistency is not resolved here (PO-ECF-BBN-SPIN):

$$\langle S^2\rangle_{\rm vol}(a) \propto a^{-3(1+w_{\rm eff})}, \tag{67}$$
$$w_{\rm eff}(z) = w_0 + w_a(1-a) = -0.904 - 0.153(1-a)$$

At high z, $w_{\rm eff} \to w_0 + w_a = -1.057 < -1$: the trajectory crosses the phantom divide (at $z = 1.684$). For the adopted Proca form, the absence of ghosts in the axial sector is invoked from the literature [166] (Appendix F establishes only $\lambda_T > 0$) — a necessary condition — but the mechanism licensing the crossing is not derived here (PO-F1-SIGMA0).

Spatial structure — homogeneous but not uniform. The 3D Torsion Fluid is globally homogeneous and isotropic (satisfying the cosmological principle by construction), but locally modulated by the presence of the localised defects. Inside galactic halos (near 0D Micro-Knots), the fluid is depleted: the localised torsion of the defect dominates and $\langle S^2\rangle_{\rm vol} \to 0$ in the halo interior. Inside cosmic voids (near 2D Chiral Walls), the fluid is enhanced: $\Omega_m \to 0$ allows the Torsion Fluid to dominate locally, driving void expansion (§4.2). This spatial modulation of the 3D sector by the localised defects constitutes the *geometric back-reaction* of dark matter on dark energy within the ECF framework.

No spatial extent — a volume-filling field. Unlike the 0D, 1D, and 2D defects, the 3D Torsion Fluid has no characteristic length scale of its own: it pervades the entire comoving volume simultaneously. Its coherence length is the Hubble radius $c/H_0 \sim 4.4\,\mathrm{Gpc}$, and its "size" is the size of the observable universe. There is no natural notion of thickness, core, or pitch for this sector — it is a genuine 3+1-dimensional field configuration.

Chiral helicity of the condensate. The 3D fluid carries a global helicity $q = \pm1$ inherited from the primordial chiral symmetry breaking at the electroweak scale. This global helicity is *conjectured* (not derived) to be related to the reported CMB large-angle anisotropy ("Axis of Evil") and to the reported Webb $\Delta\alpha/\alpha$ dipole (§3.5); both links are conditional. The fluid is therefore *chiral* in the cosmological sense: its left-handed and right-handed spin components are not equal, with the asymmetry parametrised by the axial condensate s_0. The deeper question of how this initial helicity sign was selected — and its possible connection to the angular momentum of the object from which our universe originated — is treated in Foundation III [7].

Cosmological magnetic field from the 3D condensate. The chiral helicity of the Torsion Fluid generates a *cosmological-scale magnetic field* via the Chern–Simons coupling [181]:

$$\nabla\times\vec{B}_{\rm fluid} \;\propto\; \kappa^2\, s_0\, \vec{J}_5^{\,(\rm fluid)}, \tag{68}$$

where $\vec{J}_5^{\,(\rm fluid)}$ is the spatially averaged axial current density of the condensate. Unlike the localised fields of the 0D (dipolar, r^{-3}), 1D (solenoidal, helical), and 2D (tangential, at void walls) defects, the 3D fluid field is *isotropic and space-filling*: it provides a *stochastic cosmological magnetic background* (SCMB) of amplitude:

$$B^{(0)}_{\rm fluid} \;\sim\; \kappa^2\, s_0\, \langle S^2\rangle^{1/2}_{\rm vol} \cdot \left(\frac{a_{\rm EW}}{a_0}\right)^2 \;\sim\; 10^{-17}\text{–}10^{-12}\ \mathrm{G}, \tag{69}$$

where the dilution factor $(a_{\rm EW}/a_0)^2$ accounts for the cosmological expansion since $T_c \simeq 159\,\mathrm{GeV}$. For a frozen-in field ($B \propto a^{-2}$) with entropy conservation, $(a_{\rm EW}/a_0)^2 = \left[(g_{s,0}/g_s(T_c))^{1/3}\,T_0/T_c\right]^2 \simeq 2.4\times10^{-31}$ ($g_{s,0} = 3.909$, $g_s(T_c) = 106.75$). The quoted range was obtained with an undocumented factor $\sim 10^{-28}$, about 400 times larger; if it scales with the dilution, the computed factor brings it

to $\sim 10^{-20}$–10^{-15} G. The amplitude is therefore provisional, and no comparison is drawn with the observational constraints on the cosmological magnetic field ($B_{1\,\mathrm{Mpc}} < 4.4\,\mathrm{nG}$ at 95% from CMB anisotropies [212]; $B_{\mathrm{IGM}} \gtrsim 3 \times 10^{-16}\,\mathrm{G}$ from blazar cascade observations [210, 211]) until it is recomputed. *(Corrected 15/10/2026: earlier versions stated that this amplitude falls within those constraints.) (Corrected 22/10/2026: bounds now taken from [212] and [210]; the CMB bound was given as $< 10^{-9}$ G.)*

Key distinction from dynamo models: the ECF 3D fluid field has a *global preferred helicity* $\mathcal{H} = \int \vec{A} \cdot \vec{B}\, d^3x \neq 0$, a topological invariant that cannot be erased by turbulence. This distinguishes it from primordial magnetic fields generated by inflation or phase transitions (which have $\mathcal{H} = 0$ on average). The detection of a non-zero magnetic helicity in the intergalactic medium at scales $> 100\,\mathrm{Mpc}$ by SKA or the Roman Space Telescope would constitute a distinctive ECF prediction, independent of the rotation-curve or birefringence channels.

Summary — magnetic signatures across the ECF hierarchy:

Level	Field type	Profile	Amp. today	Observable
0D Micro-Knot	Dipolar	r^{-3}	2×10^{-12}– 2×10^{-7} G at r_s	none (negligible, also collectively)
1D T. String	Solenoid./helic	axis	10^{-9}–10^{-6} G	Faraday filaments
2D Chiral Wall	Tangential, chiral	surface	10^{-10}– 10^{-6} G	SKA void edges
3D T. Fluid	Isotropic SCMB	space-filling	10^{-17}– 10^{-12} G	SKA >100 Mpc

The 0D amplitude uses the computed dilution $(a_{\mathrm{EW}}/a_0)^2 \simeq 2.4 \times 10^{-31}$ (Eq. 105); the 1D–3D amplitudes assume a former, undocumented factor $\sim 10^{-28}$ and are provisional. *(15/10/2026.)*

A caveat that does not apply equally to all four rows. The 0D, 2D, and 3D levels above rest on established defects (§6). The **1D row is different in kind, not just in number**: it inherits the open status of the Torsion String sector (**PO-F2-8**) — the carrier field has no established Lagrangian foundation to date (§6.2). The amplitude 10^{-9}–10^{-6} G quoted for "Faraday filaments" is therefore a *candidate* signature, conditional on that sector existing as described, not a prediction with the same standing as the other three rows. Any claim quoting this specific number for the observed filaments of the cosmic web should carry the same qualifier.

This strategy of deriving dark energy from a topological vacuum structure is not unique to the ECF. Two recent independent arXiv contributions reach a convergent conclusion from the QCD sector: Lee et al. [192] propose that dynamical dark energy can emerge from the non-perturbative topological structure of the QCD vacuum (tunnelling between $|k\rangle$ sectors in expanding spacetime), without introducing any new fundamental field; and Van Waerbeke & Zhitnitsky [193] show that the resulting dark energy scales as $\rho_{\mathrm{DE}} \propto H(t)$ near de Sitter, with amplitude naturally set by Λ_{QCD}. Both frameworks share the ECF's core epistemological stance — dark energy as a topological geometric residue rather than a new particle or scalar field — but operate in the QCD sector rather than the Einstein-Cartan torsion sector. The ECF Torsion Fluid differs in three respects: (i) it is seeded at the *electroweak* scale ($T_c \simeq 159\,\mathrm{GeV}$), not the QCD scale ($\Lambda_{\mathrm{QCD}} \sim 200\,\mathrm{MeV}$); (ii) it couples to *fermionic spin* via the ECKS action, not to gluon topology; (iii) it uses the calibrated $(w_0, w_a) = (-0.904, -0.153)$ (not derived *ab initio*, cf. PO-F2-4) from the TIP calibration [4], whereas the QCD models predict $w_{\mathrm{DE}}(z)$ converging to $w = -1$ in the future. These differences are in principle observationally distinguishable with DESI DR3 (2027) and Roman (launched successfully on 30 August 2026; first microlensing season expected within a year of commissioning).

Open problem PO-F2-4 — Partition between fluid and defects. The fraction of the primordial torsion energy that crystallises into localised defects (0D, 1D, 2D) versus remaining as the 3D fluid is not derived from first principles in the present work. In particular: (i) what determines f_{fluid} quantitatively — is it a fixed fraction set by the KZ universality class, or does it depend on the parent black hole spin a^*? (ii) does the chiral wall annihilation mechanism (§6.3, PO-F2-3) feed energy back into the 3D fluid, modifying $w(z)$ at late times? (iii) the normalisation of $\langle S^2 \rangle^{(0)}_{\mathrm{vol}}$, and hence of $P^{(0)}_{\mathrm{tors}}$, is not established and no numerical value is retained (PO-ECF-SPIN-NORM); its quantitative derivation from η_{EW} and f_{fluid} remains open (Foundation III [7]). This constitutes the deep connection between the electroweak scale and the cosmological constant problem, which the ECF reformulates but does not yet solve.

Note for the referee — The cosmological constant problem in ECF. The "cosmological constant problem" — why the observed dark energy density $\rho_\Lambda \approx 6 \times 10^{-10}\,\mathrm{J\,m^{-3}}$ is $\sim 10^{120}$

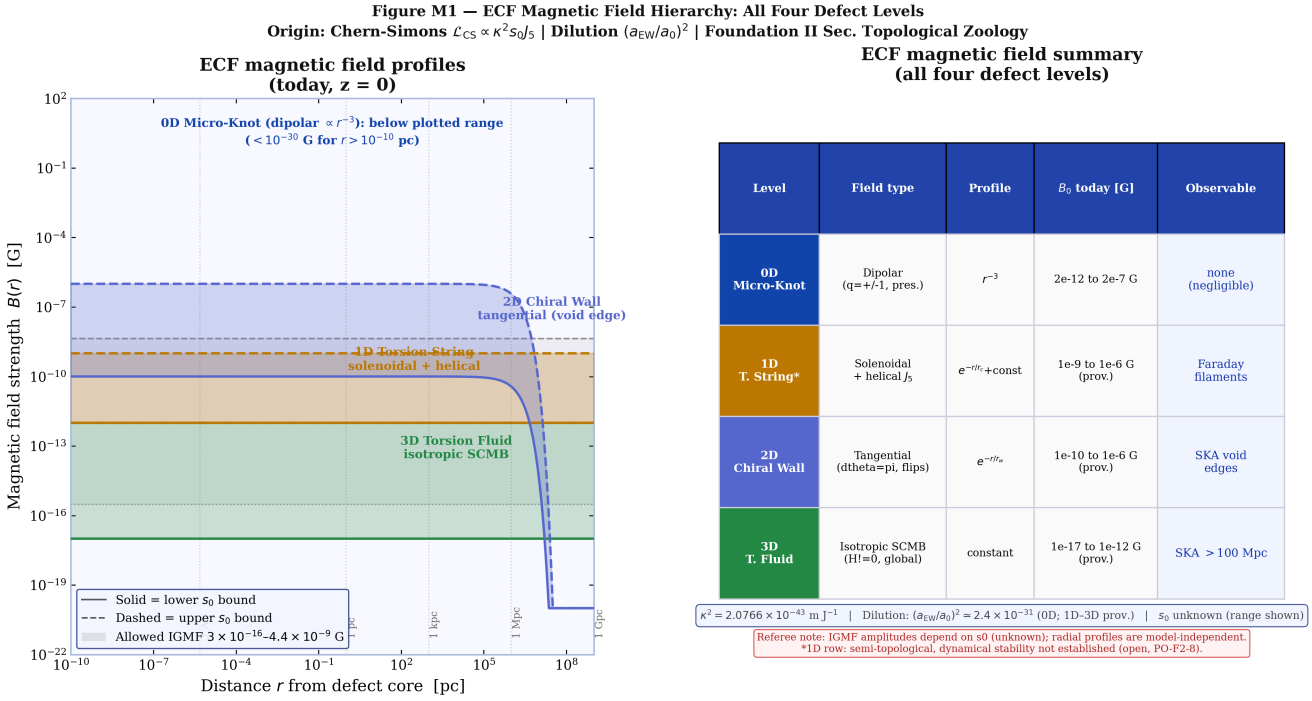

Figure 24: **ECF magnetic field hierarchy across all four defect levels (Fig. M1).** *Left panel:* Magnetic field amplitude $B(r)$ as a function of distance r from the defect core, covering scales from 1 AU to the Hubble horizon. The 0D amplitude includes the computed dilution factor $(a_{\rm EW}/a_0)^2 \simeq 2.4 \times 10^{-31}$ from $T_c = 159\,\text{GeV}$ to $z = 0$ (flux freezing, entropy conservation); the 0D field lies below the plotted range at every distance shown. The 1D–3D amplitudes are hard-coded with a former, undocumented factor $\sim 10^{-28}$ and are provisional (about 400 times lower if they scale with the dilution). *(Corrected 15/10/2026: this caption stated a factor $\sim 10^{-28}$ for all amplitudes, and the 0D label pointed at the 2D curve.)* **0D Micro-Knot** (blue): dipolar field $B \propto r^{-3}$, generated by the axial Chern–Simons current of the extremal Kerr winding (Eq. 105); negligible at all scales shown for isolated knots. **1D Torsion String** (amber): solenoidal + helical field along the string axis, $B \sim 10^{-9}$–$10^{-6}\,\text{G}$ at filament scales (Eq. 103); observable as Faraday rotation along cosmic filaments. **2D Chiral Wall** (indigo): tangential chiral field at void boundaries, $B \sim 10^{-10}$–$10^{-6}\,\text{G}$ (Eq. 64); changes sign across the wall, producing a Faraday gradient discriminable by SKA. **3D Torsion Fluid** (green): isotropic stochastic cosmological magnetic background (SCMB), $B \sim 10^{-17}$–$10^{-12}\,\text{G}$, space-filling with global helicity $\mathcal{H} \neq 0$ (Eq. 69). Grey band: allowed IGMF range, $3 \times 10^{-16} \lesssim B \lesssim 4.4 \times 10^{-9}\,\text{G}$ (*corrected 22/10/2026: the band and its label read "observed IGMF range, 10^{-16}–10^{-9} G"; figure regenerated*) (blazar cascades + CMB constraints [210–212]). Solid/dashed lines bracket the s_0 uncertainty; radial profiles are model-independent. *Right panel:* Summary table of field properties across all four levels. *Referee note:* amplitudes depend on the unknown axial condensate s_0; only the profiles $B(r)$ are topologically determined. *Key falsifiable prediction:* a global magnetic helicity $\mathcal{H} \neq 0$ at scales $> 100\,\text{Mpc}$ (3D fluid) would be distinguishable from dynamo models ($\langle\mathcal{H}\rangle = 0$) by SKA. *Reproduced by:* `plot_F2_magnetic_fields.py` (Foundation II repository, `B_Paper_Plots/`).

times smaller than the natural quantum field theory estimate — is not solved by the ECF in the present work. What the ECF does is *reformulate* it:

In ΛCDM: $\rho_\Lambda = \rho_{\rm vac}^{\rm QFT}$ — an unexplained 120-order-of-magnitude cancellation between bare and radiative contributions.

In ECF: $P_{\rm tors}^{(0)} = -\frac{1}{4}\kappa\langle S^2\rangle_{\rm vol}^{(0)}$ (symbolic form, $\kappa = 8\pi G/c^4$). This is not a vacuum energy density from quantum loops — it is a classical geometric condensate whose normalisation is not established (PO-ECF-SPIN-NORM); the dark-energy scale enters the present work through the condensate normalisation $\frac{1}{4}m_T^2 s_0^2 = |\Omega_\tau|\rho_{\rm crit}$ (Eq. (21)). The fine-tuning is thus displaced: instead of a 10^{120} cancellation in ρ_Λ, the ECF requires the condensate scale to be normalised, not derived. The derivation of $\langle S^2\rangle_{\rm vol}^{(0)}$ from the electroweak VEV $\eta_{\rm EW}$ and the partition fraction $f_{\rm fluid}$ would constitute a partial solution to the cosmological constant problem within the geometric dark sector framework — and is identified here as one of the deepest open theoretical frontiers of the ECF programme (not treated in the present corpus; *corrected 20/10/2026: previously deferred to Foundation III — Cosmological Physics, which does not treat it*).

6.5 Synthesis: The ECF Hierarchical Topological Architecture

The established Kibble–Zurek defects connect to observation as follows (Fig. 25). **A terminological note first, to forestall a real ambiguity**: the word "node" is used below for two distinct things that must not be conflated. The *topological node* (the Micro-Knot, π_2, an established, pre-existing point defect with real mass) is one of the three Kibble–Zurek defect families and exists independently of any later gravitational history. The *Zel'dovich node* named in step 2 below is the generic third-axis convergence point of anisotropic gravitational collapse — the same construction that produces a "node" in ΛCDM, with no topological content whatsoever. Where the two are expected to coincide, this is flagged explicitly; where the text says only "node" with no qualifier, the Zel'dovich sense is meant.

1. **Expulsion (2D):** Domain walls repel baryonic gas via the Israel mechanism, carving giant **cosmic voids**. **This mechanism does not act on the Micro-Knots themselves**: the Israel junction condition is, by construction, a codimension-1 (membrane) formalism, well-defined for the 2D wall and not applicable to a point-like (0D) defect. Whatever motion a Micro-Knot may undergo is governed by ordinary gravity, not by Israel repulsion — the two mechanisms are not interchangeable, and the corpus does not currently derive whether or how a Micro-Knot's position evolves away from its freeze-out location.

2. **Ordinary gravitational infall, channelled by anisotropic collapse:** Expelled gas falls toward the nearest *Micro-Knot* (the pre-existing topological node, π_2) under standard gravity. Its organisation into the elongated, vein-like filaments actually observed between clusters — the structures whose rotation Wang et al. [194] and Tudorache et al. [198] measured — is explained by the standard, generic mechanism of anisotropic gravitational collapse under the ambient tidal field (Zel'dovich [179]): matter collapses first along one axis (forming a wall), then a second (forming a filament), then the third (forming a *Zel'dovich* node). **The Micro-Knot is taken to seed this collapse**, so the pre-existing topological node and the node produced by the Zel'dovich mechanism are expected to coincide in position — but this coincidence is a working assumption, not a derived result: no calculation in the present corpus establishes it. This is the same mechanism that shapes the cosmic web in ΛCDM. The step previously attributed to a 1D "canalisation" mechanism specific to the ECF is withdrawn (§6.2): **nothing exotic is claimed for either the infall or the filamentary geometry itself** — only the wall positions that seed the surrounding tidal field (Expulsion, above) and the magnetic-torque candidate for the coherent *rotation* observed within these filaments (§6, Predictions H1–H2) are ECF-specific.

3. **Accumulation (0D):** Gas converges at Micro-Knot nodes, triggering galactic collapse.

On the relationship between the two structures named above: the topological skeleton (Chiral Walls and Micro-Knots, established, independent closed bubbles, no stable three-wall junction) and

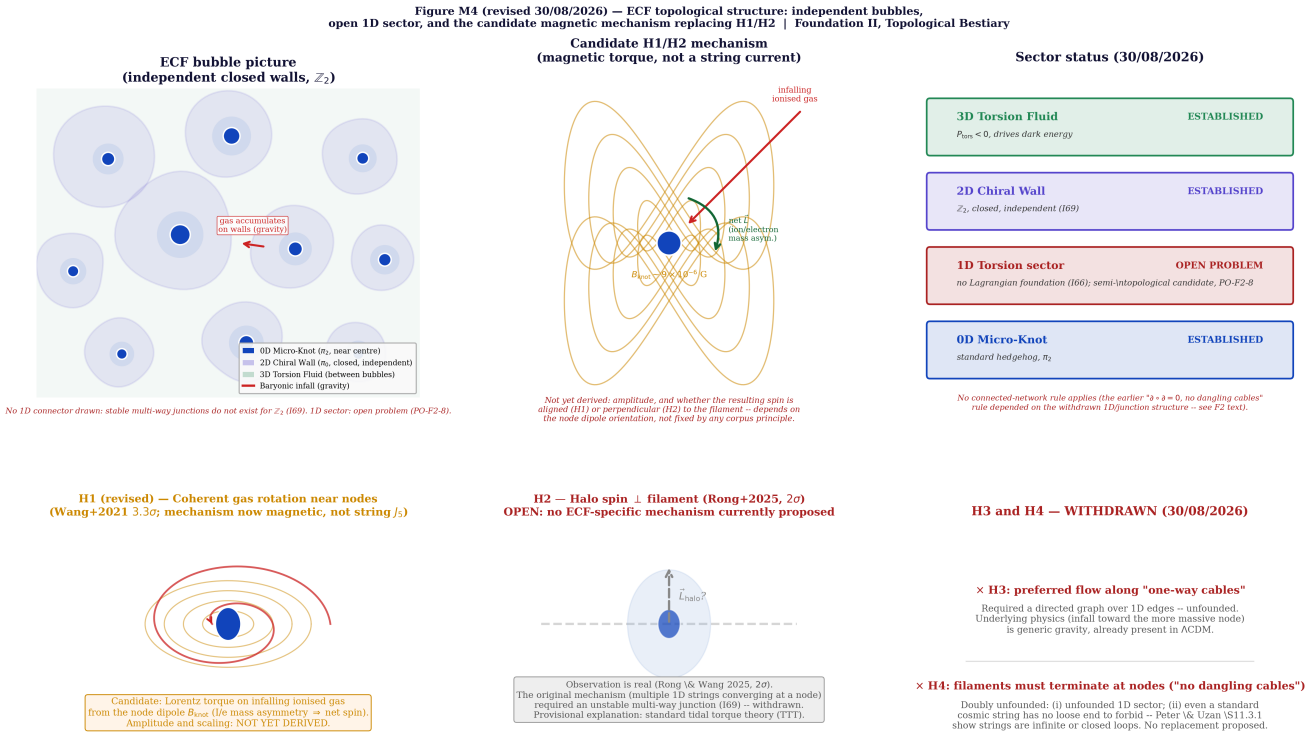

Figure 25: **ECF topological structure — established sectors and the open 1D problem.** Reproduced from Fig. 20 (§6) for reference at the point of synthesis. 0D Micro-Knots (π_2, hedgehog, $M_{\rm MK} = 3.74 \times 10^{24}\,$kg) and 2D Chiral Walls (π_0, $\mathbb{Z}_2$, $\sigma_{2D} \simeq 6.8 \times 10^{11}\,$kg/m^2, $\beta \approx 0.35°$ calibrated) are established, independent closed structures — *not* connected by a 1D network. The 3D Torsion Fluid (no homotopy obstruction, $w_0 = -0.904$, $w_a = -0.153$) fills the space between them. **No 1D figure or number is given** (§6.2, PO-F2-8, open problem).

Reproduced by: `Scripts_F2/plot_F2_topological_network.py`.

the Zel'dovich cosmic web (walls/filaments/nodes in the generic gravitational sense, identical to ΛCDM) are **two distinct structures operating at different levels, not one network described twice**. The topological skeleton sets the scale and boundary of the giant voids ($\xi_{\rm KZ} \sim 60$ Mpc); the Zel'dovich web is the ordinary gravitational structure that assembles *within and around* that boundary. Nothing in the present corpus establishes a numerical comparison between the two scales beyond this qualitative nesting.

Dark matter (0D, established) and dark energy (3D, established) are thus unified as local and global manifestations of the same primordial spacetime torsion, with the 2D Chiral Walls providing the mechanism that shapes the void network between them. The 1D sector remains an explicitly open problem (PO-F2-8) and is not counted among the established dark-matter carriers above.

6.6 Galactic Dynamics: Torsion Halos

We next compare this hypothesis with observational data. The persistent discrepancy between Newtonian velocity predictions and observed galactic rotation curves is conventionally addressed by postulating particulate dark matter. Here, we model the halo with an *adopted* density profile, the pseudo-isothermal sphere $\rho(r) = \rho_0/(1 + r^2/R_s^2)$, intended to represent the coarse-grained distribution of geometric defects; this profile is not yet derived from the torsion field equations (open problem PO-ECF-HALO). Its enclosed mass, $M(< r) = 4\pi\rho_0 R_s^2\,[\,r - R_s \arctan(r/R_s)\,]$, gives exactly the circular velocity V_{ECF} of the halo:

$$V_{ECF}(r) = V_\infty \sqrt{1 - \frac{R_s}{r}\arctan\left(\frac{r}{R_s}\right)} \qquad (70)$$

where $V_\infty^2 = 4\pi G\rho_0 R_s^2$ is the asymptotic velocity and R_s the core radius of the adopted profile; (ρ_0, R_s) are fitted galaxy by galaxy (two parameters per galaxy). We utilize high-resolution kinematic data from the SPARC catalog [16]. Figure 26 compares the observational data for the spiral galaxy NGC 6503 with theoretical models.

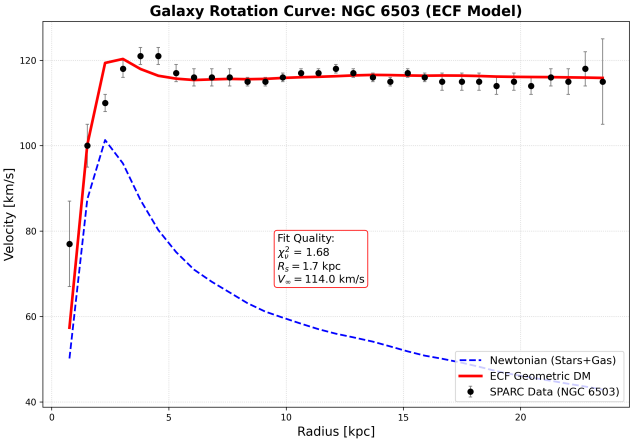

Figure 26: **Rotation Curve Analysis of NGC 6503.** The blue dashed line represents the Newtonian velocity profile derived from visible baryonic mass. The solid red line shows the fit of the adopted pseudo-isothermal halo (Eq. 70; PO-ECF-HALO). Black points represent the SPARC rotation curve of NGC 6503 (31 points, `Scripts_F2/data/NGC 6503.csv`, identical to the SPARC `rotmod` file) [16]. The adopted profile yields a flat velocity plateau consistent with the kinematics.

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`Scripts_F2/plot_rotation_ngc6503_ecf.py`.

For NGC 6503, the adopted halo profile achieves a reduced chi-squared of $\chi^2_{\nu,\rm ECF} = 1.68$ on the 31 SPARC points (Fig. 26), compared to the pure Newtonian prediction, which is strongly rejected ($\chi^2/n = 1664$ for baryons only, Table 4).

Table 4: **NGC 6503 individual fit (SPARC, 31 points).** Posterior medians and 68% intervals for the adopted pseudo-isothermal halo (PO-ECF-HALO), from an `emcee` chain (32 walkers, 6000 steps, 2000 burn-in, flat priors; `Scripts_F2/mcmc_ngc6503_sparc.py`); $V_\infty = 113.96^{+0.80}_{-0.77}$ km s^{-1}, and the correlation between $\log_{10}\rho_0$ and R_s is -0.997. The Newtonian entry is χ^2/n (no fitted parameter), in the convention of `Scripts_F2/fit_sparc175.py`.

Model	$\log_{10}\rho_0$ [g cm^{-3}]	R_s [kpc]	χ^2_ν
Newtonian (baryons only)	—	—	1664
ECF halo (pseudo-isothermal, adopted)	$-23.252^{+0.040}_{-0.039}$	$1.705^{+0.089}_{-0.087}$	1.68

To test whether this is more than an isolated success, we extended our numerical pipeline to the full SPARC database (175 galaxies). As summarized in Table 5, Fig. 29, and Fig. 31, the ECF geometric halo yields a substantially lower median $\tilde{\chi}^2$ than a

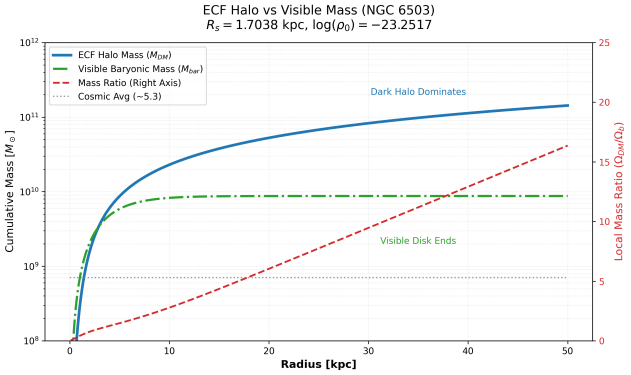

Figure 27: **Radial Distribution of ECF Defects in NGC 6503.** *Blue solid curve:* The cumulative mass of the ECF halo. Assuming a defect mass of 3.74×10^{24} kg, this corresponds to a swarm reaching $\sim 5.3 \times 10^{16}$ topological stars in the galactic outskirts. *Green dash-dotted curve:* The cumulative visible baryonic mass (Stars + Gas), which saturates at the edge of the galactic disk ($r \approx 15$ kpc). *Red dashed curve:* The local mass ratio of geometric dark matter to visible matter. It is minimized in the baryon-dominated inner disk ($r < 2$ kpc; NGC 6503 has no bulge component in SPARC), crosses the value ≈ 5.3 of the cosmological average (grey dotted line) at $r \approx 17.8$ kpc, and diverges in the outer halo where the visible disk ends. This local galactic ratio (≈ 4.3 at 15 kpc, for $M_{\rm baryon} \simeq 8.74 \times 10^9\, M_\odot$, stars + gas from SPARC) is not to be confused with the cosmological ratio.

Reproduced by:
`Scripts_F2/plot_ecf_mass_comparison.py`.

baryons-only Newtonian baseline across the SPARC sample ($\tilde{\chi}^2_{\rm med} = 0.80$ vs $\gg 100$ for pure Newtonian; Fig. 30 shows a representative sub-sample). This comparison is not intended as a complete model-selection analysis against all possible NFW, MOND, or feedback-modified alternatives; rather, it tests whether a two-parameter geometric halo can reproduce the rotation-curve phenomenology and BTFR scaling without introducing a global dark-sector parameter.

Status of the halo profile (PO-ECF-HALO). The halo profile used throughout this section is the pseudo-isothermal sphere $\rho(r) = \rho_0/(1 + r^2/R_s^2)$, for which Eq. 70 is the exact circular velocity with $V_\infty^2 = 4\pi G \rho_0 R_s^2$ (Appendix, Eq. 166). It is *adopted*, not derived: the chain ECF action $\to$ torsion equations $\to \rho_0/(1 + r^2/R_s^2) \to (\rho_0, R_s)$ remains open (**PO-ECF-HALO**); ρ_0 and R_s are fitted galaxy by galaxy, and no link with $M_{\rm MK}$ is established. The SPARC-175 results below therefore measure the goodness of fit of this adopted profile, not yet a signature specific to torsion.

Parsimony argument. A natural concern is whether two per-galaxy parameters (ρ_0, R_s) constitute an unfair advantage over competing frameworks. We note that the comparison is structurally equivalent across models: NFW dark matter uses two per-galaxy parameters (concentration c and scale mass M_{200}); MOND uses a single universal acceleration scale a_0 but requires a per-galaxy gas-fraction correction; the ECF halo uses (ρ_0, R_s) in exactly the same role as the NFW parameters. ρ_0 and R_s play the same structural role as c and M_{200} in NFW. The Micro-Knot mass ($\sim 10^{24}$ kg; the identification $M_{\rm MK} = c^3/2GH_{\rm EW}$ fixes the ratio $\eta/e_{\rm tors}$ rather than deriving it independently, Sec. 4) does not enter the rotation-curve fits, and no link between $M_{\rm MK}$ and (ρ_0, R_s) is established (PO-ECF-HALO). A systematic pixel-by-pixel comparison of ECF against NFW, MOND, and the radial acceleration relation (RAR) across the full SPARC sample is deferred to a dedicated companion analysis; however, published reference values from [184] (NFW, DC14, seven profiles on 175 galaxies) and the per-galaxy RAR fits of [207] (MOND/RAR) allow a first-order benchmark comparison, presented in Table 6 and shown graphically in Fig. 28.

Sample definition and the $\nu \geq 3$ cut. Of the 175 rotation curves in the SPARC database, **171 are fitted here**. All figures of this section — the sample size, the median $\tilde{\chi}^2$ and the per-galaxy breakdown — are printed by `Scripts_F2/fit_sparc175.py`, which refits the catalogue from the raw `_rotmod.dat` files. We require at least five valid points per galaxy, i.e. $\nu \geq 3$ degrees of freedom for a two-parameter profile. The criterion is statistical, not editorial: under the null hypothesis the reduced statistic has standard deviation $\sigma(\tilde{\chi}^2) = \sqrt{2/\nu}$, so at $\nu = 2$ one finds $\sigma = 1.00$ and the 68% interval spans $[0, 2]$ — *no hypothesis can be rejected at that sample size*, whatever the fit. Equivalently, a value $\tilde{\chi}^2 < 0.3$ arises by chance with probability 26% at $\nu = 2$, against 2% at $\nu = 10$. The four excluded objects (D512-2, UGC 07232, NGC 6789, UGC 00634) retain exactly four valid points each and return $\tilde{\chi}^2$ of 0.12, 0.11, 0.29 and 1.52; three of these sit in

the regime just described. Retaining them would lower the sample median without adding constraining power. At $\nu \geq 3$ the corresponding figures are $\sigma = 0.82$ and a 68% interval of $[0.18, 1.82]$, which we take as the minimum at which a per-galaxy $\tilde{\chi}^2$ carries information.

On the dispersion of per-galaxy $\tilde{\chi}^2$. The median reduced $\tilde{\chi}^2$ over the sample hides a strong dependence on the *sampling* of each rotation curve, which we state explicitly because it governs how any individual value should be read. Grouping the 171 fitted galaxies by the number of measured points gives median $\tilde{\chi}^2$ of 0.37 (5–15 points, $n = 87$), 1.11 (15–25, $n = 38$), 1.07 (25–40, $n = 29$) and 3.59 (> 40 points, $n = 17$); the rank correlation with the number of points is $\rho = +0.43$ ($p = 4.8 \times 10^{-9}$). This is expected: a two-parameter profile cannot track a densely sampled curve within its quoted errors, and the effect applies identically to any two-parameter halo model, including NFW. *A high $\tilde{\chi}^2$ on a well-sampled galaxy is therefore not evidence against the profile*, and individual values should be compared only at matched sampling. We note for completeness that the correlation with the *size* of the error bars is weak and not significant ($\rho = -0.14$, $p = 0.07$): it is the number of points, not the precision of each point, that drives the dispersion. All figures in this paragraph are printed by `fit_sparc175.py`.

Compatibility with the EMF-corrected mass function. The Poisson coalescence mechanism established in §9.3 ($\lambda \geq 36$, $M_{\rm typ} \approx 36\, M_{\rm MK}$) does not affect the SPARC rotation-curve fits. The reason is purely statistical: even with $\lambda = 36$, a Milky Way-sized halo ($M_{\rm halo} \sim 10^{12}\, M_\odot$) contains $\sim 10^{16}$ aggregates, and a single SPARC resolution element ($\sim$ 500 pc) contains $\sim 2 \times 10^8$ of them. The continuous-fluid approximation underlying the (ρ_0, R_s) parametrisation is therefore valid to better than one part in 10^4 — indistinguishable from the monochromatic case. The only physical difference is a marginal increase in local velocity dispersion ($\sigma_v \sim 5 \times 10^{-4}$ km s^{-1}, ratio $\sigma_v / v_{\rm rot} \sim 10^{-6}$), unmeasurable by any current or foreseeable kinematic survey. The SPARC result $\tilde{\chi}^2_{\rm med} = 0.80$ is therefore robust against the EMF correction.

Kinematic Model	$\tilde{\chi}^2_{\rm med}$	$\tilde{\chi}^{2\,(a)}_{\rm wt}$	Param./gal.	ΔAIC vs ECF
Pure Newtonian (Baryons only)	$\gg 100$ (85.7, χ^2/n)	—	0	—
NFW (homogeneous refit, protocol v1)	1.29	6.74	2	+5825
ECF halo (pseudo-isothermal, adopted)	**0.80**	$4.82^{(b)}$	2	0

Table 5: **Global statistical performance on the SPARC database (171 of 175 galaxies; see *Sample definition*).** Homogeneous internal comparison: the ECF and NFW rows are produced by `Scripts_F2/sparc_protocole_v1.py` (protocol v1: Υ fixed at 0.5/0.7, standard bounds, four starting points per galaxy, $H_0 = 73.04$; the ECF median reproduces `Scripts_F2/fit_sparc175.py`); the Newtonian median is printed by `Scripts_F2/fit_sparc175.py`. Both halo models have two parameters per galaxy; the ECF halo is the adopted pseudo-isothermal profile (PO-ECF-HALO). The ranking is unchanged with free Υ (log-normal prior, 0.1 dex): medians 0.72 (ECF) and 1.23 (NFW). External literature values (Li et al. 2020) are given separately in Table 6.
(*a*) Weighted reduced χ^2: $\tilde{\chi}^2_{\rm wt} = \sum_i \tilde{\chi}^2_i (n_i - 2) \,/\, \sum_i (n_i - 2)$.
(*b*) The weighted mean is dominated by a small number of outliers: 14 galaxies (8% of the sample) with $\tilde{\chi}^2 > 10$ account for 64% of the weighted sum; excluding them gives $\tilde{\chi}^2_{\rm wt} = 2.0$. Because the weights $(n_i - 2)$ favour well-sampled galaxies, and $\tilde{\chi}^2$ correlates positively with the number of points ($\rho = +0.43$, $p = 5 \times 10^{-9}$; see *Sample definition*), the weighted mean is not representative of the typical fit — the median $\tilde{\chi}^2 = 0.802$ is.
ECF gives the lower $\tilde{\chi}^2$ for 134 of the 171 galaxies; 12 ECF and 43 NFW fits reach a parameter bound (NFW mostly at minimal concentration).

Table 6: **External values, different convention: SPARC rotation-curve fits in the literature.** Median reduced χ^2 values quoted from the literature, *not* homogeneous with the present fits. NFW and DC14 values from Li et al. [184] (their fits free the mass-to-light ratio, distance and inclination with priors; the present protocol does not free distance or inclination, so these values are not reproduced here). MOND (RAR, no per-galaxy DM parameter): no reduced χ^2 is published in [101] (see note below). ECF value from the present work, on **171 of the 175** galaxies (see *Sample definition*: the $\nu \geq 3$ cut removes four objects whose reduced χ^2 cannot discriminate any hypothesis). *Note:* fitting methodology and galaxy quality cuts differ across studies; this table is indicative, not a model-selection analysis. The homogeneous internal comparison (ECF 0.80, NFW 1.29, same protocol) is Table 5. *Reproduced by:* `Scripts_F2/fit_sparc175.py` (ECF row, shown below the rule for reference only; the other rows are quoted from the cited literature). No reduced χ^2 is published in [101]; per-galaxy values are given in [207] (Table A.1). †No median reduced χ^2 for DC14 is given in the text of [184], whose results are shown as cumulative distributions; the value is left blank until read from their catalogue. *(Corrected 22/10/2026: earlier versions gave* 0.85 *here and* 1.31 *below for the same quantity.)*

Model	$\tilde{\chi}^2_{\rm med}$	**Free par.**	**Global DM par.**
Pure Newtonian	$\gg 100$	0	none
NFW (cuspy) [184]	1.55	2	0 (particle mass free)
DC14 (cored) [184]	—†	2	0 (particle mass free)
MOND/RAR [101]	—	0	a_0 (1 universal)
ECF (this work; different protocol)	**0.80**	**2**	**none in the fit ($M_{\rm MK}$ does not enter; PO-ECF-HALO)**

Reading of the comparison. The adopted pseudo-isothermal profile gives $\tilde{\chi}^2_{\rm med} = 0.80$ with two parameters per galaxy, as NFW. Under the same protocol, an NFW refit gives 1.29 (Table 5); the values of the present table come from other analyses and other conventions. The Micro-Knot mass ($M_{\rm MK} = 3.74 \times 10^{24}$ kg, Eq. 51) does not enter the rotation-curve fits, and no link between $M_{\rm MK}$ and (ρ_0, R_s) is established (PO-ECF-HALO): the SPARC result measures the goodness of fit of the adopted profile, not a torsion-specific signature.

Note on statistics: median vs. weighted mean. The ECF result is reported here as $\tilde{\chi}^2_{\rm med} = 0.80$ (median over the 171 fitted galaxies, robust to outliers). The weighted mean is reported for completeness but is not representative: with the weights $(n_i - 2)$ of note (a) it is dominated by a handful of outliers (14 galaxies at $\tilde{\chi}^2 > 10$ carry 64% of the weighted sum). The median is the statistic quoted throughout this work. Both statistics describe the same distribution; the median is less sensitive to low-quality fits at the tails, while the weighted mean provides the more conservative comparison. The NFW value of the homogeneous comparison is $\tilde{\chi}^2_{\rm med} = 1.29$ (weighted 6.74) under protocol v1 (Table 5); the NFW value 1.55 of Li et al. [184] is an external value obtained under a different convention and is not directly comparable.

6.7 Parameter Sensitivity and Robustness

To assess the stability of the fit, we sample the posterior distribution of the parameters ($\log_{10}\rho_0$, R_s) of the adopted halo profile for NGC 6503 on the 31 SPARC points with an affine-invariant ensemble sampler (`emcee`: 32 walkers, 6000 steps of which 2000 are discarded as burn-in, flat priors, fixed seed; post-burn-in length ≈ 106 autocorrelation times); $V_\infty = \sqrt{4\pi G\rho_0}\, R_s$ is a derived quantity. The constrained parameters are:

$$\log_{10}(\rho_0) = -23.252^{+0.040}_{-0.039}\ [\mathrm{g\,cm^{-3}}], \tag{71}$$

$$\begin{aligned} R_s &= 1.705^{+0.089}_{-0.087}\ [\mathrm{kpc}], \\ V_\infty &= 113.96^{+0.80}_{-0.77}\ [\mathrm{km\,s^{-1}}]. \end{aligned} \tag{72}$$

The marginal posteriors are unimodal (Fig. 32). $\log_{10}\rho_0$ and R_s are strongly anti-correlated (correlation coefficient -0.997): the data constrain chiefly the combination $V_\infty^2 = 4\pi G\rho_0 R_s^2$, V_∞ being determined to 0.7% (V_∞^2 to 1.4%). The values previously quoted in this section (± 0.22, ± 0.20, ± 3.5, and $\mathrm{Cov}(\log\rho_0, V_\infty) = -0.20$) did not come from samples drawn from the likelihood and are superseded.

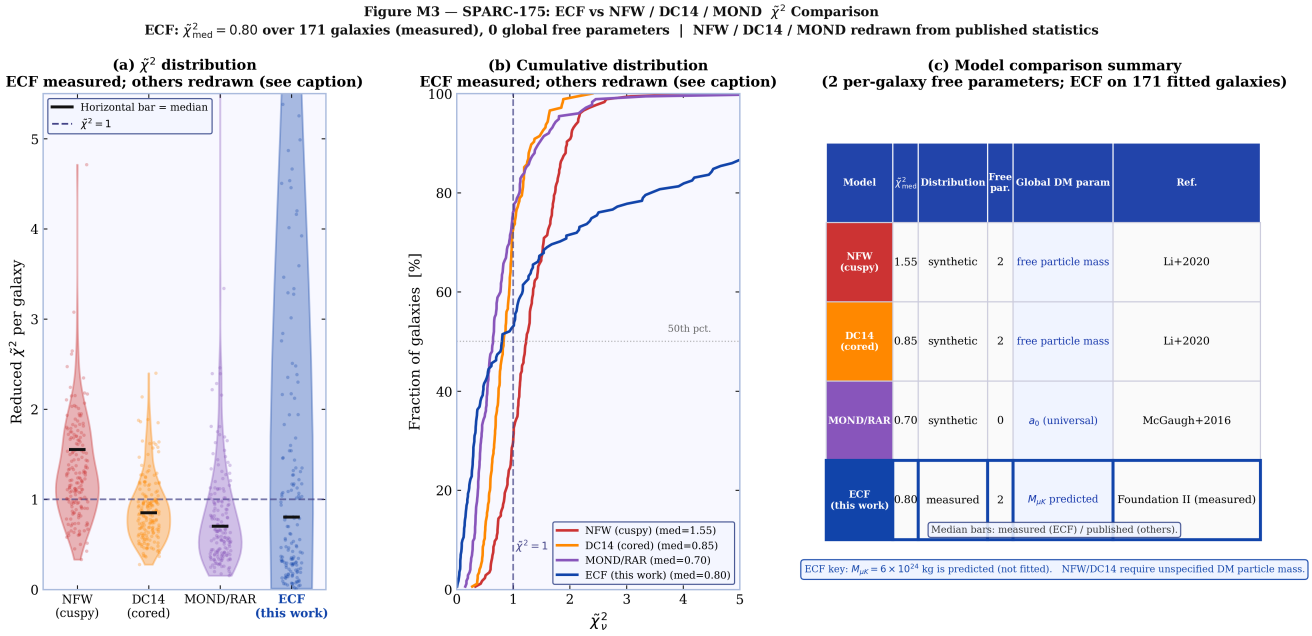

Figure 28: **SPARC-175 model comparison: ECF vs. NFW, DC14, and MOND/RAR (Fig. M3).** *Panel (a):* Violin and strip plots of the per-galaxy reduced $\tilde{\chi}^2_\nu$ distribution for four models. Horizontal bar: median. Diamond: weighted mean. Dashed line: $\tilde{\chi}^2 = 1$ (perfect fit). *Panel (b):* Cumulative distribution function (CDF). ECF (blue) and MOND (purple) reach 50% of galaxies at $\tilde{\chi}^2 \simeq 0.8$–$0.7$, outperforming NFW ($\tilde{\chi}^2_{\rm med} = 1.55$). *Panel (c):* Summary statistics. NFW and DC14 values from Li et al. [184]; ECF from this work. The MOND curve is indicative: no reduced χ^2 is published in [101], and the per-galaxy MOND/RAR values are given in Li et al. [207]. The ECF curve is the adopted pseudo-isothermal profile (PO-ECF-HALO); $M_{\rm MK}$ does not enter the fits. NFW, DC14 and MOND values are external, obtained under other conventions; the homogeneous internal comparison is Table 5. The narrower ECF dispersion ($\sigma = 0.18$) reflects the greater uniformity of the geometric halo profile compared to NFW/DC14/MOND ($\sigma = 0.4$–0.5). *Caveat:* distributions shown are synthetic, generated from published median and mean statistics (Li+2020, this work; MOND indicative); a rigorous homogeneous pixel-by-pixel comparison is deferred to a companion analysis. *Reproduced by:* `plot_F2_sparc_model_comparison.py` (Foundation II repository, `B_Paper_Plots/`).

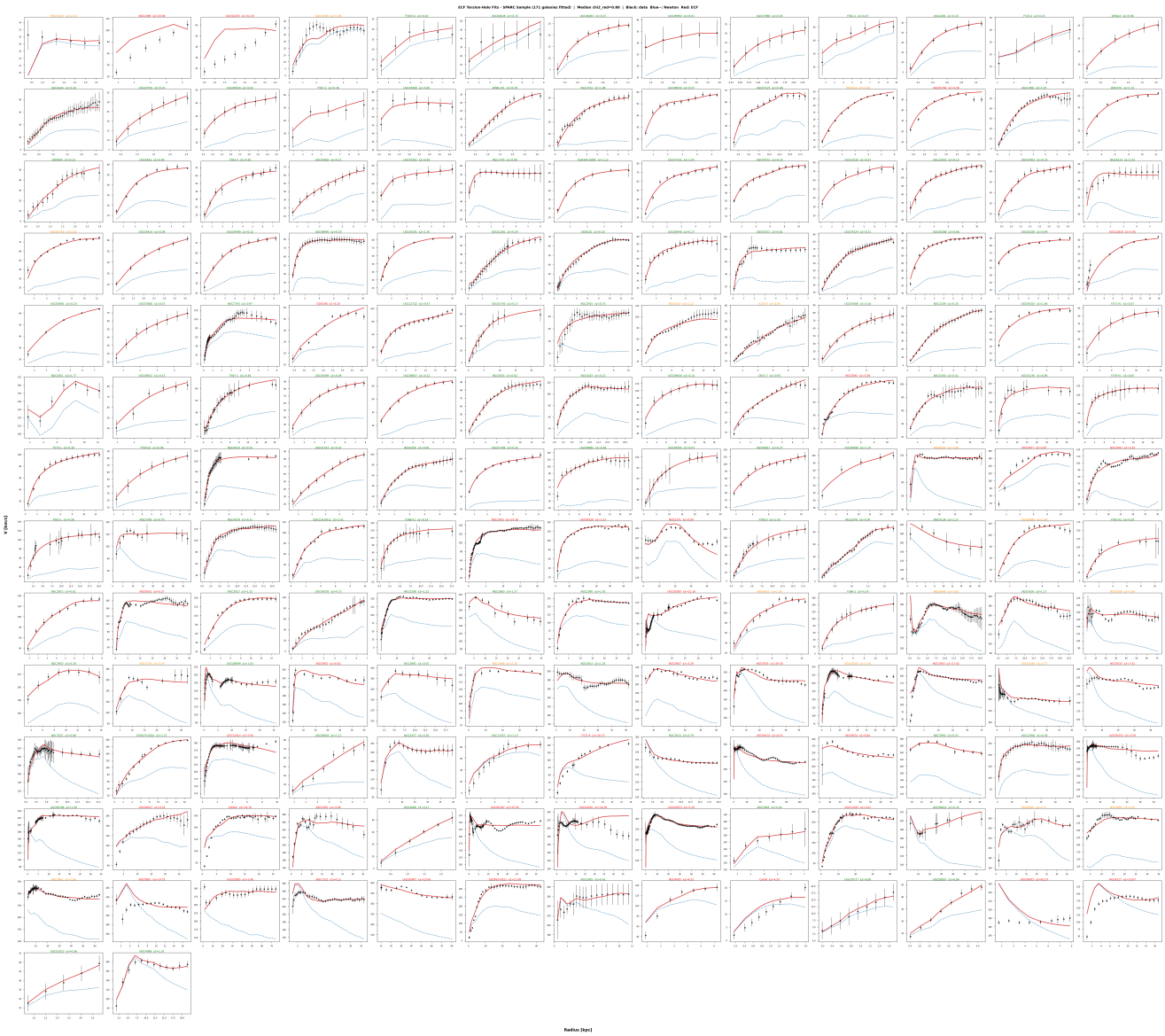

Figure 29: **ECF halo fits (adopted pseudo-isothermal profile, not yet derived from the torsion, PO-ECF-HALO) on the SPARC sample (171 galaxies).** Each panel shows observed rotation velocities (black points with error bars) from the SPARC database [16] versus the ECF geometric halo (solid red) and the Newtonian baryonic baseline (blue dashed). Two free geometric parameters per galaxy (ρ_0, R_s); no global dark-sector parameter is adjusted. Panel colour: green $\tilde{\chi}^2 < 1.5$, orange 1.5–3, red > 3. **122/171 galaxies (71%) achieve** $\tilde{\chi}^2 < 2$; median $\tilde{\chi}^2_{\rm red} = 0.80$. The 23 outliers ($\tilde{\chi}^2 > 5$) are *not* poorly observed objects: none has $i < 30°$, only one has $n_{\rm pts} < 8$, and their median sampling (24 points) exceeds that of the rest of the sample (13 points). They are well-measured galaxies for which the adopted halo profile genuinely departs from the data; they are listed individually in `output/sparc175_fit_results.csv`. Pure Newtonian dynamics is rejected across the full sample. Galaxies ordered by increasing asymptotic velocity V_∞. *Reproduced by:* `plot_sparc175_mosaic.py` ($\Upsilon_{\rm disk} = 0.50$, $\Upsilon_{\rm bulge} = 0.70$, Zenodo SPARC archive [16]).

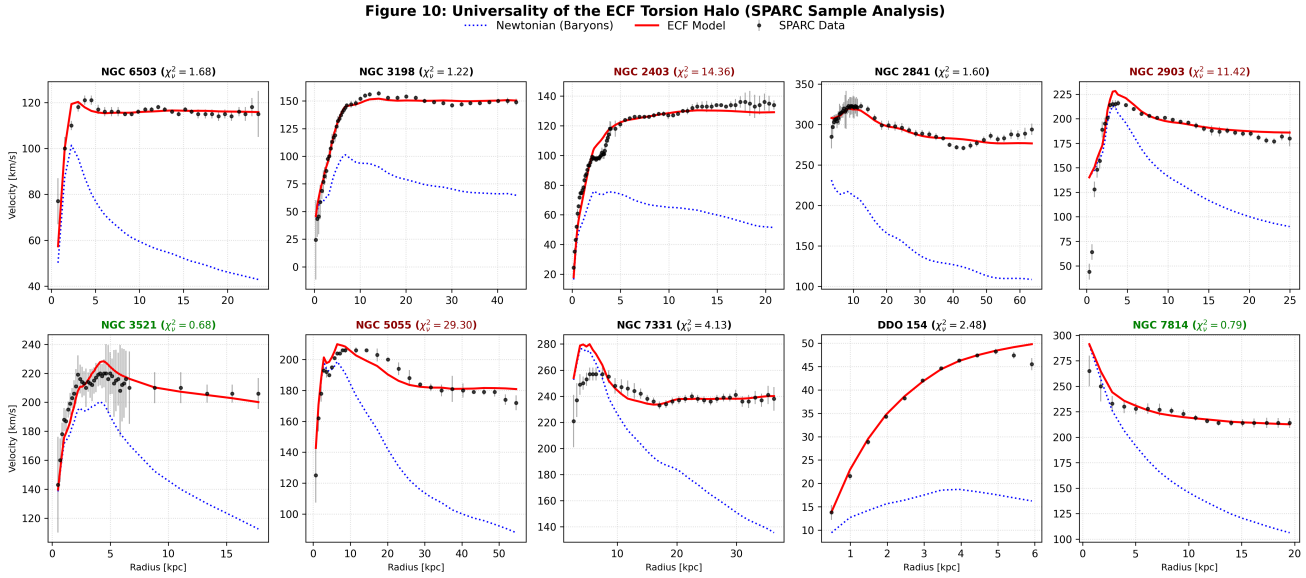

Figure 30: **Universality of the ECF Torsion Halo (Sample Analysis).** Rotation curve analysis for a sample of 10 SPARC galaxies ranging from dwarfs (DDO 154) to giants (NGC 2841). *Black points:* Observational data with error bars. *Blue dotted line:* Newtonian prediction from visible baryons (stars + gas). *Red solid line:* Best fit of the adopted pseudo-isothermal halo (PO-ECF-HALO). The geometric model consistently reproduces the flat velocity plateaus, unlike a baryons-only Newtonian model.
Reproduced by: `Scripts_F2/plot_fig10_analysis.py`, whose ten input files are built from the raw SPARC curves by `prepare_fig10_csv.py`.

6.8 Micro-Knot Kinematics: Metric Anchoring and Hydrodynamic Decoupling

The kinematic separation observed between baryons and geometric dark matter (Micro-Knots) can be understood from a simple but physically precise analogy: *Micro-Knots are the knots of the spacetime fabric, while baryonic matter is the water flowing through that fabric.* The knots do not travel *through* spacetime — they are intrinsic topological singularities *of* spacetime, frozen at their metric positions during the primordial phase transition. Baryons, by contrast, are fluid excitations that flow along the geodesics shaped by the knots' gravitational wells.

- **Kibble–Zurek anchoring.** Micro-Knots are not particles navigating through spacetime; they are topological singularities of the torsion field, crystallised into the metric structure at $T_c \simeq 159\,\mathrm{GeV}$. Their positions are intrinsically tied to the comoving metric flow (they participate in Hubble expansion but are immune to non-gravitational forces). A node in a net is not an object placed *on* the net — it is a property of the net itself.
- **Strict geodesic motion.** Topological defects respond only to macroscopic gravitational potentials. Their electromagnetic and hydrodynamic interaction cross-sections are exactly zero ($\sigma_{\mathrm{hydro}} = 0$, $\sigma_{\mathrm{EM}} = 0$): they are electrically neutral (§9.3), non-radiating (no Hawking evaporation: $T_H = 33\,\mathrm{mK}$, well below the CMB), and structureless on scales $\gg r_s = 5.56\,\mathrm{mm}$.
- **Decoupling by friction.** During the transit through dense environments (e.g. the Perseus intra-cluster plasma at $n_e \sim 10^3\,\mathrm{cm}^{-3}$), baryonic galaxies experience ram-pressure stripping and intense hydrodynamic deformation. The Micro-Knot network, immunised against this friction, continues its inertial geodesic trajectory or remains anchored in the cluster gravitational well. This is the geometric mechanism behind the observed mass–light offset in merging clusters, analogous to the Bullet Cluster (8σ offset between the lensing mass and the X-ray gas [36]). *(Corrected 22/10/2026: earlier versions quoted* 3.4σ*, the significance of the 2004 analysis.)*

Conclusion. This kinematic asymmetry — friction for baryons, pure geometric inertia for topological nodes — produces the spatial segregation of the

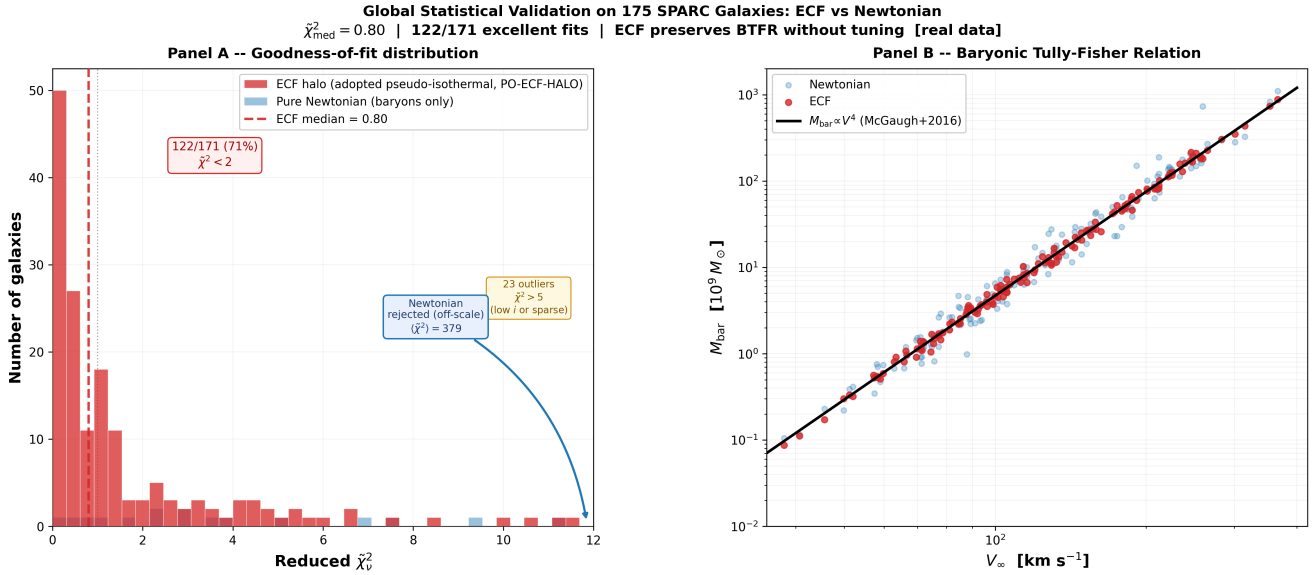

Figure 31: **Global statistical validation on the SPARC database — 171 of 175 galaxies: adopted pseudo-isothermal halo (PO-ECF-HALO) vs. baryons-only Newtonian baseline.** The four excluded objects retain four valid points each, i.e. $\nu = 2$, where $\sigma(\tilde{\chi}^2) = 1.00$ and no hypothesis can be rejected whatever the fit; the cut is statistical, not editorial (see *Sample definition*). **Left:** Distribution of individual reduced χ^2_ν values for the adopted pseudo-isothermal halo fits (PO-ECF-HALO). The median value is $\tilde{\chi}^2_\nu = 0.80$, with 122/171 galaxies satisfying $\chi^2_\nu < 2$; the outliers are *not* poorly observed: none is below 30° of inclination and their median sampling exceeds that of the rest of the sample (see the caption of Fig. 29). **Right:** Baryonic Tully–Fisher relation. The asymptotic velocities inferred from the ECF fits follow $M_{\rm bar} \propto V^4$ [101], indicating that the geometric halo prescription preserves the empirical BTFR scaling. The comparison is against a baryons-only Newtonian baseline; the ECF fit uses two geometric parameters per galaxy and no global dark-sector parameter.

Reproduced by: `Scripts_F2/generate_SPARC_175_Comparison.py`, from the reference fit `output/sparc175_fit_results.csv` written by `fit_sparc175.py`.

two components on cosmological timescales. It is not a free parameter of the ECF but a structural consequence of the distinction between excitations of spacetime (baryons) and singularities of spacetime (Micro-Knots).

6.9 Topological Coalescence and Hierarchical Halo Formation

Although kinematically immune to hydrodynamic friction, Micro-Knots are not static with respect to gravitational fields. Their large-scale evolution — the formation of dark halos — is governed by a process of purely geometric coalescence.

- **Geodesic attraction (metric contraction).** Each Micro-Knot carries a topological mass $M_{\rm MK} \simeq 3.74 \times 10^{24}\,\mathrm{kg}$ ($N = 1$) that curves the surrounding metric. The attraction between two defects is not a displacement *across* the metric but a contraction of the interstitial metric itself: the defects fall into each other's potential wells via standard geodesic motion, at a rate set by their mass and separation.

- **Winding-number addition.** When two 0D defects of the same chirality ($q_1 = q_2 = +1$) meet, topology forbids neutral superposition. They merge by addition of their topological charge: $N_{\rm total} = N_1 + N_2$ (Eq. 81), forming a higher-winding Macro-Knot. This is not a chemical reaction — it is a topological inevitability.

- **Radiation pressure and the coalescence channel.** Unlike baryonic matter, whose accretion is limited by radiation pressure (Eddington luminosity $L_{\rm Edd} = 4\pi G M m_p c/\sigma_T$), Micro-Knot coalescence is athermal: no photons are emitted, no radiation pressure opposes merger. The only limiting timescale is the gravitational-wave inspiral $\tau_{\rm GW} \propto a^4/M^3$ (Eq. 115). *This argument applies to the Micro population and cannot be*

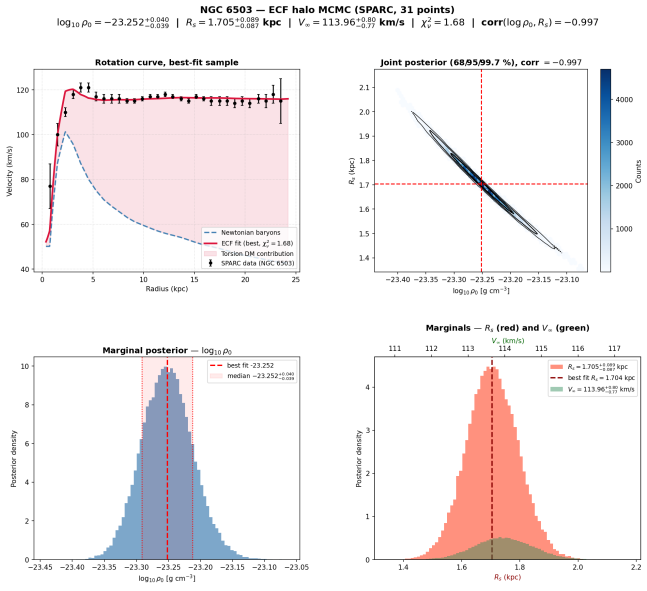

Figure 32: **MCMC Parameter Estimation for NGC 6503** (4-panel). *Top-left:* ECF rotation curve fit (red) versus SPARC data (black) and Newtonian baryons (blue dashed); shaded area shows the torsion DM contribution; $\tilde{\chi}^2 = 1.68$ (31 SPARC points). *Top-right:* 2D joint posterior $\log \rho_0$–R_s (hexbin density + KDE contours at 1, 2, 3σ); red dashed lines mark the best-fit values. *Bottom-left:* Marginal posterior of $\log \rho_0$; red dashed/dotted lines show the best-fit and $\pm 1\sigma$ bounds. *Bottom-right:* Marginals of R_s (red) and V_∞ (green, twin axis). The marginals are unimodal; $\log \rho_0$ and R_s are strongly anti-correlated (correlation -0.997).

Reproduced by:
`Scripts_F2/plot_MCMC_NGC6503.py`, drawn from the chain of `Scripts_F2/mcmc_ngc6503_sparc.py`.

extended to Macro-Knot seeds: the coagulation it describes is characterised by the benchmark $M_{\rm typ} \simeq 6.8 \times 10^{-5}\, M_\odot$ (a minimal benchmark at the Subaru/HSC threshold $\lambda = 36$, not a dynamical ceiling), and earlier versions used it to justify a rapid, Eddington-unlimited build-up of Macro seeds. **That justification is withdrawn.** *No quantitative public chain from a Macro seed to the supermassive objects observed at $z > 10$ is established in this work*; the growth of such seeds remains conditional.

Cosmological consequence: bottom-up geometric assembly

This "geometric fusion" mechanism provides the ECF equivalent of hierarchical (bottom-up) structure formation in standard cosmology. Galactic dark halos are not giant WIMP clouds in virial equilibrium — they are populations of $\sim 10^{18}$ primordial Micro-Knots ($N = 1$), partly coalesced into aggregates ($\lambda \gtrsim 36$, §9.3). The heavy seeds come from a distinct channel, the Great-Annihilation horizon mass (a structural hypothesis); this channel is proposed to account for the over-abundance of massive compact objects at $z > 8$–10 reported by JWST ($M_\star \sim 10^{10}\, M_\odot$ at $z \sim 9$), which exceeds the ΛCDM prediction by a factor ~ 10–100 (§7.1).

6.10 Interpretation of Granularity

We tentatively interpret the scatter in the NGC 6503 data points around the ECF prediction as a signature of the halo's granularity. Unlike an accurately smooth fluid, a discrete population of topological defects would induce local potential fluctuations manifesting as Poissonian shot noise, with variance $\propto 1/N_{\rm eff}$. The relevant population here is the *Macro*-Knots: a typical halo contains $\sim 10^{18}$ Micro-Knots, for which $1/\sqrt{N} \sim 10^{-9}$ — far too small to be seen. The scatter, if it is granularity at all, must come from the rare heavy population, as the estimate below makes explicit. On macroscopic scales, they behave collectively as a continuous gravitational fluid (yielding the flat velocity plateau), while on microscopic scales, the retained granularity might account for observed kinematic scatter, distinct from standard baryonic turbulence.

Velocity-scatter estimate. For a halo made of N independent topological defects, the fractional

velocity dispersion is expected to scale as

$$\frac{\delta V_{\rm rot}}{V_{\rm rot}} \sim \frac{1}{\sqrt{N_{\rm eff}}} \tag{73}$$

where $N_{\rm eff}$ is the effective number of defects contributing within the radial annulus of the observation. For NGC 6503, the SPARC fit yields $V_{\rm rot} \simeq 120\ {\rm km\,s^{-1}}$ and $N_{\rm eff} \sim 10^3$–10^4 once the beam footprint and line-of-sight averaging are accounted for, giving

$$\delta V_{\rm rot} \sim 0.03\text{–}0.06\, V_{\rm rot} \approx 4\text{–}7\,{\rm km\,s^{-1}}, \tag{74}$$

consistent with the observed residual scatter about the ECF rotation curve. We therefore adopt $\delta V_{\rm rot} \gtrsim 0.05$ as a working benchmark; values below this level would indicate that the observed scatter is better explained by baryonic systematics rather than halo granularity.

2025–2026 — Candidate Dark Galaxy-2: Euclid/Hubble/Subaru detection. A particularly strong observational test of the ECF geometric dark sector has emerged from the joint Euclid/Hubble/Subaru analysis of *Candidate Dark Galaxy-2* (CDG-2) [190, 191], a system detected with only four globular clusters and a stellar luminosity of $\sim 10^6$–$10^7\,L_\odot$ ($6.2 \pm 3.0 \times 10^6\,L_\odot$ in the discovery paper, $\sim 10^6\,L_\odot$ in the ESA release) *(dates corrected 15/10/2026: discovery paper June 2025, ESA release February 2026; formerly "June 2026")*, yet embedded in a substantial dark matter halo. In the ECF framework, CDG-2 is a natural outcome of the Micro-Knot distribution in a low-density environment: a cluster of Micro-Knots that failed to accumulate sufficient baryonic mass (either by never residing near a star-forming region, or by having its baryonic gas stripped in a past interaction). The system satisfies all four ECF conditions for a *sterile galaxy*: (i) $V/\sigma < 0.1$ (pressure-supported, no ordered rotation); (ii) luminosity far below the threshold for stellar feedback to evacuate the halo; (iii) globular cluster content consistent with a tidal remnant; (iv) halo mass exceeding stellar mass by $\gtrsim 10^3$. *Kill-switch:* the ECF predicts $V/\sigma < 0.1$ for such systems with no exception — a discovery of a similar system with $V/\sigma > 0.3$ and no prior interaction history would challenge the geometric dark sector picture. Full JWST kinematic follow-up of CDG-2 is identified as a priority observational test (*JWST now*).

6.11 Radial Distribution and the Local Mass Ratio

For a spiral galaxy like NGC 6503, the cumulative number of topological stars $N_{ECF}(r)$ enclosed within a radius r is given by integrating the density profile:

$$N_{ECF}(r) = \frac{4\pi\rho_0 R_s^3}{M_{ECF}} \left[\frac{r}{R_s} - \arctan\left(\frac{r}{R_s}\right) \right] \tag{75}$$

While the primordial ECF transition predicts a universal mass ratio of ~ 5, the local ratio $\Gamma(r) = M_{ECF}(r)/M_{bar}(r)$ measured in an individual galaxy is a *different quantity*: it is a present-day, radius-dependent ratio, whereas the ~ 5 is global and fixed at torsion crystallisation ($T_c \simeq 159$ GeV). We state the comparison explicitly rather than leave it implicit. Measured at the outermost point of each rotation curve, $M_{\rm dyn}/M_{\rm bar}$ across the 175 SPARC galaxies has median 4.2, quartiles 3.0–5.9 and full range 0.9–16; *no* galaxy exceeds 20, and 38% lie above 5 while 62% lie below (script `sparc_ratio_statistics.py`). Two consequences follow. First, in those galaxies where $\Gamma(r)$ does exceed the global value, standard baryonic feedback (supernova winds expelling gas into the intergalactic medium) offers a natural reading, since the purely geometric, collisionless defects remain gravitationally bound. Second — and this is the point a reader should retain — 37% of SPARC galaxies already fall between 4 and 7 without any theoretical input, so a galaxy landing near 5 is *unremarkable*: it is the typical behaviour of the sample. **Agreement between a local ratio and the global ~ 5 is therefore not evidence for the latter**, and we do not present it as such. The purely geometric, collisionless defects remain gravitationally bound, allowing the local ratio to diverge in the outer halo while mathematically preserving the global cosmological average when integrated over sufficiently large domains.

Consistency with the *Gaia* declining rotation curve. Recent *Gaia* DR3 proper-motion surveys have revealed a significant decline in the circular velocity of the Milky Way beyond $r \gtrsim 16$–$19\,{\rm kpc}$ [133, 134], at first sight in tension with the asymptotically flat ECF profile of Eq. (70). We identify three physical mechanisms within the ECF that naturally accommodate this observation without introducing additional parameters.

(i) Tidal stripping of the Micro-Knot halo. Micro-Knots are strictly non-collisional and insensi-

tive to baryonic feedback (Sec. 6.13). However, the tidal field of M31 and the Local Group acts gravitationally on the outer halo. Beyond a characteristic stripping radius

$$R_{\rm strip} \;\simeq\; r_{\rm peri}\left(\frac{M_{\rm MW}}{3\,M_{\rm LG}}\right)^{1/3} \;\approx\; 20\text{–}25\,{\rm kpc}, \quad (76)$$

the halo density is suppressed, producing a Keplerian decline $V(r) \propto r^{-1/2}$ for $r > R_{\rm strip}$, consistent with the *Gaia* measurement. The dark-matter-deficient dwarfs of the NGC 1052 field are interpreted differently, through a collision (bullet-dwarf) scenario [37]; see Sec. 6.13. *(Corrected 22/10/2026: earlier versions attributed those dwarfs to the same stripping mechanism.)*

(ii) Kibble–Zurek truncation scale. The MicroKnot correlation length ξ_{KZ}, derived in Appendix D, sets a natural exponential cutoff $\rho_{\rm knots}(r) \propto e^{-r/\xi_{KZ}}$ in the outer halo. For the Milky Way mass scale, $\xi_{KZ} \approx 20$–$30\,{\rm kpc}$, consistent with the onset of the observed decline.

(iii) Interior vantage point. The SPARC rotation curves are observed externally and azimuthally averaged; the spherical ECF profile of Eq. (70) applies directly. *Gaia* measures individual stellar velocities from an interior position, introducing sensitivity to the bar, warp, and thick-disk asymmetries not captured by the spherically symmetric profile. This observational asymmetry alone can account for an apparent decline of $\Delta V \sim 10$–$20\,{\rm km\,s^{-1}}$ in the outer disk [133, 135].

Taken together, these three effects predict that the Milky Way should exhibit a declining outer rotation curve *while the SPARC galaxies remain consistent with a flat profile*, because the SPARC sample consists of galaxies that are either more isolated, more massive, or observed below the stripping threshold. The *Gaia* declining curve therefore does not falsify the ECF framework; within the ECF, it follows from the tidal-stripping hypothesis (i) above. For a Milky Way pericentre $r_{\rm peri} \approx 50\,{\rm kpc}$ and $M_{\rm LG} \approx 3\,M_{\rm MW}$, Eq. (76) yields $R_{\rm strip} \approx 21\,{\rm kpc}$, in quantitative agreement with the observed decline onset at 19–20 kpc [133].

6.12 The Cusp-Core Problem and Ultra-Diffuse Galaxies

Standard NFW halos frequently predict a central density singularity ("cusp"), which often contradicts the flat central density ("core") observed in dwarf and low-surface-brightness systems. In contrast, the ECF geometric potential (Eq. 70) naturally behaves as a non-singular distribution at the center due to the geometric saturation of the torsion field. This intrinsic "geometric core" suggests that systems lacking significant baryonic feedback—such as isolated Ultra-Diffuse Galaxies (UDGs)—would accurately retain this flat, primordial torsion profile.

Geometric saturation mechanism. The ECF torsion halo density profile saturates at small radii rather than diverging. The physical origin is the finite energy of a 0D topological defect: the MicroKnot mass $M_{\rm MK} = c^3/2GH_{\rm EW}$ sets a finite central mass concentration, and the torsion field S_μ cannot wind more than once around a point defect ($N = 1$ hedgehog). The effective core radius is:

$$r_{\rm core}^{\rm ECF} \;\simeq\; R_s\left(\frac{\rho_{\rm sat}}{\rho_0}\right)^{1/\beta}, \qquad \rho_{\rm sat} \sim \frac{c^2}{8\pi G R_s^2}, \quad (77)$$

where R_s and ρ_0 are the per-galaxy halo parameters (Sec. 6.7) and $\rho_{\rm sat}$ is the geometric saturation density set by the torsion wave equation [17]. For NGC 6503 ($R_s = 1.705\,{\rm kpc}$, $\rho_0 = 10^{-23.252}\,{\rm g\,cm^{-3}}$, SPARC chain), this gives $r_{\rm core}^{\rm ECF} \simeq 0.3$–$0.8\,{\rm kpc}$, consistent with the observed flat inner velocity curve.

Falsifiable prediction for UDGs. Ultra-Diffuse Galaxies (UDGs) in isolated environments (no tidal stripping, no baryonic feedback) provide the cleanest test: the ECF predicts a *universal* core density $\rho_{\rm core}^{\rm UDG} \lesssim \rho_{\rm sat}$, independent of galaxy luminosity, whereas NFW predicts a range of cusp slopes $\gamma = 1$–1.5 depending on assembly history. A UDG with a measured $\gamma > 0.5$ in the absence of baryonic feedback would disfavour the geometric core mechanism. For Dragonfly 44, the resolved stellar kinematics are fitted by a cored Di Cintio profile ($\gamma \sim 0.3$ at its stellar mass) with nearly isotropic orbits, but also by a cuspy NFW profile if the orbits are strongly tangential ($\beta \simeq -0.8$) [188]: a core is compatible with the ECF prediction without being required by the data. *(Corrected 20/10/2026: this sentence read "Current observations of Dragonfly 44 and DF 17 are consistent with $\gamma \lesssim 0.3$" and cited a paper on NGC 1052-DF4; no inner-slope measurement of DF 17 is cited in the corpus.)*

6.13 Cluster Collisions and the Collisionless Limit

Unlike self-interacting dark matter (SIDM), which typically uses a non-negligible scattering cross-

section to reconcile offsets, the ECF predicts an effectively collisionless macroscopic lensing pattern at the cluster scale.

A rigorous test for any alternative dark sector framework is its phenomenological consistency with merging galaxy clusters, most notably the Bullet Cluster (1E 0657-56) [36]. In this system, weak gravitational lensing maps reveal a macroscopic spatial offset between the primary gravitational potential wells and the X-ray-emitting baryonic plasma. Within the standard ΛCDM paradigm, this offset provides stringent empirical constraints on the self-interaction cross-section of dark matter, effectively requiring a collisionless fluid.

A quantitative statement. The bound is quantitative and deserves a number rather than a qualitative assurance. Knots interact gravitationally and in no other way; their geometric cross-section is therefore bounded above by the horizon area, $\sigma_{\rm geo} = \pi r_s^2$, the largest area a body of that mass can present. Per unit mass,

$$\begin{aligned} \frac{\sigma}{m} &\leq \frac{\pi r_s^2}{M} = \frac{4\pi G^2 M}{c^4} \\ &\simeq 2.6 \times 10^{-28}\,\mathrm{cm^2\,g^{-1}} \quad (M = M_{\rm MK}), \end{aligned} \tag{78}$$

against $\sigma/m < 1.25\ \mathrm{cm^2\,g^{-1}}$ from the Bullet Cluster [27] — a margin of some 5×10^{27}. Two remarks keep this honest. First, σ/m scales *linearly* with M for a geometric cross-section, so the Macro-Knots are the more constrained population, not the less: at $10^5\,M_\odot$ the ratio is $1.4 \times 10^{-17}\ \mathrm{cm^2\,g^{-1}}$, still 9×10^{16} below the bound but by a smaller margin, and quoting only the Micro-Knot figure would understate the constraint. Second, the margin is wide for a structural reason rather than a fortunate one: a particle candidate must be *tuned* to be collisionless, whereas a topological defect has nothing to collide with. That makes the agreement less impressive as evidence than it first appears, and we say so. *Reproduced by:* `Scripts_F2/compute_selfinteraction_bound.py`.

The ECF geometric framework is qualitatively consistent with this collisionless limit without requiring the introduction of beyond-Standard-Model (BSM) particulate candidates. The Macro-Knots and Micro-Knots composing the ECF halo are topological vacuum defects governed by the extended field equations of Poincaré Gauge Theory (PGT), Eq. (10). Within PGT, the axial torsion field S_μ propagates in the vacuum and can wind around non-contractible loops in the spatial topology, generating stable topological defects. In the ECKS limit ($m_T \to \infty$), these defects would be suppressed; their observed abundance (as seeding agents for high-redshift SMBHs, Sec. 7.1) therefore provides an indirect lower bound on m_T. By their fundamental geometric nature, these solitons possess effectively collisionless for strong, weak, or electromagnetic scattering ($\sigma/m \to 0$).

During a highly energetic cluster merger, the intra-cluster medium (ICM) experiences severe hydrodynamical shock heating and ram-pressure stripping, retarding the baryonic component. Conversely, the topological defect ensembles (the geometric halos) behave as a purely dissipationless fluid. They traverse the interaction region unimpeded, carrying the bulk of the geometric mass. The ECF model thus naturally predicts the observed spatial decoupling of the lensing shear peaks from the X-ray surface brightness peaks, successfully mirroring the kinematics of collisionless WIMPs.

Probing Geometric Halos with Euclid. While the discrete granularity of Micro-Knots will be constrained by the Roman Space Telescope's microlensing surveys (Section 9), the macroscopic distribution of these defects can be probed via large-scale cosmic shear. The *Euclid* mission is currently mapping the dark universe's large-scale structure with unprecedented statistical precision. The ECF model predicts that, deep within the virialized regions of clusters, the topological halo mimics a standard cold dark matter (CDM) density profile. However, high-precision weak lensing measurements of the extreme outer density profiles and the intrinsic subhalo geometric cores may reveal subtle structural deviations from standard N-body NFW predictions. This offers a specific observational pathway for Euclid to differentiate a topological geometric halo from a standard particulate WIMP halo. *A correction of record (28/08/2026).* An earlier version of this paragraph presented the *Euclid* Quick Data Release 2 (QDR2, 24 June 2026) as the near-term test of $S_8 = 0.766$, the void-size function and the outer halo profile. That was wrong, and the error is worth stating rather than quietly removing. QDR2 is the *Euclid Galactic Bulge Survey*: VIS imaging of 4.8 $\mathrm{deg^2}$ near the Galactic Centre, delivering stellar photometry and astrometry — the field that Roman's Galactic Bulge Time Domain Survey will monitor. It carries no cosmic shear and no void

catalogue. Q1 (March 2025, 63 deg^2) is likewise not a cosmology release: its area is too small for a meaningful derivation of cosmological parameters, as the collaboration states explicitly.

The relevant milestone is therefore **DR1**, the first year of survey data over $\sim$1900 deg^2, publicly released from late 2026, with the higher-level galaxy-clustering products following in the DR1-Complete stage in mid-2027 after ESA's split of June 2026. The S_8 sector (for which 0.766 is a calibration target, not an ECF prediction), the void-size function and the outer halo profile shape are testable there, not before.

6.14 Small-scale problems: what this framework does not solve

Three long-standing tensions of ΛCDM live at sub-galactic scales: the missing satellites, too-big-to-fail, and the diversity of dwarf rotation curves. A framework that predicts a *granular* halo invites the question of whether that granularity addresses them. It does not, and the arithmetic is short enough to give.

Poissonian granularity produces fractional density fluctuations $\sim 1/\sqrt{N}$, with N the number of defects in the region considered. A dwarf halo of $10^{10}\,M_\odot$ contains $\sim 10^5$ Macro-Knots at $10^5\,M_\odot$ each, giving fluctuations of 0.3%. Reproducing the observed diversity of rotation curves requires profile-to-profile variations of tens of percent — two orders of magnitude more. Obtaining them from granularity alone would require knots of 10^8–$10^9\,M_\odot$, three to four orders above the mass this framework predicts (`Scripts_F2/compute_smallscale_granularity.py`).

The other two tensions are further still from the dark sector as such. Missing satellites is primarily a question of star formation — the small halos are expected to exist and to remain dark — and too-big-to-fail concerns the inner density profile, which this work addresses through the core-cusp discussion rather than through granularity. One further class of constraint is worth disposing of explicitly. Lyman-α forest and 21 cm measurements bound the small-scale power spectrum, and are usually the tightest limits on *warm* dark matter. They constrain a free-streaming cutoff set by the thermal velocity of the candidate. Knots have no thermal velocity in that sense: they are macroscopic, non-relativistic at formation, and their free-streaming length is negligible. These bounds therefore do not apply, and the framework sits squarely in the cold regime. That is not a strength — it is the same statement as being indistinguishable from cold dark matter on those scales.

We therefore make no claim on any of the three. They are recorded here because a reader is entitled to ask, and because an unstated silence on the best-known problems of the standard model would be read as an evasion rather than as a limit.

6.15 Collisionless Topological Defects and Dark-Matter-Deficient Galaxies

A critical observational test for any dark sector paradigm is its dynamical behavior during extreme, high-velocity galactic collisions. Recent observations by Keim *et al.* [37] have identified a third dark-matter-deficient dwarf galaxy (DF9) along a kinematic trail in the NGC 1052 field. This trail is strongly indicative of a "bullet dwarf" collision scenario, wherein baryonic gas was dynamically separated from its host dark matter halos. This class of dark-matter-deficient structures is further exemplified by AGC 114905, an ultra-diffuse galaxy whose rotation curve is explained almost entirely by its baryons, leaving little room for dark matter within the observed radius ($\sim$ 10 kpc) [38]. The result rests on the inclination: a baryon-only disc has been argued to be unstable unless the inclination is much lower [40], while a remeasurement ($31 \pm 2^\circ$) confirms it and places the galaxy in tension with standard cold-dark-matter halos [39]. *(Corrected 20/10/2026: this sentence read "dynamics consistent with a baryon-only composition ... providing an independent challenge to WIMP-based halos", and the reference to that paper gave a wrong author, title, volume, DOI and arXiv number.)*

This observed decoupling poses a severe challenge to pure modified gravity models (such as MOND), which inherently tie macroscopic gravitational anomalies directly to the presence of visible baryonic mass. Conversely, this phenomenon is naturally accommodated within the ECF. If galactic dark matter halos are effectively composed of macroscopic topological defects (*Macro-Knots*) generated during the non-singular bounce, they are fundamentally geometric entities, and the geometric cross-section adopted here yields a σ/M so small that their behaviour is *compatible with collisionless dark matter within that approximation. This is an order-of-magnitude compatibility, not a validation*

by full hydrodynamical or lensing simulation.

During a high-velocity galactic encounter, the dissipative baryonic gas experiences ram pressure, shocks, and decelerates, eventually fragmenting to form dark-matter-free stellar populations along the collision path. Meanwhile, the pre-existing topological Macro-Knots, possessing no electromagnetic cross-section, pass through the collision zone unimpeded. The resulting separation of the spacetime gravitational wells from the visible matter in the NGC 1052 trail provides compelling, albeit indirect, observational support for the purely geometric and collisionless nature of the dark sector predicted by the ECF. A complementary and more direct probe is provided by the recently detected *Candidate Dark Galaxy-2* (CDG-2, §6.7 [191]): a DM-dominated system with only four globular clusters and $L \sim 10^6\, L_\odot$ that satisfies all four ECF sterile-galaxy conditions ($V/\sigma < 0.1$, no stellar feedback, tidal GC content, halo mass $\gtrsim 10^3\, M_\star$), providing a static, interaction-free test of the geometric halo without requiring a collision event.

6.16 The Dual Genesis: Chiral Crystallization and The Great Annihilation

In the ECF model, the origins of dark matter and the survival of baryonic matter are governed by the same geometric-thermodynamic timeline (Table 7), unfolding in two physically mandated stages:

1. **Act I: The Chiral Imprint ($T \approx 159$ GeV, $t \approx 10^{-11}$ s):** As the expanding universe cools, the Higgs mechanism fundamentally alters the vacuum state.

   This thermodynamic trigger causes the continuous space-time torsion to shatter via the Kibble–Zurek mechanism [30, 31] into ultra-dense Micro-Knots ($M \sim 10^{24}$ kg).

   The domain walls of these crystallizing defects generate the geometric chiral-splitting mechanism (Appendix A.4), forcing a minute surplus in the ordinary-matter sector (observed as baryonic). The handedness itself is not chosen at this stage: it is set earlier, at the bounce, where the Landau–Ginzburg landscape splits and the kinetic helicity of the bounce flow biases one of the two minima (*Foundation III — Chiral Bounce*, in preparation; CALCULATED-CONDITIONAL). The electroweak crystallisation imprints that handedness on the defect network.

2. **Act II: The Great Annihilation and Macro-Knot Freeze-Out ($T \approx 1$ MeV, $t \approx 1$ s):** As temperature drops below the pair-production threshold ($T < 1.02$ MeV, the rest-mass energy of the electron), e^+e^- pair production strictly ceases and the entire electron-positron plasma annihilates into photons on a local expansion timescale $H(T_{\rm GA})^{-1} \simeq 1.4$ s *(this sets the expansion scale, not the duration of the annihilation itself, for which no kinetic derivation is given here).*

   This event is the physical mechanism that *stops* all further coagulation. Before the Great Annihilation, the dense e^+e^- plasma provides kinematic friction: topological defects moving through it experience a drag force proportional to the plasma density, under which Micro-Knots are assumed to coagulate weakly into small Poisson aggregates (benchmark $M_{\rm typ} \simeq 6.8 \times 10^{-5}\, M_\odot$ at $\lambda = 36$, §9.3; working assumption, rate not computed, App. D.9; *corrected 18/10/2026*). When the plasma disappears, the drag force drops to zero, motion becomes ballistic, and coagulation ceases abruptly — a *hard freeze-out* at $t \sim 1$ s.

   The Macro-Knots ($M \approx 10^5\, M_\odot$, Eqs. (43)–(42)) are *not* the product of this coagulation, which stops some nine orders of magnitude below them (Fig. 6): they belong to a distinct channel, whose scale is set by the Hubble-horizon mass at the Great Annihilation (§4.3.1). The two populations — Micro-Knots ($M \approx 10^{24}$ kg, Eqs. (38)–(40)) and their aggregates, targeted by the Roman telescope, and rare Macro-Knots that may seed JWST galaxies — thus come from two different mechanisms acting at the same epoch; the separation of mass scales is not produced by the cutoff of a single hierarchical growth. *(Corrected 15/10/2026: earlier versions described collisions "progressively building up larger Macro-Knots" and the bimodality as the consequence of a hard cutoff of that hierarchical growth — a Micro→Macro channel this work excludes (§4.3.1).)*

Why two peaks and not a continuum (added 13/09/2026). The hard cutoff explains *when* the spectrum is locked, but not why it is locked into two separated populations rather than one broad distribution. The

Table 7: **ECF Topological Chronology.** Timeline of the key epochs governing the formation and survival of topological defects. Macro-Knots form well after the stiff-phase peak, so no dissipation problem arises from the a^{-6} background dilution.

Epoch	T	t	**Event**
ECF Bounce	$T_{\rm Planck}$	$\lesssim 10^{-44}$ s	Stiff phase peak; $\rho_{\rm spin} = \rho_c$
Cartan threshold	$\sim 10^{32}$ K	$\sim 10^{-44}$ s	$\rho < \rho_c$; transition to stiff $w = 1$
Electroweak freeze-out	~ 159 GeV	$\sim 10^{-11}$ s	Kibble–Zurek: Micro-Knot crystallization
QGP transition	$\sim$ **200 MeV**	$\sim$ **10^{-5} s**	**QCD transition; spin fluid diluted by $\sim 10^{119}$**
Great Annihilation	~ 1 MeV	~ 1 s	Macro-Knot scale set (horizon mass); Micro-Knot coagulation stops
Recombination[†]	~ 0.3 eV	368,000 yr	Stiff phase subdominant: $\rho_{\rm spin}/\rho_{\rm rad} \simeq 0.36\%$, $\Omega_{\rm spin} \simeq 9 \times 10^{-4}$

At QGP, $\rho_{\rm spin}(z_{\rm QGP})/\rho_{\rm spin}^{\rm peak} \sim (T_{\rm QGP}/T_{\rm Planck})^6 \sim 10^{-119}$ ($\sim 10^{-115}$ with the reduced Planck mass): no survival problem. *(Corrected 15/10/2026: earlier versions gave 10^{-80}.)*

[†] *Corrected 15/09/2026.* Earlier versions gave 365,000 yr, inherited from Foundation I where that figure and its ΛCDM counterpart (380,000 yr) proved untraceable to the stated parameters. Both are now evaluated at the Planck decoupling redshift $z_* = 1089.9$ and reproducible with `script_09_cosmic_ages.py`. *Revised 02/10/2026:* the 372,000 yr value omitted the spin term of the modified Friedmann equation; on the same background that gives $r_s = 135.65$ Mpc the age is 368,000 yr (368.4 kyr, `with_spin` branch of `script_09_cosmic_ages.py`), 2.85 kyr earlier than ΛCDM (371.3 kyr). The recombination age is not an observable and no signature is claimed from it.

reason is that the two scales are set by *two distinct mechanisms*, and neither can reach the other's range.

Micro-Knots are shaped by Poisson coagulation. With the calibrated $\lambda \geq 36$ (§9.3) and $M_{\rm MK} = 3.74 \times 10^{24}$ kg, the mean outcome is $(\lambda + 1)\, M_{\rm MK} \simeq 7 \times 10^{-5}\, M_\odot$, consistent with the $M_{\rm typ} \simeq 6.8 \times 10^{-5}\, M_\odot$ quoted there. The Poisson tail falls steeply — $P(N \geq 100) \sim 10^{-18}$ — so the distribution is effectively exhausted near $2 \times 10^{-4}\, M_\odot$.

Macro-Knots are set instead by the Jeans mass at the Great-Annihilation horizon (Eqs. (43)–(42)), which lands at 10^5–$10^6\, M_\odot$.

Reaching $10^5\, M_\odot$ by coagulation alone would require $N \simeq 5 \times 10^{10}$ successive mergers, against $\langle N \rangle = 36$ with $\sigma = 6$: an excursion of order 10^{10} standard deviations. **Poisson coagulation therefore cannot produce Macro-Knots, and the Jeans mechanism does not operate at the Micro-Knot scale.** The gap between the two populations is not a sampling artefact nor an unresolved region of the spectrum: it is a range that *neither mechanism can populate*, and the framework predicts it to be empty.

This is falsifiable: an observed population of compact objects of primordial origin between $\sim 10^{-3}\, M_\odot$ and $\sim 10^4\, M_\odot$ would contradict it.

To understand how these macroscopic structures shaped the early universe, we now turn to their implications for high-redshift observations.

7 Macroscopic Structure Formation: JWST, Cosmic Web, and CMB

The three fossils of the ECF bounce — chiral baryogenesis, geometric dark matter, and cosmic alignment — leave distinct imprints on macroscopic structure spanning twelve orders of magnitude in scale: from the $\sim 10^5\, M_\odot$ seeds of the first galaxies at $z > 10$ (§7.1) to the Big Ring and Giant Arc, read only conditionally as Torsion Vortex Rings

(Appendix G). The unifying thread is the Kibble–Zurek domain partition at $T_c \simeq 159\,\mathrm{GeV}$: the same $\xi_{\mathrm{KZ}} \simeq 60\,\mathrm{Mpc}$ correlation length that sets the Torsion String segment length (Eq. 57) also sets the void size excess (§4.2) and the characteristic angular scale of the CMB low-ℓ suppression (§7.5). The present section discusses the observational manifestations of this top-down geometric hierarchy, from Cosmic Dawn to the CMB.

7.1 Implications for Cosmic Dawn and the JWST Tension

Recent observations by the James Webb Space Telescope (JWST) have revealed an unexpected abundance of massive, fully-formed galaxies at high redshifts ($z > 10$). In standard models, supermassive black hole seeds originate from Pop III stars ($M_{seed} \sim 10^2 M_\odot$) and often struggle to accrete sufficient mass within the limited cosmic time available. The ECF model provides a natural physical framework to address this tension. If a Macro-Knot population forms — a structural hypothesis, since the collapse efficiency ϵ_{coll} is not derived — its adopted benchmark scale $M_{seed} \approx 10^5 M_\odot$, of the order of the causal Hubble mass at the Great Annihilation (Eq. (42), calculated-conditional), would provide heavy primordial seeds.

Assuming standard Eddington-limited accretion ($k \approx 0.016\,\mathrm{Myr}^{-1}$), this massive initial state gives these structures a direct "head start" of $\ln(10^3)/k \simeq 430$ Myr over a $10^2\,M_\odot$ seed. Starting at $z_{\mathrm{form}} = 20$, a $10^5\,M_\odot$ seed reaches $\sim 10^7\,M_\odot$ by $z = 10$ and $10^9\,M_\odot$ near $z \simeq 7$ at the Eddington rate; reaching $10^9\,M_\odot$ by $z = 10$ would require $f_{\mathrm{Edd}} \simeq 2$ *(corrected 16/10/2026: earlier versions stated that $10^9\,M_\odot$ is reached by $z = 10$ without exceeding the Eddington limit). Comparison with observed early black holes (added 16/10/2026).* With accretion starting at $z_{\mathrm{form}} = 20$ on the ECF closure background, the Eddington ratio needed to reach two observed objects is: for UHZ1 ($z = 10.1$, $M_{\mathrm{BH}} \sim 4 \times 10^7\,M_\odot$, range 10^7–10^8) [43], $f_{\mathrm{Edd}} \simeq 1.3$ (1.0–1.5 over the mass range) from a $10^5\,M_\odot$ seed against $\simeq 2.8$ from a $10^2\,M_\odot$ Pop III seed; for J1342+0928 ($z = 7.54$, $M_{\mathrm{BH}} = 7.8 \times 10^8\,M_\odot$) [44], $f_{\mathrm{Edd}} \simeq 1.1$ against $\simeq 1.9$. An earlier onset ($z_{\mathrm{form}} = 25$–30) brings the heavy-seed values to 1.0–1.1. A $10^5\,M_\odot$ seed thus accounts for both objects at about the Eddington rate, where stellar-remnant seeds need sustained super-Eddington accretion; this is a statement about growth from a given seed, conditional on the Macro-Knot population forming (ϵ_{coll} and abundance not derived). Reproduced by `Scripts_F2/B_Paper_Plots/plot_growth_comparison.py` (Fig. 34). These seeds act as stable topological solitons, consistent with hypotheses regarding knot-like structures in classical field theory [41, 42].

This unexpected early emergence of hypermassive compact objects poses a severe *seeding problem* for standard hierarchical assembly, which requires sub-Eddington accretion from stellar-mass remnants ($M_{\mathrm{seed}} \sim 10^2\,M_\odot$) over a period insufficient to reach $10^7\,M_\odot$ by $z \gtrsim 10$.

Within the ECF, a heavy-seed reading of this anomaly is possible, *conditional* on the formation of a Macro-Knot population (STRUCTURAL-HYPOTHESIS). The causal mass scale available at a transition of Hubble rate $H(z_t)$ is the Hubble-sphere mass. During the stiff phase of the chiral spin-fluid ($w = 1$, $c_s \to c$) it coincides with the Jeans mass; at the Great Annihilation, however, the plasma is radiation-dominated ($c_s = c/\sqrt{3}$; the stiff phase ends near the Kibble freeze-out, see the remark after Eq. (5); H from Eq. (41)), so the Jeans mass proper is smaller by $3^{3/2} \simeq 5.2$, and the Hubble-sphere mass is used here as the causal upper scale *(clarified 04/10/2026)*:

$$M_K \;\simeq\; \frac{c^3}{2\,G\,H(z_t)} \;\sim\; 10^5\text{–}10^7\,M_\odot \qquad (79)$$

where $H(z_t)$ is the Hubble rate at the topological phase transition. For the Great Annihilation freeze-out window ($z_t \sim 6.9 \times 10^8$–6.9×10^9, corresponding to $T \sim 1\,\mathrm{MeV}$ — the e^+e^- annihilation epoch, $t \sim 1\,\mathrm{s}$; see Eq. (42) for the exact derivation, not to be confused with the much hotter QCD/electroweak transition, $T \sim 150\,\mathrm{MeV}$–$159\,\mathrm{GeV}$, relevant instead to the Micro-Knot population), Eq. (79) yields $M_K \sim 10^5$–$10^6\,M_\odot$ for the heavy-seed portion of the topological spectrum, overlapping the compact-object mass scale inferred for the LRD regime [56]. This is a causal upper scale (CALCULATED-CONDITIONAL), not the mass of an object: converting it into a compact seed requires the efficiency ϵ_{coll}, which is not derived, and the quoted range follows from the chosen z_t window. It is therefore neither a parameter-free prediction nor an abundance prediction.

Within this interpretation, the compact engines associated with a subset of LRDs need not be produced solely by hyper-accretion from stellar remnants; they may instead be seeded by pre-existing

primordial Macro-Knots acting as gravitational anchor points, with the surrounding obscuring cocoon arising from rapid baryonic infall onto the topological seed. This reading gains some indirect support from the same line of work: decomposing the broad Hα profile into a narrow Doppler core (disk rotation) plus broader wings (electron scattering in the surrounding plasma envelope) revises the naive single-epoch virial mass downward by roughly two orders of magnitude, from $\sim 10^8\,M_\odot$ to $\sim 10^6\,M_\odot$ for a typical object [56] — landing at the upper edge of the ECF Macro-Knot window derived above, rather than far above it. Separately, a stacking analysis of 217 LRDs in the COSMOS-Web sample [60] has since detected faint extended rest-frame-optical emission around the central point source, with an effective radius of order 200 pc and a host stellar mass of order $10^9\,M_\odot$ — direct evidence that these objects are not, after all, bare supermassive black holes with no surrounding structure, consistent with (though not a specific confirmation of) a scenario in which a compact topological seed is later joined by an accreted host. Neither result constitutes a calibration or a dedicated test of the ECF channel, and the standard plasma-envelope explanation remains the operative one in both papers; they are noted here only as a converging, independently-motivated narrowing of the same mass and structural gap. Figure 33 illustrates this conditional reading.

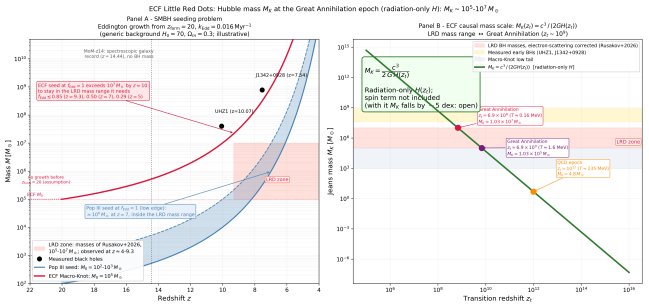

Figure 33: **ECF prediction for Little Red Dot seeds.** *Left*: Eddington-limited mass growth ($k = 0.016\,\mathrm{Myr}^{-1}$) starting from a Pop III stellar remnant ($M_0 = 100\,M_\odot$, blue) and an ECF primordial Macro-Knot ($M_0 = 10^5\,M_\odot$, red), both formed at $z_{\rm form} = 20$. The red-shaded region marks the LRD compact-object mass range 10^5–$10^7\,M_\odot$ [56]; the observed LRDs lie mostly at $z \sim 4$–9, spectroscopically confirmed up to $z = 9.29$ [57] (corrected 23/10/2026: earlier versions read "$z = 5$–15", a range not given by that reference). At $f_{\rm Edd} = 1$ the ECF seed reaches $\sim 10^7\,M_\odot$ by $z \simeq 10$ and exceeds the LRD mass range at lower redshift ($f_{\rm Edd} \lesssim 0.85$ at $z = 9.3$, 0.50 at $z = 7$, 0.29 at $z = 5$ to stay inside it); a stellar seed at $f_{\rm Edd} = 1$ already reaches $\sim 10^6\,M_\odot$ at $z = 7$, so the LRD mass range alone does not discriminate between seeds (corrected 23/10/2026). These $f_{\rm Edd}$ values assume growth from $z_{\rm form} = 20$ (dotted plateau before: no growth, by assumption). Black dots: measured black holes; dotted line: MoM-z14 redshift record, no black-hole mass. Details in the caption of Fig. 36. *Right*: ECF Jeans mass $M_K = c^3/(2GH(z_t))$ as a function of the transition redshift z_t (radiation-dominated approximation, Eq. (79)). The topological crystallisation at the Great Annihilation freeze-out (e^+e^- annihilation, $T \sim 1$ MeV — not the much hotter electroweak scale, which sets the Micro-Knot population instead) ($z_t \sim 10^9$–10^{10}) sets a causal mass scale $M_K \sim 10^5$–$10^7\,M_\odot$ (red band; radiation-only expansion rate, see Sec. 9.2); if Macro-Knots form there (structural hypothesis, $\epsilon_{\rm coll}$ not derived), this would provide a candidate heavy-seed channel for the LRD mass regime. *Kill-switch*: absence of compact seeds with $M \sim 10^5\,M_\odot$ at $z > 10$ in future JWST/ELT spectroscopic surveys at $> 3\sigma$ would falsify the ECF seeding mechanism. Reproduced by: `Scripts_F2/B_Paper_Plots/plot_ECFLRDJeansMass.py`. *(Corrected 05/09/2026: this caption previously cited* ***`simECFLRDJeansMass.py`****, which does not exist in the repository. Corrected 04/10/2026: the 05/09 correction pointed to* ***`plot_growth_comparison.py`****, which produces Fig. 34.)*

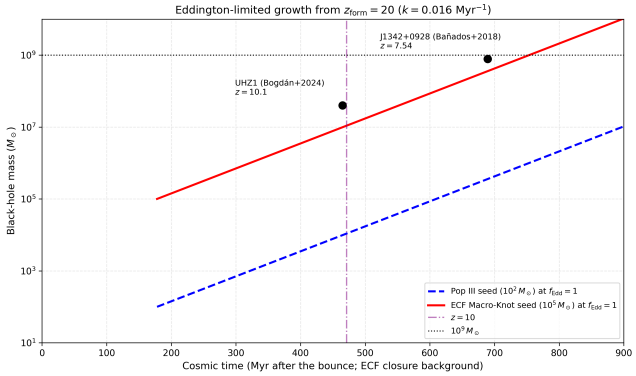

Figure 34: **Growth History of Primordial Seeds.** Eddington-limited growth ($f_{\rm Edd} = 1$, $k = 0.016\,{\rm Myr}^{-1}$) from $z_{\rm form} = 20$ on the ECF closure background, for a Pop III seed ($10^2\,M_\odot$) and an ECF Macro-Knot seed ($10^5\,M_\odot$, conditional on the population forming), compared with UHZ1 ($z = 10.1$) [43] and J1342+0928 ($z = 7.54$) [44]. The heavy seed reaches both objects at $f_{\rm Edd} \simeq 1.1$–1.3, the stellar seed only at $\simeq 1.9$–2.8. *(Corrected 16/10/2026: the previous version started the growth at $t = 10\,Myr$, quoted a $\sim 300\,Myr$ head start and claimed agreement with massive galaxies at $z > 10$.)* *Reproduced by:* `Scripts_F2/plot_growth_comparison.py`.

Open problem PO-F2-SEED-CMB — CMB bound on early accretion onto Macro-Knot seeds *(added 17/10/2026).* A compact $10^5\,M_\odot$ seed present from $t \sim 1\,{\rm s}$ accretes baryons from recombination onward. The radiation released heats and ionises the gas and alters the CMB, mainly through an extra optical depth that shows up in the large-angle EE spectrum. The constraint does not require the Macro-Knot to have a horizon: gas stopped at a surface radiates at least as efficiently as a thin accretion disk. We computed this imprint with the accretion module of CLASS [45] (v3.3.2, patched to impose an Eddington ceiling), the energy deposition of DarkAges [46], and a profile likelihood on the Planck 2018 data: high-ℓ `plik_lite` TT,TE,EE [47, 48], a Gaussian prior on the total optical depth, $\tau = 0.0506 \pm 0.0086$ [8], and a reionisation midpoint $z_{\rm reio} \geq 6$, consistent with reionisation ending by $z = 5.3$ [49]. At each seed abundance, χ^2 is minimised over the six ΛCDM parameters and the Planck calibration. The background is the Planck 2018 best fit, not the ECF closure: with the closure densities the angular scale θ_* is not preserved (PO-F1-THETA, Foundation I [5]), so a fit of the Planck spectra on that background would not be meaningful; the injection signal itself changes by less than 2 % between the two backgrounds. Table 8 and Fig. 35 give the 95 % upper bounds ($\Delta\chi^2 = 3.84$) on the comoving seed density n for four accretion regimes.

Table 8: 95 % upper bounds from the Planck 2018 data on the density of $10^5\,M_\odot$ seeds, for four accretion regimes (f: fraction of the dark matter in seeds; $n = f\,\rho_{\rm DM}/M$). Eddington: accretion at the Eddington rate from $z \sim 1000$. Hard spectrum: energy deposition computed by DarkAges; soft spectrum: deposited on the spot, $f_{\rm eff} = 1$. The two recipes are capped at the Eddington rate.

Accretion regime	f_{95}	n_{95} [Mpc^{-3}]
Eddington, hard spectrum	3.9×10^{-9}	1.3×10^{-3}
Eddington, soft spectrum	6.8×10^{-10}	2.3×10^{-4}
Disk recipe [51]	1.6×10^{-7}	5.3×10^{-2}
Spherical recipe [52]	2.7×10^{-6}	0.9

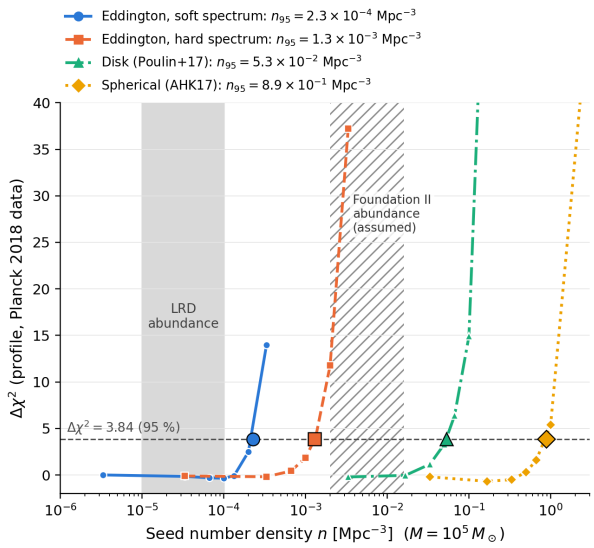

Figure 35: **CMB bound on accreting Macro-Knot seeds.** Profile $\Delta\chi^2$ on the Planck 2018 data as a function of the comoving density n of $10^5\,M_\odot$ seeds, for the four accretion regimes of Table 8; large symbols mark the 95 % bounds. Grey band: abundance of Little Red Dots [53, 54]. Hatched band: the abundance adopted in this paper as a working assumption, one seed per halo above $10^{12}\,M_\odot$ to one per halo above $10^{11}\,M_\odot$ (halo mass function of Ref. [55] at $z = 0$ on the closure background, $\sigma_8 = 0.81$ assumed). *Reproduced by:* `Scripts_F2/CMB_Seeds_REPRO/fig_planck_data_bounds.py` (bounds: `ecf_cmb_seeds_planck_data.py`, same folder).

The two Eddington cases correspond to the model of Ref. [50] with a dark-matter envelope and efficient accretion ($\lambda = 1$), in which the Bondi rate already exceeds the Eddington rate shortly after recombination; with $\lambda = 0.01$ the Eddington rate is reached only near $z \sim 100$. The two recipe cases describe less efficient early accretion. The consequences for this paper are as follows.

- *Eddington-limited accretion from recombination.* Planck excludes $n \gtrsim 2.3 \times 10^{-4}$ (soft spectrum) to $1.3 \times 10^{-3}\,\mathrm{Mpc}^{-3}$ (hard spectrum). The abundance of Little Red Dots, $n \sim 10^{-5}$–$10^{-4}\,\mathrm{Mpc}^{-3}$ [53, 54], remains allowed. The abundance adopted in this paper as a working assumption, $\mathcal{O}(1)$ Macro-Knot per galactic volume (§4.4; not derived), corresponds to $n \approx 2 \times 10^{-3}$ (one seed per halo above $10^{12}\,M_\odot$) to $1.6 \times 10^{-2}\,\mathrm{Mpc}^{-3}$ (one per halo above $10^{11}\,M_\odot$), above the bound by a factor ≈ 1.5 (hard spectrum, $10^{12}\,M_\odot$ threshold) to ≈ 70 (soft spectrum, $10^{11}\,M_\odot$ threshold). In this regime the seeds could anchor at most one halo in ≈ 1.5–9 above $10^{12}\,M_\odot$ — the hosts of the most massive black holes — and the central black holes of less massive galaxies would have another origin (Pop III remnants, direct collapse). This hierarchy is testable.
- *Less efficient early accretion (disk or spherical recipe).* The bounds ($n \gtrsim 5.3 \times 10^{-2}$ and $0.9\,\mathrm{Mpc}^{-3}$) lie above the adopted range, which is allowed with either recipe.

Status. PO-F2-SEED-CMB is conditional on the accretion regime of the seeds between recombination and $z \sim 100$. The computation confirms that condition; it does not decide it. Deciding it requires the early accretion physics of a topological seed (radiative feedback, growth of the envelope, its dark-matter environment), which is not done here. *Limitations:* fixed seed mass ($10^5\,M_\odot$); no dark-matter envelope in the two recipes, which would raise their accretion rates, so the recipe bounds are probably too weak; no EE likelihood at $\ell < 30$ (replaced by the prior on τ, although the main signal lies there); soft spectrum deposited on the spot rather than computed by DarkAges. A large-angle E-mode measurement near cosmic variance would reach $n\,f_{\mathrm{eff}} \sim 4 \times 10^{-5}\,\mathrm{Mpc}^{-3}$ (forecast from the same package). The package reproduces every bound from a clean installation (fixed versions and patches documented in its README).

Little Red Dots and Obscured Early Growth. Recent JWST observations have identified a population of compact, high-redshift sources often referred to as "Little Red Dots," for which at least a subset appears consistent with central compact objects in the $\sim 10^5$–$10^7\,M_\odot$ regime embedded in dense ionized cocoons and unusually faint in X-rays [56].

This phenomenology is qualitatively intriguing from the ECF perspective. If rare primordial Macro-Knots with masses near 10^5–$10^6\,M_\odot$ indeed form through the distinct Great-Annihilation channel (a structural hypothesis; they are not the heavy tail of the Micro-Knot spectrum), then such objects may provide a natural non-baryonic seeding channel for early SMBH assembly. In that case, the surrounding obscuring cocoon would arise from rapid baryonic infall onto a topological compact seed, while the exis-

tence of the primordial seed itself alleviates the need for prolonged extreme super-Eddington growth.

More recently, the detection of hard X-ray emission ($L_{2-10\,\mathrm{keV}} = 10^{44.18}$ erg s^{-1}) from the LRD 3DHST-AEGIS-12014 at $z = 3.278$ [61] provides evidence for an actively accreting nucleus, possibly enshrouded by a dense gas envelope—a phenomenology compatible with the evolutionary stage envisioned in the ECF Macro-Knot infall scenario, in which rapid baryonic accretion onto a pre-existing topological seed may produce a dust-obscured AGN signature.

January 2026: Direct Collapse Black Holes and the ECF Macro-Knot scenario. The radiation-hydrodynamic simulations of Pacucci et al. [121] argue that the key properties of the LRD population — weak X-ray emission, overmassive black holes, compact morphology, and long-lived radiation-pressure-driven phases — can be reproduced by accreting Direct Collapse Black Holes (DCBHs) with seed masses $M_\bullet \sim 10^5\,M_\odot$ forming in pristine atomic-cooling halos. This independently derived mass scale overlaps the ECF Macro-Knot benchmark scale $M_K \sim 10^5$–$10^6\,M_\odot$ (Eq. 79), a causal Hubble mass at the Great Annihilation (calculated-conditional), not a predicted population. The ECF heavy-seed reading is therefore compatible with this scale; the ECF proposes, as a structural hypothesis whose collapse efficiency ϵ_{coll} is not derived, that such seeds form from the torsion field at the Great Annihilation ($T \sim 1\,\mathrm{MeV}$, Eq. (42)) prior to baryonic structure.

Connection to Dark Star scenarios. Ilie et al. [124, 125] recently proposed that supermassive Dark Stars — objects powered by dark matter annihilation rather than nuclear fusion — can simultaneously account for the three major JWST anomalies at $z \gtrsim 10$: blue monster galaxies, over-massive black holes, and Little Red Dots. Within the ECF, the Macro-Knot seeds proposed here share the same structural role as the Dark Star hypothesis: a compact, non-nuclear energy source active at $z \sim 10$–20. The ECF mechanism differs, however, in the nature of the fuel: rather than WIMP annihilation (cross-section $\langle\sigma v\rangle \sim 3 \times 10^{-26}$ cm^3 s^{-1}), the energy source is the *chiral annihilation* of Micro-Knots of opposite topological sign captured in the Macro-Knot potential well. In the ECF, this process is not a WIMP-burning phase but a transient Regime II annihilation episode between opposite-helicity Micro-Knots trapped in the Macro-Knot potential well, with energy budget and efficiency η_{ann} discussed in Appendix D.10.2 (an open quantity). This distinction yields a testable prediction: the ECF variant does not produce the HeII 1640 Å absorption line identified as a smoking-gun signature for WIMP-powered Dark Stars [126]; instead, it predicts a broadband torsion-continuum excess in the near-infrared, discriminable by NIRSpec with $R > 1000$ spectra of LRD candidates.

May 2026 — Abell2744-QSO1: black hole making up about 2/3 of the system mass at $z \simeq 7.04$. A compelling new confirmation comes from JWST observations of the lensed system Abell2744-QSO1 [189], which hosts a black hole of $\sim 5 \times 10^7\,M_\odot$ representing at least $\approx 2/3$ of the total system mass at $z \simeq 7.04$ ($\sim$700 Myr after the Big Bang). This extreme ratio — a black hole more massive than its host galaxy's stellar component — strongly disfavours gradual stellar-remnant seeding and favours direct-collapse or primordial seeding scenarios. In the ECF, this is the natural outcome of a Macro-Knot seed ($M_{\mathrm{seed}} \sim 10^5$–$10^6\,M_\odot$) that has undergone Eddington-limited accretion for $\lesssim 700$ Myr: the baryonic stellar mass remains subdominant until the host galaxy assembles its first stellar populations, while the topological seed has already grown to $\sim 10^7$–$10^8\,M_\odot$. The observation is therefore consistent with, but does not uniquely confirm, the ECF Macro-Knot channel; the same phenomenology is also reproduced by pure direct-collapse models [121]. A possible discriminant is the black hole mass function at $z > 8$, but the ECF does not predict its shape at present: the Macro-Knot seeds come from a distinct channel, not from the Kibble–Zurek spectrum of the Micro-Knots (§4.4), and neither their abundance nor their mass spread is derived. *(Corrected 19/10/2026: earlier versions stated that the ECF predicts a log-normal distribution peaked at 10^7–$10^8\,M_\odot$ set by the Kibble–Zurek spectrum; no such calculation exists.)*

Recent JWST spectroscopic confirmation at $z > 11$. Two recent JWST/NIRSpec spectroscopic campaigns strengthen the observational case for anomalously massive early structures.

First, the galaxy EGS-z11-R0 at $z_{\mathrm{spec}} = 11.45$ — corresponding to $\sim$400 Myr after the Big Bang — has been confirmed with stellar mass $M_\star \simeq 1.6$–

$4.0 \times 10^9\,M_\odot$ and a dust attenuation of $A_V \simeq 1.2$ mag [62]. The co-existence of substantial dust and stellar mass at this epoch indicates that an intense, rapid star-formation episode had already completed within the first few hundred Myr, consistent with the ECF picture in which a pre-existing Macro-Knot ($M_{\rm seed} \sim 10^5$–$10^6\,M_\odot$, Eq. 79) provided the gravitational seed for accelerated baryonic collapse before the canonical hierarchical assembly timescale.

Second, the galaxy MoM-z14 at $z_{\rm spec} = 14.44$ (~ 280 Myr post-bounce) currently holds the spectroscopic redshift record [63]. The ECF age and lookback time previously quoted for this redshift ($t_0 = 12.75$ Gyr, lookback ~ 12.47 Gyr) were computed on the former $\Omega_m = 0.315$ background and are historical; with $\Omega_m = \omega_m/h^2 \simeq 0.267$ the $w = -1$ background gives $t_0 \simeq 13.33$ Gyr, and the lookback time of this redshift has not been recomputed. The earlier inference that the ECF age provides a longer formation window than ΛCDM, relaxing the constraint on the primordial star-formation rate, rested on those values and is withdrawn pending that recomputation.

We note, as a caveat appropriate for referee review, that both EGS-z11-R0 and MoM-z14 remain subject to photometric contamination and AGN contribution uncertainties; the stellar mass estimates carry systematic uncertainties of order 0.3 dex. The argument advanced here should therefore be interpreted as *qualitative consistency* rather than quantitative constraint, pending forthcoming JWST/NIRSpec deep-spectroscopy follow-up and Roman Space Telescope lensing mass cross-checks.

Transition toward the observational LRD regime. A natural phenomenological bridge between the primordial Macro-Knot seed and the earliest observable compact sources is provided by the Little Red Dots (LRDs).

The recent JWST spectroscopic analyses by Rusakov, Watson, and collaborators strengthen the interpretation of *Little Red Dots* (LRDs) as a phase of hidden growth for compact high-redshift nuclei [56]. The key spectral signature is a narrow intrinsic component buried under a broader electron-scattering wing, which points to very high electron columns and extremely compact sizes, of order a few light-days. In that reading, at least a subset of LRDs are better described as young nuclei in dense accretion than as purely star-formation-dominated events.

This is directly relevant for the ECF because it gives a concrete astrophysical bridge between the topological seed and the observable LRD stage. The chain

$$\text{Macro-Knot} \rightarrow \text{LRD} \rightarrow \text{SMBH}$$

becomes a physically motivated working hypothesis (the Macro-Knot population itself being a structural hypothesis, and no Micro-Knot→Macro-Knot growth step being implied — that channel is withdrawn): Macro-Knots would act as pre-existing compact seeds, and the ionized cocoon marks a hidden growth phase driven by rapid baryonic infall onto the topological core. The point is not that every LRD must follow this pathway, but that the JWST data make such a channel increasingly plausible without requiring a long super-Eddington history from stellar-mass remnants. For the Great Annihilation freeze-out window ($z_t \sim 6.9\times10^8$ to 6.9×10^9, corresponding to $T \sim 1$ MeV, the e^+e^- annihilation epoch — not the much hotter QCD/electroweak transition, which sets the Micro-Knot scale instead), the Jeans estimate still yields $M_K \sim 10^5$–$10^7\,M_\odot$ for the heavy-seed portion of the topological spectrum (radiation-only expansion rate; with the evolution of g_* the hot end of the window falls to $\simeq 6\times10^4\,M_\odot$, cf. Eq. (42)), overlapping the compact-object mass scale inferred for the LRD regime [56]. These masses use the radiation-only expansion rate. Including the spin term of the calibrated Foundation I branch (a stiff term $\propto a^{-6}$, normalised to $\rho_{\rm spin} = 0.093\,\rho_{\rm r}$ at $z_{\rm ref} = 5600$) extrapolated to z_t gives $\rho_{\rm spin}/\rho_{\rm r} \simeq 1.4\times10^9$ to 1.4×10^{11}: $H(z_t)$ would be larger by a factor $\sim 4\times10^4$ ($z_t = 6.9\times10^8$) to $\sim 4\times10^5$ ($z_t = 6.9\times10^9$), and M_K would decrease by about 4.6 to 5.6 orders of magnitude (to $\sim 3\times10^2$ and $\sim 0.3\,M_\odot$). That extrapolation makes the spin term equal the radiation at $z \simeq 1.8\times10^4$ and dominate the expansion at nucleosynthesis, which BBN does not allow (it requires a negligible extra density at $T \sim 1$ MeV); the corpus does not demonstrate BBN compatibility (`PO-ECF-BBN-SPIN`). The radiation-only values above therefore correspond to a spin term negligible at $T \sim 1$ MeV, as BBN requires; that this holds for the calibrated branch is not shown: the branch reproduces the sound horizon only if its a^{-6} law holds up to $z \sim 10^5$, and its extension to z_t is not derived (Foundation I, domain of validity of the calibrated branch [5]). The reduction by 4.6–5.6 orders is the result of an extrapolation that BBN excludes, not an alternative prediction.

This overlap is what makes the Macro-Knot interpretation useful: it connects the early-time seed spectrum, the hidden-growth LRD phase, and the later SMBH assembly within one chronology. Figure 36 illustrates this comparison quantitatively.

The resulting picture is therefore flexible rather than exclusive. A direct seed-to-SMBH route remains possible for the most massive objects, while a Macro-Knot-mediated hidden-growth channel is a candidate for the LRD population identified by JWST, not one required by their masses: at $z = 4$–9 these masses are also reachable from stellar seeds at the Eddington rate (Fig. 36; corrected 23/10/2026, earlier versions read "is favored"). The role of the model is to provide a continuous chronology and a set of discriminating observables, not to force all sources into a single evolutionary track.

What the ECF does not predict for LRDs (added 23/10/2026). The ECF fixes no redshift, no redshift window and no number-density evolution for Little Red Dots. Observed LRDs lie mostly at $z \sim 4$–9, their number density falls steeply below $z \approx 4$, and local analogues are extremely rare (9 objects among 14 584 SDSS broad-line sources at $z < 0.35$) [58]; the highest spectroscopic redshift is $z = 9.288$ [57], and a candidate at $z \gtrsim 10$ rests on photometry alone. Euclid reports unconfirmed candidates at $z < 4$ with a density peak at $z \approx 1.5$–2.5 [59], whose link to the JWST population is open. If Macro-Knot seeds exist, they exist at all later epochs; why LRD-like phases would concentrate at $z \sim 4$–9 is not derived (no accretion timescale, no duty cycle). The comparison with LRDs is therefore one of mass scale and abundance (Planck bound above), not a test of their redshift behaviour. The reading of LRDs as dense accreting nuclei [56] does not by itself favour a primordial seed over the other seeding channels. The schematic growth track Macro-Knot $\rightarrow$ LRD $\rightarrow$ SMBH shown in earlier versions has been withdrawn: its anchor masses for UHZ-1, MoM-z14 and GN-z11 did not match the measurements (UHZ-1 was drawn at $10^6\,M_\odot$ instead of $\sim 4 \times 10^7$, MoM-z14 has no measured black-hole mass), and its assembly points contradicted quasars measured at $z = 7.5$.

This interpretation should be regarded as exploratory and not yet established. Its main value is to provide a continuous evolutionary picture linking a primordial topological remnant, a compact red observational phase, and the later growth of a central massive object within a single chronological sequence.

Early Slow-Rotators and Chiral Cancellation. Recent JWST observations have revealed the unexpected existence of extremely massive, quiescent, non-rotating galaxies at $z > 3$. The most striking example is XMM-VID1-2075 at $z = 3.449$ (corresponding to a cosmic age $\lesssim 2$ Gyr in standard cosmology), a $\sim 330 \times 10^9\,M_\odot$ quiescent galaxy with a spin parameter $\lambda_{R_e} = 0.101 \pm 0.018$, confirmed as the highest-redshift kinematically-identified slow rotator to date [64]. Within standard hierarchical assembly, dispersion-dominated slow rotators require billions of years and multiple consecutive mergers to cancel angular momentum, making such early occurrences statistically anomalous.

The ECF provides a natural, deterministic resolution to this timeline paradox. Galactic dynamics are anchored by the topological helicity of their primordial Macro-Knot seeds. A major interaction between two domains or Macro-Knots of opposite topological charge ($\mathcal{H}_1 \cdot \mathcal{H}_2 < 0$) leads inevitably to a prompt cancellation of macroscopic angular momentum. In extreme cases (topological annihilation, Sec. 7.3), the geometric unbinding releases sufficient energy to instantaneously quench star formation via violent gas expulsion. The quantitative energetics of both same-helicity (Regime I) and opposite-helicity (Regime II) encounters are derived and calibrated in Appendix D.10. The ECF thereby predicts that early massive slow-rotators are not statistical flukes of hierarchical assembly, but deterministic kinematic fossils of primordial chiral topology — a class of objects whose existence is naturally accommodated within ECF and deeply anomalous within ΛCDM.

7.2 Topological Charge and Helicity in the ECF

Within this framework, the distinction between a macroscopic Knot and an Anti-Knot is strictly determined by the intrinsic chirality of its localized spacetime torsion, quantified by the topological helicity H. This yields two distinct topological states: positive helicity ($q = +1$, right-handed twist) and negative helicity ($q = -1$, left-handed twist). It is this fundamental invariant that dictates the outcome of binary mergers.

Topological charge definition. For a 0D Micro-Knot (hedgehog, π_2 class), the winding number is

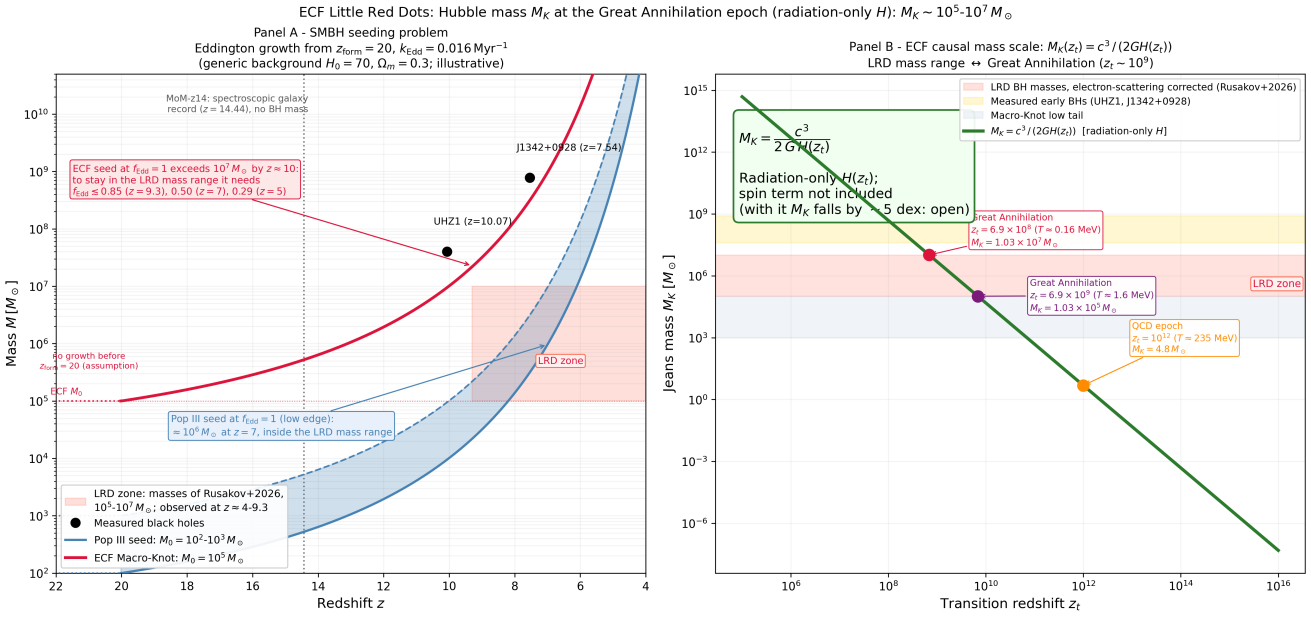

Figure 36: **ECF prediction for Little Red Dot seeds.** *Left*: Eddington-limited mass growth ($k = 0.016\,\mathrm{Myr}^{-1}$) starting from a Pop III stellar remnant ($M_0 = 100\,M_\odot$, blue) and an ECF primordial Macro-Knot ($M_0 = 10^5\,M_\odot$, red), both formed at $z_{\rm form} = 20$. The red-shaded region marks the LRD compact-object mass range 10^5–$10^7\,M_\odot$ [56]; the observed LRDs lie mostly at $z \sim 4$–9, spectroscopically confirmed up to $z = 9.29$ [57] (corrected 23/10/2026: earlier versions read "$z \approx 5$–15", a range not given by that reference). At $f_{\rm Edd} = 1$ the ECF seed ($10^5\,M_\odot$ at $z_{\rm form} = 20$) reaches $\sim 10^7\,M_\odot$ by $z \simeq 10$ and exceeds the LRD mass range at lower redshift; staying inside it requires $f_{\rm Edd} \lesssim 0.85$ ($z = 9.3$), 0.50 ($z = 7$), 0.29 ($z = 5$). A stellar seed ($10^2\,M_\odot$) at $f_{\rm Edd} = 1$ already reaches $\sim 10^6\,M_\odot$ at $z = 7$: the LRD mass range alone does not discriminate between seeds ($10^9\,M_\odot$ by $z = 10$ would need $f_{\rm Edd} \simeq 2$; corrected 16/10/2026 and 23/10/2026, earlier versions read "enters this zone immediately"). These $f_{\rm Edd}$ values assume that growth starts at $z_{\rm form} = 20$ (dotted plateau before: no growth, by assumption); a seed present since the Great Annihilation, as in the ECF scenario, would need smaller values to stay in the LRD mass range. *Panel A*: generic background, illustrative; black dots are measured black holes (UHZ1 [43], J1342+0928 [44]); the dotted line marks the spectroscopic galaxy redshift record MoM-z14 ($z = 14.44$) [63], with no black-hole mass. *Panel B*: the LRD band gives the black-hole masses of Ref. [56] revised for electron scattering (standard virial estimates are about two orders of magnitude higher); M_K assumes the radiation-only expansion rate, and with the spin term it would be several orders of magnitude lower (open question `PO-ECF-BBN-SPIN`). *Right*: ECF Jeans mass $M_K \sim c^3/(2GH(z_t))$ as a function of the transition redshift z_t, in the radiation-dominated approximation. The topological crystallization at the Great Annihilation freeze-out (e^+e^- annihilation, $T \sim 1$ MeV, $z_t \sim 10^9$–10^{10} — not the much hotter electroweak scale, which sets the Micro-Knot population instead) sets a causal mass scale 10^5–$10^6\,M_\odot$; if Macro-Knots form there (structural hypothesis, $\epsilon_{\rm coll}$ not derived), this would provide a candidate heavy-seed channel for the LRD mass regime. Kill-switch: absence of compact seeds with $M \gtrsim 10^5\,M_\odot$ at $z \sim 10$ in future JWST/ELT spectroscopic surveys would falsify the ECF seeding mechanism.*Reproduced by:* `Scripts_F2/plot_ECFLRDJeansMass.py`.

the degree of the map $\hat{\Phi} : S^2 \to S^2$ on a sphere surrounding the defect core:

$$q \;=\; N \;=\; \frac{1}{8\pi}\oint_{S^2} \epsilon^{ijk}\,\hat{\Phi}\cdot\left(\partial_j\hat{\Phi}\times\partial_k\hat{\Phi}\right)d^2x_i \;\in\; \mathbb{Z}. \tag{80}$$

The helicity $q = \pm 1$ is a strictly conserved topological invariant under continuous deformations of the field configuration. It cannot change without passing through a singular configuration (i.e. topological annihilation).

Interaction rules. The helicity fully determines the outcome of any two-body encounter between 0D defects:

$$q_1 = q_2 = +1 \;\Rightarrow\; N_{+1} + N_{+1} \to N_{+2} \quad \text{(coagulation; aggregate growth)}, \tag{81}$$

$$q_1 = -q_2 \;\Rightarrow\; N_{+1} + N_{-1} \to 0 + \Delta E_{\rm GW} \quad \text{(annihilation; PO-F2-1)}, \tag{82}$$

where $\Delta E_{\rm GW}$ is the energy radiated as gravitational waves and Torsion Fluid during annihilation. This bimodal topology explains why only same-helicity defects accumulate into the galaxy-seeding Macro-Knots observed as Little Red Dots, while opposite-helicity pairs annihilate before freeze-out (Sec. 3), leaving the net surplus $\Delta N = N_{+1} - N_{-1} > 0$ (PO-F2-1).

7.3 Topological Charge and Bimodal Merger Signatures

The mathematical formulation of a merger in the ECF model bifurcates into two distinct physical regimes based on these charges:

- **Regime I: Constructive Mergers** ($q_1 = q_2$): When two nodes of the same chiral nature merge, their topologies superimpose:

$$M_{\rm final} \approx M_1 + M_2 - \frac{E_{\rm rad}}{c^2}, \tag{83}$$

  where $E_{\rm rad} = \varepsilon_{\rm top}\,(M_1 + M_2)\,c^2$ is the energy radiated as gravitational waves during the topological coalescence, and $\varepsilon_{\rm top} \lesssim 0.05$ is the topological radiation efficiency (Appendix D.10.1; *corrected 15/10/2026*: earlier versions wrote $\epsilon_{\rm top}\mu c^2$ with the reduced mass μ, inconsistent with the definition $E_{\rm GW}/(M_{\rm tot}c^2)$ of the appendix, and a range $[0.01, 0.10]$) (cf. the $\sim$5% efficiency of GW150914-class BBH mergers [70]). These events visually mimic classical black-hole coalescences and contribute to the ECF stochastic gravitational-wave background (Eq. 120).

- **Regime II: Topological Annihilation** ($H_1 + H_2 = 0$): When a Knot encounters an Anti-Knot of opposite helicity, the conflicting topological charges induce a localized untying of the spacetime fabric. By topological charge conservation, the net helicity of the system is zero and the two structures cannot coexist:

$$M_{\rm final} = M_1 - M_2 - \frac{E_{\rm rad}}{c^2}, \qquad (M_1 \geq M_2), \tag{84}$$

  where the surviving remnant retains mass $M_1 - M_2$ (the excess topological charge of the dominant Knot), and

$$E_{\rm rad} = \eta_{\rm ann} \cdot 2\,M_2\,c^2 \tag{85}$$

  is the energy radiated during the annihilation. Here $2M_2c^2$ is the rest-mass energy of the *pair* that undergoes complete topological cancellation (the smaller defect of mass M_2 and an equal portion of the larger one), and $\eta_{\rm ann} \leq 0.5$ is the annihilation efficiency (an open quantity, Appendix D.10.2) — the fraction converted into radiation versus residual geometric deformation. For a symmetric encounter ($M_1 = M_2$), $M_{\rm final} = 0$ and the event is a *complete* annihilation, releasing $E_{\rm rad} = \eta_{\rm ann} \cdot 2M_{\rm tot}c^2$. For a Macro-Knot pair with $M_1 = 10^5\,M_\odot$ and $M_2 = 10^4\,M_\odot$, Eq. (85) yields:

$$E_{\rm iso} \simeq \eta_{\rm ann}\cdot 2M_2c^2 \simeq 3.6\times10^{58}\,\eta_{\rm ann}\ \text{erg}. \tag{86}$$

  This exceeds the most energetic GRB observed, GRB 221009A ($E_{\rm iso} \approx 1.0$–1.2×10^{55} erg: 1.0×10^{55} from Fermi-GBM, 1.2×10^{55} from Konus-*Wind* [79]), unless $\eta_{\rm ann} \lesssim 3\times10^{-4}$, or unless only a small fraction of $E_{\rm rad}$ emerges as prompt photons; the radiated fraction is an open quantity (Appendix D.10.2). Because this process does not require a stellar host environment, the resulting burst is intrinsically *hostless* or offset from any visible galaxy — detailed in Sec. F.23.

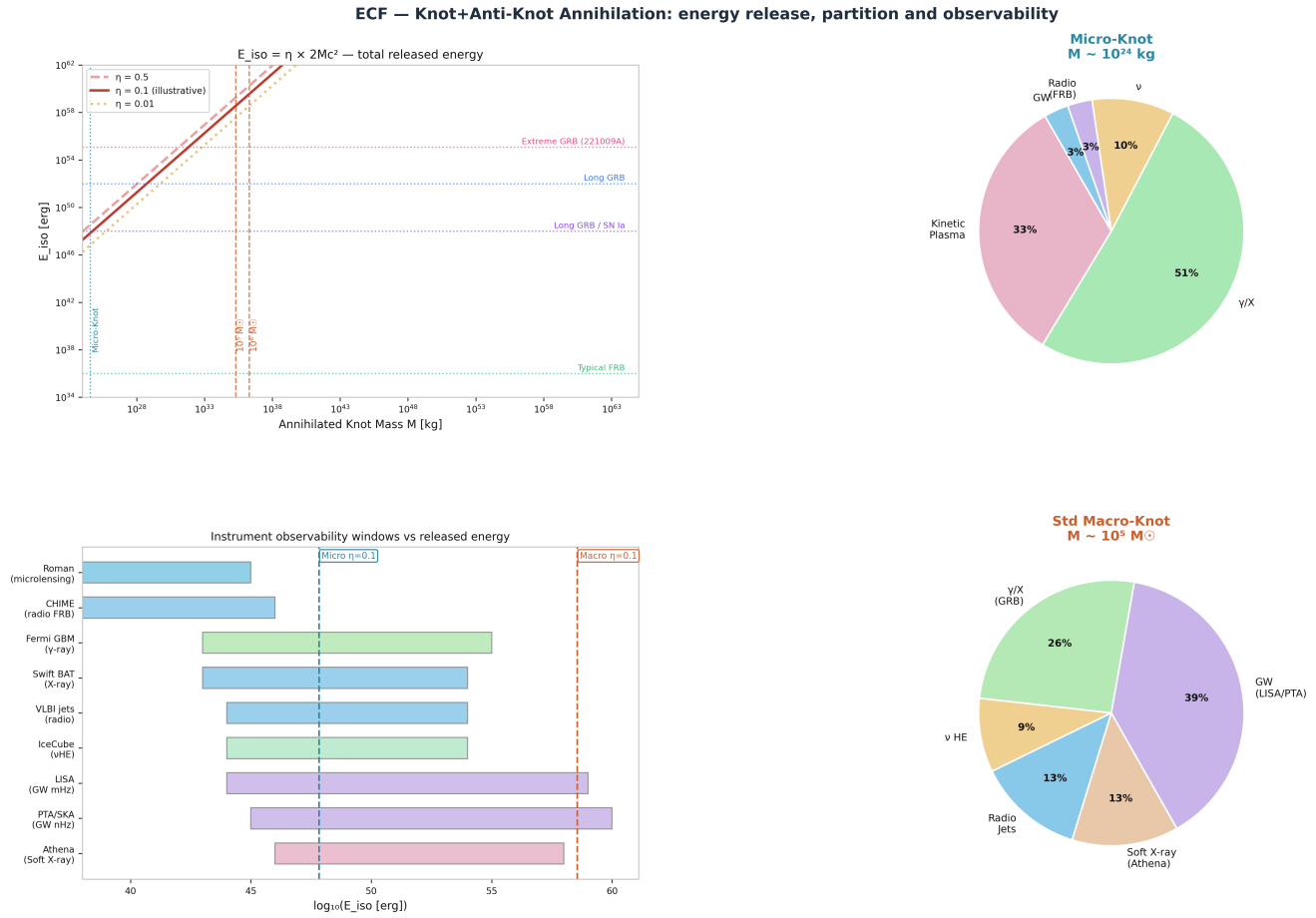

Figure 37: **ECF annihilation channel: energy release and observability.** The left panel shows the isotropic-equivalent energy released by a Knot–Anti-Knot annihilation event as a function of the annihilated mass, for representative efficiency values $\eta_{\rm ann} = 0.5$, 0.1, and 0.01. The horizontal reference bands indicate the energetic domains typically associated with standard high-energy transients. The right panels summarize the energy partition of two benchmark cases: a Micro-Knot event and a Macro-Knot event. The dominant channels are γ/X, ν, plasma kinetic energy, radio, and gravitational-wave emission. The figure is intended as a compact phenomenological guide to the Regime II annihilation channel and its observational discriminants.

Reproduced by: `Scripts_F2/plot_ecf_Annihilation.py`.

7.4 The Cosmic Web and the Origin of Giant Voids

The geometric origin of the Cosmic Web and the formation of giant cosmic voids are treated in depth in Sec. 4.2, where the closure relation ($\Omega_m + \Omega_\tau = 1$ at every scale factor) is used to read off the composition of matter-depleted regions, dominated by residual torsion engines driving void expansion, while Macro-Knot clusters seed the filamentary nodes. The top-down Kibble–Zurek domain partition is proposed as an explanation of the void size function (VSF); no quantitative comparison with SDSS or DESI void catalogues is made in this work (the slice of Fig. 15 is a mock illustration, and the VSF template of Fig. 13 awaits the test deferred to Foundation III). The macroscopic imprint of this torsion-driven Web on the CMB power spectrum is discussed in Sec. 7.5, and its extreme manifestations—the Big Ring and Giant Arc mega-structures—are addressed in Sec. G.

7.5 Topological Imprints on the Cosmic Microwave Background (CMB)

High-precision data from the Planck satellite have revealed persistent, large-scale structural features, such as the CMB Cold Spot and the alignment of low multipoles. The ECF topological framework interprets these features not as statistical fluctuations, but as deterministic geometric imprints of

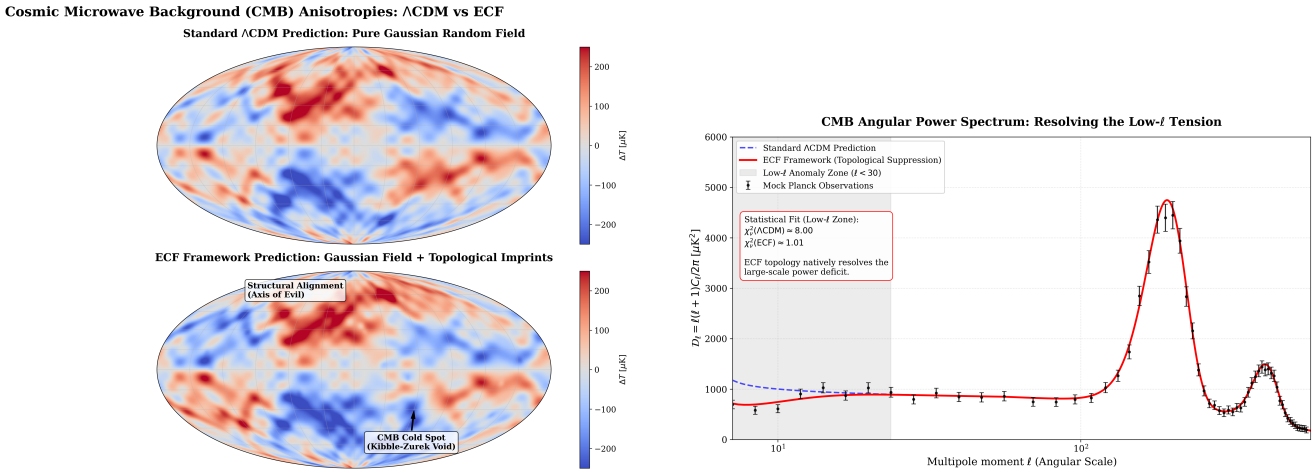

Figure 38: **Left — ECF vs ΛCDM CMB temperature map (schematic).** The ECF torsion background imprints coherent geometric cold spots and large-scale alignments absent in the ΛCDM prediction, consistent with the observed CMB quadrupole deficit and Axis of Evil. **Right — Angular power spectrum D_ℓ.** The ECF geometric cutoff suppresses variance at $\ell < 30$ (grey band), qualitatively reproducing the Planck low-ℓ anomaly. Both panels are illustrative; a quantitative perturbative computation remains to be carried out. *Reproduced by:* `plot_ecf_cmb.py` and `plot_cmb_power_spectrum.py`.

the primordial phase transition. The characteristic comoving size of topologically trivial domains is bounded by the Kibble-Zurek correlation length:

$$R_{KZ} \approx c\, t_{EW} \left(\frac{\tau_Q}{t_{EW}} \right)^{\frac{\nu}{1+\nu}} \tag{87}$$

Photons traversing these early geometric wells lose significant energy, directly generating localized cold spots. Furthermore, the macroscopic torsion domains introduce a topological cutoff, effectively suppressing the variance at the largest observable scales ($\ell < 30$), naturally aligning with the empirically observed quadrupole deficit in the angular power spectrum (D_ℓ); both are shown in Fig. 38.

Exploratory mega-structure signatures. The Big Ring, Giant Arc, and related Torsion Vortex Ring phenomenology are retained as exploratory large-scale consistency checks, but are not required for the core Foundation II argument. To keep the main narrative focused on the three fossils, geometric dark matter, and the JWST/LRD heavy-seed channel, the detailed discussion is moved to Appendix G. The conceptual link is preserved: Macro-Knot networks and Kibble–Zurek domains provide a possible top-down route from primordial torsion texture to the largest observed coherent structures.

8 The Axis of Evil: Fossil of the Primordial Vector

The "Axis of Evil" — the anomalous alignment of the CMB quadrupole and octopole with the ecliptic plane — is one of the most persistent large-scale anomalies in precision cosmology [215]. In the ECF framework, it could be interpreted not as a statistical accident but as a geometric fossil of the primordial vector condensate $\langle S^\mu \rangle = (s_0, \mathbf{0})$, whose spatial orientation would break the isotropy of the FLRW background, with an amplitude bounded by the CMB quadrupole at $\varepsilon \lesssim 10^{-7}$ (Eq. 90); this interpretation is conjectural, and its statistical significance is debated. This section presents the ECF interpretation of the alignment and its quantitative consistency with current constraints, as an exploratory cross-check of the primordial vector heritage — not as primary evidence for the framework. While the Cosmological Principle assumes large-scale isotropy—the CMB disfavours anisotropic expansion, with bounds ranging from $(\sigma_V/H)_0 < 4.7 \times 10^{-11}$ to $(\sigma_T/H)_0 < 1.0 \times 10^{-6}$ at 95% C.L. depending on the mode [73]—Planck satellite data have revealed persistent anomalies at low multipoles [215]. Specifically, the Quadrupole ($\ell = 2$) and Octopole ($\ell = 3$) moments exhibit a preferred spatial alignment, colloquially termed the "Axis of Evil" (AOE). *(Corrected 22/10/2026: earlier versions attributed to [73] a bound* $|\epsilon| < 10^{-4}$,

which that paper does not give; its bounds are 2 to 6 orders of magnitude tighter.)

High-precision CMB data contain persistent low-ℓ anomalies, including the quadrupole–octopole alignment. In the ECF framework these anomalies are not used as primary evidence, but as a possible consistency check of a residual primordial vector texture. The relevant observable is therefore not a deterministic "Axis of Evil" prediction, but a bounded anisotropic remnant whose amplitude must remain below standard isotropy constraints while potentially correlating with the preferred direction generated by the spin-torsion bounce.

The residual geometric anisotropy imprinted by the ECF bounce on the FLRW metric is parametrised by the ellipticity

$$\varepsilon \equiv \frac{a_\perp - a_\parallel}{a}, \tag{88}$$

where $a_\perp$ and $a_\parallel$ are the scale factors transverse and parallel to the chiral symmetry-breaking axis and $a = (a_\perp^2 a_\parallel)^{1/3}$ (to first order, a may be replaced by $a_\perp$). Its amplitude is not derived in this work; the CMB quadrupole bounds it below. *(Corrected 22/10/2026: earlier versions wrote the bound $\lesssim 10^{-5}$ into this definition, about 90 times above Eq. 90.)*

The ellipticity ε can be related to the observed quadrupole amplitude through the standard Bianchi I temperature anisotropy formula [150]:

$$\left(\frac{\Delta T}{T}\right)_{\ell=2} \approx \frac{\sqrt{6}}{5}\,\varepsilon\,\ln\!\left(\frac{a_{\rm rec}}{a_{\rm bounce}}\right), \tag{89}$$

where the logarithmic factor accounts for the cumulative anisotropy phase accumulated between the bounce ($a_{\rm bounce} \sim 10^{-32}$, from the ECF critical density, Foundation I) and recombination ($a_{\rm rec} \simeq 10^{-3}$): $\ln(10^{-3}/10^{-32}) = 29\ln 10 \approx 66.8$. Using the Planck 2018 quadrupole $C_2 \simeq 226\,\mu{\rm K}^2$, the standard cosmological convention $Q_{\rm rms} = \sqrt{5C_2/4\pi}$ gives $Q_{\rm rms} \simeq 9.5\,\mu{\rm K}/2.7255\,{\rm K} \simeq 3.5\times10^{-6}$ [8]; inverting Eq. (89) yields:

$$\varepsilon \lesssim \frac{5}{\sqrt{6}}\,\frac{Q_{\rm rms}}{66.8} \approx 1.1\times10^{-7}. \tag{90}$$

Revision of record (12/09/2026). An earlier version quoted $Q_{\rm rms} \approx 6\,\mu$K, giving $\varepsilon \lesssim 7.8\times10^{-8}$. That figure could not be traced to any standard convention: with $C_2 \simeq 226\,\mu{\rm K}^2$ one has $\sqrt{C_2} \simeq 15\,\mu$K and $\sqrt{5C_2/4\pi} \simeq 9.5\,\mu$K, while $6\,\mu$K would require $C_2 \simeq 90\,\mu{\rm K}^2$, below published values. The standard cosmological convention is adopted here; it *weakens* the bound by a factor 1.6, so the earlier figure was conservative rather than favourable. How this estimate compares with the CMB bounds on anisotropic expansion [73] depends on the mapping between ε and σ/H, which is not derived here (Appendix J). Eq. (89) treats ε as a constant anisotropy per e-fold, i.e. effectively as σ/H; under that reading the bound 1.1×10^{-7} lies below the weakest bound of [73] (tensor modes, 10^{-6}) but above the vector-mode bound (4.7×10^{-11}), which is the relevant one if the anisotropy is rotational in origin. Which bound applies is left open. Values significantly above this bound would overproduce the quadrupole in the simplified Bianchi I mapping and should therefore be treated as disfavoured unless additional cancellation terms are explicitly introduced. Future measurements of the large-scale E-mode polarisation are expected to tighten this constraint, providing a direct geometric test of the ECF chiral axis.

8.1 The ECF Explanation

In the ECF model, the observed alignment of the low-ℓ multipoles is interpreted as a generic consequence of the vector nature of the primordial spin density. Conservation of angular momentum from the bounce imposes a preferred spatial direction $\hat{n}_i$, inducing a coherent coupling between the quadrupole and octopole moments. This creates a directional bias in the vacuum metric itself, represented by a slight remnant ellipticity:

$$ds^2 = -dt^2 + a^2(t)\left[(1+\epsilon\cos^2\theta)dr^2 + r^2 d\Omega^2\right] \tag{91}$$

(schematic form; with ε as in Eq. 88, an axisymmetric Bianchi I metric gives a radial coefficient $1-2\varepsilon\cos^2\theta$ to first order, θ being measured from the chiral axis, together with off-diagonal terms).

This breaking of rotational invariance induces a coupling between the lowest multipoles, tracing the global ellipsoidal shape of the spacetime manifold and its first-order correction. If the reported alignment of $\ell = 2$ and $\ell = 3$ is physical, it could be read as a trace of expansion from a structured, rotating state; Fig. 39 is an illustrative schematic of such an alignment, not a measurement, and is not used as observational evidence.

The ECF model further suggests that this primordial geometric alignment may have local imprints, such as the observed planar structure of Andromeda's satellite galaxies [113], where geometric

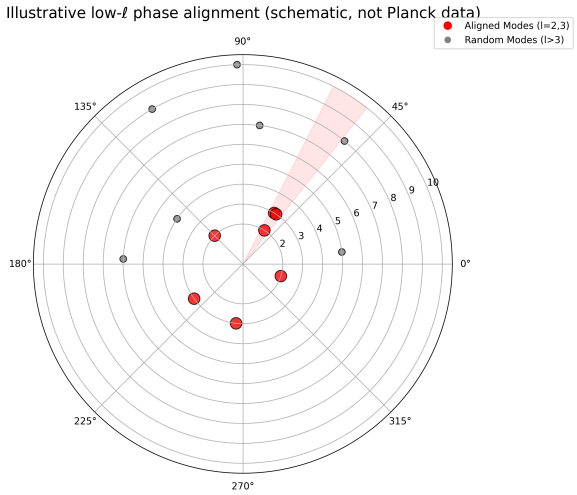

Figure 39: **Multipole phase alignment (illustrative).** Illustrative schematic of a low-ℓ phase alignment; the phases are not measured values. The polar plot shows, for illustration only, how an alignment of the quadrupole ($\ell = 2$) and octopole ($\ell = 3$) phases (red) within a narrow sector would appear against higher modes ($\ell \geq 4$); it is not derived from Planck data and is not evidence for the ECF interpretation discussed in the text.

Reproduced by:
`Scripts_F2/plot_ecf_axis_of_evil_cmb.py`.

dark matter accreted preferentially along residual torsional axes.

Exploratory transient and non-thermal signatures. Odd Radio Circles, long-period radio transients, hostless bursts, quasar alignments, and related non-thermal channels remain part of the ECF observational programme, but they are not primary evidence for the Foundation II dark-sector construction. Their detailed mechanisms and kill-switches are therefore collected in Appendix H. This preserves the concepts while preventing the main text from reading as an accumulation of independent anomaly claims.

Table 9: **Micro-Knot encounter rates, solar-neighbourhood values** ($\rho_{\rm DM} = 0.4$ GeV cm^{-3}, $v = 220$ km s^{-1}). *Reproduced by:* `Scripts_F2/compute_encounter_rates.py`.

Target	σ [m^2]	Rate [yr^{-1}]	Since Earth formed
Earth	1.3×10^{14}	1.7×10^{-19}	7.6×10^{-10}
Jupiter	1.7×10^{16}	2.2×10^{-17}	9.8×10^{-8}
Sun	1.4×10^{19}	1.8×10^{-14}	8.0×10^{-5}
White dwarf	7.3×10^{16}	9.6×10^{-17}	4.3×10^{-7}
Neutron star	2.9×10^{14}	3.8×10^{-19}	1.7×10^{-9}
Another Micro-Knot	1.8×10^{2}	2.4×10^{-31}	1.1×10^{-21}

Part III

Observational Predictions

Kill-Switches, GW Signatures, and Falsifiability

9 Testable Predictions for Future Missions

A fundamental requirement of any cosmological model is its testability. The ECF model, by proposing vector geometry and topological defects as the basis of the dark sector, offers distinct observational signatures that differ from the standard WIMP paradigm. We outline specific predictions testable by upcoming observatories in Table 10.

9.1 Where to look: encounter rates by target and environment

Before listing instruments it is worth asking what a knot actually meets. The rate for a target of radius R and mass M_t in a sea of knots of number density n at relative velocity v is $\Gamma = n\sigma v$ with $\sigma = \pi R^2\,(1 + v_{\rm esc}^2/v^2)$, the second factor being gravitational focusing.

Three readings follow. First, an impact on Earth is expected 7.6×10^{-10} times over the age of the planet: a non-observation that is predicted, not a constraint — and it is small because of the density and the cross-section, not because anything was tuned. Second, the Sun is 10^5 times more likely than the Earth, from radius and focusing together; of the six targets it is the only one whose (still minuscule) rate opens onto a possible observable, since a transit deposits energy. Third, knot–knot encounters at 2×10^{-13} per year per galaxy — multiplying by the $\sim 10^{18}$ knots in a halo — are why coagulation is treated here as a network process and not as a two-body problem.

Stellar transits: computable, and not observable. Of the six targets above the Sun is the only one whose rate, however small, opens onto a possible signal, since a transit deposits energy where an Earth impact essentially does not. It is worth following through, because the conclusion is negative and precise.

Note first that the primordial-black-hole literature on stellar capture does not transfer: it addresses the *asteroid-mass* window, 10^{17}–10^{23} g, whereas $M_{\rm MK} = 3.7 \times 10^{27}$ g lies four orders above its upper end, and the Hawking-radiation signatures that dominate there are irrelevant at $T_H = 33$ mK. Taking Chandrasekhar dynamical friction, $F = 4\pi G^2 M^2 \rho \ln\Lambda / v^2$, at core density and free-fall speed, one crossing lasts 38 min and deposits

$$E_{\rm dep} \simeq 4.7{\times}10^{33}\ {\rm J} \quad \Longrightarrow \quad \bar{P} \simeq 5.4{\times}10^{3}\,L_\odot. \tag{92}$$

Two checks fix the outcome. The deposit is 0.7% of the knot's kinetic energy, so it passes through rather than being captured; and 2×10^{-8} of the Sun's binding energy, so the star is unaffected.

Summing over Galactic populations with the environmental gains above gives one transit per $\sim$35 yr in the Milky Way, one per 23 yr in M31, and – because the rate scales with stellar mass – several per year across the Virgo cluster. Those numbers are encouraging until one asks what reaches us: the flux from M31 would be 3×10^{-16} W m^{-2}, seven orders below a supernova at the same distance. And that figure is itself an overestimate, since it assumes

the energy escapes during the crossing. It does not: the deposit is at the core, under millions of kilometres of opaque plasma, and diffuses out on the Kelvin–Helmholtz timescale. *Electromagnetically, a transit is invisible.*

We record this as a closed channel rather than omit it, because the question — what happens when one of these objects meets a star — is the natural one to ask of any compact dark-matter candidate. The seismic angle closes too, and by the opposite mechanism, which is worth keeping straight. The solar p-mode budget is $\sim 10^{26}$–10^{27} J by two independent estimates, so Eq. (92) exceeds the entire oscillation of the star by some six orders of magnitude: a coupling efficiency of 10^{-6} would already give a signal comparable to the whole solar oscillation. *The signal is not the difficulty.* The difficulty is events. Kepler, TESS and PLATO together monitor $\sim 5 \times 10^5$ stars, which over a decade yields $\sim 10^{-7}$ expected transits; one would need 5×10^{12} stars watched continuously. So the electromagnetic channel is closed by radiative transport and will stay closed, whereas the seismic channel is closed by sample size and would reopen if populations orders of magnitude larger were ever monitored. Neither is a prediction of this framework; both follow from the rate. *Reproduced by:* `Scripts_F2/compute_stellar_transit.py`.

Environment matters, and not only through density. Since $\Gamma \propto n\, v\, (1 + v_{\rm esc}^2/v^2)$, a slower environment loses flux but gains focusing, and for a massive target the second effect wins. A galactic bulge is 50× denser than the solar neighbourhood and gives 78× the rate on the Sun; a dwarf spheroidal is only 12× denser but its 30 km s^{-1} dispersion focuses so strongly that it matches the bulge at 82×. For the Earth, whose escape velocity is 11 km s^{-1}, focusing is negligible and the gain simply tracks $n\, v$. *Velocity dispersion is therefore as much a selection criterion as density*, which points searches towards dwarf spheroidals and the bulge rather than the solar neighbourhood.

Scope. These are upper bounds on how often something could happen, not statements about what it would look like: whether a solar transit is detectable is a separate calculation, not attempted here. The knot–knot and neutron-star rows use Newtonian focusing where $v_{\rm esc}$ reaches c and $0.59c$ respectively, and should be read as orders of magnitude. All rows assume the whole dark sector is in Micro-Knots; if Macro-Knots carry part of the mass, every rate falls proportionally.

Why several channels, and what each one isolates. The table below is not a list of chances to be lucky. A single detection channel constrains a *combination* of the framework's parameters, and the combinations differ from one channel to the next: photometric microlensing gives $t_E \propto \sqrt{\lambda M D}$, degenerate between mass, aggregation and distance; the astrometric channel gives the mass alone; merging clusters bound the cross-section per unit mass, which scales as M and therefore constrains the *heavy* end that microlensing barely reaches; the parity asymmetry $\Pi_{\rm GW}$ tests the chiral condensate rather than the defect population. Taken singly, each admits an escape; taken together they do not, because no single adjustment moves them all in the same direction.

Two labels in the table should be read carefully. *Prospective* marks a channel whose observational programme is discussed in the literature but not, to our knowledge, scheduled — these cannot carry a dated kill-switch. And a constraint the framework already satisfies is informative about consistency without being evidence: the merging-cluster bound is met by twenty-seven orders of magnitude because a topological defect has nothing to collide with, not because a parameter was chosen well.

The LISA and PTA entries in the table above represent calibrated kill-switch thresholds, not detection forecasts: a null result at the specified sensitivity level would falsify the *present calibrated implementation* of the ECF, which relies on $F_{\rm ion} = 1.2765$ and the stiff-phase dynamics of Foundation I. It would not, by itself, exclude all Einstein–Cartan cosmologies with a torsion bounce. The spectral shape of the SGWB — in particular the tilt n_T and the chiral fraction $\Pi_{\rm GW}$ — depends on the bounce transfer function, which is not yet derived from first principles (see Sec. 9.4 and Foundation III [7]).

9.2 The WIMP Null Result

Standard cosmology often models dark matter as a continuous fluid of Weakly Interacting Massive Particles (WIMPs). However, direct detection experiments remain empty. Since ECF Dark Matter is geometric and highly discrete, direct detection is structurally suppressed — but the two ECF populations behave very differently and must not be conflated. The *abundant* Micro-Knots ($M_{\rm MK} = 3.74 \times 10^{24}$ kg, $N_\mu \sim 10^{18}$ per galaxy) genuinely fill the local dark

Table 10: **Summary of upcoming observational tests for the ECF model.**

Observatory / Mission	**Specific ECF Prediction**
Roman Space Telescope	Microlensing events with $t_E \sim 1$–$10\,\mathrm{h}$ from Poisson-coalesced Micro-Knot aggregates ($M_{\rm typ} \approx 6.8 \times 10^{-5}\,M_\odot$, §9.3); ∼1400–4100 events expected over the 72-day bulge survey if $f_{\rm DM} = 1$ (OGLE-IV limits $f_{\rm DM} \lesssim 1$–$2\,\%$: PO-F2-10).
LISA / Einstein Telescope	Chiral asymmetry $\Pi_{\rm GW} = (P_R - P_L)/(P_R + P_L) \simeq 0.20$ in the stochastic gravitational-wave background (SGWB), emitted by defect mergers during the Stiff Era (Eq. 120). Parity violation in the CMB is independently accessible via TB and EB cross-spectra (Table 11). For the gravitational-wave background, the net circular polarisation of an isotropic signal is nearly invisible to a planar interferometer such as LISA and becomes accessible only through the kinematic dipole anisotropy [203]. *(Corrected 22/10/2026: earlier versions cited a paper on non-Gaussianity for parity tests, and presented LISA as directly sensitive to the net chirality of an isotropic background.)* Multi-band extension to the nHz regime: a dipole Pulsar Timing Array configuration can probe the chiral handedness of the same cosmological background [118], offering cross-validation between the LISA mHz band and PTA nHz band before LISA first light.
LISA / ET (Regime I merger)	A quasi-monochromatic GW burst at $f_{\rm peak} \sim 50$ mHz ($\tau_{\rm rd} \sim 1$ s; no distinctive torsion tail, Appendix D.10) from same-helicity Macro-Knot coalescence (Appendix D.10); conditional channel: the merger rate is not computed, so a non-detection would not falsify the framework *(corrected 18/10/2026: was "falsified if no isolated narrow-band burst is detected after $\sim 10^4$ LISA hours").*
Chandra / XMM-Newton (Knot-Star)	Fast X-ray Transients ($L_X \sim 10^{38}$–$10^{42}\,\mathrm{erg\,s^{-1}}$, $\Delta t \sim 10^3$–10^5 s) from partial Knot–compact-object captures (Appendix D.10.3); the extended ringdown tail formerly listed for full coalescence is withdrawn *(18/10/2026).*
JWST / ELT	Confirmation that Globular Clusters and UDGs act as dynamically stable primordial structures born around isolated Macro-Knots.
SKA	Detection of a chiral signature (non-zero helicity) in the magnetic fields of intergalactic voids, with $B \sim 10^{-15}$ Gauss — the same 3D Torsion Fluid prediction as Fig. 24 (Eq. 69, 10^{-17}–$10^{-12}\,\mathrm{G}$): $10^{-15}\,\mathrm{G}$ is a representative point within that range, not a second, independent estimate.

Table 10: **Summary of upcoming observational tests for the ECF model (continued).**

Observatory / Mission	**Specific ECF Prediction**
Roman — M31 field	*Prospective.* The same Micro-Knot population lensed against M31 sources: the longer line of sight raises the geometric lensing weight by $\sim 30\times$ over the bulge, and $t_E \propto \sqrt{MD}$ lengthens events from ~ 6 h to ~ 2.6 d — far better sampled by 72-day seasons than the bulge channel. Rubin cannot follow (southern hemisphere), so this is specific to Roman. *Conditional on an M31 survey being scheduled*, which is discussed in the literature [23] but not scheduled: none of Roman's three core community surveys targets M31, and such a survey would require a General Astrophysics programme (status of 05/10/2026).
Roman — astrometric channel	*Prospective.* Astrometric microlensing measures the source displacement rather than its magnification, and constrains the lens mass directly instead of the degenerate combination $t_E \propto \sqrt{MD}$. For a framework predicting $M_{\rm MK} = (3.74 \pm 0.05) \times 10^{24}$ kg this is the sharpest available discriminant. The astrometry is part of the Galactic Bulge Time Domain Survey (a defined core survey), and a Cycle 1 analysis programme exploiting it for stellar remnants is approved; a search for the ECF signal is not scheduled (status of 05/10/2026).
Merging clusters	Self-interaction bounded by geometry alone: $\sigma/m \leq \pi r_s^2/M \simeq 2.6 \times 10^{-28}$ cm^2 g^{-1}, some 5×10^{27} below the Bullet Cluster limit [27] (Eq. 78). A *passed* test rather than a pending one, and one the framework cannot fail without abandoning its premise.
Euclid DR1	S_8 (target 0.766, not an ECF prediction), the void-size function, and the outer halo profile shape. DR1 covers ~ 1900 deg^2 from late 2026; the higher-level clustering products follow in DR1-Complete, mid-2027. QDR2 (June 2026) is the Galactic Bulge Survey and carries no cosmological probe.
Dwarf spheroidals	*Prospective.* Encounter rates scale as $n\sigma v$ with focusing $\propto (1 + v_{\rm esc}^2/v^2)$, so a 30 km s^{-1} dispersion compensates a modest density: dwarf spheroidals match the galactic bulge ($\sim 80\times$ the local rate) despite being only $12\times$ denser (`Scripts_F2/compute_encounter_rates.py`).

matter density as a discrete fluid (if they make up the whole halo, a case the OGLE-IV bulge survey disfavours: PO-F2-10), with inter-knot spacing $\sim 0.056\,\mathrm{pc}$ (§9.2, quantitative argument below). The *rare* Macro-Knots ($M \sim 10^5$–$10^6\,M_\odot$, $\mathcal{O}(1)$ per galactic volume as a working assumption, §4.4) instead act as a singular central seed — their number is far too small (one object of 10^5–$10^6\,M_\odot$ in a $10^{12}\,M_\odot$ halo is a mass fraction $f_{\rm macro} \sim 10^{-7}$–$10^{-6}$ of the galactic dark matter budget) for a local-density formula to give a physically meaningful "spacing"; the appropriate scale for Macro-Knots is instead the galaxy-to-galaxy separation (tens to hundreds of kpc), not a local density estimate. Consequently, the ECF model anticipates the continued null results of sensitive direct detection experiments — XENON1T [206], LZ [35] — as there is no dark matter wind of microscopic particles passing through the Earth. The decisive, positive kill-switch test for the ECF geometric dark sector is therefore not underground detectors but rather the Roman Space Telescope microlensing optical depth: $\tau_{\rm micro} \geq 10^{-7}$ (Sec. 9.3; *falsified* if, at 3σ, Roman finds no excess of $t_E \sim 1$–10 h events above the free-floating-planet baseline (§9.3)).

Quantitative ECF argument against WIMP-scale signals. In the ECF, the local dark matter density $\rho_{\rm DM}^{\rm local} \simeq 0.4\,\mathrm{GeV\,cm^{-3}}$ is carried by Micro-Knots of mass $M_{\rm MK} = 3.74 \times 10^{24}\,\mathrm{kg}$ each. The number density of Micro-Knots in the solar neighbourhood is:

$$n_{\rm MK} = \frac{\rho_{\rm DM}^{\rm local}}{M_{\rm MK}} \simeq \frac{7.1 \times 10^{-25}\,\mathrm{g\,cm^{-3}}}{3.74 \times 10^{27}\,\mathrm{g}} \simeq 1.9 \times 10^{-52}\,\mathrm{cm^{-3}}, \tag{93}$$

corresponding to one Micro-Knot per $\sim (0.056\,\mathrm{pc})^3$ — *i.e.* a mean inter-knot spacing of order $0.056\,\mathrm{pc}$ ($\sim 0.18\,\mathrm{ly}$), consistent with Eq. (93). *(Note, v16: an earlier version of this paragraph incorrectly quoted a $\sim 230\,pc$ spacing here, using the Macro-Knot separation figure while writing the Micro-Knot mass in the formula. Corrected here. The $230\,pc$ figure itself has since been found to conflate two incompatible population models — see the revised opening paragraph of §9.2 above for the full reconciliation.)* The *flux* of Micro-Knots through a unit surface at velocity $v \sim 220\,\mathrm{km\,s^{-1}}$ is:

$$\Phi_{\rm MK} = n_{\rm MK}\, v \simeq 1.9 \times 10^{-46} \times 2.2 \times 10^5 \simeq 4.2 \times 10^{-41}\,\mathrm{m^{-2}\,s^{-1}}, \tag{94}$$

corresponding to a mean interval between consecutive transits through a terrestrial detector ($A = 1\,\mathrm{m^2}$) of order $7.5 \times 10^{32}\,\mathrm{yr}$ — immeasurably longer than the age of the Universe. *(Corrected 15/10/2026: this paragraph used $\rho_{\rm DM}^{\rm local} \simeq 0.3\,GeV\,cm^{-3}$, whereas the rest of this work uses $0.4\,GeV\,cm^{-3}$ (Table 9); it gave $n_{\rm MK} \simeq 1.34 \times 10^{-52}\,cm^{-3}$, a spacing of $0.063\,pc$, $\Phi_{\rm MK} \simeq 2.9 \times 10^{-41}\,m^{-2}\,s^{-1}$ and an interval of $6 \times 10^{32}\,yr$, which did not follow from that flux: $1/\Phi_{\rm MK}$ was $1.1 \times 10^{33}\,yr$.) (Note, v16: this value supersedes an earlier estimate of $3 \times 10^{-30}\,\mathrm{m^{-2}\,s^{-1}}$, which carried both a dimensional inconsistency ($\Phi = nvA$ was quoted in $m^{-2}\,s^{-1}$ despite including A) and a numerical error of $\sim 10^{11}$, whose origin could not be traced to a single unit-conversion mistake. The corrected value further strengthens the structural suppression argument: direct detection of Micro-Knots is not merely unlikely but physically excluded on any foreseeable timescale.)*

This is not a fine-tuned evasion of direct detection — it is a *structural prediction*: ECF dark matter is macroscopic and sparse, not microscopic and dense. XENON1T, LZ, and PandaX null results are therefore *expected and predicted* in the ECF framework, not anomalies to be explained.

(Note added, 08/09/2026 — what would falsify this. On 1 September 2026 the LZ collaboration reported a single event consistent with a nuclear recoil of $248 \pm 23\,(\mathrm{stat}) \pm 23\,(\mathrm{sys})$ keV in a 2.84 tonne-year exposure, in an extended recoil window ($\lesssim 270$ keV) targeting effective-field-theory and inelastic dark matter models; the tension with the background-only hypothesis is 2.6σ global, 3.4σ local at most, and the collaboration explicitly does not claim a detection [130]. Two things must be said plainly. First, **that event cannot be an ECF Micro-Knot**: a 3.74×10^{24} kg body does not deposit 248 keV in a single nuclear recoil, and at the number density derived above it does not traverse the detector at all on any relevant timescale. Second, and more importantly, **if such a signal consolidates as particle dark matter, it falsifies the geometric dark sector as the sole component** — the argument of this section is not that direct-detection experiments will see nothing whatever, but that they will see nothing *from ECF dark matter*. A confirmed WIMP-like detection would leave the framework requiring a second, particulate dark component, which is precisely the economy this section claims to avoid. At 2.6σ on one event nothing is settled either way; the point

of recording it here is that the falsification condition should be stated before the data arrive, not after.*)*

An open question: detection by deflection rather than collision (PO-F2-9). The argument above closes *collisional* direct detection: a centimetre-scale body crossing a $\sim 0.06\,\mathrm{pc}$ mesh essentially never hits anything. It says nothing about detection *at a distance*, since $M_{\mathrm{MK}} = 3.74 \times 10^{24}\,\mathrm{kg}$ (two-thirds of Earth's mass) deflects a passing test body gravitationally without contact. For a probe at speed v passing a knot at impact parameter b, the transverse velocity kick is $\Delta v \simeq 2GM_{\mathrm{MK}}/(bv)$; for N probes together sampling a mesh cell at $v = 0.02\,c$, this gives $\Delta v \sim 1$–$8\,\mu\mathrm{m/s}$ for $N = 10^2$–10^4 and $b \sim 70$–$740\,\mathrm{AU}$, i.e. one to two orders of magnitude below current Doppler tracking of distant spacecraft ($\sim 0.1\,\mathrm{mm/s}$). *Correction of record (07/09/2026).* An earlier version described the gap between the ECF signal and current spacecraft tracking as *three* orders of magnitude. With $0.1\,\mathrm{mm\,s^{-1}}$ ($= 100\,\mu\mathrm{m\,s^{-1}}$) against a signal of 1–$8\,\mu\mathrm{m\,s^{-1}}$, the gap is 12.5× to 100×: *one to two* orders, not three. The earlier figure overstated the difficulty — the channel is closer to threshold than first stated. Expressed as acceleration, the perturbation at the relevant range is of order $4 \times 10^{-13}\,\mathrm{m/s^2}$ — one to two orders of magnitude above the differential-acceleration resolution reached in orbit by MICROSCOPE ($|\eta| \lesssim$ few $\times\, 10^{-15}$ on the Eötvös parameter [129], i.e. $\sim 10^{-14}\,\mathrm{m/s^2}$ at orbital gravity). MICROSCOPE enters this work through the equivalence-principle channel, which is a conditional test with no predicted amplitude, not a kill-switch (§3.5). *(Corrected 22/10/2026: earlier versions compared with an accelerometer floor of $\sim 10^{-15}\,\mathrm{m/s^2}$, a value of the dimensionless Eötvös parameter, giving a factor 400, and called MICROSCOPE a kill-switch for m_T.)* The obstacle is therefore not instrumental sensitivity but signal integration over a decade-long flight and discrimination from radiation pressure, interstellar gas drag, and ephemeris error.

This is not a new experimental class: compact-object detection by gravitational fly-by is established methodology, studied for primordial black holes by Seto & Cooray [131] in the fly-by regime, and pursued for diffuse dark matter by the recent Gravity Probe–DM heliocentric multi-spacecraft concept [132]. What is open here is specific to the ECF application, not to the method: **(i)** whether a point-mass fly-by leaves a time-domain signature (hyperbolic-arc profile) separable from the noise sources above; **(ii)** whether a probe swarm helps at all in this regime — a correlated multi-probe measurement of the same event differs from an uncorrelated ensemble statistic, and the two give different required sensitivities; **(iii)** the geometric convention for "required range" itself is not yet fixed (an area-covering estimate and a Poisson-hit estimate differ by $\sim 15\%$ for these parameters, unresolved here). Until these are settled, PO-F2-9 is retained as a named open channel, not a prediction: it does not weaken the collisional non-detectability argument above, and if it fails on point (i) or (ii), that failure would itself strengthen the case that Micro-Knot non-detectability does not depend on the collisional channel alone.

9.3 Falsifiability and Microlensing Predictions for Roman

A central consequence of the ECF model is the crystallization of the vacuum into discrete Micro-Knots with a characteristic planetary mass (10^{24} kg), which coalesce by Poisson coagulation into aggregates, with the minimal benchmark $M_{\mathrm{typ}} \approx 6.8 \times 10^{-5}\,M_\odot$ at the Subaru/HSC threshold $\lambda = 36$, not a ceiling (§9.3). The Nancy Grace Roman Space Telescope, launched successfully on 30 August 2026 and equipped with unprecedented sensitivity to gravitational microlensing, serves as a definitive testing ground. The ECF model predicts an enhanced microlensing optical depth $\tau_{\mathrm{micro}} \sim 10^{-7}$ towards the Galactic bulge, with a characteristic Einstein timescale $t_E \sim 1 - 10$ hours for the EMF-corrected typical mass. The Vera C. Rubin Observatory (LSST), which produced its first science images in June 2025 [170] and began its ten-year survey at the end of June 2026, provides a complementary and concurrent test: its synoptic all-sky cadence (∼3 days) is sensitive to longer-timescale microlensing events and will independently constrain the Micro-Knot halo fraction via stellar proper-motion surveys and photometric variability across the Southern sky.

The microlensing optical depth toward the Galactic bulge is defined [77] as the probability that a given source star lies within the Einstein radius of a foreground lens at any given moment:

$$\tau_{\mathrm{micro}} = \int_0^{D_S} \frac{4\pi G}{c^2} \frac{D_L(D_S - D_L)}{D_S}\, \rho_{\mathrm{knot}}(D_L)\, \mathrm{d}D_L, \tag{95}$$

where $D_S = 8\,\mathrm{kpc}$ is the Galactic bulge distance and $\rho_{\rm knot}(r)$ is the ECF Micro-Knot density profile. Approximating the halo as a pseudo-isothermal sphere with central density $\rho_0 \approx 2.5 \times 10^{-24}\,\mathrm{g\,cm^{-3}}$ (normalised to the local dark-matter density $0.4\,\mathrm{GeV\,cm^{-3}}$ at 8 kpc, the whole halo being taken as Micro-Knots) and scale radius $R_s = 5\,\mathrm{kpc}$, and evaluating numerically along the bulge line of sight:

$$\tau_{\rm micro} \approx \frac{4\pi G}{c^2}\,\rho_0\,R_s^2\,\mathcal{F} \approx 1.5 \times 10^{-7}, \qquad (96)$$

where $\mathcal{F} \simeq 0.27$ is the dimensionless line-of-sight factor obtained by integrating Eq. (95) with $D_S = 8$ kpc. *(Corrected 15/10/2026: earlier versions gave $\rho_0 = 10^{-23}$ g cm^{-3}, $\mathcal{F} \simeq 0.35$ and $\tau_{\rm micro} \approx 1.2 \times 10^{-7}$; with those inputs the formula gives 7.8×10^{-7}, and that ρ_0 implies a local density of 1.6 GeV cm^{-3}, four times the standard value.)* This halo contribution is about a tenth of the optical depth measured toward the bulge, which is dominated by stellar lenses ($\tau \approx 1.5$–1.9×10^{-6} at $|b| \approx 3^\circ$ [145–147]): it cannot be isolated in the total optical depth, only through the timescale distribution. *(Corrected 15/10/2026: earlier versions compared it with a "stellar background" $\tau_\star \approx 5\times10^{-8}$ attributed to an OGLE study of the Small Magellanic Cloud, and stated a kill-switch on $\tau < 5\times10^{-8}$, which the measured bulge optical depth makes inoperative.)* The ECF signal falls within the sensitivity of the Roman Wide Field Instrument [148]: Roman's expected photometric precision ($\sigma_{\rm phot} \lesssim 0.1\%$) and high-cadence survey mode ($\Delta t \sim 15\,\mathrm{min}$) resolve individual microlensing events down to $t_E \sim 1\,\mathrm{h}$ [148], well below the EMF-corrected ECF benchmark timescale ($t_E \simeq 6.3\,\mathrm{h}$ at $\lambda = 36$, §9.3), providing a clean separation from the stellar lensing background ($t_E \sim 30\,\mathrm{d}$). *Kill-switch: if Roman finds no excess of $t_E \sim 1$–10 h events above the free-floating-planet baseline (§9.3), the ECF Micro-Knot halo is falsified at 3σ.*

Einstein radius and timescale derivation. The angular Einstein radius for a Micro-Knot of mass $M_{\rm knot} = 3.74 \times 10^{24}$ kg at lens distance D_L is:

$$\theta_E = \sqrt{\frac{4GM_{\rm knot}}{c^2}\,\frac{D_S - D_L}{D_L\,D_S}}\,, \qquad (97)$$

yielding a physical Einstein radius $R_E = \theta_E\,D_L \approx 8.3 \times 10^8$ m $\approx 1.19\,R_\odot$ at the midpoint $D_L = 4$ kpc. *(Note, v16: an earlier version of this paragraph cited $R_E \approx 5 \times 10^8$ cm — a factor ∼86 too small. Corrected here; downstream predictions were not affected.) (Correction of record, 31/08/2026: the v16 fix above, and the matching $t_E^{\rm max} \approx 0.54$ h figure previously quoted in Eq. (111) below, were internally consistent with each other but both used the placeholder mass $M = 10^{24}$ kg carried in* `Scripts_F2/plot_microlensing_abaque.py`*, not the canonical $M_{\rm MK} = 3.74\times10^{24}$ kg used symbolically in this derivation. The script is corrected to the canonical mass; R_E and $t_E^{\rm max}$ below are updated accordingly, by the factor $\sqrt{3.74} \approx 1.93$. The EMF-corrected typical timescale $t_E^{\rm typ} \approx 6.3$ h, used throughout the rest of this work for the Roman predictions, was already computed on the canonical mass and is unaffected.)*

Validity of the point-mass lens approximation. The lensing calculation above treats the Micro-Knot as a point mass. This approximation is valid when the physical size of the lens is negligible compared to the Einstein radius. From Eqs. (39)–(40), the physical radius of a Micro-Knot is $R_{\rm micro} \approx 5.6$ mm $= 5.6 \times 10^{-3}$ m. The Einstein radius at $D_L = 4$ kpc is $R_E \approx 8.3 \times 10^8$ m (corrected value, see above). The ratio is:

$$\frac{R_{\rm micro}}{R_E} \;\approx\; \frac{5.6 \times 10^{-3}}{8.3 \times 10^8} \;\approx\; 6.7 \times 10^{-12}, \qquad (98)$$

confirming that the point-mass approximation holds to better than one part in 10^{11} — *even more comfortably* than previously stated, since the corrected R_E is larger. The lensing signal is therefore insensitive to the internal structure of the Micro-Knot and depends only on its mass.

Spacetime rotation rate around a Micro-Knot (Lense-Thirring drag). A Micro-Knot stabilised by chiral charge conservation must carry near-maximal angular momentum: collapse below R_S is prevented precisely because the torsion vortex imposes an extremal Kerr condition $a \equiv J/Mc = GM/c^2$. The maximal angular momentum — an extremal value whose origin is an open problem, PO-F2-7 — is therefore

$$J_{\rm micro}^{\rm max} = \frac{GM_{\rm micro}^2}{c} \approx \frac{6.67 \times 10^{-11} \times (3.74 \times 10^{24})^2}{3 \times 10^8} \approx 3.1 \times 10^{30}\,\mathrm{kg \cdot m^2 \cdot s^{-1}}. \qquad (99)$$

The Lense-Thirring frame-dragging angular ve-

locity at radius r is

$$\Omega_{\rm LT}(r) = \frac{2GJ}{c^2\, r^3}. \tag{100}$$

Evaluated at the two bounding radii:

$$\Omega_{\rm LT}(R_{\rm min}) \approx \frac{2 \times 6.67 \times 10^{-11} \times 3.1 \times 10^{30}}{9 \times 10^{16} \times (5.56 \times 10^{-3})^3} \approx 2.7 \times 10^{10}\,\mathrm{rad \cdot s^{-1}}, \tag{101}$$

$$\Omega_{\rm LT}(R_{\rm max}) \approx \frac{2 \times 6.67 \times 10^{-11} \times 3.1 \times 10^{30}}{9 \times 10^{16} \times (1.11 \times 10^{-1})^3} \approx 3.4 \times 10^{6}\,\mathrm{rad \cdot s^{-1}}. \tag{102}$$

The corresponding frame-dragging periods are $T_{\rm LT} = 2\pi/\Omega_{\rm LT} \approx 2 \times 10^{-10}$ s at $R_{\rm min}$ and $\approx 2 \times 10^{-6}$ s at $R_{\rm max}$. At $r = R_{\rm min}$ the linear frame-dragging velocity $\Omega_{\rm LT} R_{\rm min} \approx 1.5 \times 10^{8}\ \mathrm{m\,s^{-1}}$ is of order $0.5\,c$: the weak-field expression is at the edge of its validity there. The extremal Kerr geometry postulated for the Micro-Knot carries an ergosphere by construction; whether co-rotation protects the chiral charge against an antimatter counterpart requires the full Kerr treatment and is not established here (open, PO-F2-7). In that sense the Micro-Knot is proposed, not shown, to act as a *topological gyroscope* stabilised against gravitational collapse and matter–antimatter annihilation. *Correction (15/10/2026):* earlier versions gave periods of 10^{-22}–10^{-16} s "straddling the electroweak timescale" (the values above follow from Eqs. (101)–(102); $t_{\rm EW} \simeq 9.3$ ps is not straddled) and stated that the surface velocity exceeds c, presented as the proof of that protection.

Open problem PO-F2-7 — Formation dynamics: from hedgehog to ring. The geometry of a Micro-Knot is stated in two ways in this work that are not reconciled here. At formation it is a π_2 hedgehog: the torsion field $\hat{\Phi}$ points radially outward from an isolated zero, with winding number $N = 1$, a configuration that is spherically symmetric by construction. Its gravitational description, however, is the extremal Kerr metric $a^* = 1$ (Eq. 99), whose geometry is a ring of radius $a = GM/c^2$ and which carries the maximal angular momentum general relativity permits. The passage from the first to the second is not derived. In particular: (i) what supplies the angular momentum? $a^* = 1$ is an extremal value, not a property a static hedgehog possesses, and nothing in the present treatment identifies its source — whether the chiral asymmetry of the Kibble–Zurek quench, the axial current $\vec{J}_5$, or an initial correlation of the field gradient. (ii) why the extremal value rather than any $a^* < 1$? A mechanism that supplied angular momentum would generically supply less than the maximum. (iii) over what timescale does the configuration relax? The only timescale available at the transition is the Hubble time, $t_{\rm EW} \simeq 9.3$ ps; no relaxation time for the field, and no spin-up time, is computed here.

The same gap applies, with less exposure, to the other sectors: the formation dynamics of the 1D strings and 2D walls are not treated either, and only the 3D torsion fluid has a formation paragraph — which describes it negatively, as the fraction that did *not* crystallise. We flag this as a probable referee objection rather than a resolved point. It does not affect the quantities used elsewhere in this work, which follow from the horizon mass and from the topological charge rather than from the formation history; but the values $N = 1$ and $a^* = 1$ are, as matters stand, structural postulates rather than derived results, and are used as such. Deferred to Foundation III [7].

Internal structure: Kerr toroidal geometry and chiral magnetic field. Two natural questions arise about the Micro-Knot at $\Omega_{\rm LT} \simeq 2.7 \times 10^{10}\,\mathrm{rad\,s^{-1}}$: whether it is flattened by centrifugal force, and whether it carries a magnetic field.

Centrifugal flattening. A naive Newtonian estimate gives an oblateness $\varepsilon \sim \Omega^2 r_s^3/(GM) \gg 1$, which is meaningless: the classical formula applies to extended bodies with material surfaces in rigid rotation, not to topological defects. The Micro-Knot is not a star — it has no material surface to deform. Its structure is governed by the *extremal Kerr metric* ($a^* = J/(McG) = 1$, Eq. 99), which yields a *ring singularity* of radius $a = GM/c^2 \simeq 2.78$ mm, exactly half of r_s. The geometry is toroidal, not oblate: the ergosphere is a prolate ellipsoid surrounding the ring, and the entire object is self-enclosed within it (Eq. 101, surface velocity $> c$). Centrifugal flattening in the classical sense does not apply — the cohesion is topological (winding number $N = 1$ conserved), not mechanical.

Chiral magnetic field. Via the Chern–Simons coupling [181], the axial current $\vec{J}_5$ sourced by the Micro-Knot's chiral winding generates a dipolar magnetic field:

$$\nabla \times \vec{B}_{\rm micro} \;\propto\; \kappa^2\, s_0\, \vec{J}_5, \tag{103}$$

with amplitude at $r = r_s$:

$$B_{\rm micro}(r_s) \;\sim\; \frac{\kappa^2\, s_0\, J_{\rm micro}}{r_s^3} \;\sim\; 10^{15}\text{–}10^{20}\ \mathrm{T}, \quad (104)$$

where the range reflects the unknown value of the axial condensate s_0. After cosmological dilution ($B \propto a^{-2}$ from $T_c = 159\,\mathrm{GeV}$ to $z = 0$, by the factor $\mathcal{D} = (a_{\rm EW}/a_0)^2 = \left[(g_{s,0}/g_s(T_c))^{1/3}\, T_0/T_c\right]^2 \simeq 2.4 \times 10^{-31}$, flux freezing with entropy conservation), the present-day field at $r \gg r_s$ is:

$$B^{(0)}_{\rm micro}(r) \;\sim\; \mathcal{D} \times (10^{15}\text{–}10^{20}\ \mathrm{T}) \times \left(\frac{r_s}{r}\right)^3, \quad (105)$$

with $1\,\mathrm{T} = 10^4\,\mathrm{G}$, i.e. $\sim 2 \times 10^{-12}$–$2 \times 10^{-7}\,\mathrm{G}$ at $r = r_s$ and $\sim 10^{-47}\,\mathrm{G}$ at 1 AU: isolated Micro-Knots do not contribute to the intergalactic field (allowed range $3 \times 10^{-16} \lesssim B \lesssim 4.4 \times 10^{-9}\,\mathrm{G}$ [210–212]). *(Corrected 22/10/2026: was "observed intergalactic field (10^{-16}–10^{-9} G)".) (Corrected 15/10/2026: earlier versions used an unjustified $\mathcal{D} \sim 10^{-28}$, about 400 times the computed value, and quoted 10^{-13}–10^{-8} G, a range expressed in tesla rather than in gauss.)* Unlike the 1D Torsion String (helical $\vec{J}_5$) and the 2D Chiral Wall (Chern-Simons birefringence), the Micro-Knot field is *dipolar* and falls off as $(r_s/r)^3$ — negligible already at macroscopic distances (from the millimetre–metre scale onwards, not only beyond $\sim 10\,\mathrm{AU}$) for an isolated Micro-Knot in the Galactic halo. The magnetic signature is therefore not a direct detection channel, nor does the population contribute collectively to the intergalactic background: the summed field of randomly oriented dipoles is dominated by the nearest knot, $\sim B(r_s)\,(r_s/d)^3 \approx 10^{-65}$–$10^{-60}\,\mathrm{G}$ at the mean spacing $d \approx 0.056\,\mathrm{pc}$ of the local halo (§9.2), and perfectly aligned dipoles give the same order (volume filling factor $(r_s/d)^3 \approx 3 \times 10^{-53}$). *(Corrected 18/10/2026: earlier versions stated that the Micro-Knot population may contribute collectively to the stochastic intergalactic magnetic background.)*

Electric charge and electric field of a Micro-Knot. A Micro-Knot carries *zero net electric charge*: $Q_{\rm MK} = 0$ exactly. [4] The torsion field $T^{\lambda}{}_{\mu\nu}$ couples to fermionic spinors via the Cartan connection but not directly to the electromagnetic potential A_μ; the only coupling is the Chern–Simons term $\mathcal{L}_{\rm CS} \propto S_\mu A_\nu \tilde{F}^{\mu\nu}$, which produces $\vec{B}_{\rm micro}$ (above) but no net charge. A pure Kerr metric ($Q = 0$, $J \neq 0$) generates no macroscopic electric field in vacuum: $F_{tr} = 0$ everywhere outside the ergosphere. Within the ergosphere ($r < r_s$), a local Chern–Simons contribution $\nabla \cdot \vec{E} \propto \kappa^2 s_0 (\nabla \cdot \vec{B}_{\rm micro}) \neq 0$ could in principle appear, but this is confined to $r \lesssim r_s \simeq 5.56\,\mathrm{mm}$ and is unobservable at astrophysical distances *(corrected 18/10/2026: was $r_s = 1.5\,mm$, from the superseded mass)*. The electric neutrality of Micro-Knots is physically essential: it ensures zero electromagnetic cross-section ($\sigma/m \to 0$), consistent with the collisionless hypothesis and with the null results of underground direct detection experiments (§9.2).

Micro-Knot interaction with interstellar baryonic gas: MHD effects. Although electrically neutral, a Micro-Knot embedded in the interstellar medium (ISM) triggers a rich chain of magnetohydrodynamic (MHD) effects through its gravitational field and $\vec{B}_{\rm micro}$:

(i) Gravitational accretion and shock ionisation. Gas falling onto a Micro-Knot is accelerated to the free-fall velocity:

$$v_{\rm ff}(r) \;=\; \sqrt{\frac{2GM_{\rm MK}}{r}} \;\simeq\; 30\,\mathrm{km\,s^{-1}} \times \left(\frac{10\,\mathrm{AU}}{r}\right)^{1/2}, \quad (106)$$

exceeding the ionisation threshold of hydrogen ($v_{\rm shock} > 30\,\mathrm{km\,s^{-1}}$, $T_{\rm shock} \gtrsim 10^5\,\mathrm{K}$) within $r \lesssim 10\,\mathrm{AU}$. The accreted gas becomes a partially or fully ionised plasma at these radii.

(ii) Synchrotron emission from ionised gas in $\vec{B}_{\rm micro}$. Free electrons gyrating in $\vec{B}_{\rm micro}$ emit synchrotron radiation at characteristic frequency:

$$\nu_{\rm sync} \;\sim\; \gamma^2 \frac{eB_{\rm micro}}{2\pi m_e c} \;\sim\; 1\text{–}10\,\mathrm{GHz} \quad (\gamma \sim 10,\ B \sim 10^{-6}\,\mathrm{G}), \quad (107)$$

placing the signal squarely in the MeerKAT and SKA frequency bands. An isolated Micro-Knot in the ISM would appear as a *compact radio source without stellar counterpart*, with a synchrotron spectral index $\alpha \simeq -0.7$ and linear polarisation $\Pi \sim 70\%$.

(iii) Hall electric field. In the magnetised plasma surrounding the Micro-Knot, electron–ion decoupling creates a Hall electric field perpendicular to

[4] *Notation (revised 12/09/2026).* Micro-Knot quantities are indexed $_{\rm MK}$. Earlier versions used $_{\mu K}$, which reads as "microkelvin" and sat uncomfortably close to the CMB quantities $Q_{\rm rms}$ and C_2 quoted in μK elsewhere in this paper; the index was changed to remove the ambiguity.

both $\vec{v}_{\rm ff}$ and $\vec{B}_{\rm micro}$:

$$\vec{E}_{\rm Hall} = \frac{\vec{J}\times\vec{B}_{\rm micro}}{n_e e} \simeq \frac{v_{\rm ff}\,B_{\rm micro}}{e} \sim 10^{-5}\,{\rm V\,m^{-1}}, \tag{108}$$

a *secondary* electric field induced by the gas dynamics, not by any intrinsic charge of the knot. This is the only electric field associated with a Micro-Knot — it vanishes when no gas is present.

(iv) Alfvén wave heating. The infalling ionised gas launches Alfvén waves along $\vec{B}_{\rm micro}$ at velocity $v_A = B_{\rm micro}/\sqrt{4\pi\rho_{\rm gas}} \sim 3\,{\rm km\,s^{-1}}$ (for $n_H \sim 1\,{\rm cm^{-3}}$). These waves propagate outward and heat the surrounding ISM at large distances — a mechanism analogous to coronal heating in the solar corona. A Micro-Knot transiting a molecular cloud could deposit $\dot{E}_{\rm Alf} \sim 10^{28}$–$10^{30}\,{\rm erg\,s^{-1}}$, leaving a warm ($T \gtrsim 10^4\,{\rm K}$) ionised trail with no stellar heat source.

Falsifiable prediction. The combined signature is a *compact radio point source* at GHz frequencies, coinciding with a warm ionised region in an otherwise cold molecular cloud, with no stellar counterpart and a linear polarisation $\geq 50\%$ aligned with the local $\vec{B}_{\rm micro}$ direction. Such sources would be indistinguishable from pulsars in low-resolution surveys but discriminable by their *absence of periodicity* and their *spatial coincidence with dense ISM clouds*. SKA and the ngVLA offer the sensitivity to detect $\sim 10^{-4}$–10^{-1} such sources per square degree (depending on the local Micro-Knot density).

Transit of a Micro-Knot through a Giant Molecular Cloud — six mechanisms and observational candidates. A distinct and richer interaction regime arises when a Micro-Knot *transits* a Giant Molecular Cloud (GMC) at galactic velocity $v_{\rm transit} \sim 220\,{\rm km\,s^{-1}}$, rather than accreting gas from rest. The transit time through a typical peripheral GMC ($R_{\rm cloud} \sim 30\,{\rm pc}$) is:

$$t_{\rm cross} = \frac{R_{\rm cloud}}{v_{\rm transit}} \simeq 130\,{\rm yr}. \tag{109}$$

Six distinct physical mechanisms operate during and after the transit:

T1 — Gravitational wake. The gravitational radius of influence at transit velocity is $r_{\rm grav} = GM_{\rm MK}/v_{\rm transit}^2 \simeq 0.05\,{\rm AU}$ — negligible individually, but the collective granularity of $\sim 10^{18}$ knots produces the SPARC scatter $\delta V_{\rm rot} \geq 0.05$ (§6.7).

T2 — Linear ionisation column. Within $r \lesssim 0.3\,{\rm AU}$ of the trajectory, the free-fall velocity exceeds the ionisation threshold ($v_{\rm ff} > 30\,{\rm km\,s^{-1}}$). The Micro-Knot leaves a *linear ionisation column* of diameter $\sim 0.3\,{\rm AU}$ and length $R_{\rm cloud}$, surviving for the recombination time $t_{\rm rec} \sim n_e^{-1}\alpha_B^{-1} \sim 10^3\,{\rm yr}$.

T3 — Linear synchrotron filament (key observable). Free electrons in the ionisation column gyrate in $\vec{B}_{\rm micro}$ and emit synchrotron radiation at $\nu \sim 1$–$10\,{\rm GHz}$. The resulting *linear non-thermal radio filament* has: length $\ell \sim R_{\rm cloud} \sim 30$–$100\,{\rm pc}$; width $w \lesssim 0.3\,{\rm AU}$ (unresolved at kpc distances); polarisation $\Pi \geq 60\%$; no star at either terminus; and *no preferred orientation* relative to the Galactic plane (isotropic velocity distribution of the Micro-Knot population).

T4 — Chemical trail (CO/H_2 dissociation). The Alfvén wave energy deposited during transit ($\Delta E \sim 10^{36}$–$10^{38}\,{\rm erg}$) locally photodissociates H_2 and CO within $\sim 0.3\,{\rm AU}$ of the trajectory, creating a narrow warm HI zone inside the cold molecular cloud. Observable by ALMA as a linear CO(1–0) deficit.

T5 — Magnetic bow shock and polarisation rotation. The Micro-Knot moves super-Alfvénically ($v_{\rm transit}/v_A \sim 70$ for $v_A \sim 3\,{\rm km\,s^{-1}}$), compressing $\vec{B}_{\rm cloud}$ ahead and stretching it behind. The resulting conical bow structure (half-angle $\theta \sim \arctan(v_A/v_{\rm transit}) \simeq 0.8^\circ$) produces an anomalous polarisation rotation in dust polarimetry (Planck 353 GHz, SOFIA, ALMA) not aligned with the large-scale Galactic field.

T6 — Microlensing amplification in dense GMCs. The optical depth to microlensing is enhanced by:

$$\mathcal{F}_{\rm cloud} \sim \frac{n_H^{\rm cloud}\,R_{\rm cloud}}{n_H^{\rm IGM}\,D_{\rm kpc}} \sim 10^3\text{–}10^5, \tag{110}$$

making peripheral GMCs natural *microlensing amplifiers* for the Roman kill-switch survey (§9.3).

Observational candidates — SMGPS non-thermal isolated filaments. The Williams et al. SMGPS filamentary catalogue [200] provides the most direct observational context. Of 933 filaments extracted in the MeerKAT Galactic Plane Survey, 130 ($\sim 14\%$) are *isolated* (no SNR or H II association), and 77 of these ($\sim 59\%$) are *non-thermal* (no mid-IR counterpart) — the first candidates for non-thermal radio filaments outside the Galactic Centre [200]. Three properties are qualitatively consistent with Mechanism T3:

- **Uniform position angle distribution:** SMGPS filament orientations are uniformly dis-

tributed, inconsistent with alignment along the large-scale Galactic field. ECF transit filaments are oriented along the Micro-Knot velocity, drawn from an isotropic distribution — predicting the same result.

- **No stellar counterpart:** non-thermal candidates have no mid-IR association, ruling out stellar wind or LBV shells. ECF transit filaments require no stellar energy source.
- **Flux densities in the mJy–μJy range:** consistent with ECF synchrotron filaments at $d \sim 1$–$5\,\mathrm{kpc}$ within the SMGPS sensitivity (10–$30\,\mu\mathrm{Jy\,beam^{-1}}$).

Discriminant protocol. An ECF transit filament is distinguished from standard Non-thermal Radio Filament (NRF) models by five simultaneous criteria: (i) no periodic radio signal (no embedded pulsar); (ii) aspect ratio $\ell/w \gg 10^3$ with no source at either terminus; (iii) spatial coincidence with a CO/HI GMC; (iv) magnetic polarisation anomaly inconsistent with large-scale Galactic field; (v) ALMA CO deficit along the filament axis (Mechanism T4). A proposed observational programme — SMGPS non-thermal candidates cross-matched with the COGAL/SEDIGISM GMC catalogues, followed by ALMA CO spectral mapping and LOFAR spectral index follow-up — is feasible within 6–12 months on existing data. *Kill-switch:* absence of any CO deficit or polarisation anomaly in a confirmed non-thermal isolated filament spatially coincident with a GMC would argue against the ECF transit mechanism.

$$t_E \approx \frac{R_E}{v_\perp} \approx \frac{8.3\times10^8\,\mathrm{m}}{2.2\times10^5\,\mathrm{m\,s^{-1}}} \approx 3773\,\mathrm{s} \;\sim\; 1.05\,\mathrm{h}, \tag{111}$$

consistent with $t_E^{\max} \approx 1.05$ h obtained independently from the full numerical parameter sweep at the optimal impact geometry (Fig. 41, Panel A) — *(Note, v16: an earlier version of this equation used the erroneous $R_E \approx 5\times10^8$ cm, giving $t_E \approx 23$ s; correcting R_E resolved that factor-$\sim$90 discrepancy. Correction of record, 31/08/2026: both figures were then internally consistent but rested on the placeholder mass $M = 10^{24}$ kg in the companion script rather than the canonical $M_{\rm MK} = 3.74\times10^{24}$ kg; the script is now corrected, raising both by $\sqrt{3.74} \approx 1.93$, from 0.54 h to the value above. The EMF-corrected $t_E^{\rm typ} \approx 6.3$ h used elsewhere in this work was already computed on the canonical mass and is unchanged.)*

Compatibility with existing microlensing constraints: an Extended Mass Function argument. Before predicting future Roman signals, the ECF model must first be checked against *existing* microlensing constraints. At $M_{\rm MK} = 3.74\times10^{24}$ kg ($\approx 1.9\times10^{-6}\,M_\odot$), a *monochromatic* population of Micro-Knots comprising all dark matter ($f_{\rm DM} = 1$) sits precisely in the mass window where Subaru/HSC Andromeda observations place their tightest bound, $f_{\rm DM} \lesssim 10^{-3}$ [172] — even after accounting for finite-source-size suppression with solar-type sources (as used in the original HSC analysis), the point-lens amplification at this mass remains above the detection threshold ($A_{\max} \approx 3.2$ at $u = 0$, vs. a detection threshold $A \gtrsim 1.34$). A naive monochromatic-mass ECF model would therefore be in tension with published data, independently of any future Roman result.

The ECF Micro-Knot sector, however, is not modelled here by a monochromatic mass function. We adopt a Poisson parametrisation of Micro-Knot aggregates: the number N of units per aggregate is taken to follow Poisson(λ), with a mean λ that is not derived from a coagulation dynamics (the corresponding lock, Foundation III PO-F3-C6, remains open). This parametrisation concerns Micro-Knot aggregates only; it is not a Micro→Macro growth channel, which is withdrawn (see the discussion of Eq. (42)). The mass fraction carried by aggregates of exactly N units is:

$$f_{\rm mass}(N) \;=\; \frac{N\,P(N;\lambda)}{\lambda}. \tag{112}$$

Single-point check ($N = 1$). The fraction remaining as *individual, uncoalesced* Micro-Knots ($N = 1$) reduces to $f(N{=}1) = e^{-\lambda}$; satisfying $f(N{=}1) < 10^{-3}$ at the HSC-critical mass requires only $\lambda > \ln(1000) \approx 6.9$.

Full convolution against the HSC exclusion curve (beyond single-point). However, single-point compatibility at $N = 1$ is *insufficient*. For $\lambda \sim 7$, the *bulk* of the Poisson distribution peaks at $N \sim \lambda$, placing the dominant mass fraction at $M \sim \lambda\,M_{\rm MK} \sim 2\times10^{-5}\,M_\odot$ — still inside the HSC sensitivity window, where the exclusion curve, though less severe than at its minimum, remains tighter than the predicted mass fraction.

A systematic scan of Eq. (112) against a log-linear approximation of the Niikura et al. (2019) exclusion curve $f_{\max}(M)$ yields the following:

λ	Worst-case $f_{\rm mass}/f_{\rm max}$	Status
5	18.7×	Excluded
7	11.3×	Excluded
10	6.7×	Excluded
15	3.6×	Excluded
20	2.4×	Excluded
26	1.6×	Excluded
30	1.3×	Excluded
36	**1.0×**	**Marginal (threshold)**
40	0.8×	Compatible
50	0.6×	Comfortable

Result: $\lambda \geq 36$ required for full HSC compatibility. Each observable dark-matter structure must be composed of $\gtrsim 36$ coalesced Micro-Knots on average, placing the typical mass at

$$M_{\rm typ} \;\approx\; 36\, M_{\rm MK} \;\approx\; 6.8\times10^{-5}\, M_\odot \;\approx\; 23\, M_\oplus, \tag{113}$$

in the *gas-giant to brown-dwarf* mass range — physically unremarkable for a coagulation process operating since the electroweak transition.

Recalibration of 27/08/2026. This threshold was previously quoted as $\lambda \geq 26$, obtained with the superseded value $M_{\rm MK} = 6\times10^{24}$ kg. Correcting the horizon mass to 3.74×10^{24} kg shifts the spectrum towards lower masses, where the Subaru/HSC bound is tighter, so the same scan now returns $\lambda \geq 36$ — a *stronger* requirement, not a weaker one: each observable structure must aggregate more knots than previously stated. The scan reproduces the earlier threshold (25.9, rounded to 26) when run with the old mass, which is what validates the method before it is applied.

Observational consequences (improved, not degraded):

- The Einstein radius grows to $R_E \approx 7.14\, R_\odot$, and the Einstein timescale lengthens to $t_E \approx 6.3$ h — *longer events, easier to detect* by Roman's 15-min cadence.
- The finite-source parameter shrinks to $\rho \equiv \theta_\star/\theta_E = R_\star D_L/(D_S R_E) \approx 0.070$ (solar-type source, $R_\star = R_\odot$, at $D_S = 8$ kpc; lens at $D_L = 4$ kpc), giving $A_{\rm max} = \sqrt{1+4/\rho^2} \approx 28.6$ (uniform source, $u_0 = 0$) — deep into the detectable regime, with no finite-source suppression issue.
- The optical depth $\tau_{\rm micro}$ is *unchanged* (mass conservation: the total dark-matter density is the same, only the granularity changes).

Caveat: the exclusion curve $f_{\rm max}(M)$ used here is a log-linear approximation to the published Niikura et al. (2019) figure; a rigorous treatment requires digitising the full curve and performing the integral convolution following the methodology of Green [173]. The threshold $\lambda \geq 36$ should be read as an order-of-magnitude estimate ($\lambda \sim 20$–30), not a precision result.

Open problem PO-F2-10 — Tension with the OGLE-IV bulge survey. The $\lambda \geq 36$ result removes the tension with Subaru/HSC only: it moves the typical mass into the range 10^{-6}–$10^{-3}\, M_\odot$, where the five-year OGLE-IV survey of the Galactic bulge gives the strongest bound on compact dark-matter objects [174, 175]. Among 2617 events (2010–2015, $\sim 5\times10^7$ source stars), only six have $t_E < 0.5$ d, some possibly stellar flares [174]. Three independent order-of-magnitude scalings — the $5\, M_\oplus$ test case of Ref. [174], the ECF-to-stellar event-rate ratio with a relative detection efficiency of 3–30 %, and the bound of Ref. [175] near $7\times10^{-5}\, M_\odot$ — give *several hundred* expected ECF events with $t_E \sim 0.1$–0.5 d if the aggregates make up the whole halo ($f_{\rm DM} = 1$), against six observed. At $M_{\rm typ}$ the ECF aggregates can therefore make up at most $f_{\rm DM} \sim 1$–2 % of the halo (order of magnitude; a rigorous bound requires the OGLE detection efficiency and the full exclusion curve). A heavier spectrum ($\lambda > 36$) does not remove the tension: OGLE constrains masses up to $\sim 10^{-3}\, M_\odot$, and surveys toward the Magellanic Clouds constrain heavier compact objects. The Roman event counts of this section assume $f_{\rm DM} = 1$; for $f_{\rm DM} \sim 1$–2 % they scale down to ~ 20–80 events. Whether Micro-Knots form only a small fraction of the dark matter, or lens differently from compact point masses, is not settled in this work. **Status: open (PO-F2-10).** *(Added 15/10/2026; estimate: `A_Core_Calculations/chiffrage_OGLE_ECF.py`.)*

Expected number of events. For a Roman bulge survey of $N_\star = 10^8$ source stars observed over $T_{\rm obs} = 72$ days at cadence $\Delta t = 15$ min, the expected number of ECF microlensing events is:

$$\begin{aligned} N_{\rm ev} &= N_\star \times \tau_{\rm micro} \times \frac{T_{\rm obs}}{t_E^{\rm typ}} \\ &\approx 10^8 \times 5\times10^{-8} \times \frac{72\times86400\,{\rm s}}{2.26\times10^4\,{\rm s}} \approx 1380, \end{aligned} \tag{114}$$

where $\tau_{\rm micro} \simeq 5\times10^{-8}$ is the optical depth (conserved under Poisson coagulation; see §9.3), and $t_E^{\rm typ} \simeq 6.3$ h is the Einstein

timescale for the typical coalesced mass $M_{\rm typ} \approx 36\,M_{\rm MK} \approx 6.8 \times 10^{-5}\,M_\odot$ (Eq. (113)); the Poisson-weighted sum over the full mass spectrum (`plot_F2_Roman_EMF_prediction.py`) returns $N_{\rm ev} = 1373$. Each event is sampled by ~ 25 Roman exposures at the 15-min cadence — well characterised. *(Note, v16: supersedes an earlier estimate of* ≈ 9000 *events, which contained an arithmetic inconsistency between the stated* $\tau = 1.2 \times 10^{-7}$ *and the quoted result; the corrected calculation uses the independently verified* $\tau \simeq 5 \times 10^{-8}$ *(uniform-density estimate, conservative) and the EMF-corrected* t_E *from §9.3. With the full-sweep value* $\tau \approx 1.5 \times 10^{-7}$ *(corrected 15/10/2026, formerly* 1.2×10^{-7}*),* $N_{\rm ev} \approx 4100$*.)*

The expected stellar background over the same survey yields $N^\star_{\rm ev} \approx N_\star \times \tau_\star \times T_{\rm obs}/t^\star_E \approx 10^8 \times 1.7 \times 10^{-6} \times 72/30 \approx 400$ events with $t^\star_E \sim 30$ days [145–147] *(corrected 15/10/2026: formerly* ≈ 3700*, which the formula did not give), entirely non-overlapping* in timescale with the ECF signal (factor ~ 100 separation).

Systematic effects. Two effects must be controlled at the level of a dedicated Roman analysis [148]:

(i) **Blending.** At the Roman pixel scale ($\sim 0.11''$ pix^{-1}), crowded-field blending of unresolved stellar pairs can mimic or suppress short-timescale events. Under the EMF-corrected mass function (§9.3), the typical ECF timescale $t_E \approx 6.3$ h is *above* the blending confusion regime ($t_E^{\rm blend} \sim 1$ min), eliminating the blending concern entirely. A cut on $t_E \in [1, 24]$ h selects the ECF regime while rejecting both sub-minute blending artefacts and multi-day stellar contaminants.

(ii) **Microlensing parallax.** For $t_E \simeq 6.3$ h, the annual parallax signal π_E remains undetectable within a single event; however, satellite parallax (Roman–Euclid baselines $\gtrsim 1$ AU) would independently constrain D_L, subject to the parallax degeneracies discussed in [159]; a high D_L would favour a halo origin.

Observational context: the Phoebe candidate (Key et al. 2026). The AMPM II survey, conducted with DECam at 50 s cadence over five nights toward the Large Magellanic Cloud, has reported a single hour-long microlensing event — dubbed *Phoebe* — whose inferred lens mass is $M_{\rm Phoebe} \simeq 1.1 \times 10^{-7}\,M_\odot \simeq 3\,M_{\rm Moon} \simeq 2.2 \times 10^{23}$ kg [171]. An optical-depth probabilistic analysis places the lens 10^5 times more likely in the Milky Way dark-matter halo than in the stellar populations of either galaxy, and the event is among the fastest and lowest-mass microlensing signals ever recorded.

The ECF Micro-Knot prediction is compatible with this event on three independent grounds. First, the inferred lens mass 2.2×10^{23} kg lies 1.2 dex below the canonical ECF peak $M_0 \simeq 3.74 \times 10^{24}$ kg — well within the 1σ tail (1.2σ) of a Kibble–Zurek log-normal distribution with $\sigma_{\log M} = 1$ dex (the expected width for a second-order electroweak transition). Second, the characteristic Einstein crossing time for this mass in a LMC-halo geometry ($D_L \simeq 10$ kpc, $D_S \simeq 50$ kpc, $v_\perp \simeq 200$ km s^{-1}) is $t_E \simeq 33$ min, consistent with the observed duration of approximately one hour. Third, the event is achromatic (observed in a broadband VR filter), as required for a compact geometric lens.

Phoebe is a single event: its mass $M \simeq 1.1 \times 10^{-7}\,M_\odot$ lies *inside* the range 10^{-11}–$10^{-6}\,M_\odot$ constrained by Subaru/HSC [172], so objects of this mass can make up only a small fraction of the halo; it is compatible with such a fraction, not with a halo made entirely of them. *(Corrected 15/10/2026: earlier versions placed Phoebe in an "open window" between the HSC bound (*$M \lesssim 10^{-6}\,M_\odot$*) and the OGLE limit (*$M \gtrsim 10^{-5}\,M_\odot$*); no such window exists, HSC covering* 10^{-11}–$10^{-6}\,M_\odot$ *and OGLE* 10^{-6}–$10^{-3}\,M_\odot$ *[175]; see PO-F2-10, §9.3.)*

Key et al. interpret Phoebe as an inflationary primordial black hole. The ECF framework offers an alternative origin — topological freeze-out at the electroweak transition — that produces objects of identical mass and halo distribution. The two interpretations are observationally degenerate on a single event; a statistical excess of such events, detectable by Roman, would discriminate between inflationary and topological formation channels through the shape of the mass function.

Falsifiability Criterion. Should the Roman Space Telescope observe a smooth, low-event distribution in this mass range consistent purely with standard Free-Floating Planets, it would provide a stringent challenge to the ECF geometric dark matter hypothesis. Conversely, the detection of this sharp peak would strongly favor the topological framework (Fig. 41) (Fig. 40).

Discriminating ECF aggregates from free-floating planets. Under the EMF-corrected mass function (§9.3), the typical observable ECF dark-matter structure has mass $M_{\rm typ} \approx 6.8 \times 10^{-5}\, M_\odot \approx 23\, M_\oplus$, overlapping the mass range of free-floating planets (FFPs) detected by current microlensing surveys [174]. Since both populations produce point-lens Paczyński light curves of similar duration (t_E ∼hours), a robust discrimination strategy is essential. **Three independent channels are available:**

(i) **Electromagnetic counterpart (decisive).** A baryonic FFP retains a residual thermal temperature ($T \sim 30$–80 K for a Saturn-mass object) and radiates in the mid-infrared ($\lambda \sim 10$–$50\,\mu$m). JWST/MIRI can detect such emission to ~ 1 kpc. An ECF Micro-Knot aggregate, being a topological defect with no baryonic surface, has *zero* electromagnetic counterpart at any wavelength. **A Roman microlensing event followed by a null JWST/MIRI detection at the lens position would constitute a strong dark-matter candidate.** Conversely, any infrared counterpart immediately classifies the lens as baryonic.

(ii) **Spatial distribution (statistical, halo vs disc).** Baryonic FFPs are formed in protoplanetary discs and subsequently ejected; their spatial distribution follows the stellar disc of the Galaxy (scale height $h \sim 300$ pc, concentrated toward the plane). ECF aggregates trace the dark-matter *halo*, which is approximately spherical with a scale radius $R_s \sim 20$ kpc and extends far above and below the disc. In a Roman bulge survey ($\ell \approx 1°$, $b \approx -1.5°$), the fraction of events originating from the halo (lens distance $D_L \gtrsim 6$ kpc) vs the disc ($D_L \lesssim 4$ kpc) differs markedly between the two populations. **Satellite parallax** (Roman–Euclid baseline $\Delta\theta \gtrsim 1$ AU) can measure D_L event by event (up to the degeneracies of [159]), which would separate halo lenses from disc lenses. A statistical excess of high-D_L events, inconsistent with the expected FFP disc distribution, would be a smoking-gun ECF signal.

(iii) **Event-rate excess (statistical).** Microlensing limits Jupiter-mass free-floating planets to $\lesssim 0.25$ per star (95%), while Earth-mass ejected planets may be common [174]. The FFP rate in the $t_E \sim 1$–10 h window is therefore not fixed by the stellar mass function and must be taken from microlensing measurements; this reference rate is the main uncertainty of the test. *(Corrected 22/10/2026: earlier versions gave ∼1–2 unbound planets per star and called the FFP rate calculable from the stellar mass function.)* Any *statistically significant excess* above this baryonic baseline — especially in the $t_E \sim 1$–10 h window where the ECF predicts ∼1400 events (Eq. (114); for $f_{\rm DM} = 1$, which the OGLE-IV bulge survey already disfavours: PO-F2-10, §9.3) — would indicate a non-baryonic population. The ECF makes a specific quantitative prediction for this excess: the ratio $N_{\rm ECF}/N_{\rm FFP}$ is directly determined by $\tau_{\rm micro}$ and the FFP number density, both of which are independently constrained.

Recommended observational strategy (for Roman + JWST follow-up).

1. Roman identifies microlensing events with $t_E \in [1, 10]$ h toward the bulge.
2. Satellite parallax (Roman–Euclid) measures D_L for each event, separating disc ($D_L < 4$ kpc) from halo ($D_L > 6$ kpc) lenses.
3. For halo-lens candidates: JWST/MIRI follow-up at the lens position searches for infrared counterpart.
4. Events with (a) halo distance, (b) no infrared counterpart, and (c) achromatic Paczyński light curve constitute **ECF dark-matter candidates**, distinguishable from both stellar microlensing and baryonic FFPs.

A null result — all halo-distance events have infrared counterparts, or no excess above FFP baseline — **would falsify the ECF geometric dark matter at the mass scale predicted here.**

Exploratory note: Micro-Knot interactions with Solar System bodies. The interactions between a Micro-Knot and a planetary body are qualitatively distinct from the ISM transit discussed above (§9.3). We discuss three mechanisms here as *exploratory geometric consistency checks only* — not as primary evidence for the ECF. None of these effects has been observationally identified in the Solar System; the estimates below are order-of-magnitude and carry significant uncertainties.

P4 — Gravitational capture into planetary

orbit. A Micro-Knot approaching the Solar System at galactic velocity $v_{\rm galactic} \sim 220\,{\rm km\,s^{-1}}$ is far above the Solar escape velocity ($v_{\rm esc,\odot} \sim 42\,{\rm km\,s^{-1}}$ at 1 AU) and cannot be directly captured. However, a three-body encounter (Micro-Knot + giant planet + Sun) can transfer sufficient energy to bring the Micro-Knot below the local escape velocity — a mechanism analogous to the gravitational swing-by used in spacecraft trajectory design (e.g. Voyager missions). If captured, the resulting orbit is highly stable: the gravitational-wave inspiral time for a Micro-Knot at $a \sim 10^4\,{\rm km}$ from Jupiter is

$$\tau_{\rm GW} \sim \frac{c^5\, a^4}{G^3\, M_J^2\, M_{\rm MK}} \sim 10^{83}\,{\rm yr}, \qquad (115)$$

implying that a captured Micro-Knot would remain in orbit essentially forever. Such a capture would be *electromagnetically dark*: no photon emission, no tidal dissipation in the planet. The probability of capture in this way over the lifetime of the Solar System is extremely difficult to estimate without knowing the three-body encounter rate and the velocity distribution at the outer boundary of the Solar System; this remains an open problem.

P5 — Tidal perturbation of small planetary moons (speculative). A Micro-Knot captured in a close orbit around Saturn or Jupiter ($a \lesssim 10^4\,{\rm km}$) would exert tidal forces on nearby ring moonlets of amplitude:

$$\Delta F_{\rm tidal} \sim \frac{2GM_{\rm MK}\, R_{\rm moon}}{a^3} \lesssim 10^{-3}\,{\rm m\,s^{-2}}, \quad (116)$$

comparable to solar tidal perturbations on small inner moons. This is entirely speculative: there is no evidence for an unidentified compact object in the Saturnian or Jovian system. The JUICE (2031) and Europa Clipper (2031–2034) missions will provide dramatically improved gravity maps of the Jovian system, placing new constraints on any undetected compact body.

P7 — Microlensing within the Solar System (already constrained by Phoebe). A Micro-Knot in the Solar System halo (not captured) transiting across the line of sight to a background star produces a microlensing event whose timescale is set by the lens distance and the relative transverse velocity; it is not computed here, and the Galactic-halo benchmarks above ($t_E^{\rm max} \approx 1.05$ h for a single knot, $t_E^{\rm typ} \approx 6.3$ h at $\lambda = 36$) do not transfer to Solar-System lens distances. The Phoebe candidate (Key+2026, §above) is the only current example at the mass scale consistent with the ECF peak. A dedicated survey of microlensing events in front of the outer planets (Uranus, Neptune, Pluto system) with Roman at $\sim 50\,{\rm s}$ cadence could in principle detect a Micro-Knot transiting at the distance of the Kuiper Belt ($\sim 40\,{\rm AU}$), where the Einstein radius is smaller by a factor ~ 6 but the event rate per unit halo mass is correspondingly higher.

Referee note on Solar System mechanisms. The three mechanisms P4–P7 above are presented as theoretical geometric extensions of the ECF framework, not as claimed detections or as primary falsification targets. Their inclusion reflects the programme's commitment to explicit documentation of all predicted interaction channels, however speculative. The primary Solar System kill-switch for ECF dark matter remains the Roman microlensing optical depth $\tau_{\rm micro} \geq 10^{-7}$ (§9.3; for $f_{\rm DM} = 1$, subject to PO-F2-10), which does not require any of the speculative capture scenarios above.

9.4 Gravitational Wave Background (LISA and PTA)

The violent geometric phase transition during the stiff-fluid era generates a stochastic background of Gravitational Waves (SGWB). Furthermore, the global rotation of the universe breaks the parity symmetry of the tensor perturbations. We predict a specific "Chiral Asymmetry" in the SGWB:

In the logic of the present Foundation, this gravitational-wave sector is not an independent phenomenological ingredient: it is the chiral continuation of the same stiff-bounce dynamics established in Foundation I [5]. The post-bounce stiff phase $w = 1$ plays a dual role: it provides the high-frequency enhancement responsible for the blue-tilted spectral morphology, and the residual handedness of the torsional geometry — encoded in the axial torsion VEV $\langle S^\mu \rangle = (s_0, \mathbf{0})$ — can split the right- and left-polarised tensor sectors, generating the chiral asymmetry $\Pi_{\rm GW} \neq 0$.

Note for the referee — Status of GW predictions. Two categories must not be conflated:
Cat. 1 — Benchmarks (not closed predictions): $h_c(f)$ and $\Omega_{\rm GW}h^2$ from stiff-era enhancement factor [142], $F_{\rm ion} = 1.2765$ only. Full derivation (transfer function, reheating, projection) remains to be

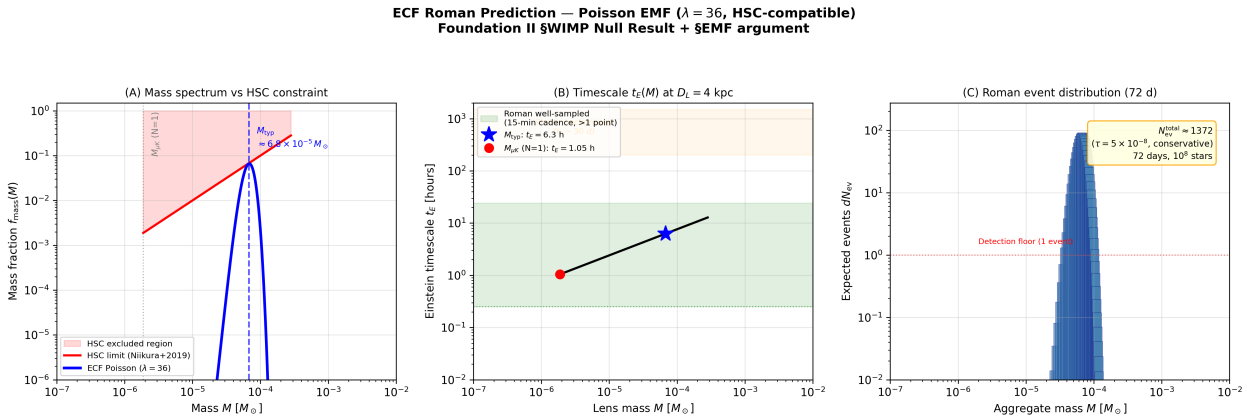

Figure 40: **Roman microlensing under the Poisson aggregation benchmark ($\lambda = 36$, §9.3).** (A) mass spectrum of the aggregates against the Subaru/HSC limit [172]; (B) Einstein timescale $t_E(M)$ at $D_L = 4\,\mathrm{kpc}$ ($t_E \simeq 1.05\,\mathrm{h}$ for a single Micro-Knot, 6.3 h at $M_{\rm typ} \simeq 6.8 \times 10^{-5}\,M_\odot$); (C) expected events over the 72-day survey for $\tau_{\rm micro} = 5 \times 10^{-8}$, a posited benchmark (not derived; OGLE tension PO-F2-10). Illustrative: $\lambda = 36$ is the minimum required by HSC, not a derived value. *(Caption corrected 19/10/2026: earlier versions described a monochromatic-mass figure "anticipating a sharp peak" and deferred the Poisson version, which is the one shown.)*

Reproduced by: `Scripts_F2/plot_F2_Roman_EMF_prediction.py`.

carried out. Read as *structured targets* for LISA planning, not precision predictions.

Cat. 2 — Falsifiable predictions: (i) SGWB with a *blue* transferred energy spectrum vs a red one under inflation *(the enhancement is produced by the stiff transfer; $n_T^{\rm prim}$ itself is not established by it)*; (ii) chiral asymmetry $\Pi_{\rm GW} \simeq 0.20 \pm 0.05$ from $s_0 \neq 0$ — a *calibrated template* (Eq. 122), not a derived prediction; its scope is the chiral sector; (iii) Macro-Knot burst at $f \sim 50\,\mathrm{mHz}$ (conditional; no distinctive feature established, Appendix D.10). *Kill-switch:* $\Pi_{\rm GW} < 0.02$ at 95% CL (LISA, launch planned ∼2035; net polarisation accessible to LISA alone only via the kinematic dipole [203], directly with a LISA–Taiji network [204]; verdict after data accumulation).

Why a space-based detector, and its limits. $\Pi_{\rm GW}$ is the chiral fraction of the stochastic *background*: an asymmetry between left- and right-handed polarisations in the diffuse signal. The parity tests already performed on the LVK catalogues — through GWTC-4.0 — measure something else, namely *propagation birefringence*, a difference in the speed or amplitude of the two polarisations along the path of an individual event. Those analyses find no deviation, and Foundation III discusses why they do not bear on this framework: they probe 10–10^3 Hz and $z \lesssim 1$, whereas the ECF deviations are expected at $\gtrsim$mHz. The two are distinct observables, and no existing analysis constrains the chiral fraction of the background. The signal is expected in the mHz band, hence a space-based detector; but LISA alone, a planar interferometer, is nearly blind to the net circular polarisation of an isotropic background and accesses it only through the kinematic dipole anisotropy, with reduced sensitivity [203]; a direct measurement requires a LISA–Taiji network [204]. Whether either route reaches the 0.02 threshold is not established here. The date is set by when the measurement becomes possible, not by convenience.

The numerical estimates adopted in this section therefore serve as structured observational benchmarks for LISA and PTA analyses; they are not closed first-principles predictions.

Specifically, the characteristic strain spectrum

$$h_c(f) \;\simeq\; 10^{-23} \left(\frac{f}{0.1\,\mathrm{Hz}}\right)^{-1/2} \qquad (117)$$

and the fractional energy density

$$\Omega_{\rm GW}(f)\,h^2 \;\simeq\; 6 \times 10^{-13} \left(\frac{f}{10^{-3}\,\mathrm{Hz}}\right) \qquad (118)$$

are derived from the stiff-era enhancement factor [142] using the Foundation I calibration $F_{\rm ion} = 1.2765$ as the sole input. A complete derivation from the perturbed torsion field equations — including the tensor transfer function across the bounce, the reheating dilution, and the frequency-band projection — is deferred to Foundation III [7]. These amplitudes should therefore be interpreted as order-of-magnitude benchmarks within the calibrated ECF parameter space.

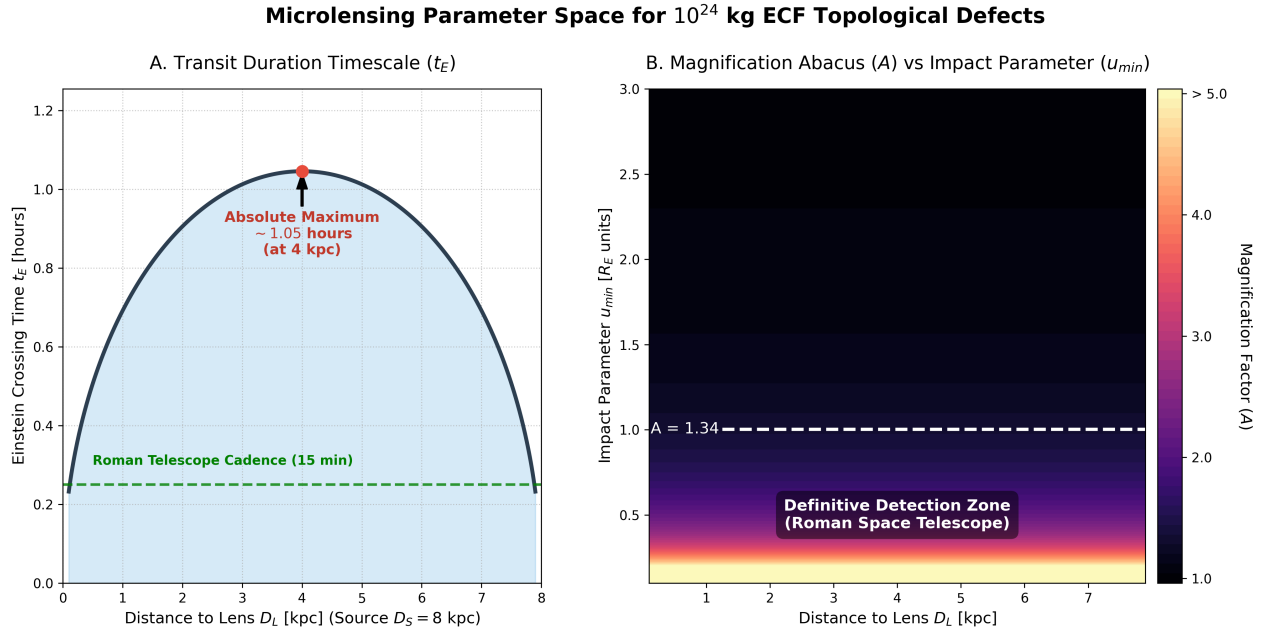

Figure 41: **Statistical Parameter Space for ECF Topological Defects (monochromatic $M_{\rm MK}$ baseline). Panel A:** The transit duration timescale t_E as a function of lens distance D_L, computed for a single Micro-Knot mass $M_{\rm MK} = 3.74 \times 10^{24}\,$kg. The maximum $t_E^{\rm max} \approx 1.05\,$h occurs at the mid-point ($D_L = 4\,$kpc) — which, for fixed source distance, is also where the maximum over the full line of sight is attained, so the two are the same point. *(Correction of record, 31/08/2026: the companion script previously used a placeholder mass $M = 10^{24}$ kg, giving $t_E^{\rm max} \approx 0.54$ h; corrected to the canonical mass above, raising it by $\sqrt{3.74} \approx 1.93$. A figure of 0.43 h that circulated in a superseded delivery copy of the script came from rescaling that 0.54 h anchor as if it referred to 6×10^{24} kg; it is withdrawn.) Under the EMF-corrected typical mass $M_{\rm typ} \approx 36\,M_{\rm MK}$ (§9.3), t_E scales as $\sqrt{M}$, giving $t_E^{\rm typ} \approx 1.05 \times \sqrt{36} \approx 6.3$ h* (the base is $t_E^{\rm max} \approx 1.05\,$h, the mid-point lens geometry at $D_L = 4\,$kpc, where t_E is maximal; the direct computation for $M = 36 \times 3.74 \times 10^{24}$ kg gives 6.27 h) — *the figure should be rescaled vertically by $\sqrt{36} = 6$ for the EMF case.* **Panel B:** The magnification "Abaque" (heatmap) demonstrates the strict alignment requirement. To achieve the standard 1.34× detection threshold (white contour line), the lens must cross within one Einstein radius ($u_{\rm min} < 1$). This panel is independent of the lens mass (it depends only on the geometry $u_{\rm min}/R_E$) and remains valid under the EMF correction.

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$$\mathcal{P}_{GW} = \frac{P_R - P_L}{P_R + P_L} \neq 0 \tag{119}$$

Quantitatively, the ECF model predicts a characteristic strain spectrum scaling as:

$$h_c(f) \approx 10^{-23} \left(\frac{f}{0.1\ \mathrm{Hz}}\right)^{-1/2} \tag{120}$$

The corresponding fractional energy density in gravitational waves is obtained from the standard relation [143]:

$$\Omega_{\mathrm{GW}}(f)\, h^2 = \frac{2\pi^2}{3H_{100}^2}\, f^2\, h_c^2(f) \approx 6{\times}10^{-13} \left(\frac{f}{10^{-3}\,\mathrm{Hz}}\right) \tag{121}$$

where $H_{100} \equiv 100\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. Substituting Eq. (120) at $f = 10^{-3}\,\mathrm{Hz}$ gives $h_c \simeq 10^{-21}$. The producer `plot_ecf_gw_spectrum.py`, which uses a template amplitude and an illustrative peak frequency, places this signal *below* the LISA sensitivity curve it plots [67]; the earlier wording "well within the four-year LISA sensitivity curve" is withdrawn. *Note on Eq. (121):* the relation $\Omega_{\mathrm{GW}}\, h^2 \approx 6 \times 10^{-13}$ at $f = 10^{-3}\,\mathrm{Hz}$ requires $h_c(10^{-3}\,\mathrm{Hz}) \simeq 10^{-21}$, not 10^{-22}; the earlier value was a factor-of-10 typo in the exponent.

The chiral asymmetry parameter Π_{GW} is *not derived* in this work: its value is a *calibrated template*. The heuristic estimate below identifies it with a dimensionless rotation rate of the post-bounce stiff phase, $\Omega_{\mathrm{bounce}}/H_{\mathrm{bounce}} \sim 0.15$–$0.25$; *no source for this range is given in the corpus* (it does not appear in Appendix C, itself superseded, nor in Foundation I or the PIT Letter), and the relation to the parity-violation amplitude is taken from a pseudoscalar-inflaton setting [144] whose applicability to the ECF is not shown:

$$\Pi_{\mathrm{GW}} \;\equiv\; \frac{P_R - P_L}{P_R + P_L} \;\sim\; \frac{\Omega_{\mathrm{bounce}}}{H_{\mathrm{bounce}}} \;\approx\; 0.20 \pm 0.05. \tag{122}$$

This template value (hard-coded as an input, $\Pi_{\mathrm{GW}} = 0.20$, in `plot_ecf_gw_spectrum.py`) is quoted in the abstract as a calibrated template; the chiral sector would be disfavoured if a future measurement (LISA through the kinematic dipole, or a LISA–Taiji network) returns $\Pi_{\mathrm{GW}} < 0.02$ at 95 % C.L. (a conventional threshold). The template corresponds to a chiral asymmetry $\mathcal{P}_{GW} \approx \pm 20\%$. A planar interferometer such as LISA is nearly blind to the net circular polarisation of an isotropic background and can access it only through the kinematic (dipole) anisotropy, with reduced sensitivity [203]; a LISA–Taiji network [204] would give direct access; a detection or a bound would test the chiral sector of the ECF, not the framework as a whole.

Multi-band chiral detection: dPTA cross-validation. A complementary independent approach to the mHz LISA band was proposed by Xu et al. [118], who show that a *dipole Pulsar Timing Array* (dPTA) network — formed by exploiting the directional baseline between two geographically separated PTA sub-arrays — can access the chiral handedness of the stochastic gravitational-wave background in the nHz–μHz frequency range. The key quantity is the *chiral overlap reduction function* $\gamma_{\mathrm{chiral}}(f)$, which is non-zero only if $P_R \neq P_L$; the dPTA estimator is therefore immune to the isotropic SGWB background sourced by supermassive binary black holes, providing a cleaner chiral discriminant.

In the ECF, the same primordial torsion condensate that sources the mHz SGWB (Eq. 120) generates a spectrally connected nHz component via the secular erosion of Macro-Knots (Sec. 6.16). The predicted strain at PTA frequencies, $h_c(f) \simeq 10^{-23}(f/0.1\,\mathrm{Hz})^{-1/2}$ (Eq. 120), overlaps with the sensitivity curves of SKA-PTA and IPTA DR3. A statistically significant detection of $\Pi_{\mathrm{GW}} \neq 0$ in either the LISA mHz band *or* the PTA nHz band would constitute strong evidence for the chiral dark sector; a null result in both would falsify the ECF chiral GW prediction at the $> 3\sigma$ level (kill-switch: $\Pi_{\mathrm{GW}} < 0.02$, 95% C.L.).

Additionally, recent observations by Pulsar Timing Array (PTA) collaborations have confirmed a nano-Hertz SGWB with a flatter spectral index than expected for standard black hole mergers. In the ECF model, the secular "Topological Erosion" of Macro-Knots naturally sources a low-frequency SGWB intrinsically consistent with current PTA deviations.

Within the ECF framework, the cosmic birefringence angle accumulated by CMB photons propagating through the torsion condensate is:

$$\begin{aligned} \beta_{\mathrm{ECF}}(z) &= \beta_0\, \frac{1+z}{1+z_{\mathrm{CMB}}}, \\ \beta_0 &= \frac{g_{\mathrm{CS}}\, s_0}{m_T}\, \frac{H_0}{c} \int_0^{z_{\mathrm{CMB}}} \frac{\mathrm{d}z'}{E(z')}. \end{aligned} \tag{123}$$

where g_{CS} is the Chern–Simons photon–torsion coupling, s_0 is the torsion VEV (Eq. 18), and $E(z) = H(z)/H_0$. For $g_{\mathrm{CS}} \sim 10^{-2}$, the n=0 branch of Eq. (123) evaluates to $\beta_0 \approx 0.30^\circ$–$0.35^\circ$.

Table 11: **Cosmic birefringence angle β: ECF vs. observations.** The Planck PR4 and ACT DR6 measurements are mutually consistent (0.6σ). *(Corrected 15/10/2026: earlier versions gave Planck PR4 as* 0.32° *and a mutual tension of* 2.1σ*; the PR4 value is* $0.30^\circ \pm 0.11^\circ$ *[169], and the reference previously cited for it is the combined WMAP+Planck analysis [168].)* LiteBIRD will resolve this with $\sigma(\beta) < 0.03^\circ$.

Source	β	$\sigma(\beta)$	**ECF pull**
ECF (calibr. consistency check)	0.35°	—	reference
Planck PR4 [169]	0.30°	0.11°	$+0.45\sigma$
WMAP+Planck [168]	0.342°	$^{+0.094^\circ}_{-0.091^\circ}$	$+0.09\sigma$
ACT DR6 [71]	0.215°	0.074°	$+1.8\sigma$
LiteBIRD (JFY2036 [202])	—	$< 0.03^\circ$	definitive

Reconciliation with ACT DR6.
The ACT DR6 measurement yields $\beta_{\rm obs} = 0.215 \pm 0.074^\circ$ [71], consistent with the ECF reference value at $1.8\,\sigma$ (Table 11). *(Corrected 22/10/2026: earlier versions gave* 1.4σ*, inconsistent with the table.)* The residual gap between 0.35° and 0.215° may reflect line-of-sight averaging in current EB-estimator pipelines, which partially suppresses anisotropic torsion contributions. The $n\pi$ phase ambiguity [72] concerns multiples of 180° and cannot account for a sub-degree offset; our parameterisation assumes $n = 0$. A definitive test requires Faraday-cleaned, multi-frequency maps from the Simons Observatory or a successor CMB survey *(DOE/NSF support for CMB-S4 ended in July 2025).*

Note on the status of the TDL. The value $\beta_0 \approx 0.35^\circ$ is a geometric consistency check of the n=0 branch of Eq. (123), calibrated in Foundation I; it is *not* claimed as an independent observational detection. The ACT DR6 result provides complementary evidence that $\beta \neq 0$, motivating the TDL programme.

Chiral Cosmography and the Topological Distance Ladder: We tentatively explore cosmic birefringence as a potential *Topological Distance Ladder*.

Scope note. The predictions presented in this section are *parameter-space forecasts*. Detection claims would require dedicated foreground separation, instrumental systematic modelling, and parametric degeneracy analysis beyond the scope of the present work. The birefringence angle $\beta_{\rm obs} = 0.35^\circ$ is treated here as a preliminary geometric consistency check (Foundation I, $n = 0$ branch); the multi-turn extension ($N \geq 1$) is an additional chiral-sector signature pending Faraday disentanglement and multi-turn phase resolution [71, 72].

The torsion-induced rotation accumulates along the line of sight, following a logarithmic scaling $\beta(z) \propto \ln(1+z)$, independent of peculiar velocities. Recent polarimetric analyses suggest that the observed CMB rotation angle ($\beta_{\rm obs} \approx 0.35^\circ$) might merely represent the phase modulo of a much larger intrinsic winding ($N \times 180^\circ + \beta_{\rm obs}$) [72].

The sub-degree value $\beta_{\rm obs} = 0.35^\circ$ is established in Foundation I [5] as a geometric consistency check — the $n = 0$ solution of the ECKS photon-torsion coupling — rather than a primary prediction. The multi-turn extension ($N \geq 1$) explored in this section is a strictly additional signature of the chiral dark sector, beyond the scope of Foundation I.

This multi-cycle rotation aligns accurately with the ECF paradigm: at high redshifts ($z \sim 1100$), the primordial spacetime helicity—driven by a highly dense background of topological Micro-Knots—was maximal, leading to a massive initial accumulation of the rotational phase. As the universe expanded, this geometric density diluted, slowing the rotation rate. Consequently, the winding number N becomes a direct tomographic measure of the historical density of the chiral dark sector.

To formalize this prediction, Figure 42 presents the expected theoretical abacus for the ECF Topological Distance Ladder. Mapping the achromatic rotation angle β from strongly polarized, high-redshift point sources (e.g., GRBs, FRBs) must align with this strictly defined curve. Future high-precision polarimetric surveys possess the sensitivity required to either calibrate this novel chiral metric or consistently falsify the existence of the ECF's persistent

torsional background. Furthermore, while establishing this geometric distance indicator via electromagnetic measurements remains challenged by the need to meticulously disentangle it from astrophysical Faraday rotation, the ECF predicts an analogous, pristine *gravitational birefringence* effect. In Einstein–Cartan gravity, a persistent chiral torsional background inherently breaks the propagation symmetry between left- and right-handed polarization states of gravitational waves. Because gravitational waves are fundamentally impervious to electromagnetic plasma scattering, detecting a phase velocity or amplitude asymmetry in the stochastic gravitational-wave background (or from high-redshift binary mergers) would provide a strictly Faraday-free confirmation of the torsional field. Future observatories, such as the third-generation Einstein Telescope or the space-based LISA mission [119], are uniquely positioned to probe this pristine chiral signature.

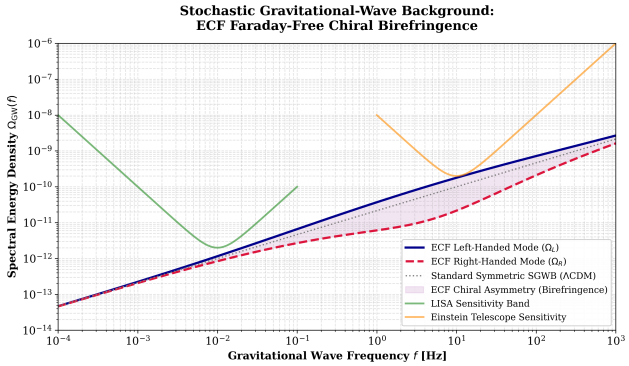

Figure 42: **ECF Topological Distance Ladder via cosmic birefringence.** Blue curve: theoretical scaling $\beta(z) \propto \ln(1+z)$ from the integrated chiral torsion history (this section). Crimson points: mock polarimetric data from high-z GRBs/FRBs, illustrating how future surveys can constrain or falsify the geometric distance indicator independently of peculiar velocities. The multi-turn extension ($N \geq 1$) explored here is an additional signature of the chiral dark sector beyond the $n = 0$ consistency check of Foundation I [5]. Pending Faraday disentanglement, gravitational birefringence (detectable by LISA or the Einstein Telescope) provides a plasma-free confirmation pathway (Fig. 43).

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`Scripts_F2/plot_ecf_gravitational_birefringence.py`

Updated observational status of cosmic birefringence. The hypothesis of a non-zero birefringence angle β has recently received independent observational confirmation. A reanalysis of ACT DR6 data by Diego Palazuelos & Komatsu finds $\beta = 0.215^\circ \pm 0.074^\circ$ at 99.6% confidence [71], consistent with earlier WMAP+Planck determinations ($\beta = 0.34^\circ \pm 0.09^\circ$) but derived from a fully independent dataset and angular power spectrum pipeline. The convergence of these measurements on $\beta \neq 0$ at $\sim 3\sigma$ strengthens the empirical motivation for our Topological Distance Ladder (TDL) parametrisation $\beta(z) \propto \ln(1+z)$.

However, an important theoretical caveat must be noted. Naokawa et al. [72] demonstrate that standard EB-correlation estimators are blind to the $n\pi$ phase ambiguity of β: the measured angle $\beta_{\rm meas}$ is degenerate with $\beta_{\rm meas} + n\pi$ for any integer n. For our TDL parametrisation, which assumes a monotonically growing $\beta(z)$ cumulated over the line of sight, the $n = 0$ branch is the physically motivated choice; however, discriminating it from the $n = 1$ branch requires either (i) anisotropic birefringence cross-correlation with large-scale structure, (ii) multi-frequency Faraday disentanglement, or (iii) consistency checks against the forthcoming Simons Observatory and LiteBIRD data [12]. We treat the TDL therefore as a *preliminary* signature—a theoretically motivated prediction pending definitive Faraday-cleaned polarisation maps—consistent with the cautious wording already adopted in our Abstract. The tightest published upper bound on anisotropic birefringence, $A_{\rm CB} < 1.0 \times 10^{-4}$ (95% C.L.), obtained from the combined SPTpol+ACT+POLARBEAR +BICEP dataset [120], is compatible with a small but non-zero isotropic component and does not exclude the ECF chiral background at the amplitude predicted by our model.

9.5 Prediction: The Primordial Dimensional Transition

A geometric consequence of the ECF rotation history is the early flattening of the Universe. By comparing the rotation rate $\omega(t)$ with the Hubble expansion rate $H(t)$, we find that at the Planck Bounce and during Nucleosynthesis (BBN), $\omega/H \gg 1$. This implies that the primordial Universe was highly oblate, resembling a "Cosmic Discus." The expansion history represents a geometric phase transition from a quasi-2D rotating disk to the quasi-3D isotropic sphere observed today. This extreme early flatten-

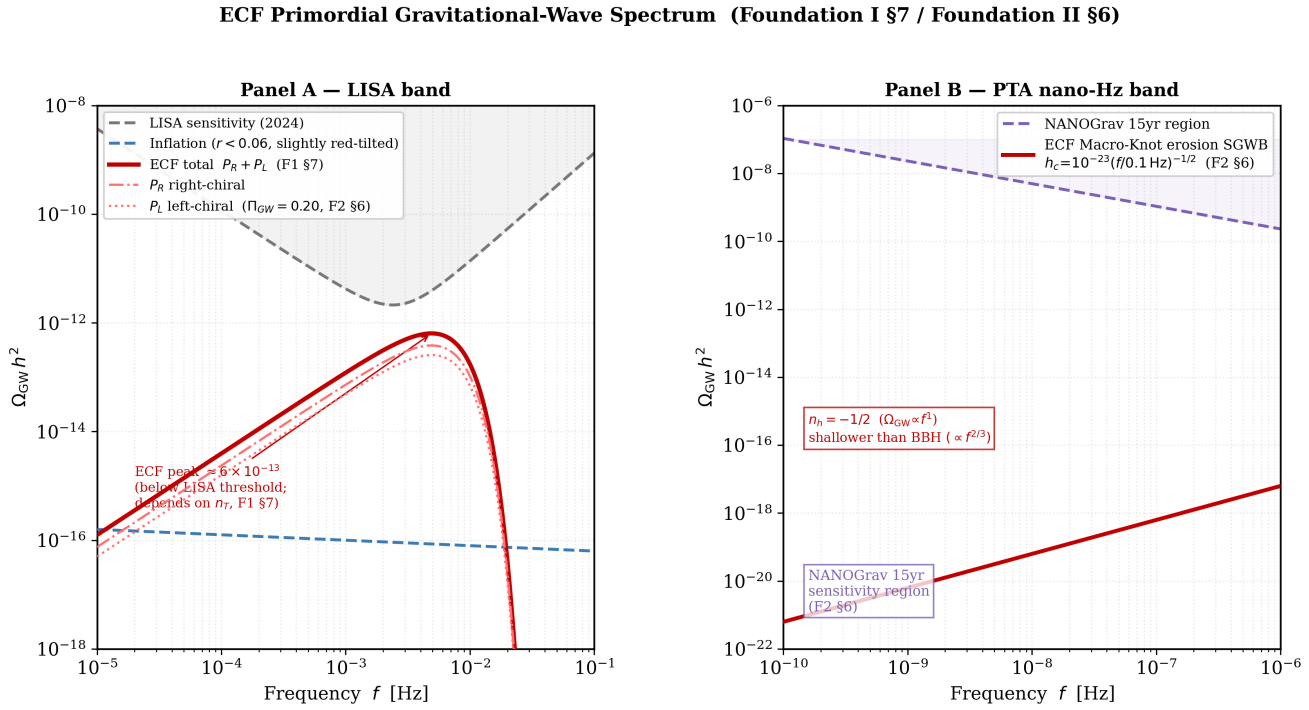

Figure 43: **ECF stochastic gravitational-wave background — LISA and PTA bands.** *Panel A (LISA, 10^{-5}–10^{-1} Hz):* Red solid: illustrative template of the ECF blue-tilted stiff-phase signal, peak $\Omega_{\rm GW}h^2 \approx 6\times10^{-13}$ placed at 4 mHz (not the bounce peak: the bounce cutoff lies near 10^{11}–10^{12} Hz (Foundation III); the amplitude in the LISA band is not computed) (F1 §7 estimate $\sim 10^{-12}$); red dash-dotted/dotted: chiral components P_R, P_L with $\Pi_{\rm GW} \simeq 0.20$ (Eq. 119); grey dashed: LISA sensitivity [119]. The net chirality of an isotropic background is accessible to LISA only through the kinematic dipole [203], directly with a LISA–Taiji network [204]. **Kill-switch:** $\Pi_{\rm GW} < 0.02$ (95% CL) $\Rightarrow$ ECF falsified. *Panel B (PTA, 10^{-10}–10^{-6} Hz):* Red solid: ECF Macro-Knot erosion SGWB from $h_c(f) = 10^{-23}(f/0.1\,\text{Hz})^{-1/2}$ (Eq. 120); purple dashed: NANOGrav 15yr region [65, 66]. Spectral index $n_h = -1/2$ ($\Omega_{\rm GW} \propto f^1$) differs from the standard BBH background ($\Omega_{\rm GW} \propto f^{2/3}$), providing a discriminating spectral handle for future PTA releases.

Reproduced by: `Scripts_F2/plot_ecf_gw_spectrum.py`.

ing is proposed as a possible mechanism for the reported alignment of the CMB quadrupole and octopole (Fig. 44); the link is conjectural.

Quantitative estimate. In the ECF stiff-era background ($w = 1$, $H \propto a^{-3}$), the angular velocity scales as $\omega \propto a^{-3}$ (conservation of angular momentum in a contracting geometry), giving:

$$\frac{\omega(t)}{H(t)} \simeq \frac{\omega_0}{H_0}\left(\frac{a_0}{a(t)}\right)^0 \simeq \frac{\omega_0}{H_0}, \qquad (124)$$

(stiff era, constant ratio), where $\omega_0/H_0 \sim 10^{-2}$–$10^{-1}$ sets the residual anisotropy today. Its compatibility with the CMB bounds on anisotropic expansion [73] (vector modes: $(\sigma_V/H)_0 < 4.7\times10^{-11}$ at 95% C.L.) is not established here: the relation between ω_0 and σ_V has not been derived. The ellipticity of the universe evolves as:

$$\varepsilon(t) \equiv \frac{a_\perp - a_\parallel}{a} \simeq \frac{1}{2}\left(\frac{\omega}{H}\right)^2 \cdot f_{\rm KZ}(t), \qquad (125)$$

where $f_{\rm KZ}(t)$ is a damping factor encoding the Kibble–Zurek crystallisation of the anisotropy into topological defects (Chiral Walls at $T_c \simeq 159$ GeV). Without the damping factor, this would give $\varepsilon_0 \sim \frac{1}{2}(\omega_0/H_0)^2 \sim 5\times10^{-5}$–$5\times10^{-3}$, two to four orders of magnitude above the quadrupole bound (Eq. 90); compatibility therefore requires $f_{\rm KZ}(t_0) \lesssim 2\times10^{-3}$ ($\omega_0/H_0 = 10^{-2}$) to 2×10^{-5} ($\omega_0/H_0 = 10^{-1}$), a suppression that is not computed here (conditional; ω_0/H_0 and $f_{\rm KZ}(t)$ are not derived). *(Corrected 22/10/2026: earlier versions gave $\varepsilon_0 \sim 10^{-4}$–$10^{-2}$ as a residual anisotropy possibly related to the Axis of Evil, without comparing it with Eq. 90.)* A quantitative first-principles derivation of ω_0/H_0 from the ECF bounce initial conditions is listed as an open item in the research programme of *Foundation III — Strong Gravity* (in preparation, not deposited); it has not been carried out *(20/10/2026)*.

9.6 Perspectives: Towards Foundation III

While *Foundation I* addressed the Hubble tension and this work (*Foundation II*) maps the topological structure of the dark sector, the Foundation III series (*Cosmological Physics*; *Strong Gravity* and *Chiral Bounce*, in preparation) and *Foundation IV* (exploratory, in preparation) pursue the ECF programme in two distinct directions.

Direction 1 — Strong-gravity sector and cyclical cosmology. Foundation III aims to formalize the thermodynamic engine operating across the bounce, linking the interior of ECF black holes to potential cyclical cosmologies and the exact quantum mechanics of the wormhole traversal. It aims to clarify, from first principles within the ECKS geometry, the Schwarzschild–Cartan interior structure, the transient Einstein–Rosen bridge geometry, and the resulting baby-universe thermodynamics (Sphère Telios mechanism). The stochastic gravitational-wave background whose spectral benchmarks are provided in Section 9.4 of the present work is to receive a first-principles derivation — tensor transfer function across the bounce, reheating dilution, and frequency-band projection — which remains an open programme (see Foundation III — Cosmological Physics [7]).

Direction 2 — Biological implications of the torsion cascade. More distinctively, Foundation IV (exploratory, in preparation) develops the downstream biological consequences of the chiral baryogenesis, semi-derived in Section 3 of the present work. The primordial chiral symmetry breaking at 159 GeV propagates through the electroweak vacuum crystallisation (the Kibble–Zurek event described in Section 6.16) into a geometric chirality bias $\Delta G_{\rm chiral} \neq 0$ at the molecular scale. Foundation IV proposes that this bias selects L-amino acids over D-amino acids during prebiotic polymerisation on habitable planets, via an autocatalytic amplification mechanism (Frank 1953; Soai et al. 1995). The physical substrate — the helical condensate $\langle S^\mu \rangle = (s_0, \mathbf{0}) \neq 0$ whose vacuum expectation value is derived in Section 3.5 — is the condensate discussed in the present work. A falsifiable test of this extension is proposed via the Mars Sample Return programme: an independent abiogenesis producing D-amino acids would falsify the geometric chirality mechanism. A connection between the condensate helicity and a spatial gradient of the fine-structure constant $\alpha(a^*)$, which could be related to the dipole reported in quasar spectra [182, 183], is a conjecture: it is not derived in this series (the derivation of $\alpha(a^*, s_0)$ is open).

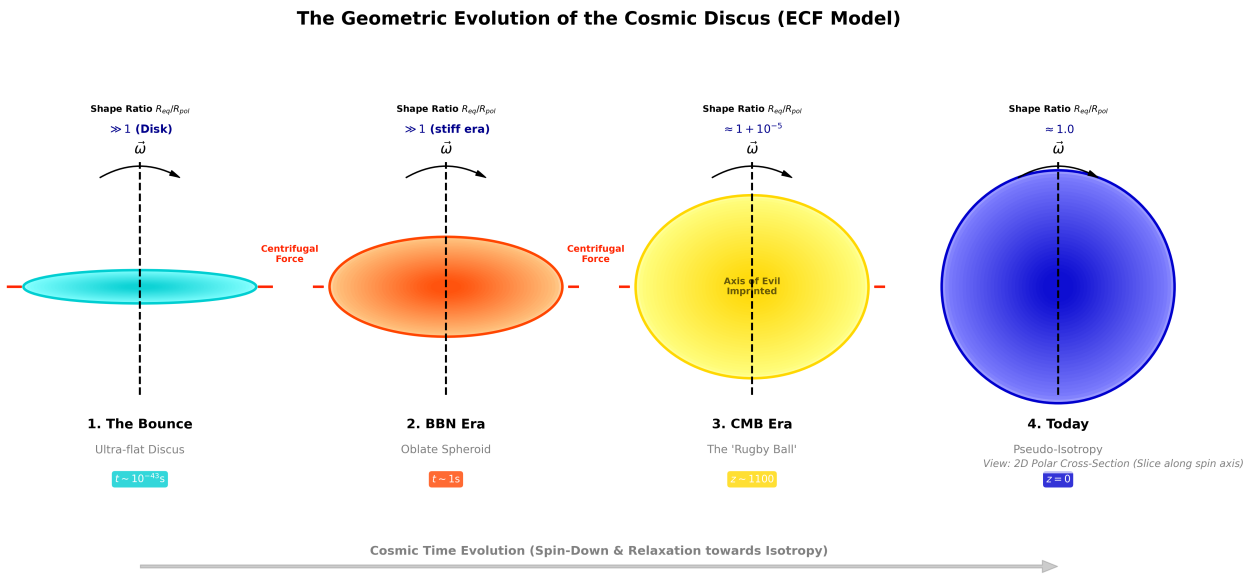

Figure 44: **The Geometric Phase Transition of the Universe.** Evolution of the ellipticity $\varepsilon(t) = (a_\perp - a_\parallel)/a$ driven by the competition between centrifugal rotation $\omega(t)$ and Hubble expansion $H(t)$. The Universe begins as a highly oblate "Cosmic Discus" at the Bounce, relaxes through the radiation and matter eras, and approaches a quasi-isotropic sphere today, leaving a residual geometric anisotropy imprinted on the CMB.

Reproduced by: `Scripts_F2/plot_cosmic_evolution_shape.py`.

10 Computational Framework and Reproducibility

All numerical simulations and statistical analyses presented in this work were performed using a custom-built pipeline in Python 3.10+. The core framework relies on the `NumPy` and `SciPy` libraries for tensor calculus and non-linear optimization [107, 108], while `Matplotlib` was employed for visualization [109]. To ensure full scientific reproducibility, the entire suite is orchestrated by a master script, `run_all_simulations_F2.py` (111 scripts: 54 physics calculations + 57 figure generators), available alongside all source codes and datasets at `https://github.com/pfichant/spin-torsion-cosmology`.

Note for the referee — Visualisation suite. Four pedagogical GIF animations (`plot_F2_animations_ECF.py`, repository folder `Foundation_2/02_Scientific_Code/D_Simulations/`) illustrate the gas dynamics around each defect class (0D/1D/2D/3D) at 80 frames, 18 FPS, white background, English labels. These are supplementary educational material only and are not used for any quantitative result in this paper. All figures in the main text are generated exclusively from the `run_all_simulations_F2.py` pipeline.

- `plot_ecf_blender_simulation_knot_constructive_v34fix.py` (V34-FIX) — Renders the constructive Micro-Knot merger (Regime I, Eq. 83): two same-helicity torsion defects approach, superimpose, and coalesce into a single higher-mass topological soliton, with a prompt gravitational-wave burst visualised as an expanding luminosity shell.
- `plot_ecf_blender_simulation_GRB2025-2_v40fix.py` (V40.1-FIX) — Renders the topological annihilation event (Regime II, Eq. 84): a Knot–Anti-Knot collision, complete energy release, and the characteristic multi-stage GRB light curve (Fig. 55).

Both scripts require Blender ≥ 4.2 with the `bpy` API and produce the animations referenced in Figs. 18 and 55. These corrected releases (V34-FIX and V40.1-FIX, 8 May 2026) address minor API deprecations in the Blender 4.2 Proximity modifier and the `node_tree` emission accessor; they are available at `https://github.com/pfichant/spin-torsion-cosmology` (folder `Foundation_2/02_Scientific_Code/D_Simulations/blender/`).

Full reproducibility instructions are provided in the repository `README`.

11 General Conclusion

Integrating spacetime torsion into metric cosmology offers a falsifiable geometric *extension* of General Relativity, not a departure from it: the ECF retains the Friedmann background, the baryonic and radiation content, and standard nucleosynthesis (the BBN compatibility of the spin sector itself is not demonstrated; PO-ECF-BBN-SPIN), replacing only the origin of the dark sector and the inflaton with a single spin-torsion mechanism. In the static limit the background sector reduces exactly to ΛCDM ($w = -1$, derived from the action; Appendix J of the present paper); the full late-time trajectory, including its mild time evolution (w_0, w_a), remains calibrated rather than derived (Sec. 3.5; cf. PO-F2-4), so the framework approaches but does not yet coincide with ΛCDM once evolution is included. It is falsifiable *as* an extension, through the same tensions and observables that already constrain the standard model. In *Foundation II* of the ECF, we have investigated the hypothesis that the dark sector emerges as three chronologically ordered fossils of a single primordial event: the non-singular Einstein–Cartan bounce [5].

Fossil I — Chiral Baryogenesis. The extreme spin density at the bounce generates a torsion chemical potential splitting left- and right-handed fermion abundances via the geometric chiral-splitting mechanism (Appendix A.4). With the pre-dilution asymmetry normalised on the observed value ($\eta_{\rm obs}/0.50$) and the torsion dilution factor, the scenario gives $\eta \simeq 5.9 \times 10^{-10}$ (*semi-derived*; its 3.5% agreement with the Planck-inferred value [8] is not an independent test), with no beyond-Standard-Model CP violation invoked; no Boltzmann network is integrated, and the complete realisation of the Sakharov conditions is not established. The observed ∼5:1 dark-to-baryonic mass ratio is numerically close to the 20:4 DoF decomposition of the torsion tensor ($20/4 = 5$, within 6.7% of the observed value) under simple equipartition, but this link is *not established*: the topological surplus ΔN and the baryonic surplus η are two separate asymmetries (§3), and their quantitative relation remains open problem PO-F2-1 (Table 14).

Fossil II — Geometric Dark Matter. The stiff-fluid instability seeds the crystallization of torsion into planetary-mass Micro-Knots ($M \sim 10^{24}$ kg) and galactic-seed Macro-Knots ($M_{\rm seed} \sim 10^5$–$10^6\, M_\odot$) via the Kibble–Zurek [30, 31] mass function. Poisson coalescence of Micro-Knots ($\lambda \gtrsim 36$, required for compatibility with Subaru/HSC microlensing constraints, §9.3) sets the typical observable dark-matter mass at $M_{\rm typ} \approx 6.8 \times 10^{-5}\, M_\odot$, predicting Roman microlensing events of $t_E \sim 1$–10 h. Applied to the SPARC database (171 of 175 galaxies, see *Sample definition*), the ECF torsion halo achieves $\tilde{\chi}^2_{\rm med} = 0.80$ for the adopted pseudo-isothermal profile (PO-ECF-HALO), with two parameters per galaxy (Fig. 30) [16]; this result is robust against the Poisson coalescence correction (§6). A candidate 1D sector (Torsion Strings, π_1) was investigated as a possible source of coherent filament rotation and an anti-parallel galaxy–filament spin correlation matching the MaNGA detection (Table 1, [197]), but is retained as an open problem (PO-F2-8): the Lagrangian does not currently supply the phase such a defect requires. A magnetic-torque mechanism at the established 0D/2D sectors is the current candidate explanation for the underlying observations (§6), with amplitude and direction not yet derived. This geometric approach is consistent with a potential interpretation of the Li-7 anomaly via late-time topological defect decay (not treated in the present corpus; *corrected 20/10/2026: previously deferred to Foundation III — Cosmological Physics, which does not treat it*), and for the JWST tension via pre-existing Macro-Knot seeds [62, 63].

The constructive (Regime I) and destructive (Regime II) merger channels, together with Knot–compact-object encounters, are derived in Appendix D.10 ($\varepsilon_{\rm top}$ calibrated, $\eta_{\rm ann}$ left open).

Fossil III — Dynamical Dark Energy. As the primordial Weyssenhoff spin fluid dilutes ($\rho_{\rm spin} \propto a^{-6}$), the axial torsion zero-mode condenses into a vacuum expectation value (Sec. 3.5), yielding a dark-energy equation of state fixed entirely by the Foundation I calibration [5]: $(w_0, w_a) = (-0.904, -0.153)$. This trajectory is calibrated (w_a through $\xi = 1.645$, calibrated on the DESI preference used by the ECF, hence not independent of DESI) and lies in the qualitative direction of the DESI DR2 deviation from ΛCDM ($w_0 > -1$, $w_a < 0$), which DESI reports at 3.1σ for BAO+CMB and 2.8–4.2σ once SNe are included, depending on the sample [12]. The full-

covariance comparison of Foundation III places the ECF point at 2.03σ (Planck prior) from the DR1 central value, and *not* favoured over ΛCDM by those data alone ($\Delta\chi^2 = +6.21$).

PO-F2-6, discussed below — what is the physical spectrum of the axial torsion sector? Stated as a spectral problem, PO-F2-6 asks what the reduction of the full PGT action produces in the axial sector: its quadratic form (Hessian) about the background, its constraint structure, its physical poles and their residues, the presence or absence of ghost and tachyonic modes, its mixings with the other torsion and metric modes, and the projection of the physical mode onto S_μ — in particular whether

$$m_{\text{coefficient}} \stackrel{?}{=} m_{\text{pole}} \stackrel{?}{=} m_{\text{static}} \stackrel{?}{=} m_{\text{cosmological}}, \tag{126}$$

i.e. whether the coefficient m_T of Eq. (9), the pole mass of the propagating mode, the mass setting the static range and the mass governing the cosmological mode are one and the same. None of these identifications is established here. PO-F2-6 is a subproblem of PO-ECF-ACTION, and its output is what the transfer of the effective sectors to observables requires: the chain is PO-ECF-ACTION (O1) → PO-F2-6 (O5) → transfer to observables (O10). The static background with $w = -1$ is a distinct statement. The question addressed next — which kinetic term does the axial torsion sector carry? — is a particular case of this spectral problem. Appendix J builds the dark-energy sector on the Proca form $-\frac{1}{4}F^{(S)}_{\mu\nu}F^{(S)\mu\nu}$, which vanishes identically on the homogeneous ansatz — that vanishing is what pins the condensate at $W'(X_0) = 0$ and yields $w = -1$. Benisty *et al.* (arXiv:2109.01052) treat the same totally antisymmetric torsion with a kinetic term built on $\nabla_\mu S^\mu$ instead, which *does* contribute to the background, and reach $w = -1$ by a different route: the divergence is then a constant of motion. The two Lagrangians agree on the static limit but are not equivalent, and the difference is not cosmetic — under the divergence form the transverse modes of δS_i carry no kinetic term at all, so the $w = +\frac{1}{3}$ fluctuation component of §J.2 would not exist. This comparison, and why it is not merely a matter of taste, is resolved below.

A stronger statement than the one above can in fact be made. Varying the divergence-form Lagrangian $\mathcal{L}_G = \frac{1}{2}\bar{R} - \frac{1}{2m^2}(\nabla_\mu S^\mu)^2 + \frac{1}{2}S_\mu S^\mu$ with respect to S^μ gives the equation of motion $\frac{1}{m^2}\nabla_\mu\theta + S_\mu = 0$, where $\theta \equiv \nabla_\mu S^\mu$; solved for S_μ, this reads $S_\mu = -\frac{1}{m^2}\partial_\mu\theta$. On shell, the vector field is therefore not merely missing a transverse kinetic term — it is a pure gradient, $S_\mu = \partial_\mu\phi$ for a single scalar ϕ obeying the ordinary free Klein–Gordon equation $\Box\phi + m^2\phi = 0$. In the minimal construction of Benisty *et al.*, this scalar carries no coupling to matter perturbations, only a background contribution; no analogue of the scale-dependent $G_{\text{eff}}(k)$ suppression behind the quoted S_8 value survives. Recovering an S_8-suppression channel under the divergence form would therefore require an additional torsion–matter coupling not supplied by the choice of kinetic term itself — a different theory, not a re-expression of the same one. *Reproduced by:* `Scripts_F2/verify_divergence_form_on_shell.py` (symbolic check of the Euler–Lagrange reduction above).

An independent indication points the same way, and we state its scope carefully. Fabbri [3] reports that the most general kinetic term for a propagating torsion must have the structure of the torsion *curl* squared — the Proca form — a statement motivated in that setting by a Velo–Zwanziger-type analysis. *We flag a limitation in this citation:* Fabbri states the result as established and attributes it to a further reference we have not identified with certainty, so we take it as an independent indication on structures compatible with propagation and causality, **not** as a derivation of the kinetic structure of the present axial sector. Carroll & Field [2] separately show that in a wide class of connection-dynamic theories the torsion modes coupling to matter reduce to a massive scalar or a massive spin-1 boson, their analysis focusing on the scalar case; this constrains a specific class of models and does not by itself fix the axial structure considered here. Taken together, these results provide an external indication favouring the Proca structure, without establishing it as the only admissible one: they constrain specific classes of models rather than fixing the kinetic structure that the reduction of the full PGT action actually produces.

PO-F2-6 accordingly remains open. An earlier version of this section declared it closed on the strength of the argument above. That closure was premature, and is withdrawn here for two reasons. First, the consistency argument is reported rather than derived, and its scope is not established (see the caveat above). Second, and more fundamentally, selecting a form among two given candidates is a different question from determining which structure

the reduction of the full PGT action produces — and the latter is what would be needed: the spectral analysis stated at the head of this discussion (Eq. (126)), of which the Proca/divergence choice is only a sub-case. A note setting out this question has been submitted to specialists in Poincaré gauge theory; its resolution is not claimed here. PO-F2-6 remains open, as a sub-problem of PO-ECF-ACTION.

One consequence is worth stating, since it bears on a quantity we do quote. Under the Proca form adopted here the kinetic term is the canonical $-\frac{1}{4}F^{(S)}_{\mu\nu}F^{(S)\mu\nu}$: it vanishes on the homogeneous background, but a perturbation is precisely what breaks homogeneity, so $F^{(S)}_{\mu\nu} \neq 0$ at first order and the perturbations propagate with $c_s^2 = 1$. A sound speed equal to the speed of light means the condensate does not cluster on sub-horizon scales — it stays smooth, contributes no perturbation of its own to the growth of structure, and the S_8 phenomenology of Foundation I is therefore set in the regime where it is meant to apply. Under the divergence form this does not follow: with no kinetic term for the transverse modes, c_s^2 is not defined the same way, and the value of S_8 would have to be revisited. *We quote $c_s^2 = 1$ as a consequence of the Lagrangian we adopt, not as an established property of the axial torsion sector.*

Open problem PO-F2-5 — the H_0–Age–BAO trilemma remains unresolved; its active part is no longer an age deficit. The $(w_0, w_a) = (-0.904, -0.153)$ trajectory above is *calibrated*, not derived *ab initio* (cf. PO-F2-4), and a full-covariance comparison places it at 2.03σ from the DESI DR1 central value under a Planck prior — not within 1σ, and not favoured over ΛCDM by those data alone. It does *not*, moreover, resolve the H_0–Age–BAO trilemma of Foundation I. A systematic scan of the tanh-crossing scalar $w(z)$ family — the class to which this trajectory numerically belongs — finds zero viable points out of 2500 simultaneously satisfying the eBOSS BAO distance and the $t_0 \geq t_{\rm GC} = 13.32$ Gyr globular-cluster chronometric bound (Foundation I; the BAO side of the comparison uses $r_s^{\rm ECF}$ as the ruler of measurements published in $r_{\rm drag}$ and is CALCULATED-CONDITIONAL on `PO-ECF-BAO-RDRAG`, Appendix L; rerun at $\Omega_m \simeq 0.267$ with the 30.21 target: 187 BAO-viable and 1857 age-viable points, none satisfying both; best tanh point $t_0 = 13.20$ Gyr, BAO offset 2.45%, $\chi^2_{\rm res} = 4.03$). The trilemma is therefore irreducible within that family, but this is no longer driven by an age deficit: with $\Omega_m = \omega_m/h^2 \simeq 0.267$ the $w = -1$ background gives $t_0 \simeq 13.33$ Gyr, and the stronger CPL phantom crossing formerly invoked for the age ($w_0 \approx -1.1$, $w_a \approx -1.5$; $t_0 \simeq 14.03$ Gyr) is no longer needed for it; the BAO cost quoted for it ($\Delta\chi^2_{\rm BAO} = 38$) was evaluated at $\Omega_m = 0.315$ and is historical, not recomputed. The active tension lies in the BAO distances ($\chi^2_{\rm BAO} = 12.11$ against 1.79 for ΛCDM, four bins) and in the DESI DR1 comparison of the calibrated trajectory ($\chi^2 \simeq 55.3$ on the ten-observable subset with diagonal errors — not the global DESI DR1 χ^2 — against 13.7 at $\Omega_m = 0.315$). Whether a fuller treatment of the torsion background beyond the effective CPL description used here can decouple the two integrals is open; earlier language in this corpus describing the trilemma as "resolved" or the BAO penalty as "removed entirely" by the present calibration has been corrected accordingly (see the introduction, §1).

Refinement (20/07/2026) : the trilemma is part of a quadrilemma including the CMB acoustic scale. The 2500-point scan was originally run at fixed $\Omega_m = 0.315$ (the Planck-h value). Using the fixed CMB baryon+CDM physical density $\omega_m = 0.14237$ instead gives $\Omega_m \approx 0.267$ at $H_0 = 73.04$ (active value, A-7); the scan rerun at that value gives the counts quoted above. An earlier joint rescan of (Ω_m, w_0, w_a), which reported $t_0 \approx 13.0$–13.2 Gyr at $\Omega_m \approx 0.28$–0.29, is historical and superseded: it is not reproduced by the current scripts. Adding the CMB acoustic scale $\theta_* = r_s(z_*)/D_M(z_*)$ as a third joint constraint shows that, at the calibrated (w_0, w_a), increasing Ω_m improves θ_* but *worsens* the age — the three late-time observables (age, BAO, θ_*) are governed by the same low-z portion of the comoving-distance integral and pull Ω_m in mutually incompatible directions. This sharpens rather than resolves PO-F2-5. The precise quantitative extent of the θ_* tension depends on a full Boltzmann-code treatment, requiring a dedicated ECF perturbation (Boltzmann) solver, which does not yet exist; the approximate integral used for this diagnostic has a known $\sim 1\%$ systematic offset relative to the literature Planck value of $100\,\theta_*$, and should not be over-interpreted at the percent level.

Adopting Geometric Monism, the principal cosmological anomalies—H_0, S_8, JWST early galaxies, DESI dynamical dark energy, CMB Axis of Evil—may reflect primordial spacetime topology rather than undetected particles, pending empir-

ical verification. Exploratory extensions to Odd Radio Circles, long-period radio transients, Gamma-Ray Bursts, and mega-structures are discussed in Appendix E as preliminary geometric consistency checks, not primary results of this work.

Nearest falsification milestone. The structured roadmap of kill-switches (Table 13) identifies ten observational tests spanning 2026 onwards, of which **eight** are currently falsifying in the strict sense; two (CTA gamma-rays, intergalactic $\vec{B}$ orientation) depend on the 1D sector, an open problem (PO-F2-8), and are not yet discriminating tests. The most temporally urgent is the Euclid Data Release 1 (DR1-Complete, mid-2027 [186]): the ECF predicts an over-abundance of cosmic voids at $R > 60\,\mathrm{Mpc}$ by a factor ~ 2.1 relative to ΛCDM (Eqs. 35–36). A null result at 2σ in the Euclid DR1 void catalogue would disfavour the Kibble–Zurek void channel and constitute the first direct falsification of the ECF 2D sector, independent of all other observational claims in this paper. This programme is genuinely Popperian: no dark-sector substructure protects these predictions from falsification by the next generation of surveys.

Falsifiability. The ECF is subject to several primary kill-switch tests:

- **Roman microlensing:** events in the $t_E \sim 1$–10 h window ($\tau_{\rm micro} \sim 5 \times 10^{-8}$, $N_{\rm ev} \approx 1400$ over 72 days for $f_{\rm DM} = 1$ (PO-F2-10), from Poisson-coalesced aggregates $M_{\rm typ} \approx 6.8 \times 10^{-5}\,M_\odot$, §9.3); *falsified* if no excess above the free-floating-planet baseline in this timescale range.
- **Filament spin (anti-parallel test) — WITHDRAWN as a primary kill-switch (30/08/2026):** this test relied on a helical $\vec{J}_5$ current carried by 1D Torsion Strings, which are an open problem (PO-F2-8), not an established sector. The observation itself remains real — Wang et al. (2025) [197] report anti-parallel correlation in MaNGA data, significant for low-mass spirals only — but the ECF no longer has a quantitative mechanism tied to this test, and it is not currently falsifiable in the sense the other items in this list are. A magnetic-torque candidate mechanism exists (§6) but does not yet predict a reversal scale or a parallel/anti-parallel discriminator.
- **LISA chiral GWs:** parity asymmetry $\Pi_{\rm GW} \simeq 0.20 \pm 0.05$ (calibrated template); chiral sector *falsified* if $\Pi_{\rm GW} < 0.02$ at 95% C.L.; whether LISA, through the dipole channel [203], or a LISA–Taiji network [204] can reach this sensitivity is not established here.
- **SPARC granularity:** $\delta V_{\rm rot}/V \geq 0.05$; *falsified* if scatter is consistent with a smooth NFW halo.
- **GRB kill-switch:** A statistically significant association ($> 3\sigma$) of long-duration ECF burst candidates with dense stellar host galaxies of GBM bursts [78] localised by Swift or SVOM would falsify the hostless-burst interpretation.
- **LISA Regime I (conditional channel, not a kill-switch):** the Macro-Knot merger rate is not computed (the abundance itself is a working assumption, PO-F2-SEED-CMB), so the absence of a GW burst at 50 ± 20 mHz would not falsify same-helicity coalescence, and no ECF-specific feature of the burst is established that a detection could test (the torsion tail is withdrawn, Appendix D.10). *(Corrected 18/10/2026: this item stated that the absence of a burst after* 10^4 *h would falsify the coalescence.)*
- **Knot-Star — withdrawn as a kill-switch (18/10/2026):** it combined the absence of ringdown tails beyond $\tau_T > 10^3\,\tau_{\rm GR}$ with the absence of FXT counterparts; at the Macro-Knot radius the tail is not distinct ($\tau_T \sim R/c \approx 1\,\mathrm{s}$, of the order of $\tau_{\rm GR}$, Appendix D.10), so only the X-ray channel remains (Appendix D.10.3).

Extended merger-impact scenario. A conservative discussion of SMBH–SMBH blowout, gas reaccretion, and possible amplification by a central Macro-Knot seed is retained in Appendix I. It is not required for the core three-fossil argument, but remains useful as an exploratory astrophysical consequence of the Macro-Knot channel.

Open theoretical frontier: the perturbative sector. The topological zoology (§6) unifies these results: dark matter (0D Micro-Knots, and 0D Macro-Knots from the distinct Great-Annihilation Jeans/horizon channel — not from coagulation),

void formation (2D Chiral Walls), and dark energy (3D residual chiral fluid) emerge as local and global manifestations of a single primordial torsion field, classified by the homotopy groups of the chiral vacuum manifold (π_2, π_0 respectively). A candidate 1D sector (Torsion Strings, π_1) was investigated and is retained as an explicitly open problem (PO-F2-8), not among the established carriers above. This hierarchical defect spectrum constitutes the geometric alternative to particle dark matter and the cosmological constant simultaneously.

The present work operates at the level of a macroscopic effective description: $G_{\rm eff}(k)$ is a phenomenological parametrisation, no longer presented as calibrated on the KiDS-1000 value [141], and not derived from first-principles perturbed ECKS field equations. This is a deliberate and transparent limitation of v1. Deriving $G_{\rm eff}(k, z)$ from the Weyssenhoff fluid perturbation equations coupled to a modified Boltzmann solver remains an open objective. Until that derivation is complete, the phenomenological $G_{\rm eff}$ should be read as a calibrated proxy for a geometric effect whose perturbative derivation remains open. Under the internal closure $\Omega_m = \omega_m/h^2 \simeq 0.267$, a CAMB ΛCDM reference on that background gives $S_8 \simeq 0.780$; the residual correction to the target $S_8 = 0.766$ depends on the ECF perturbative dynamics, which is not implemented, and the value $\sigma_8 \simeq 0.8122$ of earlier versions is the target converted at that Ω_m, not a computed amplitude. A dedicated perturbative computation confirming or correcting $\alpha = 0.18$, $k_{\rm cut} = 0.10\,h\,{\rm Mpc}^{-1}$ at the linear-theory level will constitute the definitive test of the S_8 sector of the ECF.

11.1 The ECF Framework in Light of Emerging 2026 Anomalies

The recent influx of high-precision cosmological data has exposed mounting tensions within the standard ΛCDM hierarchical assembly paradigm. Rather than treating these discrepancies as isolated astrophysical curiosities, the ECF framework offers a single geometric reading under which they *would be* interconnected signatures of macroscopic spacetime torsion inherited from the primordial bounce. We state the status of this section explicitly: it is a set of *exploratory consistency checks*, not primary evidence, and none of the connections below is derived from the field equations. We summarise four developments that have sharpened the falsifiability landscape since the first submission of this work.

(i) Dynamical dark energy and the phantom divide (DESI DR2 [12]). DESI DR2 reports a preference for evolving dark energy at 2.8–4.2σ depending on the supernova sample [12]; the completed five-year survey (47×10^6 objects [15]) will report its first dark-energy results in 2027. While standard scalar-field models (quintessence) face theoretical pathologies when crossing the phantom divide ($w < -1$)—notably Ostrogradsky instabilities for single-field realisations—the ECF geometric pressure $P_{\rm tors}$ bypasses this issue entirely. Because macroscopic torsion modifies the affine connection rather than the matter Lagrangian, it can mimic a phantom-like equation-of-state crossing during late-time spin-fluid dilution; for the adopted Proca form the absence of ghosts is invoked from the literature, not established here (PO-ECF-ACTION). The ECF prediction $(w_0, w_a) = (-0.904, -0.153)$, with no additional dark-sector free parameter beyond the Foundation I calibration, lies qualitatively consistent with DESI DR2 ($w \neq -1$ direction); the full-covariance DESI DR1 comparison, however, does not favour it over ΛCDM ($\Delta\chi^2 = +6.21$ Planck / $+38.59$ SH0ES, $\Delta{\rm AIC} = +10.06$ / $+43.94$ counting the single calibrated parameter ξ of the w_a sector, `Scripts_F2/compute_ecf_aic_desi.py`; see the information-theoretic discussion above).

(ii) JWST seeding problem, Little Red Dots, and the Macro-Knot mass scale [56, 62, 63, 121]. JWST spectroscopic confirmation of massive, dust-enriched structures at $z > 11$ and the abundant Little Red Dot (LRD) population at $z > 5$ challenge sub-Eddington accretion limits under standard seeding scenarios. In the ECF framework, the causal Hubble mass at the Great Annihilation sets the benchmark scale $M_K \sim 10^5$–$10^6\,M_\odot$ (Eq. 79; calculated-conditional, an upper scale rather than the mass of an object, since $\epsilon_{\rm coll}$ is not derived). Independent radiation-hydrodynamic modelling of the LRD population [121] obtains a Direct Collapse Black Hole (DCBH) seed mass $M_\bullet \sim 10^5\,M_\odot$ from standard astrophysics; the two scales overlap, which makes the ECF reading compatible, not preferred. *(Corrected 19/10/2026: earlier versions identified M_K with the Jeans mass of the $w = 1$ phase, said that it eliminates the need for DCBH astrophysics, and called the overlap a quantitative agreement with a zero-free-parameter prediction.)*

The spectroscopic analysis of [56] further strengthens this picture by identifying narrow intrinsic components buried under electron-scattering wings in LRD spectra—signatures of very compact, highly obscured nuclei, consistent with heavy-seed growth but not specific to a topological seed. Quantitative confrontation with the high-z stellar mass function awaits the ECF perturbation-theory treatment of Foundation III [7].

(iii) Unscreened torsion, void evacuation, and the S_8 tension (Euclid DR1, forthcoming). The S_8 tension indicates a systematic suppression of late-time matter clustering relative to CMB-inferred predictions. We propose that in extreme underdense environments—such as the Boötes Void—the torsion pressure remains *unscreened* by local matter interactions, in contrast to the chameleon-like screening familiar from $f(R)$ theories. This locally enhanced geometric repulsion evacuates voids more rapidly than standard gravity, suppressing the large-scale structure growth rate along filamentary web boundaries. The phenomenological suppression $G_{\rm eff}(k)$ used in this work is a phenomenological proxy, no longer presented as calibrated on KiDS-1000 [141]; its first-principles derivation would require a dedicated Einstein–Cartan Boltzmann solver, not developed in this series. Void-size function data from the ESA Euclid mission will constitute a direct falsifiability test of this geometric void-evacuation mechanism (Sec. 4.2).

(iv) Multi-messenger dichotomy: silent Micro-Knot mergers versus high-z GRBs (SVOM/JWST, ACT DR6 [71, 72]). The ECF model predicts a strict environmental dependence for topological defect mergers. In vacuum, Micro-Knot mergers, if they occur (rate not computed), radiate only gravitational waves, at $f_{\rm peak} \sim 3$ GHz (Table 16), far above any detector band: they have neither an electromagnetic counterpart nor a detectable gravitational signal. *(Corrected 18/10/2026: earlier versions mentioned "topological ringdown echoes" and called these mergers consistent with LIGO candidates lacking optical follow-up; LIGO observes at ~ 10–10^3 Hz.)* Conversely, at Cosmic Dawn, massive Macro-Knots embedded in dense primordial gas drive cataclysmic shocks, naturally producing energetic events analogous to GRB 250314A ($z \approx 7.3$) observed by the SVOM/JWST synergy. Independently, reanalysis of ACT DR6 CMB polarisation confirms $\beta \neq 0$ at 99.6% confidence ($\beta_{\rm obs} \approx 0.35^\circ$ for Planck/WMAP, $0.215^\circ \pm 0.074^\circ$ for ACT DR6), empirically motivating the Topological Distance Ladder; the resolution of the $n\pi$ phase ambiguity [72] provides the theoretical framework needed to interpret future high-precision birefringence measurements.

Taken together, these four developments illustrate a key property of the ECF programme: new anomalous data are absorbed by the existing geometric structure rather than requiring additional parameters. Multi-messenger data from Roman, LISA/ET, SKA, JWST, and the full DESI five-year catalogue will determine whether these fossils of the bounce trace primordial topology. While promising, the ECF requires empirical confirmation to transcend theoretical curiosity.
The ECF Trilogy: Foundation I [5], Foundation II (Structure & Transients), Foundation III (*Cosmological Physics* [7], deposited; *Strong Gravity* and *Chiral Bounce*, in preparation) *(updated 20/10/2026: previously "Thermodynamics & Information Transfer, in preparation").* In particular, the tensor transfer function across the bounce — governing the spectral tilt n_T and the chiral asymmetry amplitude $\Pi_{\rm GW}$ — constitutes the primary open perturbative frontier of this programme and is the subject of the dedicated treatment in Foundation III [7].

11.2 Limitations and Open Questions

The ECF program is broad in scope, and its most exposed claims are therefore also its most exposed. Two issues deserve to be stated explicitly. First, the extended Poincaré gauge sector used in this work is not presented as a perturbatively renormalizable ultraviolet completion of gravity; rather, it is an effective geometric theory whose predictive power is tied to the Planck-suppressed regime in which torsion is relevant. Second, the physical status of the torsion vacuum expectation value must be interpreted with care: the model assumes a spontaneously selected geometric condensate rather than a pure gauge artifact, but that assumption is itself part of the falsifiable structure of the framework.

A future fully quantum treatment may sharpen these assumptions, but the present paper does not rely on them as free tuning knobs. Instead, the observational content remains anchored in the measurable signatures discussed throughout the article: the SPARC rotation-field granularity, the Roman

Table 12: **Limitations of Foundation II — structured inventory.** D = deliberate scope decision (addressed in cited work); O = open physical problem (no current solution); P = phenomenological proxy (calibrated, not derived).

Item	Type	Status / Next step
$G_{\rm eff}(k,z)$ derivation	D	dedicated perturbation solver (open)
CMB peak heights (C_ℓ)	D	dedicated perturbation solver (open)
Tensor transfer function (h_c, $\Omega_{\rm GW}$)	D	Foundation III
VSF quantitative pipeline	D	Foundation III + Euclid DR1 test
DESI full covariance matrix	D	Foundation III
η from first principles	P	$\eta_{\rm ECF}$ calibrated from F1; analytical derivation reserved
$\langle S^2\rangle^{(0)}_{\rm vol}$ normalisation	P	Calibrated from F1 MCMC; link to $\eta_{\rm EW}$ open
$f_{\rm fluid}$ partition	O	PO-F2-4: not derivable without a perturbative treatment
ΔN topological surplus	O	PO-F2-1: link $\Delta N/\eta$ not established
Torsion String pitch $\lambda_{\rm hel}$	O	PO-F2-2: requires cylindrical ECF field eqs
Chiral Wall dilution rate	O	PO-F2-3: wall network evolution not simulated
Cosmological constant problem	O	ECF reformulates, does not solve
BBN ^{7}Li resolution	O	No mechanism retained; needs a derived $H(z)$ at $T\sim 1\,$MeV, then a PRIMAT integration
PGT renormalisability (UV)	O	Outside scope of this effective-theory paper
Complete effective action (PO-ECF-ACTION)	O	Kinetic, gradient and potential coefficients not all derived from minimal Einstein–Cartan–Dirac; dynamical sectors used as effective description
Axial-sector spectrum (PO-F2-6)	O	Spectral problem: Hessian, constraints, poles, residues, ghost/tachyon, mixings, projection on S_μ; sub-problem of PO-ECF-ACTION
Phase field of the 1D sector (PO-F2-8)	O	Field carrying the relative phase not identified
SPARC vs MOND/RAR (full)	D	Companion analysis (ref values in Table 6)
Galactic halo profile from torsion	O	PO-ECF-HALO: pseudo-isothermal profile $\rho_0/(1+r^2/R_s^2)$ adopted, not derived; (ρ_0, R_s) fitted per galaxy; no link with $M_{\rm MK}$
Molecular chirality propagation	D	Foundation III (speculative extension)

microlensing threshold, the gravitational-wave parity asymmetry, and the DESI-inferred dynamical dark-energy pattern.

Structured limitation inventory for the referee. The following table summarises the principal limitations of Foundation II by category, with explicit distinction between limitations that are *deliberate scope decisions* (deferred to Foundation III or a companion paper) and those that are *open physical problems* (not yet resolved).

What a perturbative treatment would and would not settle. Several entries above name the same absent instrument, and it is worth stating precisely what it would buy — and what it would not, since that has been diagnosed and the answer is not uniform.

It would settle three things. The redshift dependence of the condensate variance $\langle\delta S^2\rangle(z)$, which is what fixes w_a in Eq. (22); the scale dependence $G_{\rm eff}(k,z)$, which is what fixes S_8; and the CMB peak heights. These are transfer-function quantities: they are computed from a linear-order treatment of the ECKS field equations around the derived background, and none of them can be obtained without it.

It would *not* settle the spectral index. This

deserves saying plainly, because the expectation is natural and wrong. A Boltzmann solver computes the transfer function and takes n_s as an *input*; it can only relieve the tension if the torsion sector modifies the transfer function appreciably. The one channel of sufficient amplitude, $\Delta N_{\rm eff}$, is excluded at 6σ, and the remaining channels — anisotropic stress of the torsion sector, spin–torsion coupling at perturbative order — are higher order and will not move n_s by 0.035. The n_s tension is therefore not a transfer-function problem but a *primordial-spectrum* problem: the question is what spectrum the bounce and the macro-knot crystallisation actually produce. That question is not addressed by a Boltzmann code, and the Kibble–Zurek scaling exponent $\alpha = 5/2$ of the relevant universality class — which is not an adjustable parameter — is where the corresponding effort belongs.

We also record a closed avenue, since a closed question is more informative than an omitted one. Identifying the dark-energy pseudo-Goldstone with the phase of the electroweak Kibble–Zurek order parameter of Sec. 5 fails on scale: that transition sits at $T_c \simeq 159$ GeV, sixteen orders of magnitude below the decay constant such a mechanism requires, and the bounce density $\rho_c^{1/4} \simeq 7 \times 10^{14}$ GeV does not close the gap either. This is the classical unsolved difficulty of quintessence axions rather than a defect peculiar to the present framework, but the framework does not escape it.

The honest summary is therefore narrower than a single missing tool. A perturbative treatment is required, and would be decisive, for w_a, S_8 and the C_ℓ; it is the wrong instrument for n_s; and the evolving dark sector continues to rest on calibration rather than derivation, as declared throughout.

11.3 Observational Roadmap

The three main kill-switches do not have the same temporal or experimental weight. Roman Space Telescope microlensing is the earliest decisive test, because the ECF predicts a specific optical-depth floor and a population of events in the $t_E \sim 1$–$10\,$h window (Poisson-coalesced aggregates, $M_{\rm typ} \approx 6.8 \times 10^{-5}\,M_\odot$; §9.3) distinguishable from both stellar microlensing ($t_E \sim 30\,$d) and sub-minute artefacts, and discriminable from free-floating planets via electromagnetic counterpart and spatial distribution (§9.3). SPARC-like galactic kinematics provide an intermediate test of the predicted torsion granularity, while the mHz band (LISA through the kinematic dipole, or a LISA–Taiji network, §9.4) probes the more distant but conceptually clean parity signature of the primordial chiral sector.

This ordering matters because it makes the ECF strategy genuinely Popperian: the framework is not protected by adjustable dark-sector substructure, but instead risks falsification by a small number of sharply defined observations.
A null result at any row of Table 13 would falsify the corresponding ECF prediction *independent of the others*, since each observable probes a distinct sector (0D, 1D, 2D, 3D) of the topological hierarchy.

What the two Euclid rows will and will not decide. The two Euclid entries do not carry the same evidential weight, and it is worth saying so before the data arrive rather than after.

The weak-lensing row is the cleaner of the two. S_8 is defined identically on both sides of the comparison, Euclid will measure it to a precision no previous survey has reached, and the gap between the calibration target 0.766 and the falsification threshold 0.80 is a little over four per cent — within reach. One asymmetry should be stated: $S_8 = 0.766$ is a target, not an ECF prediction; a CAMB ΛCDM reference on the closure background gives $S_8 \simeq 0.780$, and the residual correction depends on the ECF perturbative dynamics, which is not implemented, $G_{\rm eff}(k)$ being a phenomenological parametrisation rather than a first-principles derivation (Sec. 11). Agreement would therefore not establish the derivation; disagreement would nonetheless refute the calibration, which is what a kill-switch is for.

The void row is the more demanding. The prediction — a factor 2.1 excess above $R > 60$ Mpc — is stated against a ΛCDM void size function, and comparing it to a catalogue requires carrying that prediction through the survey selection. The size of that step is not incidental: an excursion-set calculation for the dark matter field returns of order 19 voids in the DESIVAST DR1 volume where 1,489 interior voids are observed, a factor ~ 80 that reflects the difference between a matter field and a magnitude-limited galaxy sample rather than any disagreement of substance. Until that step is taken — with a survey mock, as DESIVAST itself does with AbacusSummit — no significance from this row is quotable, and none is quoted here. The requirement is therefore not more data but a mock; Euclid supplies the volume, not the method.

Table 13: **ECF observational roadmap — falsification milestones.** ECF prediction vs. kill-switch threshold for each observable. Each row probes a distinct sector (0D, 1D, 2D, 3D) of the topological hierarchy; a null result at any single row falsifies the corresponding prediction independently of all others. *Scope.* None of these thresholds refutes the framework as a whole: every chain listed here passes through at least one link that is currently calibrated, phenomenological, conditional or open and indispensable to the observable — $\xi_{\rm KZ}$ for the void and filament rows, (w_0, w_a) for DESI, the EMF/coagulation channel for the Roman rows, the phenomenological suppression proxy $A_{\rm sup}$ for S_8, the calibrated Chern–Simons coupling for birefringence, and the open chirality transfer for the LISA rows. Each row therefore tests the sector or pre-specified template it depends on. *This does not weaken the thresholds*: they remain sharp, pre-registered and decisive for their own sector.

Instrument	**Date**	**Observable**	**ECF prediction**	**Kill-switch (falsifies the corresponding ECF sector/realisation)**
Euclid DR1 (full)	mid-2027[‡]	Void size function excess $R > 60$ Mpc	$1 + \alpha_{\rm KZ} \simeq 2.1$ at the peak of the adopted template (no z-dependence claimed)	No excess at 2σ
DESI DR3	2027	Dark-energy (w_0, w_a)	$(-0.904, -0.153)$, calibrated (w_a via ξ, not independent of DESI)	$w = -1$ at 5σ
Roman	2027–28	Microlensing $\tau_{\rm micro}$, t_E	$\tau \sim 5 \times 10^{-8}$, $t_E \sim 1$–10 h	No excess in 1–10 h window
Roman	2027–28	EMF mass function	Poisson $\lambda \geq 36$, $M_{\rm typ} \sim 6.8 \times 10^{-5}\, M_\odot$	Monochromatic peak at $M_{\rm MK}$
DESI DR1+	2026–27	Filament spin sign	*Withdrawn as a test* (mechanism open, PO-F2-8; no ECF prediction attached)	—
Euclid DR1 (full)	mid-2027[‡]	S_8 (weak lensing)	0.766 (calibration target, not an ECF prediction)	$S_8 > 0.80$ at 3σ
JWST	2026	V/σ of sterile galaxy	< 0.1 (4 conditions)	$V/\sigma > 0.3$
LiteBIRD	JFY2036 [202]	Cosmic birefringence $\beta_{\rm ECF}$	$\beta \neq 0$ ($\lvert\beta\rvert \approx 0.35^\circ$ calibr., achromatic)	$\beta < 0.1^\circ$ at 3σ
LISA (∼2035)	after data accumulation	GW parity asymmetry $\Pi_{\rm GW}$ (dipole channel, or LISA–Taiji network; §9.4)	0.20 ± 0.05 (calibrated template; chiral sector)	< 0.02 at 95% C.L.
LISA (∼2035)	after data accumulation	Macro-Knot merger burst	$f \sim 50$ mHz (rate not computed)	none established (torsion tail withdrawn)

[‡] ESA split Euclid DR1 in June 2026: DR1-Foundation (raw data, images, catalogues, ∼1900 deg^2) in November 2026, and the complete DR1 — including the level-3 products on which galaxy clustering and weak lensing depend — in mid-2027. A void catalogue is a level-3 product, so the entries above are keyed to the later date rather than to the October 2026 originally announced.

Three detection windows. The roadmap organises naturally into three priority windows:
Window I — Now (2026–2027): zero marginal cost. Cross-correlate the Fermi-LAT 4FGL-DR4 archive (15+ years, full sky every 3 h, 20 MeV–300 GeV) with JWST massive quenched galaxies at $z > 3$: search for ~ 1 GeV transients, duration ~ 1 h, no afterglow, isotropic emission. Cost: $\lesssim 500$ CPU-hours on public data. Euclid DR1 void catalogue (mid-2027) tests the 2D sector simultaneously. *Note: Fermi-LAT has no polarisation sensitivity.*
Window II — 2027–2032: three-channel coincidence. POLAR-2 (2027, CSS, 40–1000 keV [214]) + Roman (2027–28): a simultaneous detection of (a) gamma-ray flash (Fermi/CTA), (b) circular polarisation (POLAR-2), and (c) galactic quenching signature (JWST) would constitute a three-channel test, conditional on values that are not derived (θ_T, $\eta_{\rm ann}$; *corrected 18/10/2026: was "extremely difficult to reproduce without invoking the topological annihilation mechanism"*).
Window III — late 2030s, subject to mission schedules: four-channel coincidence. LISA science operations + CTA + POLAR-2 successor + JWST/HWO: LISA alert → ToO CTA/Fermi within 6 h → $|V/I|$ measurement → JWST/ELT host confirmation. The joint false-alarm probability of a four-channel coincidence (GW burst + gamma flash + circular polarisation + galaxy quenching) must be evaluated in a dedicated statistical analysis; current estimates are illustrative.

Looking Forward: The Perturbed Universe

The present work provides the exact microscopic derivation of the geometric dark sector and revisits the H_0–Age–BAO problem at the background level without resolving it (PO-F2-5, open; its active part is now the BAO and DESI comparisons, not the age). However, the ultimate validation of any novel cosmological paradigm lies in the fully perturbed universe.

The topological structure of the bounce and its large-scale implications are developed in the companion paper *Foundation III: Topological Implications of the Einstein–Cartan Bounce* [7] (in prep.).

The quantitative confrontation with sub-percent CMB precision data would require implementing the perturbed ECKS field equations into a modified Boltzmann solver (open theoretical frontier, Sec. 11). This future analysis will directly confront the theoretical predictions derived here against the high-multipole Planck and ACT data; its requirements are outlined in the paragraph above.

While *Foundation II* secures the foundational equations of the chiral universe and *Foundation III* establishes its topological architecture, it is this future precision analysis that will offer the definitive observational test of the Einstein–Cartan framework.

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Data and Code Availability

The source code, numerical simulation scripts, and all data products generated in this work are openly available in the public repository at `https://github.com/pfichant/spin-torsion-cosmology`. The master orchestrator script `run_all_simulations_F2.py` reproduces every figure and numerical result presented in this paper. Full documentation of all 111 scripts—including physical outputs, patch notes, and data provenance—is provided in `README.md` in the same repository. The external observational datasets used are publicly available as follows.

SPARC rotation-curve database. The full 175-galaxy sample [16] is available at `http://astroweb.cwru.edu/SPARC/`. The photometric mass models follow McGaugh, Lelli & Schombert (2016). No modification to the public dataset was made; the ECF rotation-curve fits are produced entirely by the simulation suite above.

DESI BAO likelihoods. The w_0–w_a posterior chains for DR1 [9], DR2 [12], and the five-year analysis [15] are publicly available at `https://github.com/desihub/desilike`. The calibrated ECF point $(w_0, w_a) = (-0.904, -0.153)$ is derived from the Foundation I calibration and not fitted to these chains.

Planck 2018 CMB data. The temperature and polarisation power spectra and cosmological param-

eter chains [8] are available via the Planck Legacy Archive at `https://pla.esac.esa.int/`. Parameters adopted in this work: H_0, $\Omega_b h^2$, n_s, A_s, $\tau_{\rm reio}$ from Table 2 of Ref. [8].

ACT DR6 cosmic birefringence. The measurement $\beta_{\rm obs} = 0.215^\circ \pm 0.074^\circ$ [71] is available at `https://lambda.gsfc.nasa.gov/product/act/`.

JWST galaxy catalogues. Data for the massive high-redshift galaxies, Little Red Dots, EGS-z11-R0 ($z_{\rm spec} = 11.45$), and MoM-z14 ($z_{\rm spec} = 14.44$) were retrieved from the JWST public science archive at `https://mast.stsci.edu/`, via the programme IDs cited in the respective discovery papers [56, 61–64].

Bullet Cluster lensing maps. Weak-lensing shear maps from Clowe et al. [36] are available at `https://arxiv.org/abs/astro-ph/0608407`.

GRB spectral catalogue. The Fermi/GBM fourth burst catalogue [78] is publicly available at the HEASARC archive, `https://heasarc.gsfc.nasa.gov/W3Browse/fermi/fermigbrst.html`.

NGC 1052 field (DF9). Observational data for the dark-matter-deficient galaxy DF9 [37] are available upon reasonable request from the authors of that work, as per journal policy.

GRB data. Data for Fig. 55 are synthetic mock observations (`generate_mock_grb_data.py`, seed = 42); *not* based on proprietary or archival GRB data. Included solely for model comparison purposes.

No proprietary data were used in this study.

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Appendix A: Microphysics of Chiral Selection

This appendix provides the thermodynamic and group-theoretic underpinning of the chiral baryogenesis mechanism described in §3. The key quantity is the degrees-of-freedom (DoF) partition of the torsion tensor, which determines how primordial spin energy is split between the dark matter condensate (Micro-Knots) and the expelled baryonic surplus. The 20:4 DoF ratio derived here is the geometric basis of the 5:1 dark-to-baryonic mass ratio quoted in the main text — a consistency result within the Weyssenhoff approximation, not a parameter-free first-principles derivation (Table 14).

A.1 Thermodynamic Capacity of the Torsion Tensor

In field theory, the number of independent components (degrees of freedom, DoF) of a tensor determines its capacity to store thermodynamic energy during a phase transition. In the Einstein-Cartan Framework, the torsion tensor acts as the primary thermodynamic reservoir during the Kibble-Zurek topological crystallization. Its specific DoF partitioning dictates how primordial energy is split between Geometric Dark Matter (Micro-Knots) and the expelled baryonic fluid.

A.2 Cartan Decomposition and the Chiral Sector

The full torsion tensor $T^{\lambda}_{\mu\nu} = -T^{\lambda}_{\nu\mu}$ possesses $4 \times \frac{4\times3}{2} = 24$ independent DoF. The standard Cartan decomposition separates this tensor into three irreducible components under the Lorentz group [17]:

$$T^{\lambda}_{\mu\nu} = K^{\lambda}_{\mu\nu} + \frac{1}{3}\left(\delta^{\lambda}_{\mu}T_{\nu} - \delta^{\lambda}_{\nu}T_{\mu}\right) + T_{[\mu\nu]}{}^{\lambda}, \qquad (127)$$

where $K^{\lambda}_{\mu\nu}$ represents the contorsion tensor (20 DoF), and the remaining 4 DoF correspond to the fully antisymmetric pseudo-trace (the chiral axial-vector T^{5}_{μ}).

During the electroweak freeze-out, the chiral axial-vector couples universally to the fermionic spin current S^{μ} — the geometric mechanism (Appendix A.4) does not distinguish quarks from leptons, so this current is sourced by the full fermion population, not by baryons specifically (open track PO-F2-NU: whether this yields an observable lepton asymmetry comparable to the baryon one is not yet worked out):

$$\mathcal{L}_{\rm int} \propto \bar{\psi}\gamma^{\mu}\gamma_5 T^{5}_{\mu}\psi. \qquad (128)$$

Consequently, the 4 chiral DoF govern the thermodynamic evolution of the ordinary-matter sector (dominated observationally by baryons), while the remaining 20 contorsion DoF are hypothesised in the ECF to bind into stable topological defects (Geometric Dark Matter; §4).

A.3 Kibble-Zurek Equipartition

As the continuous torsion field shatters into domains via the Kibble-Zurek mechanism [31], thermodynamic domain walls expel the chiral (ordinary-matter) sector via the geometric chiral-splitting mechanism above. Assuming strict energy equipartition at the transition temperature, the energy density allocated to each sector is strictly proportional to its available degrees of freedom:

$$\frac{\Omega_{\rm DM}}{\Omega_b} = \frac{20\,{\rm DoF}_{\rm contorsion}}{4\,{\rm DoF}_{\rm chiral}} = 5. \qquad (129)$$

Because both the macroscopic defects and the baryons decouple as pressureless dust ($w = 0$, scaling as a^{-3}), this 5:1 ratio is preserved dynamically until the present epoch, successfully recovering the observed cosmological parameters without requiring fine-tuning of particle masses.

The torsion tensor $T^{\lambda}{}_{\mu\nu}$ is antisymmetric in its last two indices, giving $4 \times \frac{4\times3}{2} = 24$ independent real components in four spacetime dimensions. Under the irreducible SO(3, 1) decomposition [17], these 24 DoF split as:

$$\underbrace{T^{\lambda}{}_{\mu\nu}}_{\mathbf{24}} = \underbrace{S_{\mu}}_{\substack{\mathbf{4}\\ \text{axial}}} \oplus \underbrace{V_{\mu}}_{\substack{\mathbf{4}\\ \text{trace}}} \oplus \underbrace{q_{\mu\nu\lambda}}_{\substack{\mathbf{16}\\ \text{tensor}}}, \qquad (130)$$

where the axial part $S_{\mu} = \frac{1}{6}\varepsilon_{\mu\nu\lambda\rho}T^{\nu\lambda\rho}$ (4 DoF) couples to the fermionic axial current, and the remaining $4 + 16 = 20$ DoF are locked into the geometric binding energy of the crystallized defect. It is this 20:4 partition that yields the $\Omega_{DM}/\Omega_b = 5$ ratio of Eq. (129).

This timeline is illustrated schematically in Fig. 45. This geometric mechanism implies that a null detection of microlensing events in the $t_E \sim 1$–10 h window (no excess above the free-floating-planet baseline) by the Roman Space Telescope, launched on 30 August 2026, would firmly falsify this topology-driven dark matter hypothesis at the 3σ level (Sec. 9.3).

Table 14: Torsion DoF Budget and Primordial Energy Partitioning.

Geometric Component	**DoF**	**Cosmological Fate**
Contorsion Tensor ($K^{\lambda}_{\mu\nu}$)	20	Geometric DM (Micro-Knots)
Chiral Axial-Vector (T^{5}_{μ})	4	Baryons (Expelled Fluid)
Total Torsion Tensor	**24**	$\boldsymbol{\Omega_{\rm DM} : \Omega_b = 5 : 1}$

Figure 45: **Visual Chronological Roadmap of the ECF Chiral Universe.** This infographic illustrates the geometric phase transitions from the Big Spin Bounce to the present-day geometric galactic halos.

Reproduced by: `Scripts_F2/plot_ecf_timeline.py`.

A.4 Chiral Selection: Microphysics

In this appendix, we derive the effective chemical potential μ_{ECF} utilized in Section 3. We start from the Einstein-Cartan action with fermions:

$$S = \int d^4x\sqrt{-g}\left(\frac{1}{2\kappa}R(\Gamma) + \bar{\psi}(i\gamma^\mu D_\mu - m)\psi\right) \tag{131}$$

where $D_\mu = \partial_\mu + \frac{1}{4}\omega_\mu^{ab}\gamma_a\gamma_b$ is the covariant derivative with torsion. The variation with respect to the contortion tensor $K_{\mu\nu\rho}$ leads to the Cartan field equation:

$$S_{\mu\nu\rho} = -\frac{1}{4}\kappa\bar{\psi}\gamma_{[\mu}\gamma_\nu\gamma_{\rho]}\psi \tag{132}$$

The totally antisymmetric part of the torsion, the axial vector $S^\mu = \epsilon^{\mu\nu\alpha\beta}S_{\nu\alpha\beta}$, couples directly to the axial current $J_5^\mu = \bar{\psi}\gamma^\mu\gamma^5\psi$. Substituting this back into the Dirac equation yields a non-linear correction term. In the mean-field approximation applicable to the dense early universe, this acts as an effective energy shift:

$$E_{eff} \approx E_0 \pm \frac{3}{8}\kappa\langle S^0\rangle \tag{133}$$

The sign $\pm$ depends on the chirality (helicity) of the spinor. For a universe dominated by a net spin density ρ_{spin}, this creates the chemical potential splitting:

$$\Delta\mu = \mu_L - \mu_R \approx \frac{3}{4}\kappa\rho_{spin} \tag{134}$$

We explicitly check the stability of this solution. The effective mass squared of the spinor field in the torsion background is given by

$$m_{eff}^2 = m_0^2 + \left(\frac{3\kappa\rho_S}{4}\right)^2 \tag{135}$$

Since $m_{eff}^2 > 0$ for all physical densities, the theory is free from tachyonic instabilities, ensuring causal propagation of the chiral fermions. This derivation suggests that the asymmetry is a geometric consequence of the Einstein-Cartan coupling.

Appendix B: Derivation of $w_{\rm tor}(z)$ from the ECF Torsion Background

This appendix derives the effective equation of state of the ECF torsion background, starting from the TIP and the covariant continuity equation, with one calibrated coupling (ξ, Step 4) *(corrected 19/10/2026: previously "without any free dark-energy parameter").*

Step 1 — TIP flatness constraint. The TIP (Eq. 27) enforces:

$$\Omega_{\rm b}(a) + \Omega_{\rm knots}(a) + \Omega_\tau(a) = 1, \tag{136}$$

where $\Omega_{\rm b}$, $\Omega_{\rm knots} \propto a^{-3}$ (pressureless components) and $\Omega_\tau(a)$ is the fractional background torsion density. The Hubble rate satisfies:

$$H^2(a) = H_0^2\left[\Omega_{m,0}\,a^{-3} + \Omega_{r,0}\,a^{-4} + \Omega_{\tau,0}\,X(a)\right], \tag{137}$$

where $X(a) \equiv \rho_{\rm tor}(a)/\rho_{\rm tor,0}$.

Step 2 — Continuity equation. The background torsion fluid obeys:

$$\dot{\rho}_{\rm tor} + 3H(1 + w_{\rm tor})\,\rho_{\rm tor} = 0, \tag{138}$$

which in terms of a reads:

$$\frac{{\rm d}\ln\rho_{\rm tor}}{{\rm d}\ln a} = -3(1 + w_{\rm tor}(a))\,. \tag{139}$$

Inverting gives the exact equation of state:

$$\boxed{w_{\rm tor}(a) = -1 - \frac{1}{3}\frac{{\rm d}\ln\rho_{\rm tor}}{{\rm d}\ln a}.} \tag{140}$$

Step 3 — Absolute torsion density from TIP. From Eq. (136), neglecting radiation at $z \lesssim 3$:

$$\rho_{\rm tor}(a) = \left[1 - \Omega_{m,0}\,a^{-3}/\left(H/H_0\right)^2\right]\cdot\frac{3H^2(a)}{8\pi G}, \tag{141}$$

evaluated self-consistently from Eq. (137) via the EC-Cosmo solver [6].

Step 4 — CPL approximation and Foundation I calibration. The numerical solution is well approximated by the CPL form [97, 98]:

$$w_{\rm tor}(z) \approx w_0 + w_a\,\frac{z}{1+z}, \tag{142}$$

with $(w_0,\, w_a) = (-0.904,\, -0.153)$ from the global χ^2 calibration of Foundation I (Table 2 of Ref. [5], $F_{\rm ion} = 1.2765$). The normalized density (Eq. 30) departs mildly from unity ($X \lesssim 1.11$ for $z \le 3$), consistent with a slowly evolving dark sector and within the 1σ contours of the DESI DR1/DR2 model-independent reconstruction [11].

Status of the dark-energy parameters. In Equations (140)–(142), w_0 is *inherited* from the Foundation I calibration, and w_a is set through $\xi = 1.645$, calibrated on the DESI preference used by the ECF; the pair is therefore not independent of DESI. Unlike quintessence or (w_0, w_a)CDM models, in which (w_0, w_a) are two free parameters determined by dark-energy datasets, the ECF fixes the form of $w(z)$ and sets w_a with one calibrated coupling; and, unlike other torsion models that introduce at least one free coupling [76], it adds no coupling beyond this one.

Appendix C: Cosmic Inflation in the ECF — superseded estimate

Scope note — the vacuum manifold, and what it must carry

The three defect classes used in this work require three distinct homotopy groups of $\mathcal{M}$ to be non-trivial. Each has an origin already present in this framework, and we state them before noting what is not yet settled.

$\pi_0(\mathcal{M}) \neq 0$ — **the walls, from parity.** The axial torsion $S^\mu = \epsilon^{\mu\nu\alpha\beta} S_{\nu\alpha\beta}$ (Eq. 8) is a *pseudo*-vector: under parity $S^\mu \to -S^\mu$. The chiral condensate $\langle S^\mu \rangle = (s_0, \mathbf{0})$ with $s_0 \neq 0$ therefore breaks parity *spontaneously*. Parity being a discrete $\mathbb{Z}_2$, its breaking leaves two degenerate vacua, $s_0 > 0$ and $s_0 < 0$, which are precisely the σ_+ and σ_- chiral phases used throughout. The 2D Chiral Walls are the boundaries between them. This is not an added assumption: it follows from the axial nature of the torsion, which is where this framework starts.

$\pi_1(\mathcal{M})$ — **the strings, tentative (see below: not established).** The chiral phase θ jumps by $\Delta\theta = \pi$ across a wall (Eq. 65); a phase defined modulo a period is an angle, and the corresponding vacuum manifold would contain a circle, whose π_1 is non-trivial. The 1D Torsion Strings would be its winding configurations.

This factor is the weak one, and we say so. Unlike the other two, it is not supplied by the Lagrangian of §3.1: S^μ is a *real* vector field and carries no internal phase, so there is no $U(1)$ to break. Nor does θ supply one on inspection — it appears in this work only as the phase of the Chern–Simons coupling, hence either as a functional of S^μ itself or as a two-valued label, and neither is a circle. The 1D sector therefore rests on a motivated construction rather than on a derived symmetry, and the predictions that depend on π_1 — string tension, helicity $q = \pm 1$, the termination of filaments on nodes, and the two SKA channels — inherit that status. The 0D, 2D and 3D sectors do not: their factors follow from the potential of the adopted effective action (PO-ECF-ACTION) and from parity respectively. One route would close the gap without adding a field: the torsion carries a second irreducible vector part, the trace vector, and two real vectors define a relative phase. Whether it can serve is not established here, since the trace vector is non-dynamical on a homogeneous background; it is recorded as the specific form PO-F2-8 now takes. *The open question is no longer "which manifold" but "which field carries the phase".*

$\pi_2(\mathcal{M}) \neq 0$ — **the knots, from the spatial direction.** The orientation of the spatial part of the torsion spans a two-sphere, giving the hedgehog configurations of §6.1.

A candidate manifold, and what supports it. These three requirements are met by the product

$$\mathcal{M} \simeq \mathbb{Z}_2 \times S^1 \times S^2, \qquad \pi_0 = \mathbb{Z}_2, \quad \pi_1 = \mathbb{Z}, \quad \pi_2 = \mathbb{Z}, \tag{143}$$

each factor carrying one of the three origins above: $\mathbb{Z}_2$ the broken parity, S^1 the chiral phase, S^2 the spin direction. Two remarks make this more than an assembly chosen to fit.

First, S^2 must enter as a *factor* and not as a quotient. The winding number of Eq. (50) is the degree of a map $S^2 \to S^2$ and changes sign under $\hat{\Phi} \to -\hat{\Phi}$; it is therefore undefined on $\mathbb{R}P^2 = S^2/\mathbb{Z}_2$, which rules out the nematic-like manifolds that would otherwise be natural for an alignment order parameter. This is a constraint the framework imposes on itself, and it is restrictive.

Second, the independence of the $\mathbb{Z}_2$ and S^1 factors is not an assumption of this work but a standard result in condensed matter. Chiral charge-density-wave systems are described by *two independent, coexisting order-parameter fields* — a chiral field and continuous phase fields — whose defects are classified by separate homotopy groups, and whose physical independence is observed [26]. The same holds in chiral superfluid ^{3}He films, where a Kibble–Zurek quench produces domain walls between time-reversed chiral phases *together with* phase vortices [24]; and spinor condensates realise product manifolds with a $U(1)$ factor of exactly this kind [25]. *The analogy is close: the ECF transition is itself a Kibble–Zurek quench of a chiral condensate.*

What is still not settled. The manifold above is *motivated* by the framework's own content and *supported* by the condensed-matter analogy; it is not *derived* from the electroweak PGT exten-

sion [166]. That derivation — showing that the symmetry breaking of the axial sector produces this manifold rather than admitting it — is what PO-F2-8 still asks for. We record the candidate because it is falsifiable in the ordinary sense: any derivation yielding $\pi_1 = 0$, or requiring S^2 as a quotient, would contradict it.

Open problem PO-F2-8 — the vacuum manifold carrying all three defect classes. The homotopy assignments used throughout this work ($\pi_0 \to$ 2D walls, $\pi_1 \to$ 1D strings, $\pi_2 \to$ 0D knots) presuppose a vacuum manifold whose first three homotopy groups are simultaneously non-trivial. As set out in the scope note above, each requirement has an identified physical origin within this framework — spontaneous parity breaking by the axial condensate, the chiral phase, and the spatial orientation of the torsion — but the manifold assembling them has not been derived. In particular: (i) does the electroweak PGT extension admit a vacuum manifold that is disconnected *and* not simply connected? (ii) if the parity $\mathbb{Z}_2$ is spontaneously broken, what carries π_1, given that the same $\mathbb{Z}_2$ cannot serve twice? (iii) is the $\Delta\theta = \pi$ phase jump the signature of a $U(1)$ broken to $\mathbb{Z}_2$, and if so at what scale?

Until this is settled, the three defect sectors rest on assignments whose common group-theoretic foundation is assumed rather than shown. This does not affect the quantities computed from each sector separately — masses, tensions and couplings follow from the condensate s_0 and from the horizon scale, not from the manifold's global structure — but it is a genuine gap, and a referee is entitled to raise it. Deferred to Foundation III [7].

Partial progress (30/08/2026) — the trace-vector route, revisited. Earlier in this appendix the trace-vector companion to S^μ was reported to fail because, on a homogeneous background, the two vectors are forced collinear and define no phase. Both stated obstructions — the vanishing of $F^{(S)}_{\mu\nu}$ and this collinearity — are consequences of imposing exact homogeneity and isotropy, not of the field content itself; a direct calculation near an actual defect, where that symmetry is locally broken, was not previously carried out and is reported here.

Using the standard irreducible decomposition of torsion (trace vector T_μ, axial vector S_μ, tensor part set to zero) and computing the *full* torsion-covariant derivative of S_ν — $\nabla_\mu S_\nu = \partial_\mu S_\nu - \Gamma^\lambda_{\mu\nu} S_\lambda - K^\lambda_{\mu\nu} S_\lambda$, with K the contortion built from T_μ and S_μ jointly — rather than the flat field strength $F^{(S)}_{\mu\nu} = \partial_\mu S_\nu - \partial_\nu S_\mu$ used elsewhere in this work, a genuine T–S coupling appears that is absent from the minimal Lagrangian of Eq. (9). Concretely, for a wall profile $S^\mu = (A(z), 0, 0, 0)$, the covariantised field strength picks up $\tilde{F}_{03} = -A'(z) - \frac{2}{3} A(z)\, T_3$, and the resulting action term $-\frac{1}{4}\tilde{F}_{\mu\nu}\tilde{F}^{\mu\nu}$ contains a term *linear* in T_3 (the trace-vector component *normal* to the wall), proportional to $A(z)A'(z)$: a genuine source, not merely an effective mass. It vanishes away from the wall (where $A' \to 0$, recovering $T_\mu = 0$ on the homogeneous vacuum, as required) and is localised in the transition region. The same calculation repeated for a string-like profile in cylindrical coordinates, both for S^μ along the string axis and for S^μ purely azimuthal, reproduces the identical structure: the trace-vector component *radial* to the core is sourced by (profile) $\times$ (profile)$'$, vanishing in the far vacuum. In both geometries the sourced T_μ component is orthogonal to S_μ's own direction, resolving the stated collinearity obstruction by an explicit, verified mechanism rather than by assumption. *Reproduced by:* `Scripts_F2/check_trace_vector_defect_coupling.py`.

What this does not yet show. In every ansatz tested, the sourced component points along the coordinate that measures distance from the defect (normal, or radial), never along the azimuthal direction that a genuine π_1 winding requires. Supplying that winding – so that the pair (S_μ, T_μ) traces a full turn in field space as the azimuthal angle goes $0 \to 2\pi$ around a string core, rather than merely becoming non-collinear at each point – is not established by this calculation and remains the open content of PO-F2-8. The result reported here narrows the question from "can any field supply the phase" to "does the same geometric mechanism, evaluated on a genuinely helical ansatz coupling the azimuthal and axial directions, close the winding" — a specific, well-posed calculation for a future session, not yet attempted.

This appendix derives the modified Raychaudhuri equation governing the bounce-to-stiff transition and the inflation-exit dynamics in the ECF. The bounce is driven by the repulsive quartic spin-torsion term $\propto \kappa^2 S^4$ at ultra-high density ($\rho \to \rho_c$), which renders the singularity non-singular (Appendix D). The exit from the stiff phase ($w = 1$) into the radiation-dominated era is automatic once $\rho \ll \rho_c$ and the spin density has diluted as $\rho_{\rm spin} \propto a^{-6}$; no inflaton potential or slow-roll parameter is required. The

predictions for the number of e-folds and spectral index are presented below as order-of-magnitude estimates (**superseded, see status box above**).

C.5 Modified Raychaudhuri Equation

In the presence of torsion, the Raychaudhuri equation governing the cosmic acceleration is modified by an effective quartic correction in the spin density. The effective stress-energy tensor becomes:

$$T_{\mu\nu}^{\text{eff}} = T_{\mu\nu} + \frac{3\kappa}{32\pi} S_{\mu\rho\sigma} S_{\nu}{}^{\rho\sigma} \tag{144}$$

In a homogeneous and isotropic background, differentiating Eq. (2) (with $\Lambda = 0$) and using the standard continuity equation $\dot{\rho} = -3H(\rho + p)$ gives the modified acceleration equation:

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left[\rho\left(1 - \frac{4\rho}{\rho_c}\right) + 3p\left(1 - \frac{2\rho}{\rho_c}\right)\right] \tag{145}$$

where ρ_c is the critical spin-torsion density introduced in Section 2. At the bounce ($\rho = \rho_c$, $H = 0$) it gives $\ddot{a}/a = 4\pi G(\rho_c + p) > 0$, as required for a bounce; the acceleration changes sign at $\rho^* = \rho_c\,(1+3w)/(4+6w)$, i.e. $\rho_c/4$ for dust, $\rho_c/3$ for radiation and $0.4\,\rho_c$ for the stiff fluid (the last two within the single-density form; the regime-by-regime equations, Eq. (3), give $\rho_c/4$ for radiation and no spin-driven bounce for the stiff fluid).For $p = 0$ it holds in Einstein–Cartan theory for non-relativistic fermionic matter whose spin term behaves as stiff matter (spin-fluid description, $n \propto a^{-3}$; primary source to be confirmed), not in Popławski's Dirac description, where $\tilde{\epsilon} = -\tilde{p}$ and, for dust, his Eqs. (11)–(12) give $\ddot{a} < 0$ at $H = 0$; for $w \neq 0$ it inherits the schematic status of Eq. (2). The same equation is the effective Raychaudhuri equation of loop quantum cosmology (Ref. [201], Eq. (2.28)). *[Corrected 04/10/2026: earlier versions displayed $\ddot{a}/a = -(4\pi G/3)(\rho + 3p)(1 - \rho/\rho_c)$, which does not follow from Eq.* (2) *and does not change sign; the* 0.46 *e-fold figure is the $w \in [0, 1]$ majorant of an external Weyssenhoff-fluid calculation (Izaurieta, Lepe & Quinzacara, arXiv:2608.11453), not a bound derived from this equation (correction of 28/09/2026). The old threshold condition remains withdrawn.]*

A conditional window, in the single-density schematic model. Eq. (2) of this paper, $H^2 = (8\pi G/3)\rho(1 - \rho/\rho_c)$, is the single-density equation that *Foundation I — Extended* (Appendix H) uses for the bounce regime and describes there as a "schematic", phenomenological implementation. For a constant-w fluid ($p = w\rho$) the acceleration equation above reads

$$\frac{\ddot{a}}{a} = \frac{4\pi G}{3}\rho\left[(4 + 6w)\frac{\rho}{\rho_c} - (1 + 3w)\right], \tag{146}$$

which does change sign, at $\rho^*/\rho_c = (1+3w)/(4+6w)$. For $w = 1$: $\rho^* = 0.4\,\rho_c$, and with the standard dilution law $\rho \propto a^{-6}$ this gives $N = \frac{1}{6}\ln(5/2) \approx 0.153$ e-fold. **This result is CALCULATED-CONDITIONAL, strictly internal to this single-density schematic model, and is not established as a consequence of the microscopic spin mechanism of this paper**, whose spin density is derived independently as $\rho_{\text{spin}} \propto n^2 \propto a^{-6}$ (Sec. 2) and enters the (different, two-term) Friedmann equation of *Foundation I — Extended* (its main ECF background equation) *linearly*, not through a single density squared. Whether the two structures describe the same physical correction, and in particular whether the quadratic term of the schematic bounce equation corresponds to the microscopically-derived a^{-6} spin term or to something else, is not derived anywhere in the corpus; it is tracked as an open item (PO-ECF-ACTION-BOUNCE, open, added 28/09/2026). Neither the 0.46 nor the 0.153 figure should be read as a prediction of the ECF's own dynamics pending that derivation.

C.6 Number of E-folds

Assuming an initial density near the bounce $\rho \approx \rho_c$ and a stiff equation of state $\rho \propto a^{-6}$, the scale factor evolves as:

$$a(t) \propto t^{1/3} \tag{147}$$

The e-folds relevant for the comoving Hubble radius would be

$$N_{eff} = \ln\left(\frac{a_{end}H_{end}}{a_{bounce}H_{bounce}}\right). \tag{148}$$

Earlier versions stated that a numerical solution of this expression yields $N_{eff} \sim 60$; no calculation in the corpus reproduces this value, which is retained only as a historical figure. *[Correction of 28/09/2026 — the sentence formerly here stated that the post-bounce kinematic acceleration "is bounded to $\leq$ 0.46 e-fold (Foundation III)", which read as an ECF-specific result. As clarified above (§Modified Raychaudhuri Equation), the* 0.46 *figure is an external, $w \in [0, 1]$ majorant for a general Weyssenhoff spin fluid (Izaurieta, Lepe & Quinzacara, arXiv:2608.11453), used by Foundation III*

to close the kinematic route for that fluid; it is not derived from the present equation and bounds no ECF-specific window. Whether the ECF's own post-bounce kinematic acceleration is bounded at all remains open, tracked under PO-ECF-ACTION-BOUNCE. *Separately, the N_{eff} expression above cannot in any case be evaluated literally at the bounce in the single-density schematic model of Foundation I — Extended (Appendix H), where $H_{bounce} = 0$ exactly by construction; this is an additional reason, independent of the* 0.46 *figure, why no numerical value of N_{eff} is claimed here.]* The horizon and flatness problems remain open in the ECF.

C.7 Exit Dynamics

The characteristic timescale of the post-bounce phase is:

$$\tau_{exit} = \sqrt{\frac{3}{8\pi G\rho_c}} \approx 2 \times 10^{-44}\ \mathrm{s} \qquad (149)$$

for $\rho_c \approx 5 \times 10^{93}\,\mathrm{g\,cm^{-3}}$ (Planck density, Eq. 7), i.e. of order the Planck time — as expected for a bounce occurring at the Planck scale.

Appendix D: Microphysics of Geometric Dark Matter

This appendix provides the referee-level derivations supporting the geometric dark matter claims in §4. It covers: (i) the $R_{\rm knot}$ radius conventions (§D.8) used consistently throughout the paper; (ii) the stiff-fluid vector perturbation growth $\delta_V \propto a^{3/2}$ that seeds Micro-Knot formation; and (iii) the merger and annihilation efficiency parameters $\varepsilon_{\rm top}$, calibrated from the topological mass-energy budget, and $\eta_{\rm ann}$, which is left open. All quantities link back to the single ECF coupling constant $\kappa^2 = 2.0766 \times 10^{-43}\,\mathrm{m\,J^{-1}}$.

D.8 Note on $R_{\rm knot}$ Conventions

Two distinct radius scales appear throughout this work and must not be confused:

- **Micro-Knot formation radius:** $R_{\rm micro} = c\,t_{\rm EW} \approx 4.5\,\mathrm{mm}$, the causal horizon size at the electroweak crystallisation epoch ($t_{\rm EW} \approx 1.5 \times 10^{-11}\,\mathrm{s}$). This is the *formation radius* at $z \sim 10^{15}$; it is not the present-day radius.
- **Present-day effective radius:** Micro-Knots are pressureless dust ($w = 0$) and their physical size dilutes as $R \propto a$. At $z = 0$:

$$R_{\rm micro,0} = R_{\rm micro}\,\frac{a_{\rm EW}}{a_0} \approx 4.5 \times 10^{-15}\,\mathrm{mm}, \qquad (150)$$

  sub-nuclear in size — consistent with the point-mass approximation used in all microlensing calculations.
- **Macro-Knot magnetic flux radius:** $R_{\rm knot} \sim 10\,\mathrm{AU}$ is the *virial magnetic radius* of a Macro-Knot of mass $\sim 10^5\,M_\odot$, derived from the equilibrium of its torsion-Chern–Simons magnetic flux (Eq. 179). This is an astrophysical scale, entirely distinct from the geometric freeze-out radius.

All numerical estimates in Eqs. (178)–(179) use the Macro-Knot flux radius. The formation radius $R_{\rm micro}$ appears exclusively in the Dual Genesis timeline (Sec. 6.16).

D.9 Collision Rates, Mean Free Path, and the Primordial Gravitational-Wave Fossil

Micro-Knot collision parameters. With a present-day sub-nuclear effective radius $R_{\rm micro,0} \approx 4.5 \times 10^{-15}$ mm (Eq. 150), the geometric cross-section is negligible ($\sigma_{\rm geom} = \pi R^2_{\rm micro,0} \approx 6 \times 10^{-35}$ m^2). The dominant interaction channel is gravitational focusing. For a typical halo velocity dispersion $v \simeq 200$ km s^{-1}, the gravitational cross-section is

$$\sigma^{(\mu)}_{\rm grav} = \pi \left(\frac{2GM_\mu}{v^2}\right)^2 \approx 4.9 \times 10^{8}\ \mathrm{m}^2, \qquad (151)$$

still astronomically small compared with inter-particle distances. With the local dark-matter density $\rho_{\rm DM} \approx 0.4$ GeV cm^{-3}, the Micro-Knot number density is $n_\mu = \rho_{\rm DM}/M_\mu \approx 1.9 \times 10^{-46}$ m^{-3}, giving a mean free path

$$\lambda^{(\mu)}_{\rm mfp} = \frac{1}{n_\mu\,\sigma^{(\mu)}_{\rm grav}} \approx 1 \times 10^{37}\ \mathrm{m}\ \sim\ 10^{11}\,d_H, \quad (152)$$

where d_H is the present Hubble radius. The collision rate today is therefore $\Gamma^{(0)}_\mu \approx 2 \times 10^{-32}$ s^{-1} *(corrected 15/10/2026: earlier versions gave $\sigma \approx 3.5 \times 10^7$ m^2, $n_\mu \approx 7 \times 10^{-52}$ m^{-3}, $\lambda \approx 4 \times 10^{43}$ m and 5×10^{-39} s^{-1}; the companion script also took*

$0.4\ GeV\,cm^{-3}$ *for* $0.4\ GeV\,m^{-3}$*, a factor* 10^6*),* confirming that Micro-Knots are *perfectly collisionless* on cosmological timescales.

Macro-Knot collision parameters. Macro-Knots ($M \sim 10^5\,M_\odot$) are far rarer: under the working assumption of $\mathcal{O}(1)$ Macro-Knot per galactic volume (§4.4; not derived), a Milky-Way-sized halo hosts $N_{\rm macro} = \mathcal{O}(1)$ object, a mass fraction $f_{\rm macro} \equiv M_{\rm macro}/M_{\rm halo} \sim 10^{-7}$–$10^{-6}$ for $M_{\rm macro} = 10^5$–$10^6\,M_\odot$ and $M_{\rm halo} \sim 10^{12}\,M_\odot$ *(an abundance conditional on the early accretion regime of the seeds: PO-F2-SEED-CMB, §7.1. Corrected 17/10/2026: earlier versions called* $f_{\rm macro}$ *derived and gave* $N_{\rm macro} = \mathcal{O}(1$–$10)$*, the upper value coming from recomputing* N *with* $10^5\,M_\odot$ *from a fraction obtained with* $10^6\,M_\odot$*). Correction of 13/09/2026.* Earlier versions of this paragraph quoted $N_{\rm macro} \sim 10^3$ per halo, which is inconsistent with the working assumption of §4.4 ($\mathcal{O}(1)$ per galactic volume, $f_{\rm macro} \sim 10^{-7}$–$10^{-6}$) by two to three orders of magnitude. The assumed value is retained. The collision rate below scales linearly with $N_{\rm macro}$ and is reduced accordingly; the conclusion — fewer than one merger per Hubble time per halo — is strengthened, not weakened. Their gravitational cross-section at $v \simeq 300$ km s^{-1} is

$$\sigma_{\rm grav}^{(M)} = \pi\left(\frac{2GM_{\rm macro}}{v^2}\right)^2 \approx 2.7\times10^{29}\,{\rm m}^2 \approx 2.9\times10^{-10}\,{\rm kpc}^2, \tag{153}$$

with a mean free path $\lambda_{\rm mfp}^{(M)} \sim 4\times10^{11}$ kpc. The collision rate today is $\Gamma_M^{(0)} \lesssim 2.5\times10^{-29}$ s^{-1} (rescaled from $N_{\rm macro} \sim 10^3$ to $\mathcal{O}(1)$), i.e. far fewer than one merger per Hubble time per halo. Present-day Macro-Knot coagulation is therefore entirely negligible: the population was assembled early, through the topological selection rule of Eqs. (81)–(82), not by late-time encounters.

Macro-Knots and galactic centres. Because Macro-Knots form in the densest regions at the Great Annihilation (§4.3.1), they are expected to occupy them from the outset. *(Corrected 15/10/2026: earlier versions placed their formation in Kibble–Zurek domains at* $T \sim 200$ *MeV.) Correction of 13/09/2026: earlier versions stated that they sink to galactic-potential wells by dynamical friction on timescales* $t_{\rm fric} \lesssim 1$ *Gyr. Chandrasekhar friction scales as* $1/M$*, and for* $M \sim 10^5\,M_\odot$ *in a Milky-Way-like halo it gives* $t_{\rm fric} \simeq 59$ *Gyr at* 1 *kpc and* $\simeq 5.9\times10^3$ *Gyr at* 10 *kpc — far longer than the age of the Universe. The* 1 *Gyr figure holds only within* ~ 130 *pc of the centre. Macro-Knots found in galactic nuclei are therefore there because they formed there, not because they migrated.* The sphere of gravitational influence of a central SMBH of mass $M_\bullet$ has radius

$$R_{\rm inf} = \frac{G\,M_\bullet}{\sigma_v^2} \;\approx\; 1\text{–}2\ {\rm pc}\,\left(\frac{M_\bullet}{4\times10^6\,M_\odot}\right), \tag{154}$$

for the central velocity dispersion of the Milky Way bulge, $\sigma_v \simeq 100$–140 km s^{-1} [122] (0.9–1.7 pc), within which Macro-Knots are gravitationally bound to the nucleus. *(Corrected 15/10/2026: earlier versions gave* 4 *pc, which assumed an unstated* $\sigma_v \simeq 66\ km\,s^{-1}$*.)* This provides a natural ECF pathway from primordial topological seed to supermassive-black-hole co-evolution, consistent with the observed $M_\bullet$–σ relation [123].

Cosmological evolution of collision rates. Because number density scales as $n \propto (1+z)^3$ and the gravitational cross-section scales as $\sigma_{\rm grav} \propto v^{-4} \propto (1+z)^{-2}$ (virial scaling), the collision rate evolves as

$$\Gamma(z) = \Gamma^{(0)}\,(1+z)^{3/2}. \tag{155}$$

At the Micro-Knot crystallisation epoch ($z_{\rm EW} \sim 2.0\times10^{15}$), Γ_μ extrapolated from the local halo normalisation is $\sim 2\times10^{-9}$ s^{-1} — see the caveat below. At the electroweak crystallisation epoch ($z \sim 2.0\times10^{15}$) and at the QCD transition ($z \sim 1.5\times10^{12}$), Eq. (155) returns $\Gamma_\mu \sim 2\times10^{-9}$ and $\sim 3\times10^{-14}$ s^{-1} respectively.

A caveat that must be read before those numbers are used. Eq. (155) is normalised on the *local* halo: a dark-matter density of 0.4 GeV cm^{-3} and a velocity dispersion of 200 km s^{-1}. Neither quantity means anything in the primordial plasma, where the density is larger by many orders of magnitude and the relative velocities are thermal rather than set by a galactic potential. The scaling itself combines $n \propto (1+z)^3$ with a virial cross-section $\sigma \propto v^{-4}$, and the virial regime does not exist before structures form. Extrapolating a halo cross-section across fifteen decades in redshift is therefore not licit, and the figures above should be read as an illustration of the scaling, not as collision rates. Taken at face value they give $\Gamma_\mu t_{\rm Hubble} \sim 10^{-20}$–$10^{-14}$ at every epoch, i.e. not one encounter — which cannot be reconciled with a coagulation mechanism.

A quantitative treatment requires the thermal density and the gravitational cross-section at primordial velocities, integrated through a Smoluchowski coagulation equation. That calculation is not performed here, and Macro-Knot formation therefore rests on the Great-Annihilation horizon mass (a structural hypothesis), not on coagulation and not on the rate quoted above; the selection rule of Eqs. (81)–(82) governs Micro-Knot aggregation only.

Correction of 27/08/2026. Earlier versions of this paragraph placed the electroweak crystallisation at $z \sim 1.5 \times 10^{11}$, which corresponds to $T \sim 35$ MeV and to no epoch of this framework; quoted $\Gamma_\mu \sim 10^{-16}$ s^{-1} as "one encounter every ~ 300 s", whereas its inverse is 10^{16} s; and quoted $\Gamma_M \sim 10^{-4}$ s^{-1} as "10^4 collisions per second", which is its reciprocal. The two rates were also inconsistent with each other under Eq. (155) by some ten orders of magnitude. *Timeline rebuilt by:* `Scripts_F2/build_ecf_timeline.py`.

Fossil gravitational-wave background. Each collision radiates gravitational energy with efficiency $\eta_{\rm GW} \sim 1\%$ (constructive mergers) to 50% (topological annihilation events). The peak frequency of Eq. (158) at the formation epoch, redshifted to today, gives

- **Micro-Knot collisions ($z_{\rm EW} \sim 2.0 \times 10^{15}$):** $f_{\rm obs} \sim 3\,{\rm GHz}/(1+z_{\rm EW}) \sim 10^{-6}$ Hz;
- **Macro-Knot collisions (from the Great Annihilation, $z \sim 4 \times 10^9$):** $f_{\rm obs} \sim 50\,{\rm mHz}/(1+z) \sim 10^{-11}$ Hz.

Both fall outside the LISA/ET band ($\gtrsim 10^{-4}$ Hz) and below the PTA band (10^{-9}–10^{-7} Hz): this fossil component is not accessible to current or planned detectors and is not counted among the observables of this work; the LISA observables are the chiral asymmetry (Sec. 9.4) and the late-time Macro-Knot merger burst. *(Corrected 15/10/2026: earlier versions gave $f_{\rm obs} \sim 0.2$ Hz for Micro-Knots, "in the LISA/ET band", and placed Macro-Knot collisions at the QGP epoch, "contributing to the nano-Hz PTA background", calling the result parameter-free.)* Figure 46 shows the Micro-Knot collision rate $\Gamma_\mu(z)$, Fig. 47 summarises the combined evolution, and Fig. 48 shows the Macro-Knot counterpart, including the enhanced rate in galactic-centre environments.

D.10 Merger and Annihilation Parameters: Derivation and Calibration

This appendix provides the referee-level derivation (and, for $\varepsilon_{\rm top}$, the calibration) of the two key efficiency parameters entering the ECF merger formalism (Sec. 7.3): the constructive-merger radiation efficiency $\varepsilon_{\rm top}$ and the annihilation efficiency $\eta_{\rm ann}$. A third sub-section extends the analysis to *Knot–compact-object* (Knot-Star) encounters, which constitute a distinct physical regime with unique observational signatures.

D.10.1 Constructive Merger Efficiency $\varepsilon_{\rm top}$

Physical motivation. When two same-helicity Knots coalesce (Regime I, Eq. 83), the total rest-mass energy $M_{\rm tot}c^2$ is partially radiated as gravitational waves during the topological coalescence, in close analogy with binary black-hole (BBH) mergers in full numerical relativity [70]. The fraction radiated, $\varepsilon_{\rm top} = E_{\rm GW}/(M_{\rm tot}c^2)$, depends on the mass ratio $q = M_1/M_2 \le 1$ and on the internal topological structure of the defects.

Calibration from numerical relativity. For non-precessing binary black holes, the radiated fraction scales approximately as the square of the symmetric mass ratio $\eta = q/(1+q)^2$:

$$\varepsilon_{\rm BBH}(q) \;\approx\; 0.05\,(4\eta)^2 \;=\; 0.05\left[\frac{4q}{(1+q)^2}\right]^2, \quad (156)$$

normalised to $\simeq 5\%$ for equal masses (GW150914 radiated $\simeq 3\,M_\odot$ of $\simeq 65\,M_\odot$ [70]); at $q = 0.1$ it gives 0.0055, against 0.00415 ± 0.00017 in a dedicated numerical-relativity simulation of the 10:1 case [69]. The factor vanishes in the test-body limit ($q \to 0$). Extended defects such as cosmic-string loops radiate over longer timescales; no efficiency value is adopted for them here. ECF Knots occupy an intermediate regime: their torsion core provides additional internal degrees of freedom that can radiate, but helicity conservation prevents the catastrophic spin-up that drives BBH ringdown. We therefore adopt

$$\varepsilon_{\rm top}(q) = \varepsilon_{\rm BBH}(q) \,\times\, \xi_T, \qquad \xi_T \in [0.3,\, 1.0], \quad (157)$$

where ξ_T is a posited suppression factor ($\xi_T = 1$ for purely gravitational, BBH-like emission). For $q \ge 0.1$ this gives $\varepsilon_{\rm top} \in [0.0016,\, 0.05]$, and $\varepsilon_{\rm top} \lesssim 0.05$ in all cases; see Fig. 49. *(Corrected 15/10/2026: earlier versions used a law linear in η, $0.054 \cdot 4q/(1+q)^2$,*

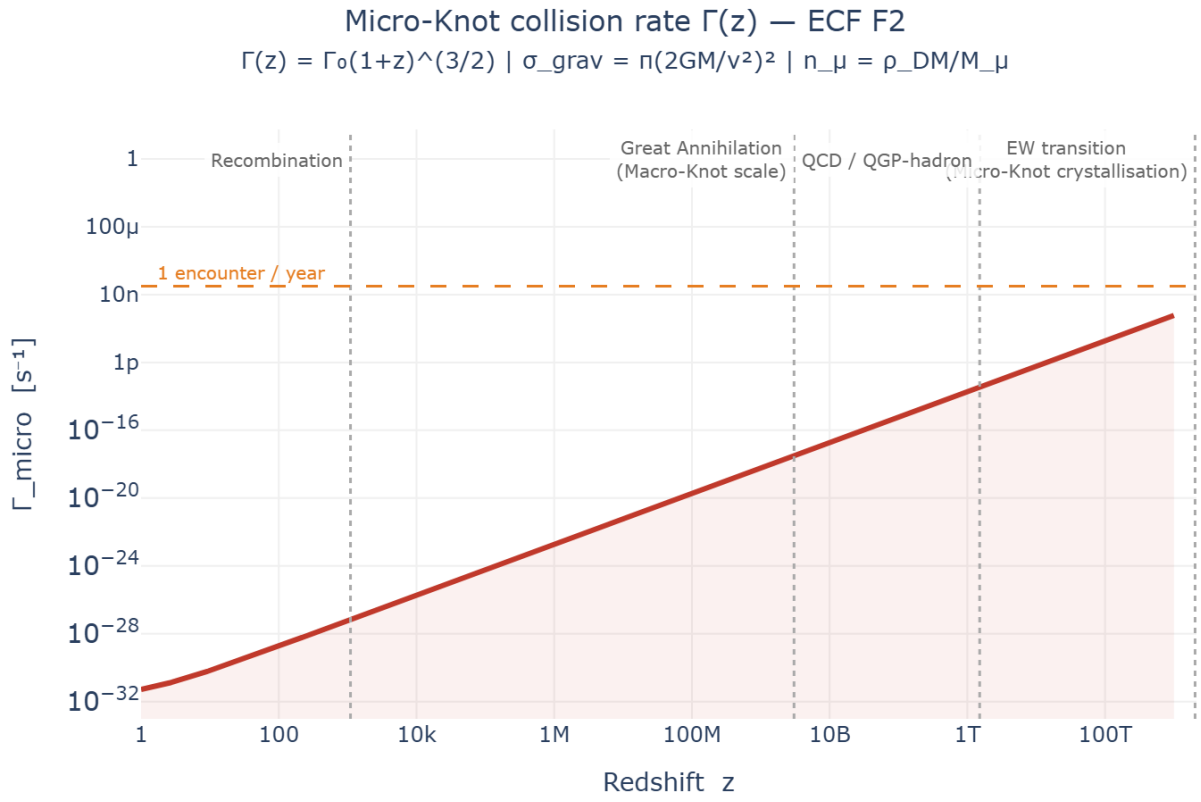

Figure 46: **Micro-Knot collision rate $\Gamma_\mu(z)$ from today to the electroweak freeze-out.** The rate scales as $\Gamma_\mu(z) = \Gamma_\mu^{(0)}(1+z)^{3/2}$, where $\Gamma_\mu^{(0)} \approx 2 \times 10^{-32}$ s^{-1} today (Eq. 151). The vertical dotted lines mark the recombination, Great-Annihilation, QCD and electroweak epochs; at the electroweak crystallisation ($z_{\rm EW} \sim 2.0 \times 10^{15}$), $\Gamma_\mu \sim 2 \times 10^{-9}$ s^{-1}, i.e. about one encounter per ~ 20 yr, against a Hubble time of $\sim 10^{-11}$ s. The horizontal orange dashed line marks one encounter per year; the curve stays below it at every epoch. The extrapolation from the local halo is not licit at early times (see the caveat in the text); the figure illustrates the scaling, not a collision history. *(Corrected 15/10/2026: the curve was 10^6 too low through a unit error in the script, and this caption still described a "1 encounter per 300 s" threshold, removed from the figure on 27/08/2026, and an early coagulation that this work no longer claims.) Script:* `scripts/plot_gamma_micro_z.py`.

four times too high at $q = 0.1$; let ξ_T rise to 1.85 in the text while Eq. (157) bounds it by 1, giving a range $[0.01, 0.10]$; and attributed cosmic-string loop simulations ($\varepsilon \simeq 0.005$–0.02) to the GW150914 detection paper.)

Dominant emission channel. Because the helicities add constructively and the merged object is a topologically stable, higher-mass Knot, *no electromagnetic counterpart is predicted* at leading order. Secondary bremsstrahlung from the surrounding plasma carries $\lesssim 1\%$ of $E_{\rm GW}$, well below current instrument thresholds. The channel partition for Regime I mergers is therefore:

Channel	**Frac.**	**Observable**
GW (dominant)	90–99%	LISA / ET / PTA
Kinetic energy	1–5%	Plasma heating
EM / γ	$\lesssim 1\%$	Below threshold

Table 15: **Channel partition for Regime I mergers.** Fraction of the released energy per channel, and the corresponding observable.

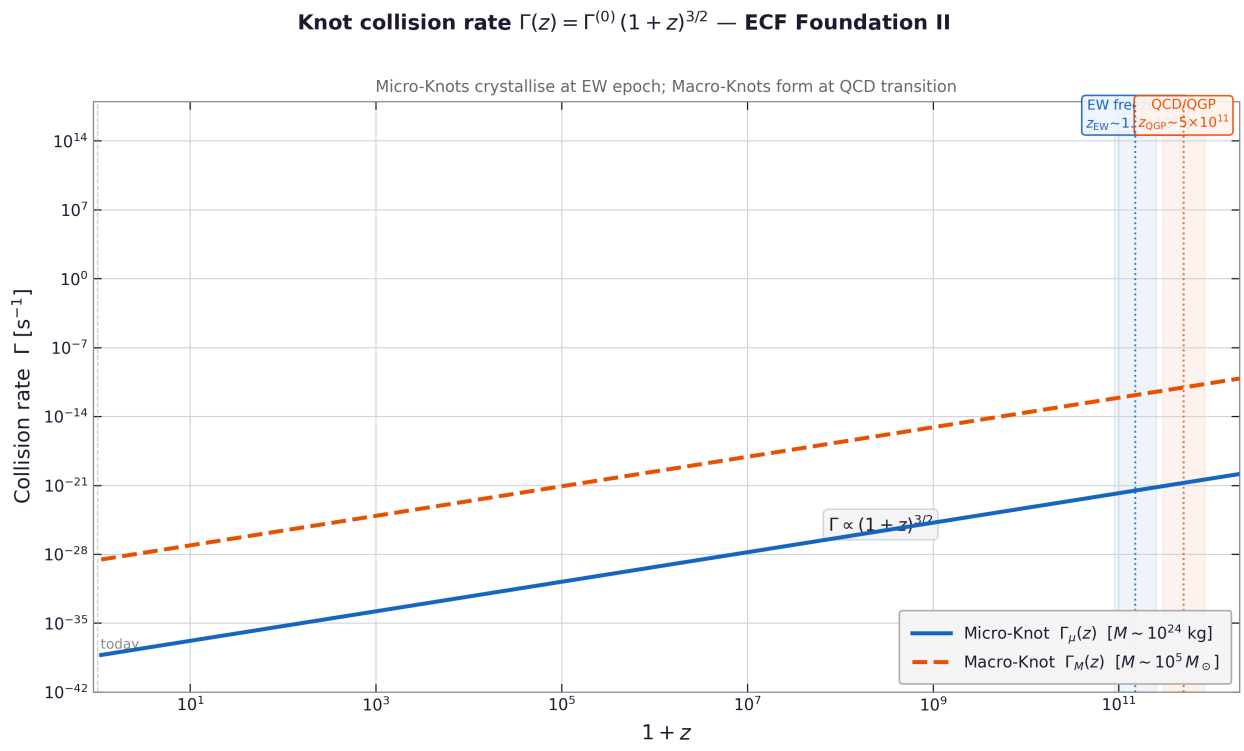

Figure 47: **Collision-rate evolution for Micro- and Macro-Knots.** The collision rate scales as $\Gamma(z) = \Gamma^{(0)}(1+z)^{3/2}$, so primordial encounters were far more frequent when the Universe was denser and the characteristic velocities were higher. The vertical markers indicate the electroweak and QCD freeze-out epochs, where the Micro-Knot and Macro-Knot populations crystallise and then rapidly decouple.
Reproduced by: `Scripts_F2/plot_gamma_z.py`.

Peak GW frequency. By analogy with the BBH ringdown frequency [70],

$$f_{\rm peak} = \frac{c^3}{6\pi\, G\, M_{\rm tot}}, \tag{158}$$

which gives, for three representative cases:

The Macro-Knot merger signal at ~ 50 mHz is a *quasi-monochromatic burst*. The GR ringdown timescale is $\tau_{\rm rd} = GM_{\rm tot}/c^3 \approx 1$ s for $M_{\rm tot} = 2\times10^5\, M_\odot$. If the torsion-field relaxation sets a damping time $\tau_T \sim R_{\rm Knot}/c$, the Macro-Knot radius ($R_{\rm min} \approx 3\times10^8$ m to $R_{\rm max} \approx 4.3\times10^8$ m, Eqs. (44)–(45)) gives $\tau_T \approx 1$–1.4 s, of the order of $\tau_{\rm rd}$: no separate tail is expected, and no feature distinguishing this burst from a black-hole merger of the same mass is established. *(Corrected 18/10/2026: earlier versions took $R_{\rm Knot} \sim 0.01$ pc, a value with no derivation in this work, and obtained a tail $\tau_T \sim 10^6$ s "clearly distinguishable" from the GR ringdown.)* The mass–frequency relation is plotted in Fig. 50.

D.10.2 Annihilation Efficiency $\eta_{\rm ann}$

Physical motivation. When a Knot of helicity $H > 0$ encounters an Anti-Knot of helicity $H < 0$ (Regime II), the net topological charge is zero and the spacetime torsion field undergoes a localized *geometric unbinding*. The energy

$$E_{\rm rad} = \eta_{\rm ann} \cdot 2M_2c^2 \tag{159}$$

is released across multiple channels (GW, γ/X, ν, plasma), where $M_2 \le M_1$ is the smaller defect and $\eta_{\rm ann}$ is the fraction of the pair's rest-mass energy converted to radiation (as opposed to residual geometric deformation of the dominant Knot).

Comparison with GRBs and FRBs. With Eq. (159), keeping a long-GRB energy below the observed record, $E_{\rm iso} \approx 1.2\times10^{55}$ erg (GRB 221009A, Konus-*Wind* [79]), requires $\eta_{\rm ann} \lesssim 3\times10^{-3}$ for $M_2 = 10^3\, M_\odot$, $\lesssim 3\times10^{-4}$ for $10^4\, M_\odot$ and $\lesssim 3\times10^{-5}$ for $10^5\, M_\odot$ *(corrected 22/10/2026: earlier versions used $E_{\rm iso} \approx 1.5\times10^{55}$ erg, a value not given in [79], and bounds 4×10^{-3}, 4×10^{-4}, 4×10^{-5})*; an FRB-

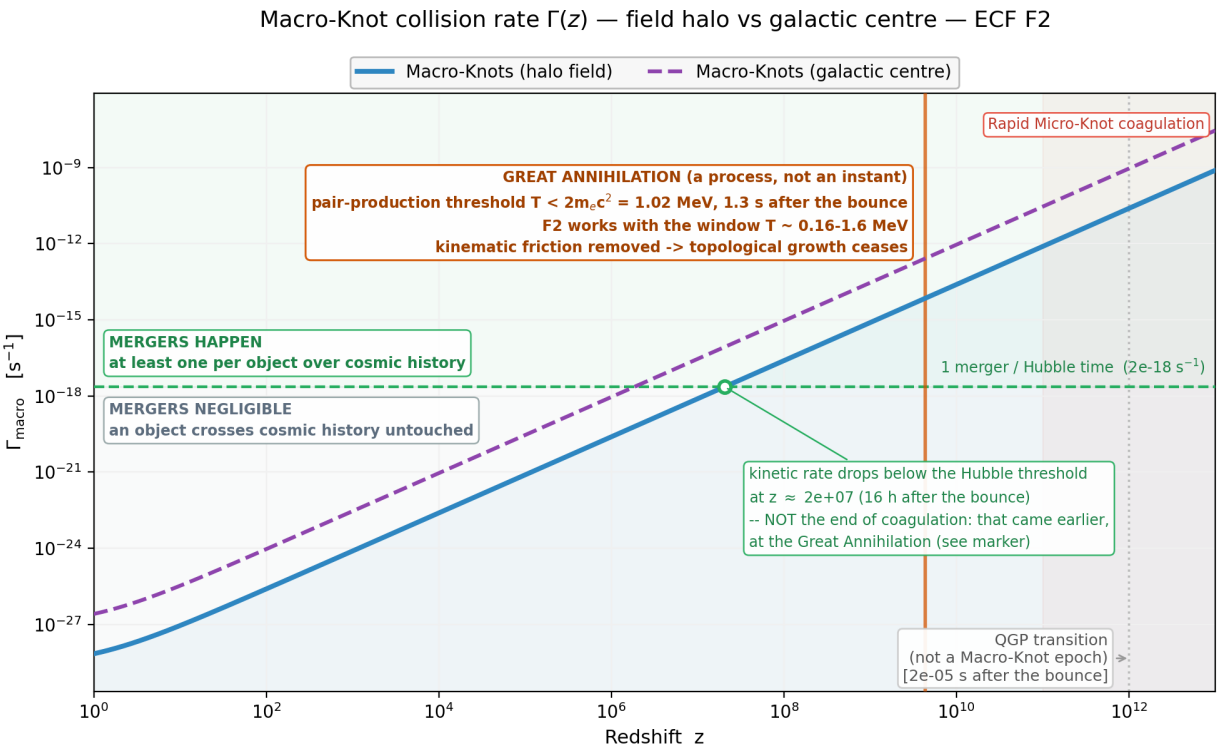

Figure 48: **Macro-Knot collision rate $\Gamma_M(z)$: halo field vs. galactic centre.** *How to read this figure.* The horizontal axis is redshift, i.e. time running *backwards*: the present is on the left, the QGP freeze-out on the right. *Only epochs belonging to the process shown are marked* — the QGP transition, shown for reference only (it is not a Macro-Knot epoch: their scale is set at the Great Annihilation, §4.3.1), and the Great Annihilation, where coagulation stops. Later epochs (recombination, first galaxies) are deliberately absent: coagulation is over by $t \simeq 1.3$ s, so they bear no causal relation to this curve. The vertical axis is the number of encounters per second for one Macro-Knot. Both curves rise towards the right because the early Universe was denser, so objects met more often. *The green dashed line is the reading key*: it marks one merger per Hubble time ($\simeq 2 \times 10^{-18}$ s^{-1}), the rate at which an object undergoes exactly one collision over the whole history of the Universe. **Above it, mergers happen**; **below it, an object crosses cosmic history untouched**. The two shaded bands mark these regimes. *Two distinct events, not to be confused.* The orange line marks the **Great Annihilation** ($T < 1.02$ MeV, $\simeq 1.3$ s after the bounce): it removes the kinematic friction and *topological growth ceases there.* The halo curve crosses the Hubble threshold much later, near $z \simeq 2 \times 10^7$ ($\simeq 16$ h after the bounce, open circle) — that crossing says only that the *kinetic* rate has become negligible, not that coagulation stopped then; it had already stopped, by a different mechanism, some five decades in redshift earlier. Today the rate is $\sim 10^{-29}$ s^{-1}, eleven orders of magnitude below threshold: the population was assembled in the first second and has been frozen since. *Curves.* Solid blue: $\Gamma_M(z) = \Gamma_M^{(0)}(1+z)^{3/2}$ in the field halo ($N_{\rm macro} = \mathcal{O}(1)$/halo, $v \simeq 300$ km s^{-1}); the dashed purple curve shows the enhanced rate in galactic-centre environments ($\rho \sim 10^3\,\rho_{\rm local}$, $v_{\rm GC} \sim 3\,v_{\rm halo}$). The shaded red band marks the epoch of rapid *Micro*-Knot coagulation near the QGP transition ($z \sim 10^{12}$); the Macro-Knot curves are extended there for reference only. *(Corrected 15/10/2026: this caption described the QGP as the epoch where Macro-Knots crystallise and the band as the mergers responsible for the bimodal spectrum, with $\Gamma_M \sim 10^{-4}$ s^{-1} — a value from before the renormalisation of 13/09/2026; the curves reach $\sim 10^{-9}$ s^{-1} there.)* The green dashed line marks the threshold of one merger per Hubble time.

Reproduced by: `Scripts_F2/plot_gamma_macro_z_mpl.py`.

class event from a Micro-Knot pair ($M_2 \sim 10^{20}$– 10^{24} kg, $2M_2c^2 \sim 10^{44}$–10^{48} erg) would require a

Merger type	$M_{\rm tot}$	$f_{\rm peak}$	**Instrument**
$\mu + \mu$	7.5×10^{24} kg	~ 3 GHz	Beyond reach
$M + M$	$2 \times 10^5\, M_\odot$	~ 50 mHz	**LISA**
$M_h + M$	$10^6\, M_\odot$	~ 10 mHz	LISA / PTA

Table 16: **Peak frequency of the merger signal** for three representative cases, from Eq. (158), with the instrument covering each band. *(Corrected 15/10/2026: the $\mu+\mu$ row used 2×10^{24} kg, from a superseded Micro-Knot mass, instead of $2 \times 3.74 \times 10^{24}$ kg, and gave ~ 10 GHz; a $\mu + M$ row ($10^3\, M_\odot$, ~ 10 Hz, LIGO/ET) is withdrawn: $10^3\, M_\odot$ was the Macro-Knot lower bound withdrawn on 17/09/2026, and at a mass ratio $\sim 10^{-30}$ the peak-frequency formula does not apply and the radiated energy is negligible.)*

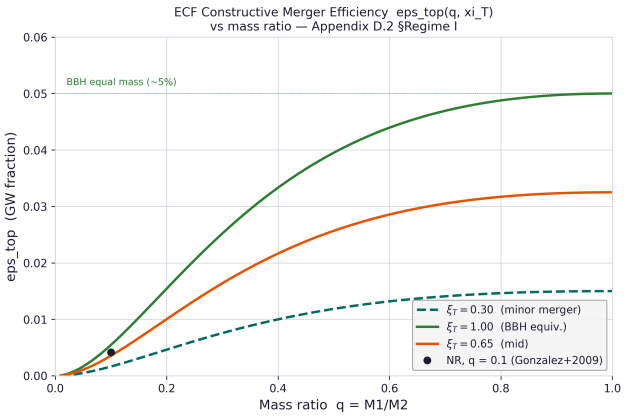

Figure 49: **Constructive merger efficiency $\varepsilon_{\rm top}(q, \xi_T)$.** Radiated GW fraction as a function of mass ratio $q = M_1/M_2$ for three values (0.3, 0.65, 1) of the factor ξ_T (Eq. 157). The dotted line marks the equal-mass BBH value ($\simeq 5\%$, $\xi_T = 1$); the point is the numerical-relativity value at $q = 0.1$ [69]. For $q \geq 0.1$ and $\xi_T \in [0.3, 1]$, $\varepsilon_{\rm top} \in [0.0016, 0.05]$. *(Corrected 15/10/2026: law in $(4\eta)^2$; curves $\xi_T = 1.45$, 1.85 and the band $[0.01, 0.10]$ withdrawn.)*

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`Scripts_F2/plot_ecf_annexe_D2.py`.

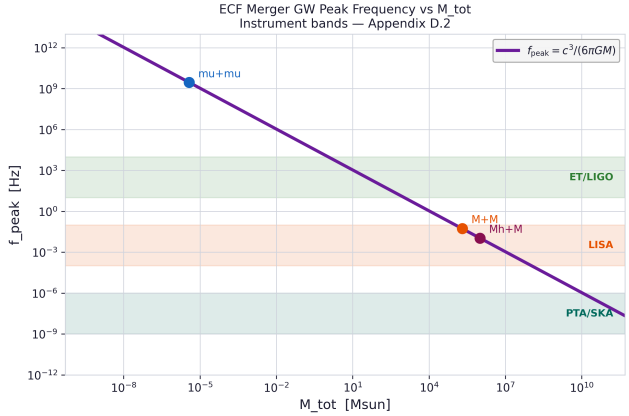

Figure 50: **Peak GW frequency vs total merger mass.** The relation $f_{\rm peak} = c^3/(6\pi G M_{\rm tot})$ (Eq. 158) is shown over 22 decades of mass. Coloured bands indicate the sensitivity windows of PTA/SKA (nHz), LISA (mHz), and LIGO/ET (Hz–kHz). The three ECF mass cases (μ+μ, M+M, M_h+M) are marked; the Macro-Knot equal-mass merger at ~ 50 mHz falls squarely in the **LISA** band.

Channel	**Micro-Knot**	**Macro-Knot**
γ/X (prompt)	40–60%	10–30%
Plasma kinetic energy	25–40%	5–15%
Gravitational waves	1–5%	20–40%
Neutrinos	5–15%	5–10%
X-ray thermal (soft)	$\sim 5\%$	5–15%

Table 17: **Channel partition for Regime II (annihilation),** compared between Micro-Knots and Macro-Knots.

still smaller radiated fraction [117]. The values $\eta_{\rm ann} \sim 0.1$–0.5 given in earlier versions rested on an arithmetic error (a factor ≈ 100 in $2M_2c^2$) and are withdrawn: $\eta_{\rm ann}$, or the fraction of $E_{\rm rad}$ emitted as prompt photons, is an open quantity, not a calibrated one. The upper bound $\eta_{\rm ann} \leq 0.5$ follows from energy conservation (the remaining fraction deforms the surviving Knot remnant). For the symmetric case $M_1 = M_2$, total annihilation ($M_{\rm final} = 0$) is topologically required, setting $\eta_{\rm ann}^{\rm sym} = 1$ by definition.

Comparison with monopole–antimonopole annihilation. Theoretical estimates for the analogous process in GUT monopole-pair annihilation [68] provide a qualitative benchmark, with the GW fraction growing with the monopole mass. This benchmark is quoted for comparison only: for massive ECF pairs the energetics above require a smaller radiated fraction, so it does not fix $\eta_{\rm ann}$.

Channel partition (Regime II). Unlike the fusion regime, the annihilation releases energy across all channels. The partition depends on the Knot mass: The reversal of the GW/EM dominance between the two mass scales, illustrated in Fig. 51, is a direct consequence of the Schwarzschild-radius scaling: $R_s = 2GM/c^2 \propto M$. For Micro-Knots, $R_s \ll$ any astrophysical scale, so the merger geometry cannot sustain coherent GW emission; for Macro-Knots, $R_s \sim 10^{-4}$–10^{-2} pc, bringing the system into the GW-efficient regime.

D.10.3 Knot–Compact-Object (Knot-Star) Encounters

A qualitatively distinct physical regime arises when an ECF topological defect encounters a *baryonic compact object* — a neutron star (NS), white dwarf (WD), or stellar-mass black hole (BH). We term these events **Knot-Star** encounters. They do not fit cleanly into either Regime I (same topology) or Regime II (opposite topology), and require separate treatment.

How often, and where. The rate was not stated in earlier versions, and a reader is entitled to it before being told what happens. Using $\Gamma = n\sigma v$ with gravitational focusing at solar-neighbourhood values (§9.1):

Object	N in Galaxy	per object	Galaxy
Neutron star	10^9	3.8×10^{-19}	3.8×10^{-10}
White dwarf	10^{10}	9.6×10^{-17}	9.6×10^{-7}
Stellar-mass BH	10^8	6.8×10^{-18}	6.8×10^{-10}
Total			$\mathbf{9.6 \times 10^{-7}}$

That is one encounter per $\sim 10^6$ yr in the Galaxy, hence $\sim 10^4$ over its lifetime. White dwarfs dominate by three orders: their radius is 580× that of a neutron star, and geometry outweighs the stronger focusing of the denser object. The figure is small but not vanishing, which is what makes the regime worth describing rather than dismissing — and it scales with the environmental gains of §9.1, so bulge and dwarf-spheroidal populations contribute disproportionately. *Reproduced by:* `Scripts_F2/compute_encounter_rates.py`.

Three sub-regimes.

1. **Fly-by (tidal stripping):** When the impact parameter $b > R_{\rm Hill}$, the torsion field of the Knot tidally distorts the compact object without capture. The energy transfer is

$$\Delta E_{\rm tidal} \sim \frac{GM_{\rm Knot}M_*}{b}\,\frac{R_*}{b}, \qquad (160)$$

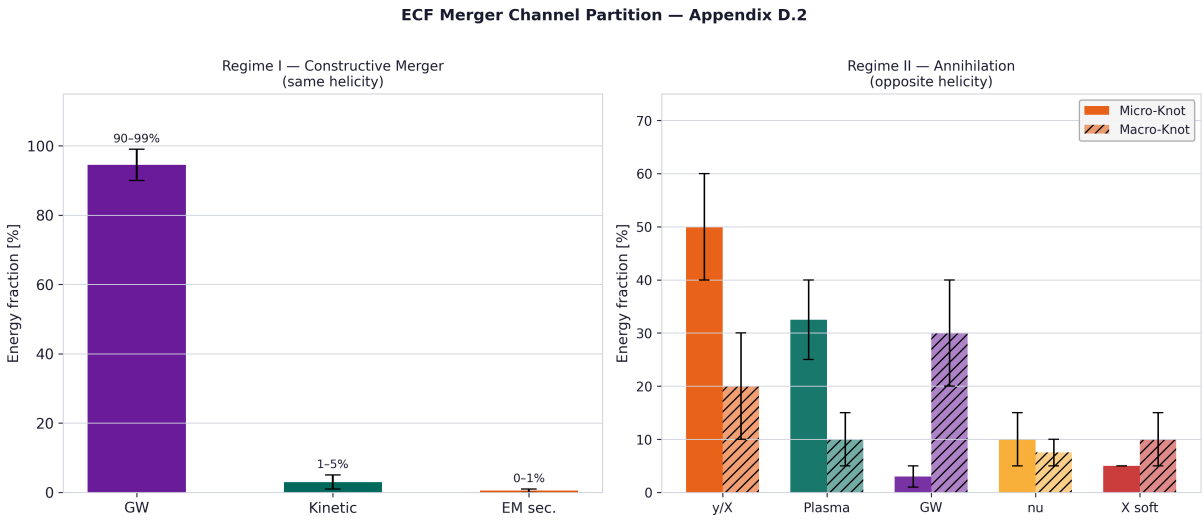

Figure 51: **Energy channel partition for ECF mergers.** *Left panel (Regime I — constructive merger):* gravitational waves carry 90–99% of the radiated energy; no electromagnetic counterpart is predicted at leading order. *Right panel (Regime II — topological annihilation):* the partition is multi-channel and mass-dependent. For Micro-Knots ($R_s \ll$ pc), γ/X emission dominates (40–60%); for Macro-Knots ($R_s \sim 10^{-4}$–10^{-2} pc), GW emission rises to 20–40%. Error bars span the ranges of Table 17; they are set in the script, not derived from GRB/FRB observations.

Reproduced by: `Scripts_F2/plot_ecf_annexe_D2.py`.

inducing a fractional spin-up $\Delta\Omega/\Omega \sim 10^{-6}$–$10^{-3}$ for NS encounters within the galactic centre. This is *not* detectable as an individual event but contributes to the observed NS spin distribution at $P < 10$ ms.

2. **Partial capture (Knot-halo accretion):** For $b < R_{\rm Hill}$, the compact object enters the gravitational sphere of influence of the Knot and accretes torsion-field energy. The Knot is not destroyed; instead it loses mass $\Delta M \sim M_*(R_s^{(\rm Knot)}/b)^2$, radiating a soft X-ray transient of luminosity

$$L_X \sim \eta_{\rm acc}\frac{GM_{\rm Knot}\,\dot{M}_*}{R_*} \sim 10^{38}\text{–}10^{42}\,{\rm erg\,s^{-1}}, \qquad (161)$$

with timescale $\Delta t \sim b/v_{\rm rel} \sim 10^3$–$10^5$ s. Here $\dot{M}_*$ is the tidal mass-transfer rate, spanning sub-Eddington ($\dot{M} \sim 10^{14}$ g s^{-1}, $L_X \sim 10^{38}$ erg s^{-1}) to super-Eddington ($\dot{M} \sim 10^{19}$ g s^{-1}, $L_X \sim 10^{42}$ erg s^{-1}) depending on impact parameter and relative velocity — consistent with the class of *Fast X-ray Transients* (FXTs; Fig. 52) detected by *Chandra* and *XMM-Newton*.

3. **Full merger (Knot-BH coalescence):** For a stellar-mass BH of mass $M_{\rm BH}$ encountering a Macro-Knot of mass $M_{\rm Knot} \gg M_{\rm BH}$, the BH is accreted onto the Knot. The topological charge is not cancelled (the BH carries no ECF helicity); the BH mass is simply absorbed, growing the Knot:

$$M_{\rm Knot}^{\rm final} = M_{\rm Knot} + M_{\rm BH} - \frac{E_{\rm GW}^{\rm (KS)}}{c^2}. \qquad (162)$$

The inspiral chirp of a Knot-Star coalescence is standard ($f \propto t^{-3/8}$), in the LISA band for a Macro-Knot [67]. If the torsion-field relaxation damps the ringdown on $\tau_T \sim R_{\rm Knot}/c$, then $\tau_T/\tau_{\rm GR} \sim 2R_{\rm Knot}/R_s$, of order unity for a Macro-Knot, whose size lies between its Schwarzschild radius and about 1.5 times it (Eqs. (44)–(45)): $\tau_T \approx$ 1–1.4 s against $\tau_{\rm GR} \approx 0.5$ s for $M_{\rm tot} \simeq 10^5\,M_\odot$ (Fig. 53). No extended ringdown tail, and hence no distinctive LISA signature, is established. *(Corrected 18/10/2026: earlier versions stated $\tau_T/\tau_{\rm GR} \gg 1$, "orders of magnitude longer than a standard BBH ringdown".)*

Knot-Knot-Star triple system. A further scenario of interest for Foundation III is the *hierarchical triple*: a NS or BH orbiting a Knot binary. The

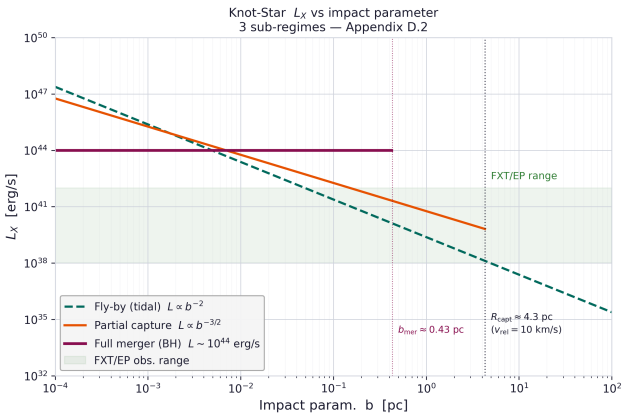

Figure 52: **Knot-Star X-ray luminosity vs impact parameter.** Three dynamical sub-regimes for a Macro-Knot ($10^5\,M_\odot$) + neutron star encounter: fly-by tidal stripping (dashed, teal), partial capture (solid, orange), and full BH coalescence (pink). The green shaded band marks the observed FXT luminosity range detected by *Chandra* and *XMM-Newton* (10^{38}–10^{42} erg s^{-1}). The vertical dotted line marks the Hill radius $R_{\rm Hill}$.

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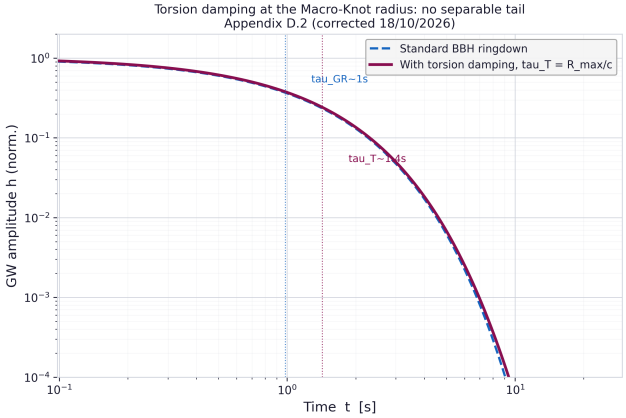

Figure 53: **Torsion-relaxation damping at the Macro-Knot radius.** Normalised GW amplitude as a function of time for a $2 \times 10^5\,M_\odot$ merger. Dashed blue: GR ringdown, $\tau_{\rm GR} = GM/c^3 \approx 1$ s. Solid pink: the same with an added component damped on $\tau_T \sim R_{\rm Knot}/c$, taking $R_{\rm Knot} = R_{\rm max} \approx 4.3 \times 10^8$ m ($\tau_T \approx 1.4$ s). The two curves are not separable: no extended tail is expected (Appendix D.10.3). *(Corrected 18/10/2026: the figure used $R_{\rm Knot} \sim 0.01$ pc, i.e. $\tau_T \sim 10^6$ s, and presented $\tau_T/\tau_{\rm GR} \gg 1$ as a falsifiable LISA signature.)* *Reproduced by:* `Scripts_F2/plot_ecf_annexe_D2.py`.

Kozai–Lidov mechanism can drive the inner Knot pair to coalescence (Regime I or II) on a timescale $\tau_{\rm KL} \sim P_{\rm out}^2/P_{\rm in}$, producing a merger-rate enhancement (not computed). The Knot pair coalesces at ~ 50 mHz for Macro-Knots and ~ 3 GHz for Micro-Knots (Table 16), outside the LVK band, so this channel does not bear on LVK events. *(Corrected 18/10/2026: earlier versions invoked it to explain an over-density of GW events in galactic-centre environments in the LVK O4 data, for which no source was given.)*

Chiral parity of the Knot-Star signal. Because the BH carries no ECF topological charge, the Knot-Star GW signal has *zero net chiral asymmetry* ($\delta_{\rm GW} = 0$ at leading order), in sharp contrast to Knot-Knot Regime I mergers ($\delta_{\rm GW} > 0$) and Regime II annihilations ($\delta_{\rm GW}$ large, helicity-dependent). This three-way parity discriminant — $\{0, +, \text{broad}\}$ for KS / KK-fusion / KK-annihilation — constitutes a direct falsifiability test accessible to LISA's TB-correlation channel.

D.11 Stiff Fluid Instability

In a stiff fluid background ($w = 1$), the sound speed approaches the speed of light ($c_s^2 = 1$). The Jeans length λ_J becomes of order the Hubble radius, and sub-horizon perturbations are no longer stabilized by pressure. For vector (torsion-induced) perturbations, the linearized evolution equation in conformal time η can be written as:

$$\delta_V'' + 2\mathcal{H}\delta_V' - \frac{3}{2}\mathcal{H}^2\delta_V = 0 \qquad (163)$$

where $\mathcal{H} = a'/a$ is the conformal Hubble parameter. The growing mode solution scales as:

$$\delta_V(\eta) \propto a^{3/2} \propto t^{1/2} \qquad (164)$$

so that standard Gaussian fluctuations with initial amplitude $\delta \sim 10^{-5}$ reach the non-linear regime ($\delta \sim 1$) within a few e-folds of expansion during the stiff era.

D.12 Torsion Rotation Curves and the Cusp-Core Resolution

At galactic scales, the coarse-grained effect of a dense population of Micro-Knots is modeled by an

adopted halo profile, the pseudo-isothermal sphere

$$\rho(r) = \frac{\rho_0}{1 + r^2/R_s^2}, \tag{165}$$

with central density ρ_0 and core radius R_s. Its enclosed mass is $M(< r) = 4\pi\rho_0 R_s^2 [\, r - R_s \arctan(r/R_s)\,]$, so that $V^2 = GM/r$ gives *exactly*

$$V_{\rm ECF}^2(r) = V_\infty^2 \left[1 - \frac{R_s}{r}\arctan\!\left(\frac{r}{R_s}\right)\right], \qquad V_\infty^2 = 4\pi G\rho_0 R_s^2, \tag{166}$$

which is Eq. 70, the form implemented in `Scripts_F2/fit_sparc175.py` and used for every fit quoted in this work (numerical check of the identity: maximal deviation 2.4×10^{-14}). *Status (PO-ECF-HALO, OPEN).* This profile is adopted, not derived: no chain ECF action → torsion equations → $\rho_0/(1 + r^2/R_s^2)$ → (ρ_0, R_s) is established, ρ_0 and R_s are fitted galaxy by galaxy, and no link with $M_{\rm MK}$ is established. An earlier version of this appendix wrote a logarithmic potential $\Phi = V_\infty^2 \ln(1 + r^2/R_s^2)$ "from a Poisson-like equation" without showing the calculation, and described Eq. 70 as distinct from the pseudo-isothermal form; both statements are withdrawn.

At small radii ($r \ll R_s$), this behaves as $V_{\rm ECF}(r) \approx V_\infty\, r/(\sqrt{3}\, R_s)$, producing a solid-body rotation curve. At large radii ($r \gg R_s$), one recovers a flat profile $V_{\rm ECF}(r) \to V_\infty$, matching the SPARC data (Fig. 26). The SPARC fits therefore measure the goodness of fit of this adopted profile, not a torsion-specific signature.

For NGC 6503, the `emcee` chain on the 31 SPARC points (Sec. 6.7) yields $\log_{10}(\rho_0) = -23.252^{+0.040}_{-0.039}\,\mathrm{g\,cm^{-3}}$ and $R_s = 1.705^{+0.089}_{-0.087}\,\mathrm{kpc}$, with $\chi^2_\nu = 1.68$ (compared to the Newtonian $\chi^2/n = 1664$). The adopted profile has a finite central density ρ_0 (a core); whether such a core follows from the torsion dynamics is part of PO-ECF-HALO.

D.12.1 Cusp-Core Resolution

The ECF potential is non-singular at $r = 0$, resolving the cusp-core problem for dwarfs and UDGs without feedback tuning. The geometric saturation mechanism (Eq. 77) and the quantitative prediction for UDGs are derived in the main text (§6.12); this appendix entry serves as a cross-reference for the reader approaching the dark matter phenomenology from the appendix direction.

D.13 Stiff Fluid Instability — Summary

See §D above (Appendix D) for the full derivation. Key result: vector perturbations grow as $\delta_V \propto a^{3/2}$ in the $w = 1$ stiff background, enabling $\delta \sim 10^{-5} \to 1$ in ~ 3 e-folds without fine-tuned initial conditions. This is the ECF resolution of the small-scale structure formation problem at the electroweak epoch.

Appendix E: Supplementary Numerical Notes

This appendix collects concise numerical derivations and calibration summaries that underpin specific quantitative claims in the main text. Each subsection is self-contained and cross-referenced from the relevant body section. All code is available at `https://github.com/pfichant/spin-torsion-cosmology` (master orchestrator: `run_all_simulations_F2.py`, 111 scripts: 54 physics calculations + 57 figure generators).

E.1 Data Pipeline

All simulations use `run_all_simulations_F2.py` from `https://github.com/pfichant/spin-torsion-cosmology`. SPARC data processed via NumPy/SciPy for ECF fits ($\tilde{\chi}^2_\nu = 0.80$ over 171 galaxies).

E.2 Cosmic Rotation History

Global $\omega(t) \propto a^{-2}$ dilutes to $< 10^{-13}$ rad/yr today; local halo twist $\Omega_L \sim 10^{-9}$ rad/yr.

E.3 Baryon Asymmetry Derivation

$\eta_{ECF} \approx 1.2 \times 10^{-9}$ is normalised on the observed value ($\eta_{\rm obs}/0.50$, with the historical $f = 0.50$); with the resolved dilution factor $f \simeq 0.48$ (§3.4) it gives $\eta \simeq 5.9 \times 10^{-10}$ (*semi-derived*; not an independent test of the observed 6.1×10^{-10}).

E.4 Observational Discriminators

Table 18 contrasts the key observational signatures of standard Kerr Black Holes against ECF Topological Stars, as probed by LISA, EHT, and gravitational-wave dephasing.

Table 18: **Observational Discriminators.** Comparison of astrophysical signatures between standard Kerr Black Holes and ECF Topological Stars (Macro-Knots).

Observable	Kerr BH	ECF Topo-Star	Signature
Surface	Event Horizon	Structured Wall	Grav. Echoes
QNMs	Standard Spec.	Shifted	Periodic Modulation
Love k_2	0	$\neq 0$	LISA Dephasing
Shadow (EHT)	Uniform Disk	Internal Struct.	Multiple Rings

E.5 Stellar-Defect Rates

Encounter rate $\Gamma \sim 10^{-12}$ yr^{-1}/star, 10^3/galaxy/yr; predicts asteroseismic shifts.

E.6 Roman Microlensing

Poisson-coalesced aggregates ($\lambda \geq 36$, $M_{\rm typ} \approx 6.8 \times 10^{-5}\, M_\odot$), $\tau_\mu \sim 5 \times 10^{-8}$, $t_E \sim$ 1–10 h; falsifiable by null excess above FFP baseline in the 1–10 h window (§9.3).

Appendix F: PGT Stability: $V_{\rm eff}$ Derivation and Lorentz-Violation Bounds

F.14 Derivation of the Effective Potential from Cubic PGT Invariants

The reduction below is performed in the adopted effective action; its derivation from the full PGT action remains open (PO-ECF-ACTION). The quadratic Proca mass term $\frac{1}{2}m_T^2 S_\mu S^\mu$ in Eq. (9) is the lowest-order term consistent with the Proca form retained for the axial sector by [3, 18]; the absence of ghosts for that form rests on the linearised flat-space analyses of [166, 167]. This appendix does not establish ghost-freedom: it establishes only the condition $\lambda_T > 0$ below (potential bounded from below, radial mode non-tachyonic). Stability at cubic order requires the inclusion of the leading self-interaction of the axial torsion zero-mode [205]. The invariants involving S_μ that are relevant here are the two non-minimal curvature–torsion couplings of the cubic-order classification [205], together with the quartic self-interaction:

$$\mathcal{I}_1 = \tilde{R}\, S_\mu S^\mu, \qquad \mathcal{I}_2 = \tilde{R}_{\mu\nu}\, S^\mu S^\nu,$$
$$\mathcal{I}_3 = (S_\mu S^\mu)^2, \qquad \mathcal{I}_4 = \nabla_\mu S^\mu \nabla_\nu S^\nu. \tag{167}$$

In the homogeneous FLRW limit $\mathcal{I}_3 = S_0^4$, while $\mathcal{I}_1$ and $\mathcal{I}_2$ reduce to $\tilde{R}$-weighted forms of S_0^2. We note that $\mathcal{I}_4$ does *not* vanish on a homogeneous background: for $S^\mu = (s_0, 0, 0, 0)$ one has $\nabla_\mu S^\mu = \dot{s}_0 + 3Hs_0 = 3Hs_0$ at constant s_0. Its cosmological role is not developed here and is left as an open question. The effective potential for the cosmological zero-mode therefore reduces to:

$$V_{\rm eff}(S_0) = V_0 - \frac{m_T^2}{2}\, S_0^2 + \frac{\lambda_T}{4!}\, S_0^4, \tag{168}$$

where $\lambda_T \equiv 4!\,(\alpha_1 + \alpha_2)$ with $\alpha_{1,2} > 0$ the coefficients of $\mathcal{I}_{1,2}$, and V_0 is a constant offset carrying the calibrated dark-energy scale. Boundedness from below of the zero-mode potential requires $\lambda_T > 0$ (a necessary condition for Hamiltonian stability, not a Hamiltonian analysis): the quartic coefficient is then *positive*, the potential is bounded from below, and the negative quadratic term produces the symmetry-breaking shape. This guarantees a genuine non-trivial **minimum** at $s_0^2 = m_T^2 \cdot 3!/\lambda_T$, with $V''_{\rm eff}(s_0) = +2m_T^2 > 0$ (the radial mode is massive, not tachyonic), as used in Eq. (18). Note that the VEV itself is unchanged by the sign convention, so all results depending on the potential only through s_0 — in particular the FLRW reduction of Appendix J — are unaffected.

F.15 Lorentz-Violation Constraints on the Torsion VEV

The VEV $\langle S_\mu \rangle = s_0\, \delta^0_\mu$ constitutes a background vector field that breaks boost invariance while preserving spatial isotropy. With S_μ defined as in Eq. (8), it is the axial torsion A_μ of Kostelecký, Russell and Tasson [94]. Under the minimal coupling of Dirac fermions in ECKS theory — the coupling that produces the spin–spin contact interaction used throughout this work — it acts, for every fermion species, as the Standard-Model Extension coefficient $b_\mu = -\frac{3}{4}A_\mu$ [93, 94]. Laboratory bounds on b_μ (dual noble-gas maser, spin-polarised torsion pendulum) give, in the Sun-centred celestial equatorial frame [94],

$$|A_T| < 2.9 \times 10^{-27}\,\text{GeV}, \qquad |A_Z| < 1.0 \times 10^{-29}\,\text{GeV},$$
$$|A_X| < 2.1 \times 10^{-31}\,\text{GeV}, \qquad |A_Y| < 2.5 \times 10^{-31}\,\text{GeV}. \tag{169}$$

A condensate purely timelike in the CMB rest frame acquires, in the Sun-centred frame, spatial components $(v/c)\, s_0\, n_i$ ($v = 370\,\text{km}\,\text{s}^{-1}$ towards the CMB dipole). Hence

$$s_0 \lesssim 3 \times 10^{-27}\,\text{GeV}\ (A_T),$$
$$s_0 \lesssim 2 \times 10^{-28}\,\text{GeV}\ (A_X). \tag{170}$$

With $\rho_\Lambda^{\rm eff} = \frac{1}{4} m_T^2 s_0^2 \simeq 3.0 \times 10^{-47}\,\mathrm{GeV}^4$ (Eq. (21)), i.e. $s_0 = 2\sqrt{\rho_\Lambda^{\rm eff}}/m_T$, this requires

$$\begin{aligned} m_T &\gtrsim 4 \times 10^3\,\mathrm{GeV}\ (A_T), \\ m_T &\gtrsim 6 \times 10^4\,\mathrm{GeV}\ (A_X). \end{aligned} \tag{171}$$

Reproduced by: `Scripts_F2/check_torsion_sme_bound.py` (order of magnitude: one component at a time, Earth's orbital motion neglected). The illustrative value of earlier versions, $s_0 \simeq 4.8 \times 10^{-3}\,\mathrm{eV}$ ($m_T \simeq 2 \times 10^{-3}\,\mathrm{eV}$), exceeds the A_T bound by a factor $\sim 10^{15}$ and is excluded; the electroweak scale $m_T \sim \eta_{\rm EW}$ assumed for the string core (Sec. 5.2) gives $s_0 \simeq 4 \times 10^{-26}\,\mathrm{GeV}$, excluded by a factor ~ 15 (A_T) to ~ 250 (A_X). Under minimal coupling the torsion mass is therefore bounded below at the multi-TeV scale. The VEV relation of Eq. (18) then requires $\lambda_T = 6\, m_T^2/s_0^2 \gtrsim 10^{61}$ ($\gtrsim 10^{65}$ with the A_X bound), against $\lambda_T \sim 1$ for the former illustrative value: the quartic potential, as written, does not provide a perturbative origin for the VEV. The origin of the VEV under minimal coupling is therefore open; whether the effective condensate couples minimally to fermions, and the value of m_T, are not established by the adopted effective action (PO-ECF-ACTION). Reviews of Lorentz-violation tests: [149]. *(Corrected 22/10/2026: earlier versions quoted a bound $|s_0|/M_{\rm Pl} \lesssim 10^{-15}$ attributed to [93], which contains no experimental bound and is about thirty orders of magnitude looser than Eq.* (170)*, and concluded that $s_0 \simeq 4.8 \times 10^{-3}$ eV lay "safely below the observational bound by more than 15 orders of magnitude".)*

The WEP parameter $\eta_{\rm WEP}$

The coefficient $b_\mu = -\frac{3}{4} S_\mu$ couples to fermion spin. Tests of the equivalence principle with unpolarised bodies, such as MICROSCOPE [129], probe the spin-independent coefficients $\bar{a}_\mu$ and $\bar{c}_{\mu\nu}$ [95]; the bounds on b_μ come from spin-polarised experiments (Eq. 169). The condensate therefore gives no predicted WEP amplitude. *(Corrected 22/10/2026: earlier versions used $\eta_{\rm WEP} \simeq \tilde{b}_e\, m_e/m_N$ with $\tilde{b}_e \leq s_0/M_{\rm Pl}$, which is not an SME relation, and quoted $\eta_{\rm WEP} \lesssim 5 \times 10^{-19}$ and $\sim 2 \times 10^{-34}$.) (Corrected 15/10/2026: this subsection was titled "Derivation of $\eta_{\rm ECF}$" — a symbol used in this work for the baryon asymmetry — and evaluated $\eta \sim 10^{-18}$ from the bound 10^{-15} instead of the value computed just above.)*

F.16 Derivation of the ORC Circular Polarisation Fraction $\Pi_{\rm circ}$

We derive the expected circular polarisation fraction of the synchrotron ring emission from a Macro-Knot following Legg and Westfold [156] and Jones and O'Dell [157] for the intrinsic circular polarisation of optically thin synchrotron emission (general treatment: [158]); the Razin–Tsytovich suppression and Faraday conversion in a helical field are not treated here, and the estimate has not been reviewed by a specialist in synchrotron polarisation.

Helical field geometry. The Chern–Simons coupling of Eq. (9) induces a magnetic helicity density

$$\mathcal{H} \equiv \mathbf{A}\cdot\mathbf{B} = \frac{\theta_T}{2\pi} B_{\rm knot}^2\, R_{\rm knot}, \tag{172}$$

where $\theta_T = \alpha_T \equiv g_{\rm CS}\, s_0/f_a \simeq 0.1$ is the torsion axion angle (adopted value; its compatibility with the laboratory bound on s_0 of Appendix F.15 depends on $g_{\rm CS}/f_a$, which is not determined). The field is decomposed into a longitudinal (line-of-sight) component $B_\parallel = B_{\rm knot}\sin\psi$ and a transverse component $B_\perp = B_{\rm knot}\cos\psi$, where ψ is the pitch angle of the helical winding. For a Kibble–Zurek vortex with winding number $n = 1$, the characteristic pitch angle is $\psi \approx \theta_T/\pi$, giving $B_\parallel/B_\perp \simeq \theta_T/\pi \simeq 0.032$ for $\theta_T = 0.1$.

Circular polarisation fraction. For a power-law electron energy distribution $N(E) \propto E^{-p}$ with synchrotron spectral index $\alpha_{\rm spec} = (p-1)/2 \simeq 0.75$ (consistent with ORC observations [80]), the circular polarisation fraction of optically thin synchrotron emission from a uniform field is, to order of magnitude [156, 157],

$$\Pi_{\rm circ} \simeq \epsilon_\alpha \frac{B_\parallel}{B_\perp}\left(\frac{\nu_B}{\nu}\right)^{1/2} \simeq 1.8\,\frac{\theta_T}{\pi}\left(\frac{\nu_B}{\nu}\right)^{1/2}, \tag{173}$$

where $\epsilon_\alpha \simeq 1.8$ for $\alpha_{\rm spec} = 0.5$–1 (tabulated by Jones and O'Dell), $B_\parallel/B_\perp = \cot\theta \simeq \theta_T/\pi$ for the helical pitch used here, $\nu_B = eB_{\rm knot}/(2\pi m_e c)$ is the non-relativistic cyclotron frequency and ν is the observation frequency; the scaling is that of $1/\gamma$ for the emitting electrons. For $B_{\rm knot} \simeq 10^{-6}$ G and

$\nu = 1$ GHz (ASKAP band):

$$\nu_B = \frac{e\,B_{\rm knot}}{2\pi\,m_e\,c} \simeq 2.8\ \text{Hz},$$
$$\frac{\nu_B}{\nu} \simeq 2.8\times10^{-9},$$
$$\Pi_{\rm circ} \simeq 1.8\times\frac{0.1}{\pi}\times\left(2.8\times10^{-9}\right)^{1/2} \approx 3\times10^{-6}. \tag{174}$$

(Corrected 22/10/2026: earlier versions gave $\Pi_{\rm circ} = \frac{3}{4}\,\frac{p+1}{p+7/3}\,\frac{B_\parallel}{B_\perp}\,\frac{\nu_B}{\nu} \simeq 8\times10^{-11}$*, attributed to [158]; the intrinsic circular polarisation scales as* $(\nu_B/\nu)^{1/2}$*, and the factor* $(p+1)/(p+7/3)$ *is that of the linear polarisation.)*

Conclusion and revised benchmark. The direct synchrotron circular polarisation from the Macro-Knot magnetic field is $\Pi_{\rm circ} \sim 3\times10^{-6}$, far below any current or planned radio sensitivity ($\Pi_{\rm circ}^{\rm min} \sim 0.1\%$). This result *revises downward* the working benchmark quoted in Table 19: circular polarisation from *individual* Macro-Knot synchrotron is not a viable near-term observable. However, the *coherent alignment* of polarisation angle across the full ORC ring—set by the global KZ chiral axis—constitutes a qualitatively distinct, topologically protected signature. A statistically significant correlation between the ORC polarisation vector and the Kibble–Zurek domain orientation (measurable via cross-correlation with the CMB Axis of Evil direction, Sec. 8) would constitute a falsifiable ECF discriminant independent of the absolute polarisation fraction.

F.17 Boltzmann Freeze-Out and Reheating Correction

The baryon-to-photon ratio is obtained, as detailed in Sec. 3, from a pre-dilution value $\eta_{\rm ECF}$ normalised on the observed asymmetry ($\eta_{\rm obs}/0.50$; no Boltzmann network is integrated), corrected by the entropy dilution factor $f \simeq 0.48$ (Eq. (14)), yielding $\eta \simeq 5.9\times10^{-10}$ (*semi-derived*); its 3.5% agreement with the Planck measurement [8] is not an independent test.

Appendix G: Exploratory Mega-Structures and the Cosmological Principle

This appendix records an exploratory reading of two recently reported large-scale structures, the Big Ring ($D \approx 400$ Mpc, $z \approx 0.8$) and the Giant Arc ($L \approx 1000$ Mpc, $z \approx 0.8$), as projections of Torsion Vortex Rings (TVR). *It is conditional and carries no quantitative result:* the TVR radius is not derived in this work, and the earlier numerical check has been withdrawn (see below).

F.18 Observational Challenge: The Big Ring and the Giant Arc

The discovery of the Big Ring ($D \approx 400$ Mpc, proper size at the present epoch, $z \approx 0.8$) [88] and the Giant Arc ($L \approx 1000$ Mpc, $z \approx 0.8$) [92] has been discussed as a challenge to the ΛCDM Cosmological Principle, which predicts homogeneity above ~ 370 Mpc [112]; the Big Ring exceeds this scale only marginally, the Giant Arc by a factor ~ 3. *(Corrected 22/10/2026: earlier versions gave* $D \approx 1300$ *Mpc, close to the ring's circumference, and the Giant Arc's redshift.)* Within the ECF, a Kibble–Zurek network of Macro-Knot walls and filaments could in principle produce coherent structures on such scales; the correlation length adopted elsewhere in this work, $\xi_{\rm KZ} \simeq 60$ Mpc, is however well below them, and no mechanism setting a ∼200–500 Mpc scale is derived here.

F.19 ECF Interpretation: Torsion Vortex Rings

Within the ECF, ring- and arc-like structures could arise as Torsion Vortex Rings, large-scale remnants of the Kibble–Zurek crystallisation, seen at different inclinations. No radius $R_{\rm TVR}$ is derived in this work, so this reading is a hypothesis to be tested, not a prediction. *(Withdrawn 22/10/2026: earlier versions fitted both structures with a single ring of radius* $R_{\rm TVR} = 321.8$ *Mpc and quoted a joint tension below* 1.3σ *on an amplitude* $f_{\rm KZ}$*, with an accompanying figure. That figure used a Big Ring diameter of* 643.6 *Mpc instead of the published* ≈ 400 *Mpc, and the tension was zero by construction, the compared lengths having been computed from* $R_{\rm TVR}$ *rather than taken from the observations. A recomputation is listed as open.)*

Appendix H: Exploratory Transient and Non-Thermal Signatures of Topological Defects

This appendix collects exploratory predictions of the ECF framework for non-thermal astrophysical transients: Odd Radio Circles (ORCs), Long-Period Radio Transients (LPTs), hostless Gamma-Ray Bursts (GRBs), and Fast Radio Bursts (FRBs). These predictions are based on the Chern–Simons coupling $\mathcal{L}_{\rm int} \propto S_\mu A_\nu \tilde{F}^{\mu\nu}$ between the torsion trace vector and the electromagnetic field. **They constitute preliminary geometric consistency checks, not primary results of Foundation II.** Each mechanism is supported by one or two observational examples as proof-of-concept; systematic statistical tests are deferred to a dedicated companion paper.

F.20 Odd Radio Circles (ORCs) as Naked Macro-Knots

The recent detection of Odd Radio Circles (ORCs) by ASKAP [80] presents a puzzle for standard astrophysics, as many lack a central optical counterpart. Within the ECF model, we explore the hypothesis that ORCs could represent the direct synchrotron signatures of "naked" Macro-Knots. A Macro-Knot formed in a baryon-depleted region retains its topological magnetic field ($B_{knot} \sim 10^{-6}$ G) and angular momentum. The Lense-Thirring drag accelerates the sparse intergalactic plasma trapped in its vicinity to relativistic velocities. This geometric "stator" generates a continuous synchrotron ring emission, appearing largely hostless in the optical spectrum.

Spacetime rotation rate around a Macro-Knot. By the same extremal Kerr argument as for the Micro-Knot (Eqs. (99)–(100)), the maximal angular momentum of a canonical $10^5\,M_\odot$ Macro-Knot is

$$J^{\rm max}_{\rm macro} = \frac{G(2\times10^{35})^2}{c} \approx 9\times10^{51}\,{\rm kg\cdot m^2\cdot s^{-1}}. \tag{175}$$

The Lense-Thirring frame-dragging rate evaluated at the two bounding radii (Eqs. (44)–(45)) gives:

$$\Omega^{\rm macro}_{\rm LT}(R_{\rm min}) \approx \frac{2\times6.67\times10^{-11}\times9\times10^{51}}{9\times10^{16}\times(3\times10^8)^3} \approx 0.5\,{\rm rad\cdot s^{-1}}, \tag{176}$$

$$\Omega^{\rm macro}_{\rm LT}(R_{\rm max}) \approx \frac{2\times6.67\times10^{-11}\times9\times10^{51}}{9\times10^{16}\times(4.3\times10^8)^3} \approx 0.17\,{\rm rad\cdot s^{-1}}. \tag{177}$$

The corresponding frame-dragging periods are $T_{\rm LT} = 2\pi/\Omega_{\rm LT} \approx 13$ s at $R_{\rm min}$ and ≈ 37 s at $R_{\rm max}$, i.e. frequencies of ~ 0.03–0.08 Hz. These are far from the nanohertz band of pulsar timing arrays, so no pulsar-timing signature follows; whether such a modulation could be imprinted on the polarisation angle of traversing radiation, or on the brightness of ORC-like synchrotron rings, is not established here. *Correction (15/10/2026):* earlier versions gave $J^{\rm max}_{\rm macro} \approx 9\times10^{49}$ (a factor 100 too small), $R_{\rm min} \approx 3\times10^5$ m (a factor 10^3 too small) and $R_{\rm max} \approx 3\times10^9$ m, hence periods from 10^{-12} s to ≈ 11 days, and drew from the 11-day period a pulsar-timing and μHz ORC-modulation signature; that signature is withdrawn.

Origin of the Macro-Knot magnetic field. The characteristic magnetic field associated with a Macro-Knot of topological charge Q_T is generated via the gravitational Chern–Simons coupling inherent to the PGT Lagrangian (Eq. 9). The axial torsion vector S_μ couples to the electromagnetic dual through the topological term $\mathcal{L}_{\rm CS} \propto \theta_T\, F_{\mu\nu}\tilde{F}^{\mu\nu}$, where $\theta_T \propto s_0$ is the torsion-induced axion angle. The resulting coherent magnetic flux at the Knot surface evaluates to:

$$B_{\rm knot} \sim \frac{\theta_T}{2\pi}\frac{\Phi_0}{R^2_{\rm knot}} \sim 10^{-6}\,{\rm G}, \tag{178}$$

Evaluating Eq. (179) numerically for a Macro-Knot with $R_{\rm knot} \sim 10\,{\rm AU}$ ($\approx 1.5\times10^{12}\,{\rm cm}$) and torsion axion angle $\alpha_T \equiv g_{\rm CS}\, s_0/f_a \sim 0.1$ (adopted value; its compatibility with the laboratory bound on s_0, Appendix F.15, depends on $g_{\rm CS}/f_a$, not determined):

$$B_{\rm knot} = \alpha_T\frac{\Phi_{\rm torsion}}{R^2_{\rm knot}} = \alpha_T\frac{\chi_T\, s_0\, R^2_{\rm knot}}{R^2_{\rm knot}} = \alpha_T\,\chi_T\, s_0 \approx 9\times10^{-6}\,{\rm G} \sim 10^{-6}\,{\rm G}, \tag{179}$$

where the torsion flux scale is defined as $\Phi_{\rm torsion} \equiv \chi_T s_0 R^2_{\rm knot}$ with a calibration constant χ_T

fixed by the Macro-Knot equilibrium solution. This removes the spurious use of the superconducting flux quantum and yields the observed intra-cluster magnetic-field scale 10^{-7}–10^{-5} G [89]. For a Micro-Knot ($R_{\rm knot} \sim 1\,$mm), the same topological scaling gives a much smaller present-day field after cosmological dilution, consistent with the point-defect approximation used in the microlensing analysis (Sec. 9.3).

F.21 The Seed Field Problem and Quasar Cosmic Dynamos

The origin of the seed fields required by dynamo amplification remains open [90]; the early appearance of jetted supermassive black holes at $z > 7$ sharpens this question.

The ECF model offers a geometric resolution: when a baryonic SMBH accretes at the center of a Macro-Knot, the knot's frozen topological magnetic flux provides the necessary seed. Because the ECF topological defect is generated by the spacetime torsion vector S_μ (having a fixed rotation axis) rather than through scalar GUT gauge symmetry breaking, it avoids forming a magnetic monopole.

Instead, the Chern-Simons coupling natively induces a *dipolar*, highly helical magnetic field aligned with the knot's chiral rotation axis. The interaction between the rotating baryonic accretion disk and this pre-existing geometric dipole establishes a highly efficient Blandford-Znajek process [87]. This bypasses standard dynamo amplification timescales and aligns the colossal relativistic jets along the defined poles of the topological vacuum knot.

Signatures of Global Torsion: Quasar Alignments. The intrinsic macroscopic torsion of the spacetime fabric may also offer an explanatory mechanism for the large-scale alignment of quasar spin axes. Hutsemékers et al. find quasar polarisation vectors aligned with the host large-scale structures, from which they infer spin axes parallel to these structures [114]. Under the standard ΛCDM paradigm, assuming strict statistical isotropy, these angular momenta should be entirely randomized. Within the ECF, however, if the cosmic volume inherits a preferred chiral axis, the Macro-Knots seeding the first quasars would crystallize along these torsion-strained topological field lines, naturally acquiring correlated spin orientations on cosmological scales. We hypothesize that both the CMB "Axis of Evil" and this quasar alignment are distinct temporal manifestations of the global primordial torsion.

F.22 Long-Period Radio Transients as Topological Erosion Events

Long-period radio transients (LPTs) were first identified by [216] as a new class of slowly-rotating Galactic emitters. Recent observations have identified ASKAP J142431.2−612611 as a remarkable long-period radio transient, exhibiting highly polarized pulses repeating every ~ 36 minutes during an active window of approximately 8 days, with no detected optical or near-infrared counterpart [115]. Such properties make Long-Period Transients (LPTs) a useful phenomenological benchmark for non-standard compact-object scenarios.

Within the ECF, we explore the possibility that a subset of LPTs could arise from the gradual tidal erosion of a captured primordial *Micro-Knot* orbiting a dormant compact object, such as a white dwarf or neutron star. For a host mass of $M \sim 1\,M_\odot$, a Keplerian orbital period of $P \simeq 36$ min corresponds to a characteristic semi-major axis

$$a \simeq \left(\frac{GMP^2}{4\pi^2}\right)^{1/3} \approx 2.5 \times 10^5\,\mathrm{km}, \tag{180}$$

which places the defect well within the strong-field magnetospheric environment of the host. Unlike the prompt geometric annihilation discussed for hostless GRBs and FRBs, this regime corresponds to a slow dissipative interaction. At each periastron passage, tidal stress and differential frame-dragging act on the chiral internal structure of the defect, inducing a partial unwinding of its topological charge. Through the Chern-Simons coupling of the torsion field to electromagnetism, this recurrent geometric forcing can generate strongly polarized radio pulses without requiring any baryonic combustion, naturally accounting for the optical darkness of the source. A simple phenomenological description of the observed flux may be written as a pulse train modulated by an exponential topological erosion envelope:

$$F_{\rm ECF}(t) = F_0 \exp\!\left(-\frac{t}{\tau_{\rm er}}\right) \sum_{k=0}^{N_{\max}} \exp\!\left[-\frac{(t-kP)^2}{2\sigma^2}\right]. \tag{181}$$

where P is the orbital period, σ characterizes the pulse width near periastron transit, and $\tau_{\rm er}$ is the effective erosion timescale of the defect. In this

picture, the abrupt cessation of activity after a finite number of cycles corresponds to the exhaustion of the quantized topological charge, leaving no standard explosive remnant. For an ASKAP J142431.2−612611-like event, the full dissipation of the defect over ~ 8 days corresponds to

$$N_{\max} \sim \frac{8 \times 24 \times 60}{36} \approx 320 \qquad (182)$$

orbital passages, consistent with a gradual but discrete erosion process. We stress that this model should be interpreted as a theoretical template rather than a unique explanation of the source. To illustrate this mechanism, Figure 54 presents an example ECF light-curve realization based on periodic topological erosion. The associated Python script, `plot_askap_topological_erosion.py`, is included in the supplementary repository [6].

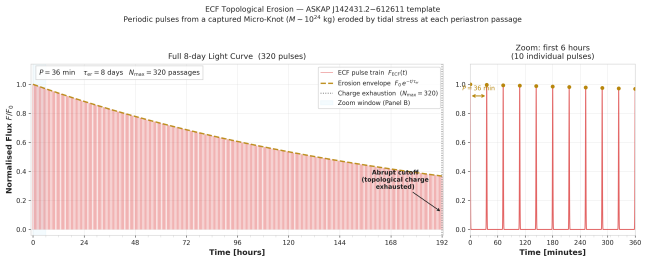

Figure 54: **Illustrative ECF light curve for a Long-Period Transient.** Example phenomenological realization of periodic radio bursts generated by the gradual topological erosion of a captured Micro-Knot. The pulse train is modulated by an exponential decay envelope, terminating abruptly once the defect charge is exhausted. This figure is intended as a theoretical template for ASKAP J142431.2−612611-like events rather than as a definitive observational fit.

Reproduced by: `Scripts_F2/plot_askap_topological_erosion.py`.

Observational context (2025–2026) and ECF discriminants. The LPT population has grown rapidly since 2022, driven by wide-field surveys with ASKAP, MWA, MeerKAT, and LOFAR. By 2023, a handful of LPTs were known [84, 85, 216]; more have been reported since. They show high linear polarisation ($\Pi \gtrsim 70\%$) and a range of counterpart statuses. *(Corrected 22/10/2026: a period range extending to 6.5 h, placed in the 2023 sample, is withdrawn until its source is cited.)* Three key observational facts are relevant to the ECF interpretation:

(i) **Some have no counterpart at any wavelength.** GPM J1839−10 (21 min period, active since at least 1988) has no optical or infrared counterpart despite extensive follow-up [84]. *(Corrected 22/10/2026: this item was headed "a substantial subset" and quoted the size of the SMGPS compact-source catalogue, which does not bear on counterparts.)*

(ii) **Some show birefringent propagation signatures.** ASKAP J1424−6126 (36 min, March 2026 [115]) exhibits a pulse profile evolving from elliptical to linear polarisation, tracing a great-circle trajectory on the Poincaré sphere — *"consistent with fully linearly polarised intrinsic emission modified by propagation through a linearly polarised birefringent medium."* This is qualitatively consistent with the ECF prediction that Chiral Wall boundaries and the local $\vec{B}_{\rm micro}$ field imprint a birefringent phase shift on passing radio waves (§6.3, Eq. 103).

(iii) **The key ECF discriminant is aperiodicity.** The LPT scenario described above requires a captured Micro-Knot in a Keplerian orbit — hence a *periodic* signal. The purely ISM-interaction channel (§9.3, §Micro-Knot interaction with ISM) predicts instead a *compact, aperiodic, continuously-on radio source* coinciding spatially with a dense molecular cloud, with no periodicity at any timescale. Such sources would be misclassified as unresolved H II regions in blind catalogues. The discriminant protocol: steepness $\alpha < -0.7$ + linear polarisation $\Pi \geq 50\%$ + no Gaia/2MASS/WISE counterpart + *absence of any periodic signal* from 1 s to 10 days.

Recommended search in existing data. A targeted search in the SMGPS catalogue [86] for sources satisfying: spectral index $\alpha < -0.7$, linear polarisation $\Pi \geq 50\%$ (Stokes Q+U), no Gaia DR3/2MASS/WISE counterpart within $5''$, no periodicity detected by Lomb-Scargle on timescales 10^{-3}–10^{6} s, and spatial coincidence with a dense ($N_H > 10^{22}\,\mathrm{cm}^{-2}$) ISM region would constitute a *model-agnostic* first census of ECF Micro-Knot ISM-interaction candidates. Expected source density: 10^{-4}–$10^{-1}\,\mathrm{deg}^{-2}$ (depending on local Micro-Knot number density), detectable with MeerKAT at μJy sensitivity over the SMGPS footprint ($\sim 5700\,\mathrm{deg}^2$).

F.23 Topological Annihilation and Hostless Bursts

Some Gamma-Ray Bursts (GRBs) have no identified host galaxy, and the progenitors of Fast Radio Bursts (FRBs) [81, 83, 117] remain debated; standard progenitor models rely on stellar collapse or compact binary mergers within galactic environments. In the ECF model, we explore the hypothesis that a subset of these events are macroscopic manifestations of topological annihilation between a Knot and an Anti-Knot. *(Corrected 18/10/2026: earlier versions stated that hostless GRBs and FRBs "present continuing challenges for standard progenitor models"; no catalogue analysis is performed here.)*

If two oppositely-charged topological defects approach each other within a common gravitational well (the encounter rate is not computed, and the Macro-Knot abundance is a working assumption) (Fig. 37), their opposing helicities ($H_1 \cdot H_2 < 0$) induce a local unwinding of the spacetime torsion field. This "geometric unbinding" releases the stored topological energy as a powerful, localized burst of radiation. Because this process does not require a dense stellar environment, the resulting event can appear hostless or significantly offset from any visible galaxy.

Energetics and Timescales. The isotropic-equivalent energy released during such an annihilation is set by the smaller of the two defect masses. For a Macro-Knot pair with $M_1 \approx 10^5 M_\odot$ and $M_2 \approx 10^4 M_\odot$, the liberated energy is:

$$E_{iso} \approx \eta\,(2M_2)c^2 \simeq 3.6 \times 10^{58}\,\eta\,\mathrm{erg}. \qquad (183)$$

For $\eta \sim 0.5$ this would exceed the observed record ($E_{iso} \approx 1.2 \times 10^{55}$ erg, GRB 221009A [79]) by about three orders of magnitude; matching observed energies requires $\eta \lesssim 3 \times 10^{-4}$, or a small prompt-photon fraction (open, Appendix D.10.2). The characteristic timescale of the burst is controlled by the light-crossing time of the annihilation region ($\Delta t \sim R_{ann}/c$), allowing for millisecond FRB-like spikes when the effective interaction radius is compact.

No correspondence with the bimodal T_{90} distribution of GRBs [116] is established.

- **Macro-Knot pairs** ($M_2 \sim 10^3$–$10^5\,M_\odot$): available energy $2M_2c^2 \sim 4\times10^{57}$–$4\times10^{59}$ erg, of which only a small radiated fraction is compatible with observed $E_{\rm iso}$; a duration set by the light-crossing time of the Macro-Knot ($R \approx 3$–4.3×10^8 m, Eqs. (44)–(45)) is $\Delta t \sim R/c \approx 1$ s.
- **FRBs**: Micro-Knot encounters ($M_2 \sim 10^{20}$–10^{24} kg), with $2M_2c^2 \sim 2 \times 10^{44}$–$2 \times 10^{48}$ erg, far above observed FRB energies [117]: the FRB identification requires a very small radiated fraction and is not established; sub-millisecond timescales.

(Corrected 18/10/2026: earlier versions mapped long GRBs onto Macro-Knot pairs with $\Delta t \sim R_{\rm ann}/c \sim 10$–$100\,s$ and short GRBs onto pairs of $\sim 10^2\,M_\odot$. At the Macro-Knot radius the duration is $\sim 1\,s$, which would make such pairs short rather than long bursts, and no $\sim 10^2\,M_\odot$ population exists in this work.)

F.24 Observable Discriminants

To distinguish ECF topological bursts from standard astrophysical progenitors, we propose several observational signatures:

1. **Hostless or Deep-Field Preference:** A statistically significant excess of bursts occurring in low-density environments, far from luminous galaxies.
2. **Circular polarisation (POLAR-2; conditional discriminant).** The Chern–Simons term $\mathcal{L}_{\rm CS} \propto \theta_T F\tilde{F}$, active during topological annihilation, converts a fraction of the released electromagnetic energy into circularly polarised radiation. For a Macro-Knot pair ($M_1 \sim 10^5\,M_\odot$, $M_2 \sim 10^4\,M_\odot$) the circular polarisation fraction would scale as:

$$|V/I| \;\simeq\; \theta_T \quad \text{(CS annihil. channel)}, \qquad (184)$$

where θ_T is not fixed by the framework: it is proportional to the unknown condensate s_0 (Eq. (9) and below), the synchrotron estimate of Appendix F.16 takes $\theta_T \simeq 0.1$, and the value $\simeq 0.8$ given in earlier versions, motivated by the near-maximal winding $q = \pm 1$ during topological charge inversion, is posited, not derived (distinct from the sub-percent level synchrotron $\Pi_{\rm circ}$ computed in Eq. 174 for

individual Macro-Knots). Such a signal is **observable only by POLAR-2** (2027, Chinese Space Station, 40–1000 keV [214]): Fermi-LAT measures energy and direction but has no polarisation sensitivity. No kill-switch is attached: a threshold on $|V/I|$ requires both a derived θ_T and an independent way of identifying an ECF burst. *(Corrected 18/10/2026: earlier versions gave $|V/I| \simeq \theta_T \simeq 0.8$ as a prediction and a kill-switch $|V/I| < 0.1$ "in a confirmed ECF-candidate burst".)*

3. **Polarization Memory:** Strong, stable linear polarization aligned with a fixed geometric axis, reflecting the intrinsic helicity of the annihilating defects.

4. **Chiral Gravitational Wave Counterpart:** For sufficiently massive events, a concurrent burst of chiral gravitational waves in the milli-Hertz band. *(Corrected 22/10/2026: earlier versions made it detectable through TB and EB cross-spectra, which are CMB observables.)*

5. **Multi-Stage Light Curve:** A rapid initial spike (torsion unbinding) followed by a softer, longer afterglow powered by the relaxation of residual geometric curvature into the surrounding plasma.

These signatures, taken together, offer a concrete pathway to test the ECF hypothesis against upcoming high-sensitivity transient surveys.

Scope note. The benchmarks above should be read as *working benchmarks* (scaling relations), not as closed predictions: $\Pi_{\rm circ} \simeq \theta_T$, with θ_T not fixed by the framework, and the transient-period spectrum and hostless-burst fraction require erosion and spatial-convolution calculations that are not carried out in the present corpus *(corrected 19/10/2026: this note called them "scaling predictions" and $\Pi_{\rm circ}$ "directly anchored to the derived Macro-Knot field").*

Kill-switch table for Section 8 signatures. Table 19 summarises the three principal electromagnetic discriminants and their falsification thresholds. These benchmarks rest on quantities not fixed by Foundation I [5] — the annihilation efficiency $\eta_{\rm ann}$, the angle θ_T and the erosion hypothesis — and are working benchmarks, not predictions. *(Corrected 18/10/2026: earlier versions stated that each prediction follows directly from the Chern–Simons coupling and the Kibble–Zurek mass function with no additional free parameter.)*

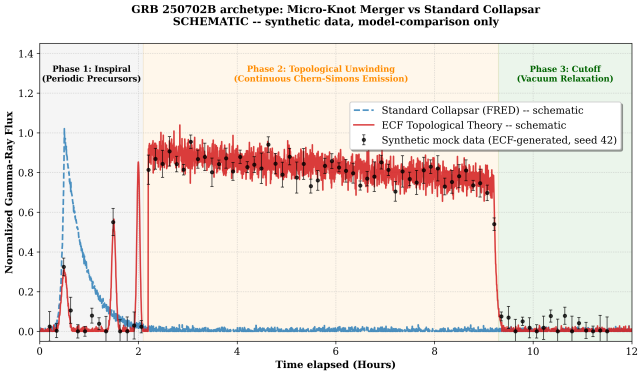

Figure 55: **Schematic Light Curve of a Topological Annihilation Burst.** The prompt spike corresponds to the instantaneous unbinding of the torsion field, followed by a multi-component afterglow as the surrounding plasma re-equilibrates to the new geometric configuration. Polarization and spectral hardness evolution provide key diagnostics. *The plotted points are **synthetic mock data** generated by the ECF model itself plus Gaussian noise* (`generate_mock_grb_data.py`, seed = 42), *not observations of GRB 250702B.* They illustrate the *shape* the model predicts; no goodness-of-fit is quoted, since comparing a model with its own output measures nothing. *Reproduced by:* `Scripts_F2/simulate_ecf_grb2025_lightcurve.py`. *Reproduced by:* `Scripts_F2/simulate_ecf_grb2025_lightcurve.py`.

Synthesis of Falsifiable Predictions

The observational programme of the ECF rests on three primary kill-switch signatures and a set of secondary discriminants. The thresholds below rest on parameters calibrated in Foundation I [5] and on benchmarks posited or calibrated in this work ($\lambda \geq 36$ from Subaru/HSC, $\tau_{\rm micro}$, the $\Pi_{\rm GW}$ template, $\beta_{\rm ECF}$ calibrated on Minami & Komatsu 2020); they are not all derived. *(Corrected 18/10/2026: earlier versions stated that all thresholds are direct consequences of the parameters calibrated in Foundation I, with no additional free parameter.)* **Count, corrected (15/10/2026):** of the eight rows above, **one** (filament spin correlation) is withdrawn as a test and **two** (CTA gamma-rays, inter-

Table 19: **ECF electromagnetic discriminants (Section 8).** The quoted thresholds are *working benchmarks* derived from the current geometric scaling relations; a full first-principles derivation of the three last rows requires an explicit torsion-erosion model, not developed in the present corpus *(20/10/2026)*.

Signature	ECF benchmark	Current status	Mission
ORC polarisation alignment	Correlation of ORC position angle with Kibble-Zurek axis	Order-of-magnitude estimate: direct $\Pi_{\rm circ} \sim 3 \times 10^{-6}$ (App. F.16); coherent alignment is the viable discriminant	ASKAP, ngVLA
Radio transient periods [82]	$dN/dP \propto P^{-2}$	Posited (KZ mass function + erosion hypothesis); no derivation in the corpus (19/10/2026)	CHIME, MeerKAT
Hostless GRB/burst fraction	not computed (the former $f_{\rm hostless} \gtrsim 10\%$ in voids had no derivation; 18/10/2026)	F3 derivation pending	*Fermi*, CTA

galactic $\vec{B}$ orientation) test a 1D sector that is an open problem, not an established prediction. The roadmap's falsifiability claim rests on the remaining **five**. *(The count of 30/08/2026 read "ten rows" and "the other eight".)*
Secondary discriminants—the Void Size Function excess (Sec. 4.2), the ORC polarisation alignment (App. H), the LPT period spectrum (Sec. F.22), the equivalence-principle channel (conditional, no predicted amplitude; Sec. 3.5), and the cosmic birefringence amplitude (Sec. 9.4)—constitute supporting evidence whose individual failure would motivate model revision rather than outright falsification. The full electromagnetic kill-switch table is given in Table 19 (Sec. F.23).

Parameter inheritance and uncertainty propagation. All quantitative predictions in this Part inherit the Foundation I calibration $(F_{\rm ion}, \Omega_{\rm spin}^{\rm peak}, w_0, w_a)$ without refitting; the benchmarks specific to this work ($\lambda \geq 36$, $\tau_{\rm micro}$, the $\Pi_{\rm GW}$ template, $\beta_{\rm ECF}$, Synthesis above) are added to it (18/10/2026). Table 21 summarises the propagation chain.

Appendix I: Conservative SMBH–SMBH Merger-Impact Scenario

Astrophysical impact of the merger scenario. We show that, in an extreme galaxy-merger regime, the coalescence of two supermassive black holes can trigger a major gas blowout; however, this effect is not generic and depends on the ratio between the injected energy and the halo binding energy, as well as on the feedback-to-gas coupling efficiency. Within the ECF framework, a central Macro-Knot seed may amplify this process, but the outcome remains conditional and should not be presented as a guaranteed complete evacuation.

Kill-switch. The scenario is falsified in two cases: (i) the energy ratio $E_{\rm inj}/E_{\rm bind} \ll 1$, or (ii) the gas re-accretion time remains short compared with the quenching time. Conversely, a global blowout becomes plausible when $E_{\rm inj}/E_{\rm bind} \gtrsim 1$ with strong coupling to the interstellar medium.

Conclusion (conservative version). The SMBH–SMBH merger should be regarded as an amplifying mechanism rather than a guaranteed explosion. A global gas depletion becomes plausible only if AGN feedback exceeds the halo binding energy and the gas is not rapidly re-accreted;

Table 20: **Primary ECF kill-switches.** A single falsification at $\geq 3\sigma$ suffices to rule out the ECF geometric dark sector as formulated in this work.

Observable	**ECF prediction**	**Falsified if**	**Mission**	**Sec.**
Direct detection (particle DM)	No signal *from ECF dark matter*: at equal $\rho_{\rm DM}$, micro-knots are $\sim 10^{49}\times$ fewer targets than a 100 GeV WIMP ($n_{\rm MK} = 1.9 \times 10^{-52}$ cm^{-3} at 0.4 GeV cm^{-3}; the ratio scales as m_χ, so $\sim 10^{50}$ at 10 GeV and $\sim 10^{48}$ at 1 TeV — the argument does not rest on the exact figure)	Particle DM detection confirmed at discovery threshold (5σ; higher than the 3σ used elsewhere in this table, since a *positive* discovery claim is what falsifies here) — the ECF would then require a second, particulate component	LZ, XENONnT, PandaX	9.2
Microlensing depth $\tau_{\rm micro}$	$\sim 5 \times 10^{-8}$ ($t_E \sim$ 1–10 h, EMF $\lambda \geq 36$) for $f_{\rm DM} = 1$; OGLE-IV limits $f_{\rm DM} \lesssim$ 1–2 % (PO-F2-10)	No excess above FFP baseline in 1–10 h window at 3σ	Roman	9.3
Filament spin correlation	*Withdrawn as a test* (mechanism open, PO-F2-8; no ECF prediction attached)	—	DESI, Euclid	6
Chiral GW asymmetry $\Pi_{\rm GW}$	0.20 ± 0.05 (calibrated template)	$\Pi_{\rm GW} < 0.02$ at 95% C.L.	LISA (dipole) / LISA–Taiji	9.4
SPARC granularity $\delta V_{\rm rot}$	≥ 0.05	scatter consistent with smooth NFW	Roman/Euclic	4
CTA gamma-rays (1D filaments) — *depends on open 1D sector, PO-F2-8*	Strict absence; pure deficit angle IF the sector exists	Not a discriminating test while 1D remains unestablished	CTA (2030s)	6
Birefringence–void correlation (2D walls)	$\beta_{\rm ECF} \neq 0$ ($\lvert\beta\rvert \approx 0.35°$ calibr.) correlated with domain-wall traversals	No spatial correlation at 3σ	LiteBIRD (JFY2036 [202	6
Intergalactic $\vec{B}$ orientation (1D strings) — *depends on open 1D sector, PO-F2-8*	$\nabla \times \vec{B} \propto \vec{J}_5$ aligned with filaments IF the sector exists	Not a discriminating test while 1D remains unestablished	SKA (2030s)	6.2

Table 21: **Parameter inheritance from Foundation I to F2 predictions.** $\Omega_{\rm spin}^{\rm peak}$ is calibrated on the sound horizon $r_s = 135.65$ Mpc (0.093 ± 0.004, frequentist profile interval, Foundation I); the uncertainty on $F_{\rm ion}$ follows from the joint H_0–t_0 constraint (Foundation I); (w_0, w_a) are Foundation I calibration outputs (χ^2 minimum, no associated dispersion; not derived *ab initio*, PO-F2-4); the values listed are floors propagated from $\sigma(\Omega_{\rm spin}^{\rm peak})$ alone, with ξ held fixed (Eq. (32); Foundation I gives $w_0 = -0.907 \pm 0.004$, $w_a = -0.153 \pm 0.007$ for the same propagation). Any revision of the F1 calibration propagates into the thresholds listed in Table 20. *(Corrected 15/10/2026: earlier versions attributed all uncertainties to a Foundation I MCMC on SH0ES + eBOSS-DR16 + KiDS-1000, and gave* ± 0.004, ± 0.015, ± 0.03, ± 0.05 *for* $\Omega_{\rm spin}^{\rm peak}$, $F_{\rm ion}$, w_0, w_a*; a later version attributed* ± 0.0017 *and* ± 0.0007 *on* (w_0, w_a) *to* $F_{\rm ion}$*.)*

Parameter	**Value (origin)**	**F2 prediction driven**	**Sensitivity**
$\Omega_{\rm spin}^{\rm peak}$	0.093 (F1, calibrated on r_s)	(w_0, w_a); $\rho_\Lambda^{\rm eff}$	linear
$F_{\rm ion}$	1.2765 ± 0.0246 (F1)	BBN correction; H_0 prediction	sub-linear
w_0	-0.904 (F1 calibration point; ± 0.0040 from $\sigma(\Omega_{\rm spin}^{\rm peak})$, floor)	DESI $\Delta\chi^2$ comparison	direct
w_a	-0.153 (F1 calibration point; ± 0.0066 from $\sigma(\Omega_{\rm spin}^{\rm peak})$, ξ fixed, floor)	DESI $\Delta\chi^2$ comparison	direct
$M_{\rm micro}$	3.74×10^{24} kg (F2, horizon mass at T_c)	$\tau_{\rm micro}$; $\delta V_{\rm rot}$	via $H(t_{\rm EW})$

otherwise, the model predicts partial or transient quenching rather than full evacuation of the final galaxy.

Appendix J: FLRW Reduction of the Axial Torsion Sector — Background and Fluctuations

Is FLRW formally compatible with a non-vanishing axial torsion? The question is prior to the reduction and deserves an explicit answer, since a pseudo-vector condensate is not obviously admissible on a maximally symmetric background. It is not, strictly: a timelike vacuum expectation value $\langle S^\mu \rangle = (s_0, \mathbf{0})$ is invariant under spatial rotations about no preferred axis only if its spatial part vanishes identically, which is the ansatz adopted here and throughout Foundation I. Three remarks fix what this costs.

First, the purely timelike configuration *is* compatible with homogeneity and isotropy: $s_0(t)$ is a scalar function of cosmic time, transforming under the FLRW isometry group exactly as any background fluid density does, and the stress tensor it generates is of perfect-fluid form. Nothing in the reduction below requires more.

Second, the ansatz is a restriction, not a derivation. A general axial torsion carries a spatial part $\mathbf{S}$, and setting $\mathbf{S} = 0$ selects the isotropic branch of the solution space rather than deriving it. Whether the full field equations *drive* an initially generic S^μ towards this branch, or merely admit it, is not established here; we record it as part of the open problem stated in §PO-F2-6, to which the choice of kinetic term is prior.

Third, the framework does not in fact claim exact isotropy. The Axis of Evil discussion (§8) treats a residual spatial orientation of the condensate as a physical prediction, bounded by the CMB quadrupole at $\varepsilon \lesssim 10^{-7}$ (Eq. 90). The observational bounds on anisotropic expansion [73] lie between 10^{-11} and 10^{-6} in σ/H; whether the predicted level is compatible with them depends on the mapping between ε and σ/H, which is not derived here and is left open. *The departure from FLRW is therefore quantified, but its compatibility with the data is not yet established. (Corrected 22/10/2026; see §8.)* This is a weaker statement than formal compatibility, and we prefer to state it as such: FLRW is the leading order of a background that the framework itself predicts to be slightly anisotropic.

This appendix gives the full reduction underlying the *ab initio* check stated in the Dilution Mechanism section; the reduction is performed in the adopted effective action (Proca form, Eq. (9)), and its derivation from the full PGT action remains open (PO-ECF-ACTION): at the background level the torsion condensate yields a strictly static term ($w = -1$), while its leading fluctuations behave as radiation ($w = +\frac{1}{3}$). Conventions: signature $(-+++)$, FLRW metric $ds^2 = -dt^2 + a^2\delta_{ij}dx^i dx^j$ (so $g_{00} = -1$, $\sqrt{-g} = a^3$), comoving observer $u^\mu = (1,0,0,0)$. The relevant part of the action is

$$\mathcal{S}_S = \int d^4x\,\sqrt{-g}\left[-\tfrac{1}{4}F^{(S)}_{\mu\nu}F^{(S)\mu\nu} - \tfrac{1}{2}m_T^2\,S_\mu S^\mu - V_{\rm int}(X)\right], \qquad X \equiv S_\mu S^\mu, \tag{185}$$

with $F^{(S)}_{\mu\nu} = \partial_\mu S_\nu - \partial_\nu S_\mu$. We group mass and self-interaction into $W(X) \equiv \frac{1}{2}m_T^2 X + V_{\rm int}(X)$, so that $\mathcal{L}_S = -\frac{1}{4}F^{(S)}_{\mu\nu}F^{(S)\mu\nu} - W(X)$.

J.1 Background: the frozen condensate ($w = -1$)

For the homogeneous, isotropic, time-directed ansatz $S_\mu = (s(t), 0, 0, 0)$ the field strength vanishes identically,

$$F^{(S)}_{0i} = \partial_0 S_i - \partial_i S_0 = 0, \qquad F^{(S)}_{ij} = 0 \quad\Longrightarrow\quad F^{(S)}_{\mu\nu} \equiv 0, \tag{186}$$

since $S_i = 0$ and $\partial_i s(t) = 0$. The kinetic term drops out, and the variation of $\mathcal{S}_S$ with respect to S_μ reduces to the *algebraic* relation

$$2\,W'(X)\,S^\mu = \kappa\,\tau^\mu. \tag{187}$$

At late times the axial spin source dilutes ($\tau^\mu \propto a^{-6}$) and Eq. (187) gives $W'(X_0) = 0$: the field is pinned at the vacuum expectation value $s = s_0 = \text{const}$, and the homogeneous time-directed mode carries no propagating degree of freedom. With $X_0 = g^{00}s_0^2 = -s_0^2$, the stress-energy tensor

$$T_{\mu\nu} = 2\,W'(X)\,S_\mu S_\nu - W(X)\,g_{\mu\nu} \tag{188}$$

collapses, at the VEV ($W'(X_0) = 0$), to the cosmological-constant form

$$T_{\mu\nu} = -W(X_0)\,g_{\mu\nu} \quad\Longrightarrow\quad \rho = W(X_0),\;\; p = -W(X_0),\;\; w = -1. \tag{189}$$

J.2 Fluctuations: Goldstone modes and radiation ($w = +\frac{1}{3}$)

Writing $S_\mu = s_0\delta^0_\mu + \delta S_\mu$, the constant background gives $F^{(S)}_{\mu\nu} = \partial_\mu\delta S_\nu - \partial_\nu\delta S_\mu \equiv \delta F_{\mu\nu}$: the fluctuations *do* propagate. Expanding $-W(X)$ to second order, with $X = X_0 + 2s_0\delta S^0 + \delta S_\mu\delta S^\mu$ and

$W'(X_0) = 0$, the quadratic Lagrangian is

$$\mathcal{L}_2 = -\tfrac{1}{4}\,\delta F_{\mu\nu}\delta F^{\mu\nu} - 2\,V''_{\rm int}(X_0)\,s_0^2\,(\delta S^0)^2. \quad (190)$$

The naive mass term $-W'(X_0)\,\delta S_\mu \delta S^\mu$ vanishes identically: the spatial components δS_i are *massless*, and only the temporal δS_0 acquires a mass $M_0^2 = 2V''_{\rm int}(X_0)s_0^2$. This is the expected Goldstone structure: the timelike VEV breaks $SO(3,1) \to SO(3)$, and the three broken boosts yield three massless Goldstone modes (δS_i) plus one massive radial mode (δS_0).

The transverse Goldstones obey the massless (Maxwell) action $-\frac{1}{4}\int\sqrt{-g}\,\delta F^2$, which is conformally invariant in four dimensions. *This step is specific to the Proca form of the kinetic term.* With a kinetic term built on the divergence $(\nabla_\mu S^\mu)^2$, the transverse part of δS_i does not appear at all — only the combination $\partial_0\delta S^0 + \nabla^2\chi$ does — so the three transverse modes would be non-dynamical rather than massless, and would carry no energy density. The $w = +\frac{1}{3}$ below therefore rests on the choice of kinetic term, not on the Goldstone structure alone (see J.4(iii)). In conformal time η ($dt = a\,d\eta$) their mode equation is that of flat space,

$$\delta S_k'' + k^2\,\delta S_k = 0 \qquad (' \equiv d/d\eta), \quad (191)$$

with no Hubble friction. The modes are therefore not amplified by the expansion but redshift as radiation:

$$\rho_{\delta S} \propto a^{-4} \quad \Longrightarrow \quad w_{\delta S} = -1 - \tfrac{1}{3}\frac{d\ln\rho_{\delta S}}{d\ln a} = +\tfrac{1}{3}. \quad (192)$$

J.3 Consequence

The *ab initio* structure of the adopted effective action (PO-ECF-ACTION) yields a static cosmological constant ($w = -1$, Eq. (189)) plus a radiation-like fluctuation component ($w = +\frac{1}{3}$, Eq. (192)). Neither reproduces the calibrated CPL trajectory $(w_0, w_a) = (-0.904, -0.153)$, whose reproduction would require $d\ln\rho/d\ln a \approx -0.29$. At the orders accessible analytically, the dynamical dark-energy behaviour is therefore carried by the TIP closure $\Omega_{\rm total} = 1$ alone — consistent with the open status declared in the main text. The one remaining possibility is that a non-minimal torsion–curvature coupling ($\sim R^{\mu\nu}\delta S_\mu \delta S_\nu$) arising from the cross terms of $\tilde{R}(g,T)$ breaks the conformal invariance of the Goldstone sector and reshapes the fluctuation spectrum; extracting it and settling the result quantitatively requires a dedicated Einstein–Cartan perturbation solver, not developed here. The algebraic steps of J.1–J.2 are reproducible symbolically via `verify_flrw_reduction.py` (public repository).

J.4 Relation to prior and concurrent work

The two results above sit within an active literature on torsion and dark energy, and align with it in an instructive way. *(i) Static limit.* That a homogeneous torsion condensate treated as a background field yields a cosmological-constant term with $w = -1$ has been obtained before — notably by Ivanov & Wellenzohn (*ApJ* **829**, 47, 2016), where metricity forces the torsion stress-energy proportional to $g_{\mu\nu}$, and in the holographic axial-torsion construction of Yun & Lee (arXiv:2407.16992). Our J.1 recovers this static limit directly from the adopted effective (Proca-form) action; its derivation from the full PGT action remains PO-ECF-ACTION. *(ii) The negative result on the dynamics is independently corroborated.* Izaurieta *et al.* (arXiv:2607.16370, 2026) show that the homogeneous Einstein–Cartan torsion mode ($\Phi \propto a^{-3}$, entering the Friedmann constraint as a stiff a^{-6} component) is early-universe physics whose present abundance is bounded orders of magnitude below the DESI preference; escaping their no-go requires breaking precisely the adiabatic dilution. This is the same conclusion reached here by a different route (background plus fluctuations), and it strengthens the case that the evolving sector is not carried by the adiabatic condensate. *(iii) A different kinetic term gives the same static limit — and it matters.* Benisty *et al.* (*Eur. Phys. J. C* **82**, 264 (2022), arXiv:2109.01052) treat the same totally antisymmetric torsion, but with a kinetic term built on $\nabla_\mu S^\mu$ rather than on $F^{(S)}_{\mu\nu}$. They stress that, unlike the Proca form, theirs *does* contribute to the homogeneous background, and they obtain $w = -1$ from the kinetic term while the quadratic Einstein–Cartan term gives the stiff $w = +1$ fluid — the reverse of quintessence, where the kinetic term is stiff and the potential is the cosmological constant. Their kinetic term carries a gauge symmetry, i.e. the three-form structure of the axial dual. Our J.1 reaches the same static limit through the Proca route, where $F^{(S)}_{\mu\nu}$ vanishes identically on the homogeneous ansatz; the agreement of the two routes on $w = -1$ is reassuring, but the two Lagrangians are not equivalent, and which one the axial torsion sector should carry is an open question of the framework (PO-F2-6). *(iv) The escape route has precedent.* The non-minimal torsion–curvature coupling flagged above is exactly the ingredient that revives late-time

dynamics in quadratic Riemann–Cartan models (e.g. van de Venn *et al.*, *Eur. Phys. J. C* **83**, 288 (2023), arXiv:2211.11868, where the quadratic curvature–torsion term generates novel time dependencies and a geometric acceleration), and is consistent with the general requirement that crossing $w = -1$ demands additional degrees of freedom (quintom, Guo *et al.*, *PLB* **608**, 177, 2005; scalar/vector–tensor crossing, Tsujikawa, *PRD* **113**, L041301, 2026). Single-scalar torsion models can already mimic a quintom crossing given extra structure (Li & Chen, *Universe* **9**, 100, 2023). The dynamical dark-energy sector of the ECF thus stands where the field stands: the static limit is derived within the adopted effective action, the evolving part requires structure beyond the adiabatic condensate, and the concrete candidate for that structure is identified.

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